

1 MATHEMATICAL MODELS IN MECHANICS

1.1

Learning objectives

After completing this chapter you should be able to:

- Understand how the concept of a mathematical model applies to mechanics → pages 2–3
- Understand and be able to apply some of the common assumptions used in mechanical models → pages 4–5
- Know SI units for quantities and derived quantities used in mechanics → pages 6–8

Prior knowledge check

Give your answers correct to 3 significant figures (s.f.) where appropriate.

1 Solve these equations:

a $5x^2 - 21x + 4 = 0$

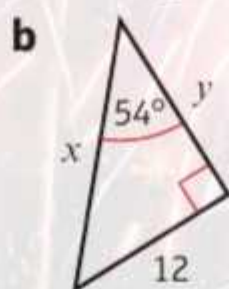
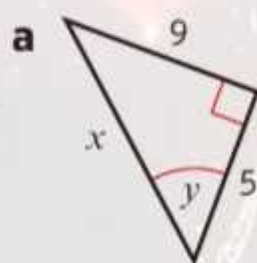
b $6x^2 + 5x = 21$

c $3x^2 - 5x - 4 = 0$

d $8x^2 - 18 = 0$

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2 Find the value of x and y in these right-angled triangles:



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3 Convert:

a 30 km h^{-1} to cm s^{-1}

b 5 g cm^{-3} to kg m^{-3}

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4 Write in standard form:

a 7 650 000

b 0.003 806

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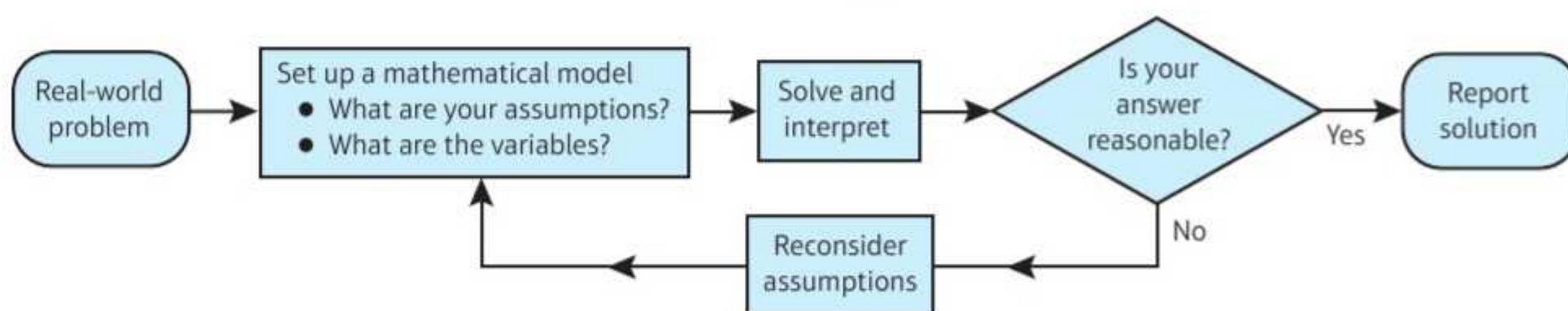
Mathematical models can be used to find solutions to **real-world problems** in many everyday situations. If you model a firework as a particle you can ignore the effects of wind and air resistance.

1.1 Constructing a model

Mechanics deals with **motion** and the action of **forces** on objects. Mathematical **models** can be constructed to simulate real-life situations (i.e. using models to create conditions that exist in real life, in order to study those conditions). However, in many cases it is necessary to simplify a problem by making one or more **assumptions**. This allows you to describe the problem using equations or graphs in order to solve it.

The solution to a mathematical model needs to be interpreted in the context of the original problem. It is possible that your model may need to be refined (improved with small changes) and your assumptions reconsidered.

This flow chart summarises the mathematical modelling process:



Example 1

1

SKILLS PROBLEM-SOLVING

The motion of a basketball as it leaves a player's hand and passes through the net can be modelled using the equation $h = 2 + 1.1x - 0.1x^2$, where h m is the height of the basketball above the ground and x m is the horizontal distance travelled.

- Find the height of the basketball:
 - when it is released
 - at a horizontal distance of 0.5 m.
- Use the model to predict the height of the basketball when it is at a horizontal distance of 15 m from the player.
- Comment on the **validity** of this prediction.

a i $x = 0$: $h = 2 + 0 - 0$
Height = 2 m

ii $x = 0.5$: $h = 2 + 1.1 \times 0.5 - 0.1 \times (0.5)^2$
Height = 2.525 m

b $x = 15$: $h = 2 + 1.1 \times 15 - 0.1 \times (15)^2$
Height = -4 m

c Height cannot be **negative** so the model is not valid when $x = 15$ m.

When the basketball is released at the start of the motion $x = 0$. Substitute $x = 0$ into the equation for h .

Substitute $x = 0.5$ into the equation for h .

Substitute $x = 15$ into the equation for h .

h represents the height of the basketball above the ground, so it is only valid if $h \geq 0$.

Exercise 1A**SKILLS** **PROBLEM-SOLVING**

- 1 The motion of a golf ball after it is struck by a golfer can be modelled using the equation $h = 0.36x - 0.003x^2$, where h m is the height of the golf ball above the ground and x m is the horizontal distance travelled.
 - a Find the height of the golf ball when it is:
 - i struck
 - ii at a horizontal distance of 100 m.
 - b Use the model to predict the height of the golf ball when it is at a horizontal distance of 200 m from the golfer.
 - c Comment on the validity of this prediction.
- 2 A stone is thrown into the sea from the top of a cliff. The height of the stone above sea level, h m at time t s after it is thrown can be modelled by the equation $h = -5t^2 + 15t + 90$.
 - a Write down the height of the cliff above sea level.
 - b Find the height of the stone:
 - i when $t = 3$
 - ii when $t = 5$.
 - c Use the model to predict the height of the stone after 20 seconds.
 - d Comment on the validity of this prediction.
- P** 3 The motion of a basketball as it leaves a player's hand and passes through the net is modelled using the equation $h = 2 + 1.1x - 0.1x^2$, where h m is the height of the basketball above the ground and x m is the horizontal distance travelled.
 - a Find the two values of x for which the basketball is exactly 4 m above the ground.
This model is valid for $0 \leq x \leq k$, where k m is the horizontal distance of the net from the player.
 - b Given that the height of the net is 3 m, find the value of k .
 - c Explain why the model is not valid for $x > k$.
- P** 4 A car accelerates from rest to 60 km h^{-1} in 10 seconds. A quadratic equation of the form $d = kt^2$ can be used to model the distance travelled, d metres in time t seconds.
 - a Given that when $t = 1$ second the distance travelled by the car is 13.2 metres, use the model to find the distance travelled when the car reaches 60 km h^{-1} .
 - b Write down the range of values of t for which the model is valid.
- P** 5 The model for the motion of a golf ball given in question 1 is valid only when h is **positive**. Find the range of values of x for which the model is valid.
- P** 6 The model for the height of the stone above sea level given in question 2 is valid only from the time the stone is thrown until the time it enters the sea. Find the range of values of t for which the model is valid.

Problem-solving

Use the information given to work out the value of k .

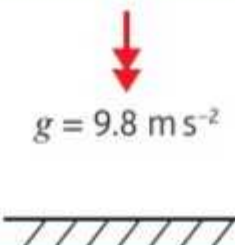
1.2 Modelling assumptions

Modelling assumptions can simplify a problem and allow you to analyse a real-life situation using known mathematical techniques. You need to understand the significance of different modelling assumptions and how they affect the calculations in a particular problem.

Watch out

Modelling assumptions can affect the validity of a model. For example, when modelling the landing of an aeroplane flight, it would not be appropriate to ignore the effects of wind and air resistance.

This table shows some common models and modelling assumptions that you need to know.

Model	Modelling assumptions
Particle – Dimensions of the object are negligible.	• mass of the object is concentrated at a single point • rotational forces (i.e. moving around a central fixed point) and air resistance can be ignored
Rod – All dimensions but one are negligible, like a pole or a beam.	• mass is concentrated along a line • no thickness • rigid (does not bend or buckle)
Lamina – Object with area but negligible thickness, like a sheet of paper.	• mass is distributed across a flat surface
Uniform body – Mass is distributed evenly.	• mass of the object is concentrated at a single point at the geometric centre of the body – the centre of mass
Light object – Mass of the object is small compared to other masses, like a string or a pulley.	• treat object as having zero mass • tension the same at both ends of a light string
Inextensible string – A string that does not stretch under load.	• acceleration is the same in objects connected by a taut inextensible string
Smooth surface – a surface on which it can be assumed there is no friction.	• assume that there is no friction between the surface and any object on it
Rough surface – a surface on which there is friction.	• objects in contact with the surface experience a frictional force if they are moving, or are acted on by a force
Wire – Rigid thin length of metal.	• treated as one-dimensional
Smooth and light pulley – All pulleys you consider will be smooth and light.	• pulley has no mass • tension is the same on either side of the pulley
Bead – Particle with a hole in it for threading on a wire or string (i.e. passing the wire or string through the hole).	• a smooth bead moves freely along a wire or string • for a smooth bead, tension is the same on either side of the bead
Peg – A support from which a body can be suspended or rested.	• dimensionless and fixed • can be rough or smooth as specified in the question
Air resistance – Resistance experienced as an object moves through the air.	• usually modelled as being negligible
Gravity – Force of attraction between all objects. Acceleration due to gravity is denoted by g . 	• assume all objects with mass are attracted toward the Earth • acceleration due to Earth's gravity is uniform (i.e. the same in all parts, at all times) and acts vertically downward • g is constant and is taken as 9.8 m s^{-2}, unless otherwise stated in the question

Example 2**SKILLS** ANALYSIS

A mass is attached to a length of string which is fixed to the ceiling.

The mass is drawn to one side with the string stretched tightly and allowed to swing.

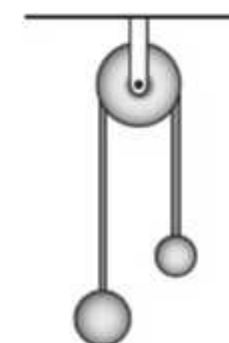
State the effect of the following assumptions on any calculations made using this model.

- a The string is **light and inextensible** (unable to be stretched further).
- b The mass is modelled as a particle.

- a Ignore the mass of the string and any stretching effect caused by the mass.
- b Ignore the rotational effect of any external forces that are acting on it, and the effects of air resistance.

Exercise 1B**SKILLS** ANALYSIS

- 1 A football is kicked by the goalkeeper from one end of the football pitch.
State the effect of the following assumptions on any calculations made using this model.
 - a The football is modelled as a particle.
 - b Air resistance is negligible.
- 2 An ice hockey puck is hit and slides across the ice.
State the effect of the following assumptions on any calculations made using this model.
 - a The ice hockey puck is modelled as a particle.
 - b The ice is smooth.
- 3 A parachutist wants to model her descent from an aeroplane to the ground. She models herself and her parachute as particles connected by a light inextensible string. Explain why this may not be a suitable modelling assumption for this situation.
- 4 A fishing rod manufacturer constructs a mathematical model to predict the behaviour of a particular fishing rod. The fishing rod is modelled as a light rod.
 - a Describe the effects of this modelling assumption.
 - b Comment on its validity in this situation.
- 5 Make a list of the assumptions you might make to create simple models of the following:
 - a the motion of a golf ball after it is hit
 - b the motion of a child on a sledge going down a snow-covered hill
 - c the motion of two objects of different masses connected by a string that passes over a pulley
 - d the motion of a suitcase on wheels being pulled along a path by its handle.



1.3 Quantities and units

The International System of Units, (abbreviated SI from the French, *Système international d'unités*) is the modern form of the metric system. These **base SI units** are most commonly used in mechanics:

Quantity	Unit	Symbol
Mass	kilogram	kg
Length/ displacement	metre	m
Time	second	s

Watch out

A common misunderstanding is that kilograms measure **weight**, not mass. However, **weight** is a **force** which is measured in **newtons (N)**.

These **derived** units are compound units built from the base units:

Quantity	Unit	Symbol
Speed/velocity	metres per second	m s^{-1}
Acceleration	metres per second per second	m s^{-2}
Weight/force	newton	$\text{N (= kg m s}^{-2}\text{)}$

Example

3

SKILLS REASONING/ARGUMENTATION

Write the following quantities in SI units.

a 4 km b 0.32 grams c $5.1 \times 10^6 \text{ km h}^{-1}$

$$\text{a } 4 \text{ km} = 4 \times 1000 = 4000 \text{ m}$$

$$\text{b } 0.32 \text{ g} = 0.32 \div 1000 = 3.2 \times 10^{-4} \text{ kg}$$

$$\begin{aligned} \text{c } 5.1 \times 10^6 \text{ km h}^{-1} &= 5.1 \times 10^6 \times 1000 \\ &= 5.1 \times 10^9 \text{ m h}^{-1} \end{aligned}$$

$$5.1 \times 10^9 \div (60 \times 60) = 1.42 \times 10^6 \text{ m s}^{-1}$$

The SI unit of length is the metre; $1 \text{ km} = 1000 \text{ m}$.

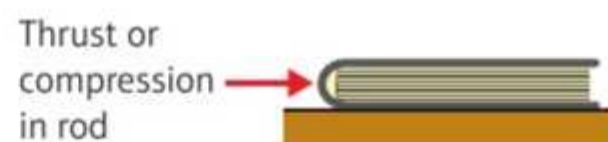
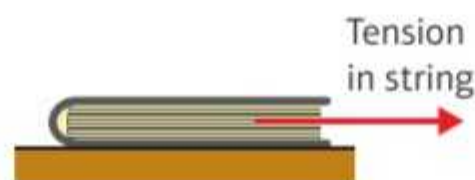
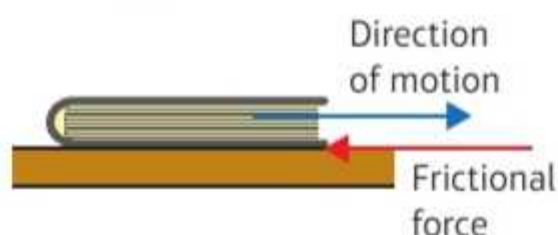
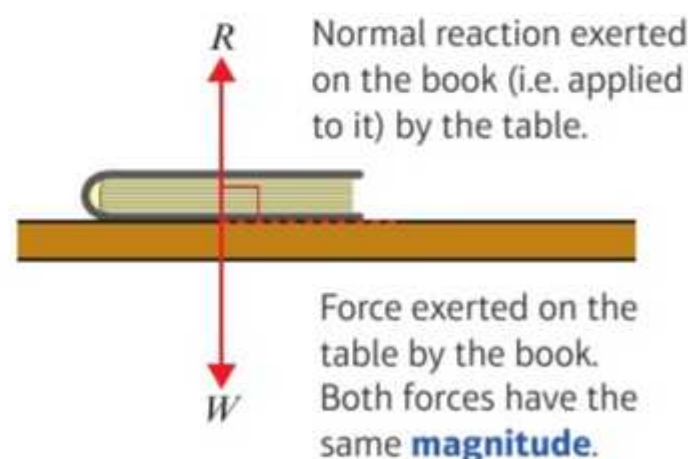
The SI unit of mass is the kg; $1 \text{ kg} = 1000 \text{ g}$.
The answer is given in standard form.

The SI unit of speed is m s^{-1} . Convert from km h^{-1} to m h^{-1} by multiplying by 1000.

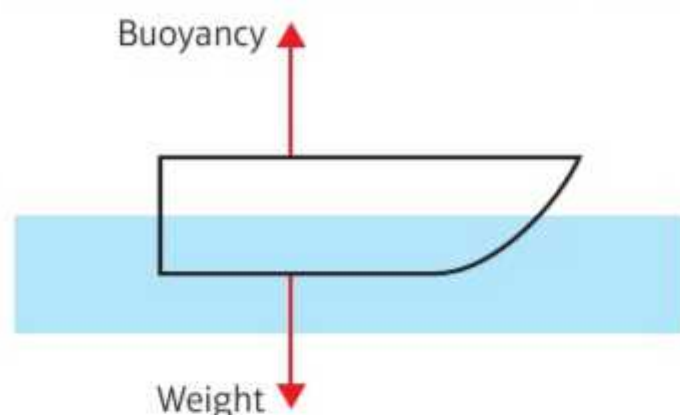
Convert from m h^{-1} to m s^{-1} by dividing by 60×60 .
The answer is given in standard form to 3 s.f.

You will encounter a variety of forces in mechanics. These **force diagrams** show some of the most common forces.

- The **weight** (or gravitational force) of an object acts vertically downward.
- The **normal reaction** is the force which acts perpendicular (i.e. at a 90° angle to it) to a surface when an object is in contact with the surface. In this example the normal reaction is due to the weight of the book resting on the surface of the table.
- Friction** is a force which opposes the motion between two rough surfaces.
- If an object is being pulled along by a string, the force acting on the object is called the **tension** in the string.
- If an object is being pushed along using a light rod, the force acting on the object is called the **thrust** or **compression** in the rod.



- **Buoyancy** is the upward force on a body that allows it to float or rise when submerged (i.e. underneath the surface) in a liquid. In this example buoyancy acts to keep the boat afloat in the water.



- **Air resistance** opposes motion. In this example the weight of the parachutist acts vertically downward and the air resistance acts vertically upward.

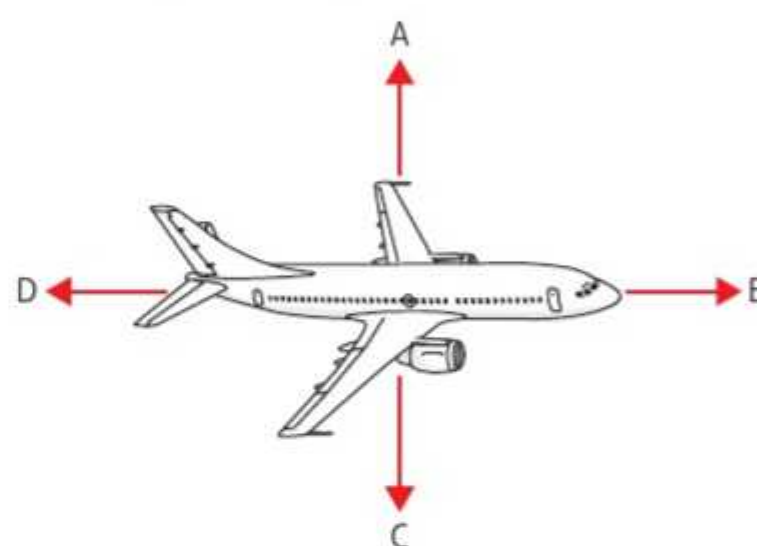


Example 4

The force diagram shows an aircraft in flight. Write down the names of the four forces shown on the diagram.

Also known as 'lift', this is the upward force that keeps the aircraft up in the air.

- A upward thrust
- B forward thrust
- C weight
- D air resistance



Also known as 'thrust', this is the force that propels the aircraft forward.

This is the gravitational force acting downward on the aircraft.

Also known as 'drag', this is the force that acts in the **opposite** direction to the forward thrust.

Exercise 1C

1 Convert to SI units:

a 65 km h^{-1}

b 15 g cm^{-2}

c 30 cm per minute

d 24 g m^{-3}

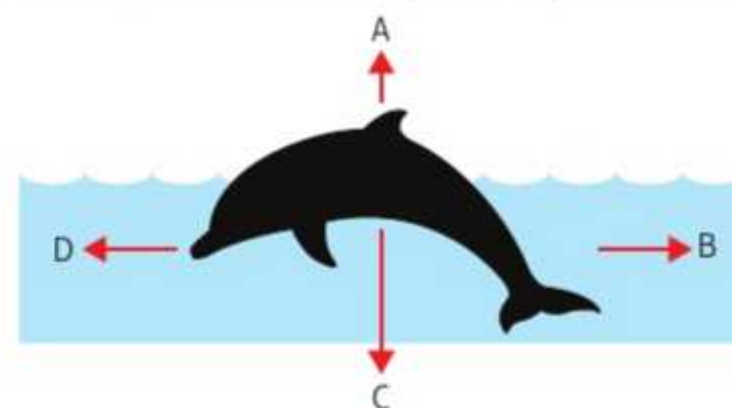
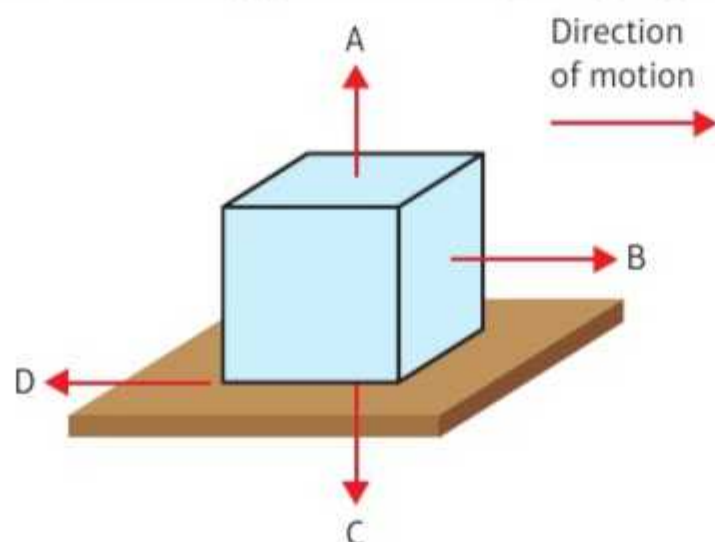
e $4.5 \times 10^{-2} \text{ g cm}^{-3}$

f $6.3 \times 10^{-3} \text{ kg cm}^{-2}$

2 Write down the names of the forces shown in each of these diagrams.

a A box being pushed along rough ground

b A dolphin swimming through the water



2 CONSTANT ACCELERATION

3.1

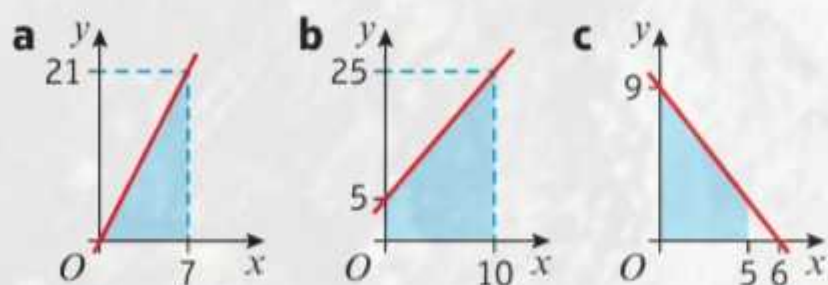
Learning objectives

After completing this chapter you should be able to:

- Understand and interpret displacement–time graphs → pages 11–13
- Understand and interpret velocity–time graphs → pages 13–16
- Derive the constant acceleration formulae and use them to solve problems → pages 17–29
- Use the constant acceleration formulae to solve problems involving vertical motion under gravity → pages 29–35

Prior knowledge check

- 1 For each graph find:
- i the gradient
 - ii the shaded area under the graph.



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- 2 A car travels for 45 minutes at an average speed of 35 km h^{-1} . Find the distance travelled.

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- 3 a Solve these simultaneous equations:

$$3x - 2y = 9$$

$$x + 4y + 4 = 0$$

- b Solve $2x^2 + 3x - 7 = 0$. Give your answers to 3 s.f.

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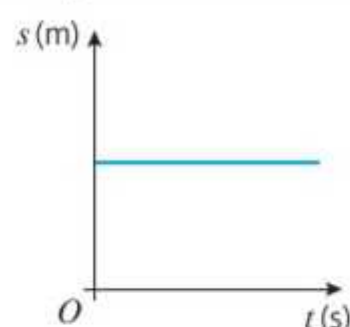


A body falling freely under **gravity** can be modelled as having **constant acceleration**. You can use this to estimate the time that a cliff diver spends in freefall.

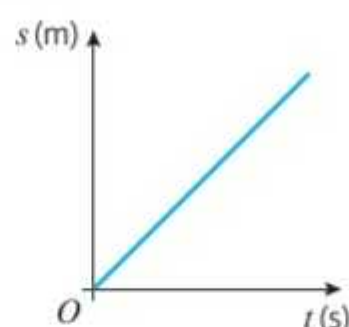
2.1 Displacement–time graphs

You can represent the motion of an object on a **displacement–time graph**. Displacement is always plotted on (i.e. marked on) the vertical axis and time on the horizontal axis.

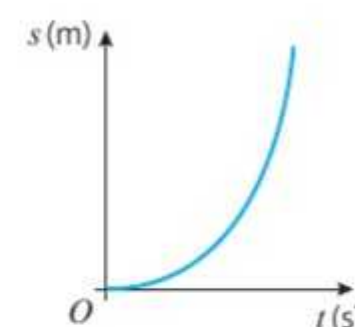
In these graphs, s represents the displacement of an object from a given point in metres and t represents the time taken in seconds.



There is no change in the displacement over time and the object is stationary (not moving).



The displacement increases at a constant rate over time and the object is moving with constant velocity.



The displacement is increasing at a greater rate as time increases. The velocity is increasing and the object is accelerating.

- **Velocity is the rate of change of displacement.**
 - On a displacement–time graph the gradient represents the velocity.
 - If the displacement–time graph is a straight line, then the velocity is constant.

■ **Average velocity** = $\frac{\text{displacement from starting point}}{\text{time taken}}$

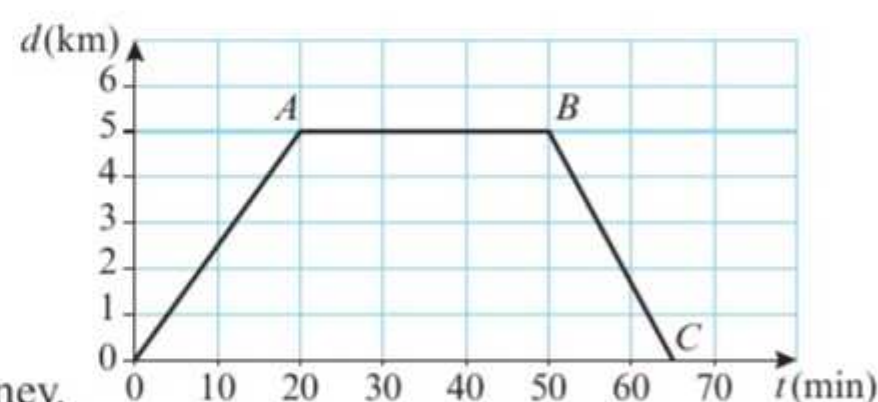
■ **Average speed** = $\frac{\text{total distance travelled}}{\text{time taken}}$

Example 1

SKILLS INTERPRETATION

A cyclist rides in a straight line for 20 minutes. She waits for half an hour, then returns in a straight line to her starting point in 15 minutes. This is a displacement–time graph for her journey.

- a Work out the average velocity for each stage of the journey in km h^{-1} .
- b Write down the average velocity for the whole journey.
- c Work out the average speed for the whole journey.



a Journey from O to A : time = 20 min; displacement = 5 km

$$\text{Average velocity} = \frac{5}{20} = 0.25 \text{ km min}^{-1}$$

$$0.25 \times 60 = 15 \text{ km h}^{-1}$$

Journey from A to B : no change in displacement so average velocity = 0

Journey from B to C : time = 15 min; displacement = -5 km

$$\text{Average velocity} = \frac{-5}{15} = -\frac{1}{3} \text{ km min}^{-1}$$

$$-\frac{1}{3} \times 60 = -20 \text{ km h}^{-1}$$

To convert from km min^{-1} to km h^{-1} multiply by 60.

A horizontal line on the graph indicates the cyclist is stationary.

The cyclist starts with a displacement of 5 km and finishes with a displacement of 0 km, so the change in displacement is -5 km, and velocity will be negative.

b The displacement for the whole journey is 0 so average velocity is 0.

c Total time = 65 min
 Total distance travelled is $5 + 5 = 10$ km
 Average speed = $\frac{10}{65} = \frac{2}{13}$ km min⁻¹
 $\frac{2}{13} \times 60 = 9.23$ km h⁻¹ (3 s.f.)

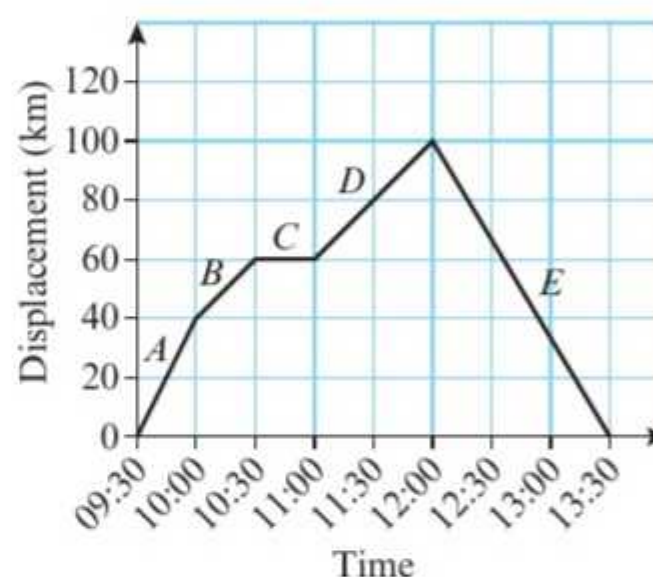
At C the cyclist has returned to the starting point.

The cyclist has travelled 5 km away from the starting point and then 5 km back to the starting point.

Exercise 2A

SKILLS INTERPRETATION

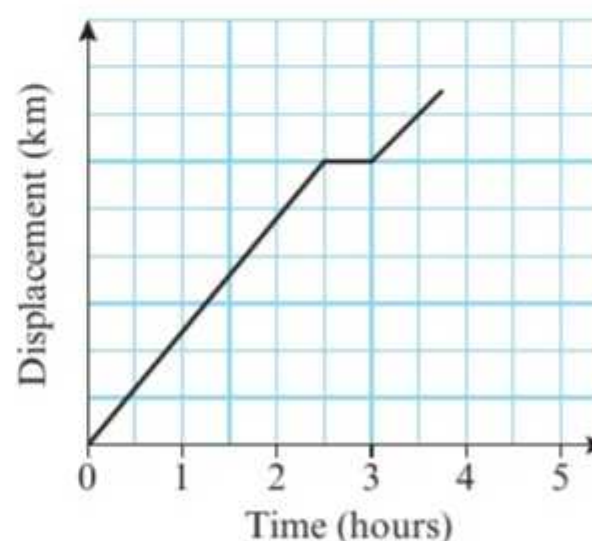
- 1 This is a displacement–time graph for a car travelling along a straight road. The journey is divided into 5 stages labelled A to E.
- Work out the average velocity for each stage of the journey.
 - State the average velocity for the whole journey.
 - Work out the average speed for the whole journey.



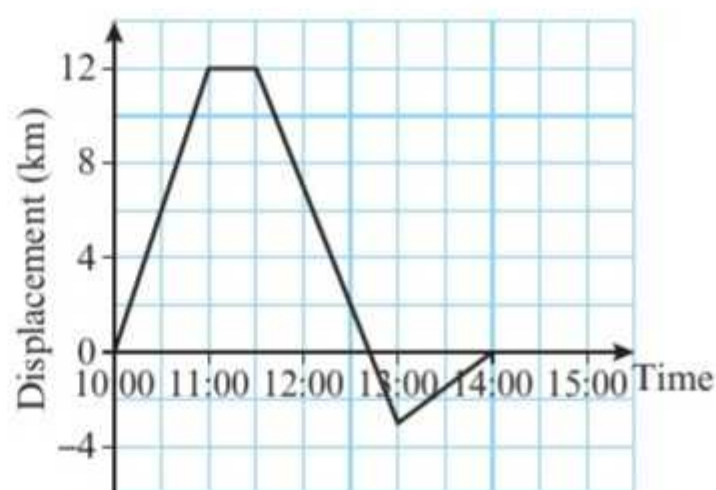
- 2 Khalid drives from his home to a hotel. He drives for $2\frac{1}{2}$ hours at an average velocity of 60 km h⁻¹. He then stops for lunch before continuing to his hotel. The diagram shows a displacement–time graph for Khalid's journey.
- Work out the displacement of the hotel from Khalid's home.
 - Work out Khalid's average velocity for his whole journey.

Problem-solving

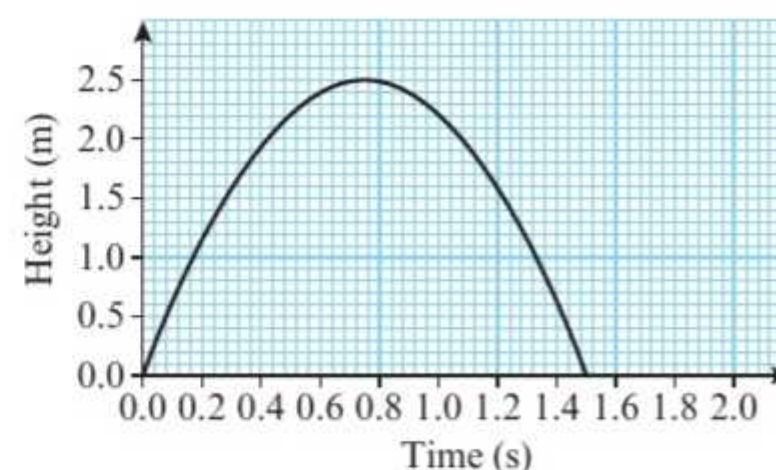
You need to work out the scale on the vertical axis.



- 3 Sarah left home at 10:00 and cycled north in a straight line. The diagram shows a displacement–time graph for her journey.
- Work out Sarah's velocity between 10:00 and 11:00. On her return journey, Sarah continued past her home for 3 km before returning.
 - Estimate the time that Sarah passed her home.
 - Work out Sarah's velocity for each of the last two stages of her journey.
 - Calculate Sarah's average speed for her entire journey.



- P 4** A ball is thrown vertically up in the air and falls to the ground. This is a displacement–time graph for the motion of the ball.
- Find the maximum height of the ball and the time at which it reaches that height.
 - Write down the velocity of the ball when it reaches its highest point.
 - Describe the motion of the ball:
 - from the time it is thrown to the time it reaches its highest point
 - after reaching its highest point.

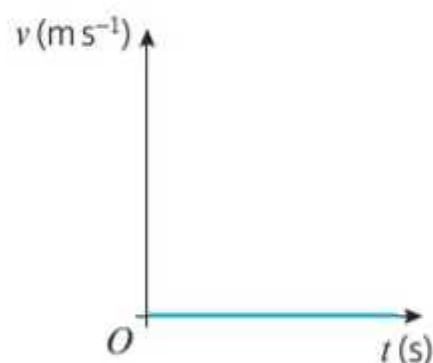


Hint To describe the motion you should state the direction of travel of the ball and whether it is accelerating or decelerating.

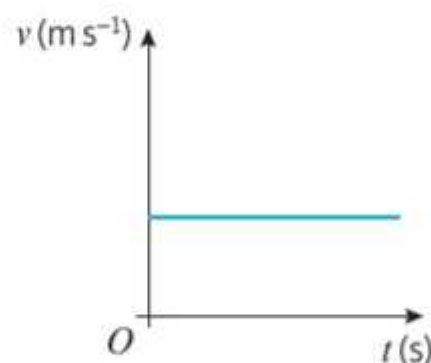
2.2 Velocity–time graphs

You can represent the motion of an object on a **velocity–time graph**. Velocity is always plotted on the vertical axis and time on the horizontal axis.

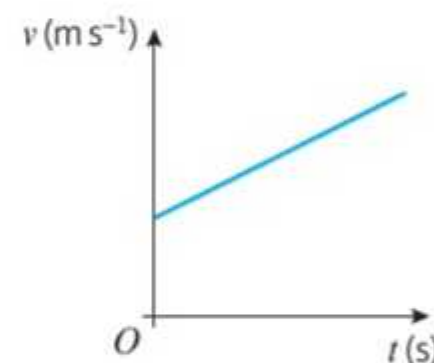
In these graphs v represents the velocity of an object in metres per second and t represents the time taken in seconds.



The velocity is zero and the object is stationary.



The velocity is unchanging and the object is moving with constant velocity.

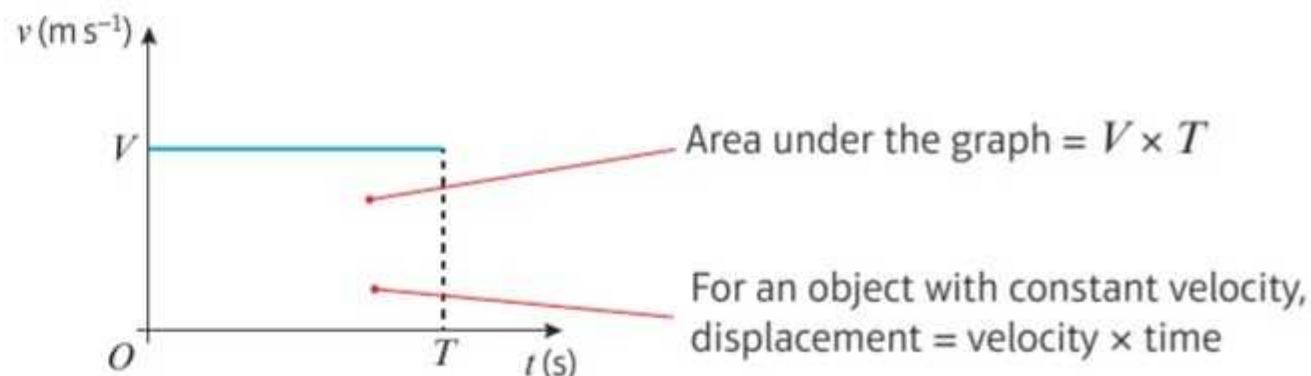


The velocity is increasing at a constant rate and the object is moving with constant acceleration.

- Acceleration is the rate of change of velocity.
 - In a velocity–time graph the gradient represents the acceleration.
 - If the velocity–time graph is a straight line, then the acceleration is constant.

Notation Negative acceleration is sometimes described as deceleration or retardation.

This velocity–time graph represents the motion of an object travelling in a straight line at constant velocity $V \text{ m s}^{-1}$ for time T seconds.



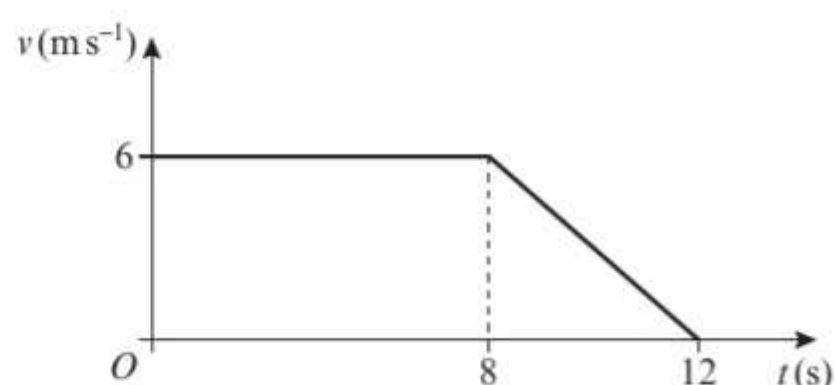
- The area between a velocity–time graph and the horizontal axis represents the distance travelled.
 - For motion in a straight line with positive velocity, the area under the velocity–time graph up to a point t represents the displacement at time t .

Example 2

SKILLS INTERPRETATION

The figure shows a velocity–time graph illustrating the motion of a cyclist moving along a straight road for a period of 12 seconds. For the first 8 seconds, she moves at a constant rate of 6 m s^{-1} . She then decelerates at a constant rate, stopping after a further 4 seconds.

- Find the displacement from the starting point of the cyclist after this 12 second period.
- Work out the rate at which the cyclist decelerates.



- The displacement s after 12 s is given by the area under the graph.

$$\begin{aligned} s &= \frac{1}{2}(a+b)h \\ &= \frac{1}{2}(8+12) \times 6 \\ &= 10 \times 6 = 60 \end{aligned}$$

The displacement of the cyclist after 12 s is 60 m.

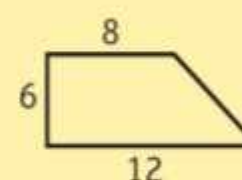
- The acceleration is the gradient of the slope.

$$a = \frac{-6}{4} = -1.5$$

The deceleration is 1.5 m s^{-2} .

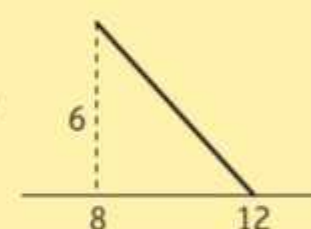
Model the cyclist as a particle moving in a straight line.

The displacement is represented by the area of the trapezium with these sides.



You can use the formula for the area of a trapezium to calculate this area.

The gradient is given by $\frac{\text{difference in the } v\text{-coordinates}}{\text{difference in the } t\text{-coordinates}}$

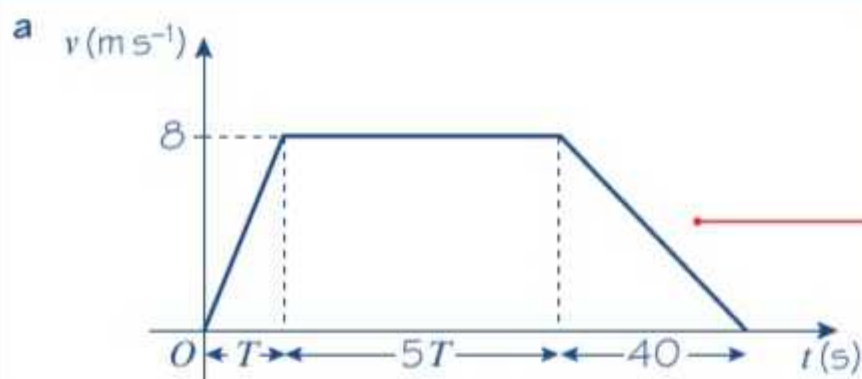


Example 3

SKILLS INTERPRETATION

A particle moves along a straight line. The particle accelerates **uniformly** from rest to a velocity of 8 m s^{-1} in T seconds. The particle then travels at a constant velocity of 8 m s^{-1} for $5T$ seconds. The particle then decelerates uniformly to rest in a further 40 seconds.

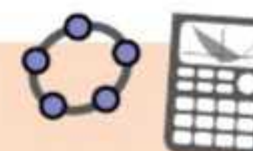
- Sketch a speed–time graph to illustrate the motion of the particle.
- Given that the total displacement of the particle is 600 m, find the value of T .



If the particle accelerates from rest and decelerates to rest this means the initial and final velocities are zero.

Online

Explore how the area of the trapezium changes as the value of T changes using technology.



b The area of the trapezium is:

$$\begin{aligned} s &= \frac{1}{2}(a + b)h \\ &= \frac{1}{2}(5T + 6T + 40) \times 8 \\ &= 4(11T + 40) \end{aligned}$$

The displacement is 600 m.

$$4(11T + 40) = 600$$

$$44T + 160 = 600$$

$$T = \frac{600 - 160}{44} = 10 \text{ s}$$

The length of the shorter of the two parallel sides is $5T$. The length of the longer side is $T + 5T + 40 = 6T + 40$.

Problem-solving

The displacement is equal to the area of the trapezium. Write an equation and solve it to find T .

Exercise 2B

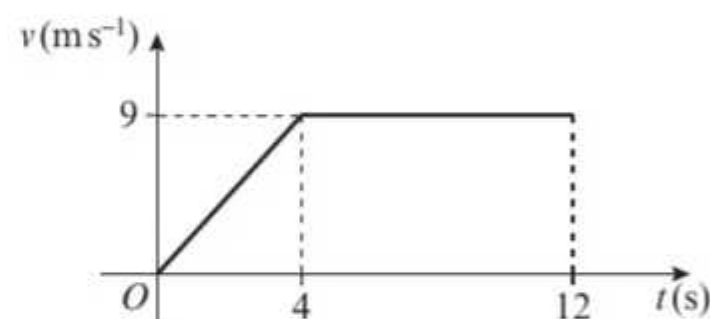
SKILLS

INTERPRETATION,
PROBLEM-SOLVING

- 1 The diagram shows the speed–time graph of the motion of an athlete running along a straight track. For the first 4 s, he accelerates uniformly from rest to a velocity of 9 m s^{-1} .

This velocity is then maintained for a further 8 s. Find:

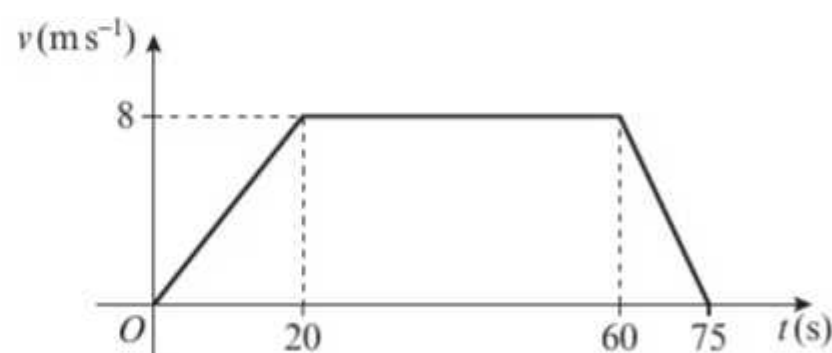
- the rate at which the athlete accelerates
- the displacement from the starting point of the athlete after 12 s.



- 2 A car is moving along a straight road. When $t = 0$, the car passes a point A with velocity 10 m s^{-1} and this velocity is maintained until $t = 30 \text{ s}$. The driver then applies the brakes and the car decelerates uniformly, coming to rest at the point B when $t = 42 \text{ s}$.

- Sketch a velocity–time graph to illustrate the motion of the car.
- Find the distance from A to B .

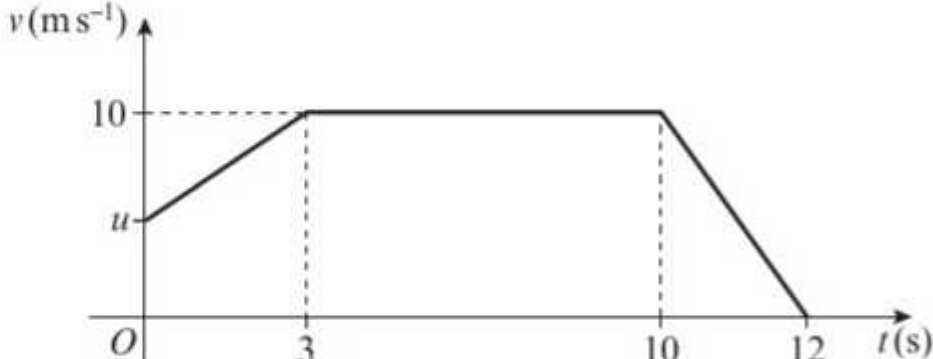
- E** 3 The diagram shows the velocity–time graph of the motion of a cyclist riding along a straight road. She accelerates uniformly from rest to 8 m s^{-1} in 20 s. She then travels at a constant velocity of 8 m s^{-1} for 40 s. She then decelerates uniformly to rest in 15 s. Find:



- the acceleration of the cyclist in the first 20 s of motion (2 marks)
- the deceleration of the cyclist in the last 15 s of motion (2 marks)
- the displacement from the starting point of the cyclist after 75 s. (2 marks)

- E** 4 A motorcyclist starts from rest at a point S on a straight race track. He moves with constant acceleration for 15 s, reaching a velocity of 30 m s^{-1} . He then travels at a constant velocity of 30 m s^{-1} for T seconds. Finally he decelerates at a constant rate coming to rest at a point F , 25 s after he begins to decelerate.

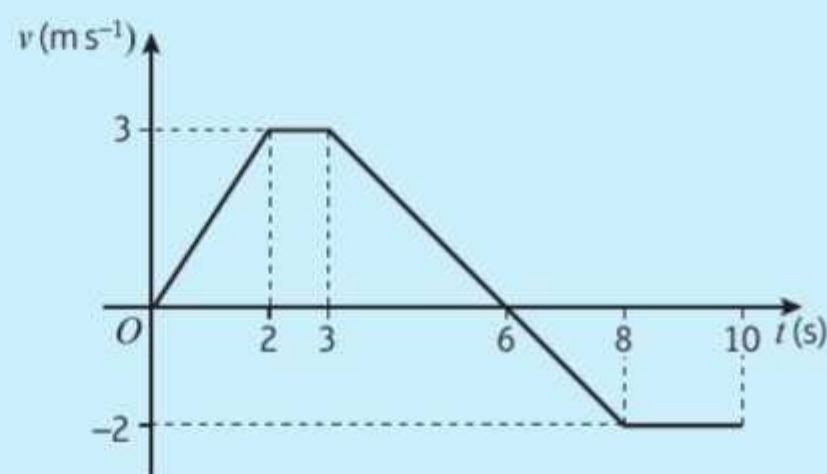
- Sketch a speed–time graph to illustrate the motion. (3 marks)
- Given that the distance between S and F is 2.4 km, calculate the time the motorcyclist takes to travel from S to F . (3 marks)

- E 5** A train starts from a station X and moves with constant acceleration of 0.6 m s^{-2} for 20 s. The velocity it has reached after 20 s is then maintained for T seconds. The train then decelerates from this velocity to rest in a further 40 s, stopping at a station Y .
- Sketch a velocity–time graph to illustrate the motion of the train. (3 marks)
- Given that the distance between the stations is 4.2 km, find:
- the value of T (3 marks)
 - the distance travelled by the train while it is moving with constant velocity. (2 marks)
- E 6** A particle moves along a straight line. The particle accelerates from rest to a velocity of 10 m s^{-1} in 15 s. The particle then moves at a constant velocity of 10 m s^{-1} for a period of time. The particle then decelerates uniformly to rest. The period of time for which the particle is travelling at a constant velocity is 4 times the period of time for which it is decelerating.
- Sketch a speed–time graph to illustrate the motion of the particle. (3 marks)
 - Given that the displacement from the starting point of the particle after it comes to rest is 480 m, find the total time for which the particle is moving. (3 marks)
- E 7** A particle moves 100 m in a straight line. The diagram is a sketch of a velocity–time graph of the motion of the particle. The particle starts with velocity $u \text{ m s}^{-1}$ and accelerates to a velocity of 10 m s^{-1} in 3 s. The velocity of 10 m s^{-1} is maintained for 7 s and then the particle decelerates to rest in a further 2 s. Find:
- 
- the value of u (3 marks)
 - the acceleration of the particle in the first 3 s of motion. (3 marks)
- E 8** A motorcyclist M leaves a road junction at time $t = 0$. She accelerates from rest at a rate of 3 m s^{-2} for 8 s and then maintains the velocity she has reached. A car C leaves the same road junction as M at time $t = 0$. The car accelerates from rest to 30 m s^{-1} in 20 s and then maintains the velocity of 30 m s^{-1} . C passes M as they both pass a pedestrian.
- On the same diagram, sketch speed–time graphs to illustrate the motion of M and C . (3 marks)
 - Find the distance of the pedestrian from the road junction. (3 marks)

Challenge

The graph shows the velocity of an object travelling in a straight line during a 10-second time interval.

- After how long did the object change direction?
- Work out the total distance travelled by the object.
- Work out the displacement from the starting point of the object after:
 - 6 seconds
 - 10 seconds.

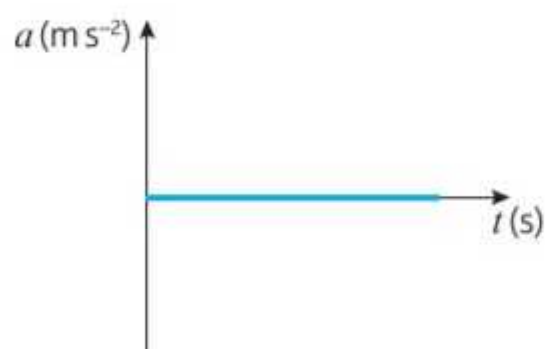


2.3 Acceleration-time graphs

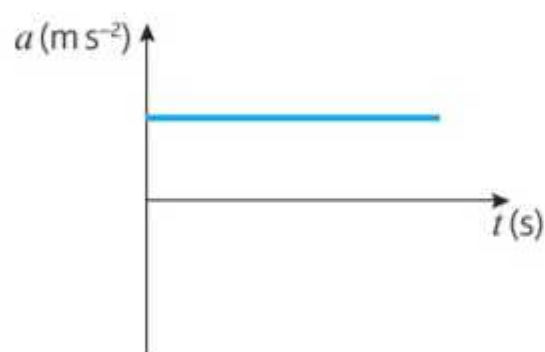
Acceleration is the rate of change of velocity.

You can represent the motion of an object on an acceleration-time graph. Acceleration is always plotted on the horizontal axis, and time on the vertical axis.

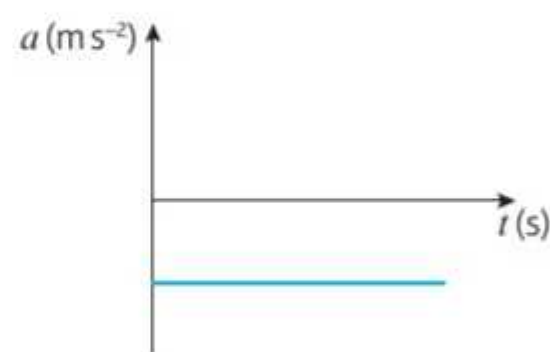
In the following graphs, a represents the acceleration of an object in m s^{-2} and t represents the time in seconds.



Zero acceleration – constant velocity



Positive acceleration; increase in velocity



Negative acceleration (deceleration); decrease in velocity

A sloping line on an acceleration-time graph represents changing acceleration and is beyond the scope of this syllabus.

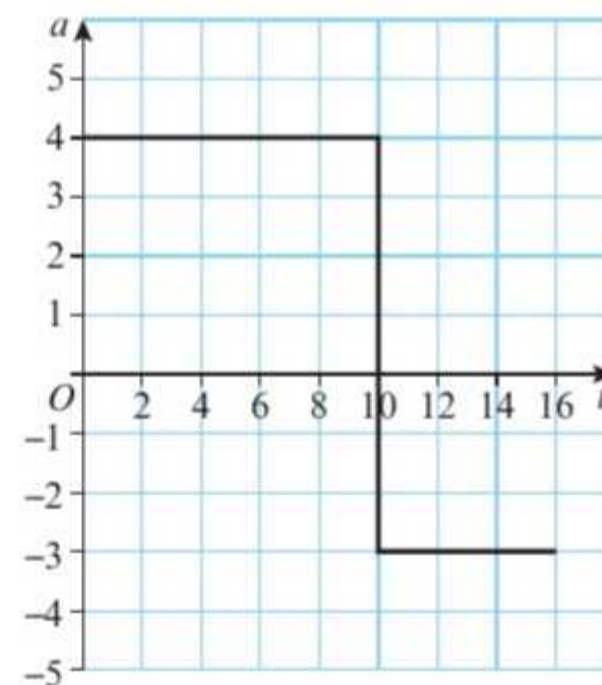
- The area between the acceleration-time graph and the horizontal axis represents the **change** in velocity. In other words, the area under the acceleration-time graph for a certain interval is equal to the change in velocity during that time interval.

Example 4

SKILLS PROBLEM-SOLVING

The acceleration-time graph shows the motion of a car over a 20 second period.

- Calculate the change in velocity between $t = 0$ and $t = 10$ seconds.
- Given that the car was initially moving at 15 ms^{-1} , calculate the velocity after 16 seconds.
- Draw a velocity-time graph of the motion of the car.
- Find the distance travelled in the first 16 seconds.



a Change in velocity $= 4 \times 10 = 40 \text{ ms}^{-1}$

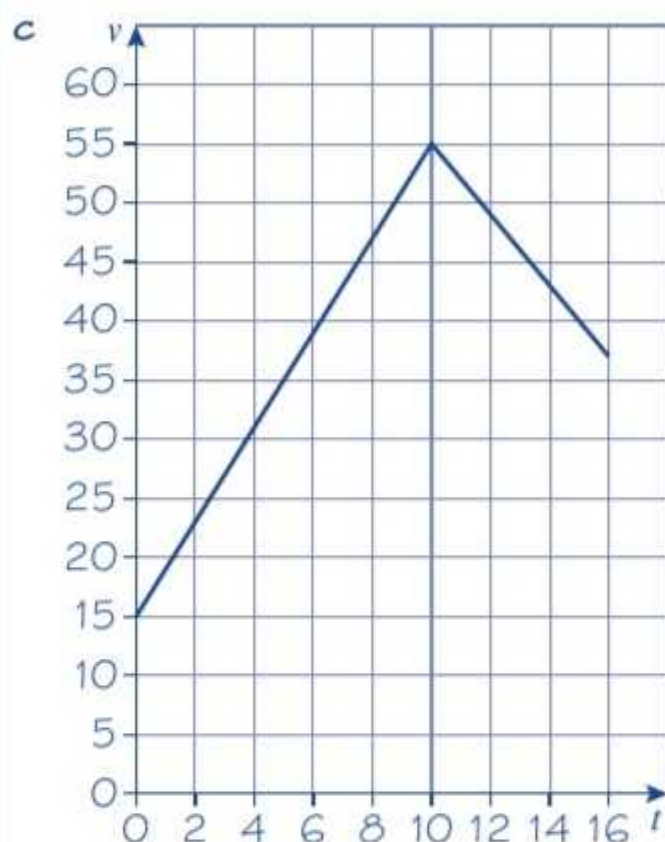
b Change in velocity from $t = 10$ to $t = 15$ is:

$$\text{Change} = -3 \times 6 = -18 \text{ ms}^{-1}$$

$$\text{Initial velocity } v_0 = 15 \text{ ms}^{-1}$$

Velocity after 15 seconds

$$= 15 + 40 - 18 = 37 \text{ ms}^{-1}$$



d Distance travelled is the area under the graph.

$$\begin{aligned} \text{Dist} &= \frac{10}{2} \times (15 + 55) + \frac{6}{2} \times (15 + 37) \\ &= 506 \text{ m} \end{aligned}$$

The change in velocity is the area under the line.

Since the car is slowing down, this area is negative.

The final velocity is the sum of the changes and the initial velocity.

The distance travelled is the area under the graph of velocity. This area has been broken down into two trapeziums.

Exercise 2C

SKILLS INTERPRETATION

- 1 The acceleration-time graph shows the motion of a racing car over 10 seconds of a race.

Describe what is happening between:

a $t = 0$ and $t = 2 \text{ s}$

b $t = 2$ and $t = 4 \text{ s}$

c $t = 4$ and $t = 8 \text{ s}$.

d Given that the initial velocity of the racing car was 44 ms^{-1} , its velocity at $t = 8 \text{ s}$.

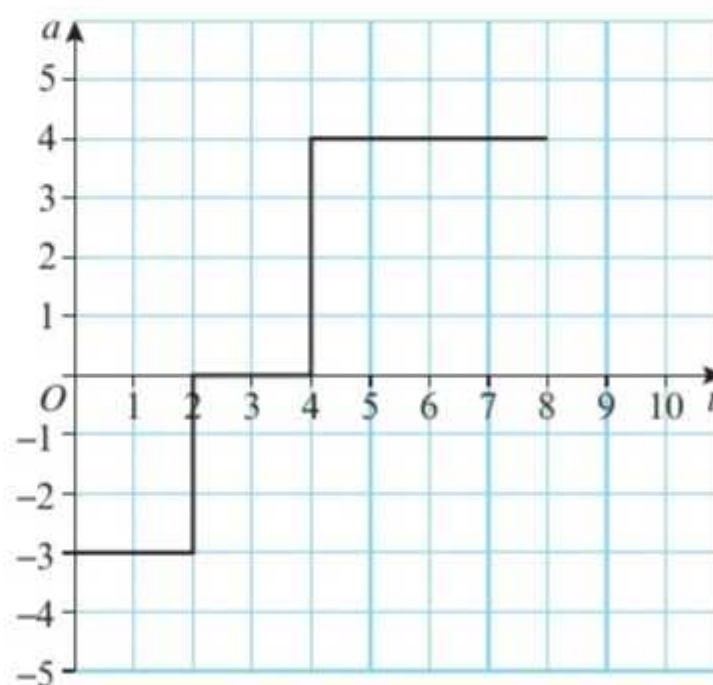
- 2 A motorcycle accelerates at a rate of 1 ms^{-2} for 4 seconds, before travelling at a constant velocity for 6 seconds. It then decelerates at a rate of 0.5 ms^{-2} for 4 seconds.

a Draw an acceleration-time graph to show the motion of the car.

Given that the initial velocity of the motorcycle was 18 ms^{-1} :

b draw a velocity-time graph of the journey of the motorcycle from $t = 0$ to $t = 14$ seconds

c hence, calculate the distance travelled by the motorcycle from $t = 0$ to $t = 14$ seconds.



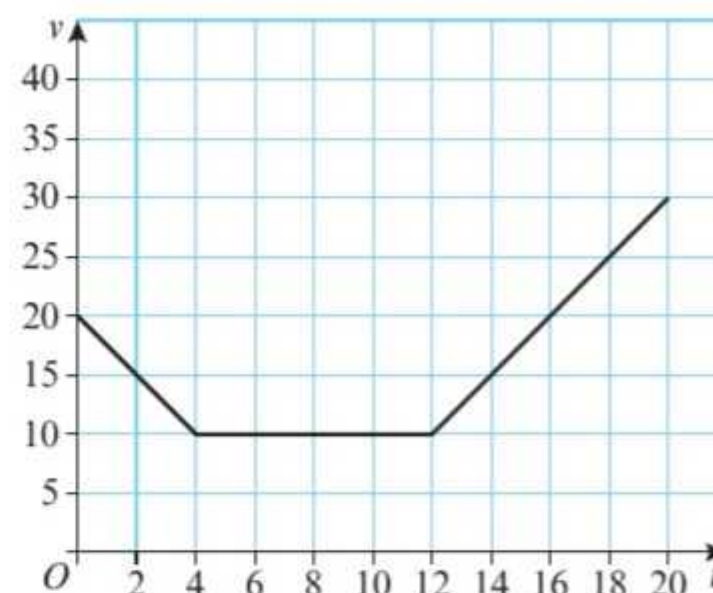
- 3 The velocity-time graph shows the motion of a car over a 20 second period. Draw an acceleration-time graph to show the motion of the car.

Challenge

A particle is travelling along the x -axis. At time $t = 0$, the velocity of the particle is 8 m s^{-1} , and the particle is accelerating at 5 m s^{-2} . After T seconds, the particle has constant velocity, and after $(T + 6)$ seconds the particle accelerates at -4 m s^{-2} .

Given that the velocity of the particle after $(T + 16)$ seconds is -12 m s^{-1} :

- draw an acceleration-time graph of the motion of the particle, giving your times in terms of T
- find the value of T .



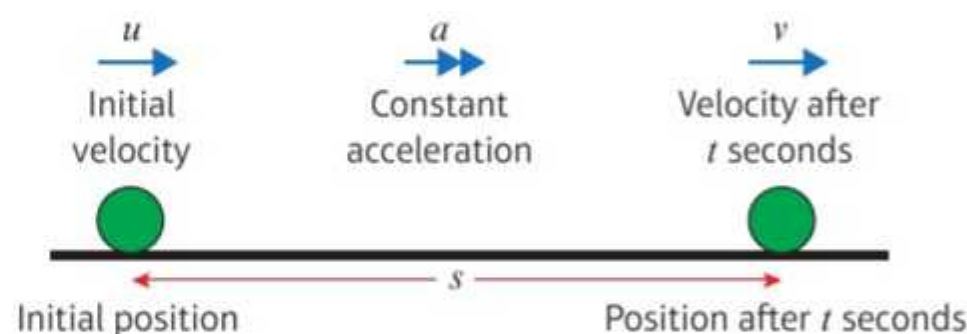
Hint

A negative acceleration means the particle is slowing down, and a negative velocity means the particle is travelling backwards.

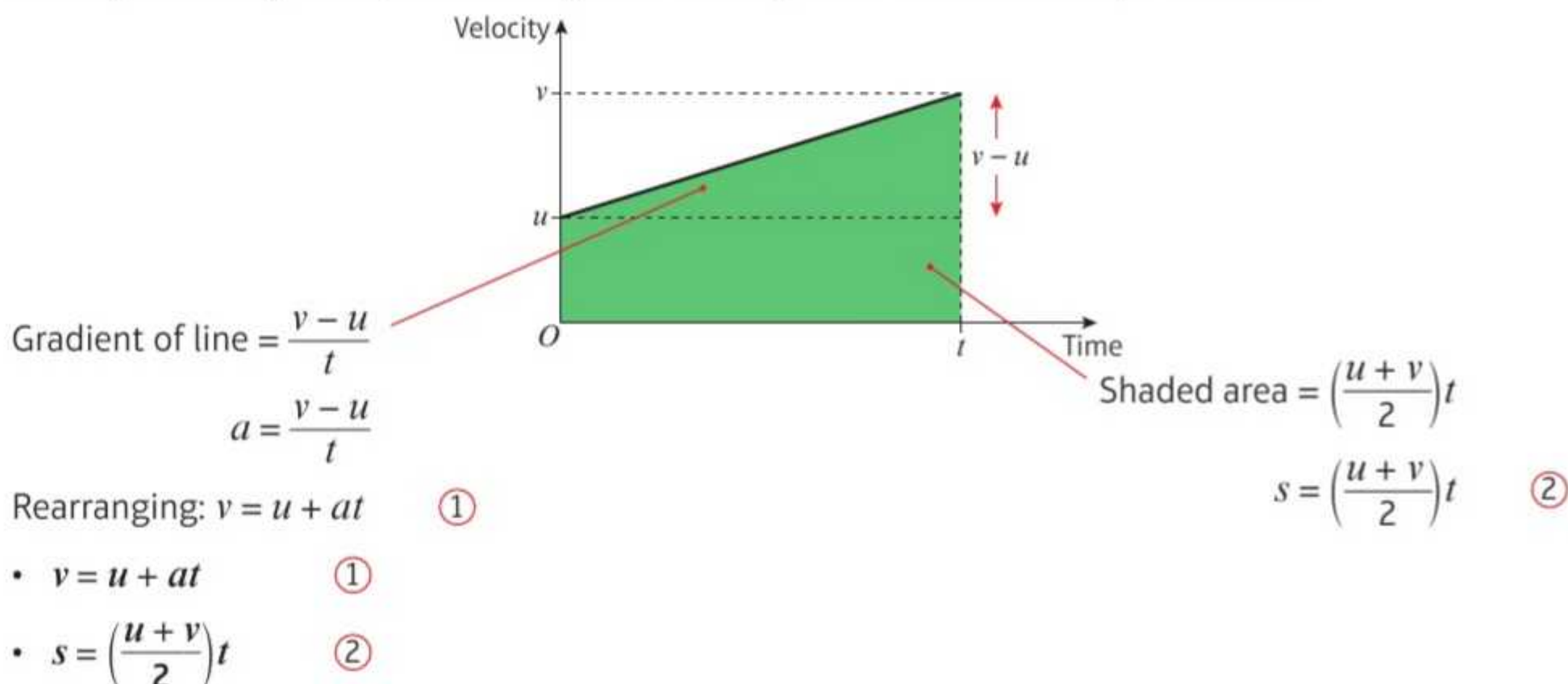
2.4 Constant acceleration formulae 1

A standard set of letters is used for the motion of an object moving in a straight line with constant acceleration.

- s is the displacement.
- u is the initial velocity.
- v is the final velocity.
- a is the acceleration.
- t is the time.



You can use these letters to label a velocity-time graph representing the motion of a particle moving in a straight line, accelerating from velocity u at time 0 to velocity v at time t .



You need to know how to derive these formulae from the velocity-time graph.

Example

5

SKILLS

PROBLEM-SOLVING

A cyclist is travelling along a straight road. She accelerates at a constant rate from a velocity of 4 m s^{-1} to a velocity of 7.5 m s^{-1} in 40 seconds. Find:

- the distance she travels in these 40 seconds
- her acceleration in these 40 seconds.



Diagram showing a cyclist accelerating from 4 m s^{-1} to 7.5 m s^{-1} over a distance s in 40 seconds.

$u = 4, v = 7.5, t = 40, s = ?, a = ?$

a $s = \left(\frac{u + v}{2} \right) t$
 $= \left(\frac{4 + 7.5}{2} \right) \times 40$
 $= 230$
 The distance the cyclist travels is 230 m.

b $v = u + at$
 $7.5 = 4 + 40a$
 $a = \frac{7.5 - 4}{40} = 0.0875$
 The acceleration of the cyclist is 0.0875 m s^{-2} .

Start by drawing a diagram.

Model the cyclist as a particle.

Write down the values you know and the values you need to find.

You need a and you know v, u and t so you can use $v = u + at$.

Substitute the values you know into the formula. You can solve this equation to find a .

You could rearrange the formula before you substitute the values:

$$a = \frac{v - u}{t}$$

In real-life situations values for the acceleration are often quite small. Large accelerations feel unpleasant and may be dangerous.

Example

6

SKILLS

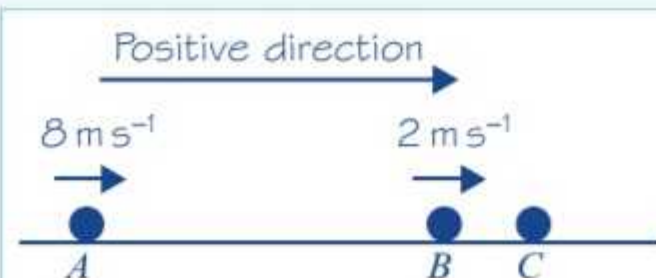
PROBLEM-SOLVING

A particle moves in a straight line from a point A to a point B with constant deceleration 1.5 m s^{-2} . The velocity of the particle at A is 8 m s^{-1} and the velocity of the particle at B is 2 m s^{-1} . Find:

- the time taken for the particle to move from A to B
- the distance from A to B .

After reaching B , the particle continues to move along the straight line with constant deceleration 1.5 m s^{-2} . The particle is at the point C 6 seconds after passing through the point A . Find:

- the velocity of the particle at C
- the distance from A to C .



$$u = 8, v = 2, a = -1.5, t = ?, s = ?$$

a $v = u + at$

$$2 = 8 - 1.5t$$

$$1.5t = 8 - 2$$

$$t = \frac{8 - 2}{1.5} = 4$$

The time taken to move from A to B is 4 s.

b $s = \left(\frac{u + v}{2}\right)t$
 $= \left(\frac{8 + 2}{2}\right) \times 4 = 20$

The distance from A to B is 20 m.

c $u = 8, a = -1.5, t = 6, v = ?$

$$v = u + at$$

$$= 8 + (-1.5) \times 6$$

$$= 8 - 9 = -1$$

The velocity of the particle is 1 m s^{-1} in the direction BA.

d $s = \left(\frac{u + v}{2}\right)t$
 $= \left(\frac{8 + (-1)}{2}\right) \times 6$

The distance from A to C is 21 m.

Problem-solving

It's always a good idea to draw a sketch showing the positions of the particle. Mark the positive direction on your sketch, and remember that when the particle is **decelerating**, your value of a will be **negative**.

Use your answer from part a as the value of t .

The velocity at C is negative. This means that the particle is moving from right to left.

Remember that to specify a velocity it is necessary to give speed and direction.

Make sure you use the correct sign when substituting a negative value into a formula.

Example 7

SKILLS PROBLEM-SOLVING

A car moves from traffic lights along a straight road with constant acceleration. The car starts from rest at the traffic lights and 30 seconds later the car passes a speed-trap where it is registered as travelling at 45 km h^{-1} . Find:

- the acceleration of the car
- the distance between the traffic lights and the speed-trap.

Hint Convert all your measurements into base SI units before substituting values into the formulae.

$$45 \text{ km h}^{-1} = 45 \times \frac{1000}{3600} \text{ m s}^{-1} = 12.5 \text{ m s}^{-1}$$



$$u = 0, v = 12.5, t = 30, a = ?, s = ?$$

Convert into SI units, using:

1 km = 1000 m

1 hour = $60 \times 60 \text{ s} = 3600 \text{ s}$

Model the car as a particle and draw a diagram.

The car starts from rest, so the initial velocity is zero.

$$\begin{aligned} \text{a} \quad v &= u + at \\ 12.5 &= 0 + 30a \\ a &= \frac{12.5}{30} = \frac{5}{12} \end{aligned}$$

The acceleration of the car is $\frac{5}{12} \text{ m s}^{-2}$.

$$\begin{aligned} \text{b} \quad s &= \left(\frac{u + v}{2} \right) t \\ &= \left(\frac{0 + 12.5}{2} \right) \times 30 = 187.5 \end{aligned}$$

The distance between the traffic lights and the speed-trap is 187.5 m.

This is an exact answer. If you want to give an answer using decimals, you should round to three significant figures.

Exercise 2D SKILLS PROBLEM-SOLVING

- 1 A particle is moving in a straight line with constant acceleration 3 m s^{-2} . At time $t = 0$, the speed of the particle is 2 m s^{-1} . Find the speed of the particle at time $t = 6 \text{ s}$.
- 2 A car is approaching traffic lights. The car is travelling with speed 10 m s^{-1} . The driver applies the brakes to the car and the car comes to rest with constant deceleration in 16 s . Modelling the car as a particle, find the deceleration of the car.
- 3 A car accelerates uniformly while travelling on a straight road. The car passes two signposts 360 m apart. The car takes 15 s to travel from one signpost to the other. When passing the second signpost, it has speed 28 m s^{-1} . Find the speed of the car at the first signpost.
- 4 A cyclist is moving along a straight road from A to B with constant acceleration 0.5 m s^{-2} . Her speed at A is 3 m s^{-1} and it takes her 12 seconds to cycle from A to B . Find:
 - a her speed at B
 - b the distance from A to B .
- 5 A particle is moving along a straight line with constant acceleration from a point A to a point B , where $AB = 24 \text{ m}$. The particle takes 6 s to move from A to B and the speed of the particle at B is 5 m s^{-1} . Find:
 - a the speed of the particle at A
 - b the acceleration of the particle.
- 6 A particle moves in a straight line from a point A to a point B with constant deceleration 1.2 m s^{-2} . The particle takes 6 s to move from A to B . The speed of the particle at B is 2 m s^{-1} and the direction of motion of the particle has not changed. Find:
 - a the speed of the particle at A
 - b the distance from A to B .

- P** 7 A train, travelling on a straight track, is slowing down with constant deceleration 0.6 m s^{-2} . The train passes one signal with speed 72 km h^{-1} and a second signal 25 s later. Find:

Hint Convert the speeds into m s^{-1} before substituting.

- a the speed, in km h^{-1} , of the train as it passes the second signal
- b the distance between the signals.

- 8 A particle moves in a straight line from a point A to a point B with a constant deceleration of 4 m s^{-2} . At A the particle has speed 32 m s^{-1} and the particle comes to rest at B . Find:

- a the time taken for the particle to travel from A to B
- b the distance between A and B .

- E** 9 A skier travelling in a straight line up a hill experiences a constant deceleration. At the bottom of the hill, the skier has a speed of 16 m s^{-1} and, after moving up the hill for 40 s, he comes to rest. Find:

- a the deceleration of the skier (2 marks)
- b the distance from the bottom of the hill to the point where the skier comes to rest. (4 marks)

- E** 10 A particle is moving in a straight line with constant acceleration. The points A , B and C lie on this line. The particle moves from A through B to C . The speed of the particle at A is 2 m s^{-1} and the speed of the particle at B is 7 m s^{-1} . The particle takes 20 s to move from A to B .

- a Find the acceleration of the particle. (2 marks)

The speed of the particle at C is 11 m s^{-1} . Find:

- b the time taken for the particle to move from B to C (2 marks)
- c the distance between A and C . (3 marks)

- E** 11 A particle moves in a straight line from A to B with constant acceleration 1.5 m s^{-2} . It then moves along the same straight line from B to C with a different acceleration. The speed of the particle at A is 1 m s^{-1} and the speed of the particle at C is 43 m s^{-1} . The particle takes 12 s to move from A to B and 10 s to move from B to C . Find:

- a the speed of the particle at B (2 marks)
- b the acceleration of the particle as it moves from B to C (2 marks)
- c the distance from A to C . (3 marks)

- E/P** 12 A cyclist travels with constant acceleration $x \text{ m s}^{-2}$, in a straight line, from rest to 5 m s^{-1} in 20 s. She then decelerates from 5 m s^{-1} to rest with constant deceleration $\frac{1}{2}x \text{ m s}^{-2}$. Find:

- a the value of x (2 marks)
- b the total distance she travelled. (4 marks)

Problem-solving

You could sketch a velocity–time graph of the cyclist’s motion and use the area under the graph to find the total distance travelled.

- E/P** 13 A particle is moving with constant acceleration in a straight line. It passes through three points, A , B and C , with speeds 20 m s^{-1} , 30 m s^{-1} and 45 m s^{-1} respectively. The time taken to move from A to B is t_1 seconds and the time taken to move from B to C is t_2 seconds.

a Show that $\frac{t_1}{t_2} = \frac{2}{3}$. (3 marks)

Given also that the total time taken for the particle to move from A to C is 50 s:

b find the distance between A and B . (5 marks)

Challenge

A particle moves in a straight line from A to B with constant acceleration. The particle moves from A with speed 3 m s^{-1} . It reaches point B with speed 5 m s^{-1} t seconds later.

One second after the first particle leaves point A , a second particle also starts to move in a straight line from A to B with constant acceleration. Its speed at point A is 4 m s^{-1} and it reaches point B with speed 8 m s^{-1} at the same time as the first particle.

Find:

- a** the value of t
- b** the distance between A and B .

Problem-solving

The time taken for the second particle to travel from A to B is $(t - 1)$ seconds.

2.5 Constant acceleration formulae 2

You can use the formulae $v = u + at$ and $s = \left(\frac{u+v}{2}\right)t$ to work out three more formulae.

You can eliminate t from the formulae for constant acceleration.

$$t = \frac{v-u}{a} \quad \text{Rearrange the formula } v = u + at \text{ to make } t \text{ the subject.}$$

$$s = \left(\frac{u+v}{2}\right)\left(\frac{v-u}{a}\right) \quad \text{Substitute this expression for } t \text{ into } s = \left(\frac{u+v}{2}\right)t.$$

$$2as = v^2 - u^2$$

$$\bullet \quad v^2 = u^2 + 2as \quad \text{Multiply out the brackets and rearrange.}$$

You can also eliminate v from the formulae for constant acceleration.

$$s = \left(\frac{u+u+at}{2}\right)t \quad \text{Substitute } v = u + at \text{ into } s = \left(\frac{u+v}{2}\right)t.$$

$$s = \left(\frac{2u}{2} + \frac{at}{2}\right)t$$

$$s = \left(u + \frac{1}{2}at\right)t \quad \text{Multiply out the brackets and rearrange.}$$

$$\bullet \quad s = ut + \frac{1}{2}at^2$$

Finally, you can eliminate u by substituting into this formula:

$$s = (v - at)t + \frac{1}{2}at^2 \quad \text{Substitute } u = v - at \text{ into } s = ut + \frac{1}{2}at^2.$$

- $s = vt - \frac{1}{2}at^2$

■ You need to be able to recall and use the five formulae for solving problems about particles moving in a straight line with constant acceleration.

- $v = u + at$

- $s = \left(\frac{u + v}{2}\right)t$

- $v^2 = u^2 + 2as$

- $s = ut + \frac{1}{2}at^2$

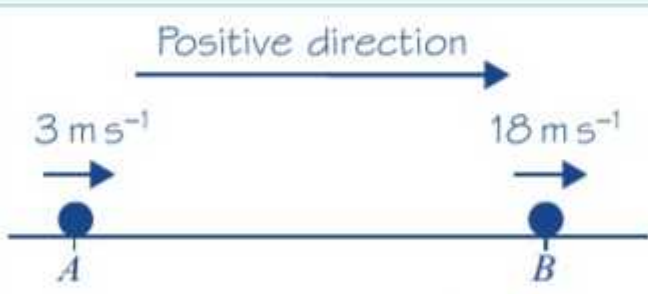
- $s = vt - \frac{1}{2}at^2$

Watch out These five formulae are sometimes referred to as the **kinematics** formulae or **suvat** formulae.

Example 8

SKILLS PROBLEM-SOLVING

A particle is moving along a straight line from A to B with constant acceleration 5 m s^{-2} . The velocity of the particle at A is 3 m s^{-1} in the direction $\overrightarrow{AB}$. The velocity of the particle at B is 18 m s^{-1} in the same direction. Find the distance from A to B .



$a = 5, u = 3, v = 18, s = ?$
 $v^2 = u^2 + 2as$
 $18^2 = 3^2 + 2 \times 5 \times s$
 $324 = 9 + 10s$
 $s = \frac{324 - 9}{10} = 31.5$
 $AB = 31.5 \text{ m}$

Write down the values you know and the values you need to find. This will help you choose the correct formula.

t is not involved so choose the formula that does not have t in it.

Substitute in the values you are given and solve the equation for s . This gives the distance you were asked to find.

Example 9

9

SKILLS

PROBLEM-SOLVING

A particle is moving in a straight horizontal line with constant deceleration 4 m s^{-2} . At time $t = 0$ the particle passes through a point O with velocity 13 m s^{-1} travelling toward a point A , where $OA = 20 \text{ m}$. Find:

- the times when the particle passes through A
- the value of t when the particle returns to O .



a $a = -4$, $u = 13$, $s = 20$, $t = ?$

$$s = ut + \frac{1}{2}at^2$$

$$20 = 13t + \frac{1}{2} \times (-4)t^2$$

$$= 13t - 2t^2$$

$$2t^2 - 13t + 20 = 0$$

$$(2t - 5)(t - 4) = 0$$

$$t = \frac{5}{2}, \text{ or } t = 4$$

The particle moves through A twice, $2\frac{1}{2}$ seconds and 4 seconds after moving through O .

- b The particle returns to O when $s = 0$.

$$s = 0, u = 13, a = -4, t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$0 = 13t - 2t^2$$

$$= t(13 - 2t)$$

$$t = 0, \text{ or } t = \frac{13}{2}$$

The particle returns to O 6.5 seconds after it first passed through O .

The particle is decelerating so the value of a is negative.

You are told the values of a , u and s and asked to find t . You are given no information about v and are not asked to find it so you choose the formula without v .

Problem-solving

When you use $s = ut + \frac{1}{2}at^2$ with an unknown value of t you obtain a quadratic equation in t . You can solve this equation by factorising, or using the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

There are two answers. Both are correct. The particle moves from O to A , goes beyond A and then turns around and returns to A .

When the particle returns to O , its displacement (distance) from O is zero.

The first solution ($t = 0$) represents the starting position of the particle. The other solution ($t = \frac{13}{2}$) tells you when the particle returns to O .

Example 10

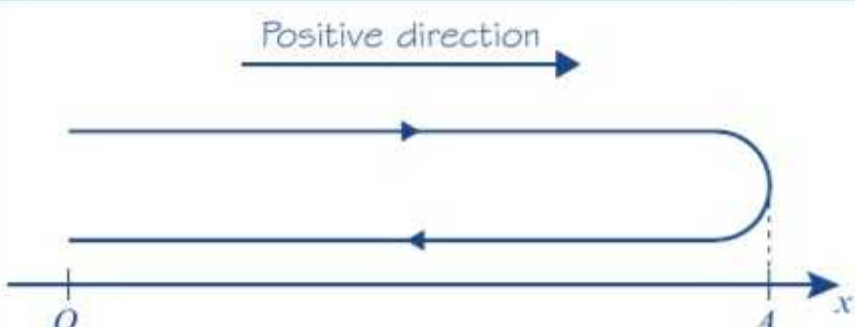
10

SKILLS

PROBLEM-SOLVING

A particle P is moving on the x -axis with constant deceleration 2.5 m s^{-2} . At time $t = 0$, the particle P passes through the origin O , moving in the positive direction of x with velocity 15 m s^{-1} . Find:

- the time between the instant when P first passes through O and the instant when it returns to O
- the total distance travelled by P during this time.



a $a = -2.5$, $u = 15$, $s = 0$, $t = ?$

$$s = ut + \frac{1}{2}at^2$$

$$0 = 15t + \frac{1}{2} \times (-2.5) \times t^2$$

$$0 = 60t - 5t^2$$

$$= 5t(12 - t)$$

$$t = 0, t = 12$$

The particle P returns to O after 12 s.

b $a = -2.5$, $u = 15$, $v = 0$, $s = ?$

$$v^2 = u^2 + 2as$$

$$0^2 = 15^2 + 2 \times (-2.5) \times s$$

$$5s = 15^2 = 225$$

$$s = \frac{225}{5} = 45$$

The distance $OA = 45$ m.

The total distance travelled by P is 2×45 m = 90 m.

Problem-solving

Before you start, draw a sketch so you can see what is happening. The particle moves through O with a positive velocity. As it is decelerating it slows down and will eventually have zero velocity at a point A , which you don't yet know. As the particle is still decelerating, its velocity becomes negative, so the particle changes direction and returns to O .

When the particle returns to O , its displacement (distance) from O is zero.

Multiply by 4 to get whole-number coefficients.

At the furthest point from O , labelled A in the diagram, the particle changes direction. At that point, for an instant, the particle has zero velocity.

In the 12 s the particle has been moving it has travelled to A and back. The total distance travelled is twice the distance OA .

Exercise 2E

SKILLS PROBLEM-SOLVING

- A particle is moving in a straight line with constant acceleration 2.5 m s^{-2} . It passes a point A with velocity 3 m s^{-1} and later passes through a point B , where $AB = 8 \text{ m}$. Find the velocity of the particle as it passes through B .
- A car is accelerating at a constant rate along a straight horizontal road. Travelling at 8 m s^{-1} , it passes a lamp post and 6 s later it passes a sign. The distance between the lamp post and the sign is 60 m. Find the acceleration of the car.
- A cyclist travelling at 12 m s^{-1} applies her brakes and comes to rest after travelling 36 m in a straight line. Assuming that the brakes cause the cyclist to decelerate uniformly, find the deceleration.
- A train is moving along a straight horizontal track with constant acceleration. The train passes a signal with a velocity of 54 km h^{-1} and a second signal with a velocity of 72 km h^{-1} . The distance between the two signals is 500 m. Find, in m s^{-2} , the acceleration of the train.
- A particle moves along a straight line, with constant acceleration, from a point A to a point B where $AB = 48 \text{ m}$. At A the particle has velocity 4 m s^{-1} and at B it has velocity 16 m s^{-1} . Find:
 - the acceleration of the particle
 - the time the particle takes to move from A to B .

- 6 A particle moves along a straight line with constant acceleration 3 m s^{-2} . The particle moves 38 m in 4 s. Find:
- the initial speed of the particle
 - the final speed of the particle.
- 7 The driver of a car is travelling at 18 m s^{-1} along a straight road when she sees an obstruction ahead. She applies the brakes and the brakes cause the car to slow down to rest with a constant deceleration of 3 m s^{-2} . Find:
- the distance travelled as the car decelerates
 - the time it takes for the car to decelerate from 18 m s^{-1} to rest.
- 8 A stone is sliding across a frozen lake in a straight line. The initial speed of the stone is 12 m s^{-1} . The friction between the stone and the ice causes the stone to slow down at a constant rate of 0.8 m s^{-2} . Find:
- the distance travelled by the stone before coming to rest
 - the speed of the stone at the instant when it has travelled half of this distance.
- 9 A particle is moving along a straight line OA with constant acceleration 2.5 m s^{-2} . At time $t = 0$, the particle passes through O with speed 8 m s^{-1} and is moving in the direction OA . The distance OA is 40 m. Find:
- the time taken for the particle to move from O to A
 - the speed of the particle at A . Give your answers to one decimal place.
- 10 A particle travels with uniform deceleration 2 m s^{-2} in a horizontal line. The points A and B lie on the line and $AB = 32 \text{ m}$. At time $t = 0$, the particle passes through A with speed 12 m s^{-1} in the direction $\overrightarrow{AB}$. Find:
- the values of t when the particle is at B
 - the speed of the particle for each of these values of t .

- E/P** 11 A particle is moving along the x -axis with constant deceleration 5 m s^{-2} . At time $t = 0$, the particle passes through the origin O with velocity 12 m s^{-1} in the positive direction. At time t seconds, the particle passes through the point A with x -coordinate 8. Find:

Problem-solving

The particle will pass through A twice. Use $s = ut + \frac{1}{2}at^2$ to set up and solve a quadratic equation.

- the values of t (3 marks)
 - the speed of the particle as it passes through the point with x -coordinate -8 . (3 marks)
- E** 12 A particle P is moving on the x -axis with constant deceleration 4 m s^{-2} . At time $t = 0$, P passes through the origin O with speed 14 m s^{-1} in the positive direction. The point A lies on the axis and $OA = 22.5 \text{ m}$. Find:
- the difference between the times when P passes through A (4 marks)
 - the total distance travelled by P during the interval between these times. (3 marks)
- E/P** 13 A car is travelling along a straight horizontal road with constant acceleration. The car passes over three consecutive points A , B and C where $AB = 100 \text{ m}$ and $BC = 300 \text{ m}$. The speed of the car at B is 14 m s^{-1} and the speed of the car at C is 20 m s^{-1} . Find:
- the acceleration of the car (3 marks)
 - the time take for the car to travel from A to C . (3 marks)

- E/P** 14 Two particles P and Q are moving along the same straight horizontal line with constant accelerations 2 m s^{-2} and 3.6 m s^{-2} respectively. At time $t = 0$, P passes through a point A with speed 4 m s^{-1} . One second later Q passes through A with speed 3 m s^{-1} , moving in the same direction as P .
- Write down expressions for the displacements of P and Q from A , in terms of t , where t seconds is the time after P has passed through A . **(2 marks)**
 - Find the value of t where the particles meet. **(3 marks)**
 - Find the distance of A from the point where the particles meet. **(3 marks)**
- Problem-solving**
When P and Q meet, their displacements from A are equal.
- E/P** 15 In an orienteering competition, a competitor moves in a straight line past three checkpoints, P , Q and R , where $PQ = 2.4 \text{ km}$ and $QR = 11.5 \text{ km}$. The competitor is modelled as a particle moving with constant acceleration. She takes 1 hour to travel from P to Q and 1.5 hours to travel from Q to R . Find:
- the acceleration of the competitor
 - her speed at the instant she passes P . **(7 marks)**

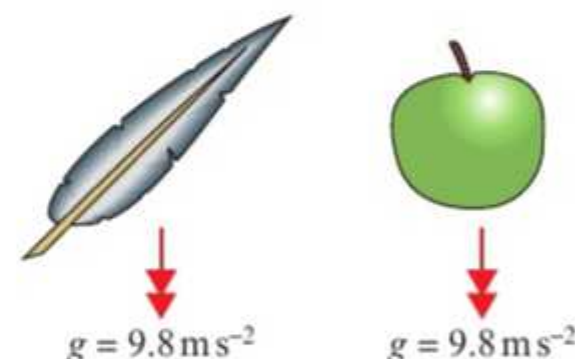
2.6 Vertical motion under gravity

You can use the formulae for constant acceleration to model an object moving vertically under gravity.

- The force of gravity causes all objects to accelerate toward the Earth. If you ignore the effects of air resistance, this acceleration is constant. It does not depend on the mass of the object.

As the force of gravity does not depend on mass, this means that in a vacuum an apple and a feather would both accelerate downward at the same rate.

On Earth, the acceleration due to gravity is represented by the letter g and is approximately 9.8 m s^{-2} .



The actual value of the acceleration can vary by very small amounts in different places due to the changing radius of the earth and height above sea level.

- An object moving vertically under gravity can be modelled as a particle with a constant downward acceleration of $g = 9.8 \text{ m s}^{-2}$.

When solving problems about vertical motion you can choose the positive direction to be either upward or downward. Acceleration due to gravity is always downward, so if the positive direction is upward then $g = -9.8 \text{ m s}^{-2}$.

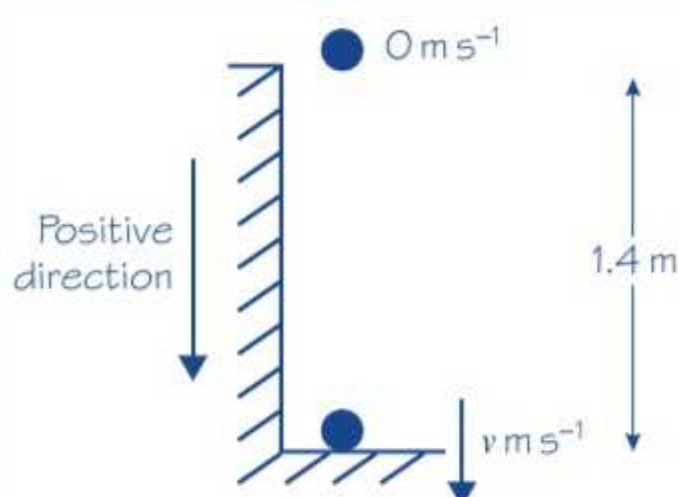
Watch out In mechanics questions you will always use $g = 9.8 \text{ m s}^{-2}$ unless a question specifies otherwise. However, if a different value of g is specified (e.g. $g = 10 \text{ m s}^{-2}$ or $g = 9.81 \text{ m s}^{-2}$) the degree of accuracy in your answer should be chosen to be consistent with this value.

Notation The total time that an object is in motion from the time it is projected (thrown) upward to the time it hits the ground is called the **time of flight**. The initial speed is sometimes called the **speed of projection**.

Example 11**SKILLS** PROBLEM-SOLVING

A book falls off the top shelf of a bookcase. The shelf is 1.4 m above a wooden floor. Find:

- the time the book takes to reach the floor
- the speed with which the book strikes the floor.



a

$$s = 1.4$$

$$a = +9.8$$

$$u = 0$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$1.4 = 0 + \frac{1}{2} \times 9.8 \times t^2$$

$$t^2 = \frac{1.4}{4.9} = 0.2857...$$

$$t = \sqrt{0.2857...} = 0.5345...$$

The time taken for the book to reach the floor is 0.53 s, to two significant figures.

b

$$s = 1.4$$

$$a = 9.8$$

$$u = 0$$

$$v = ?$$

$$v^2 = u^2 + 2as$$

$$= 0^2 + 2 \times 9.8 \times 1.4 = 27.44$$

$$v = \sqrt{27.44} = 5.238... \approx 5.2$$

The book hits the floor with speed 5.2 m s⁻¹, to two significant figures.

Model the book as a particle moving in a straight line with a constant acceleration of magnitude 9.8 m s⁻².

As the book is moving downward throughout its motion, it is sensible to take the downward direction as positive.

You have taken the downward direction as positive and gravity acts downward. Here the acceleration is positive.

Assume the book has an initial speed of zero.

Choose the formula without v .

Solve the equation for t^2 and use your calculator to find the positive square root.

Give the answer to two significant figures to be consistent with the degree of accuracy used for the value of g .

Choose the formula without t .

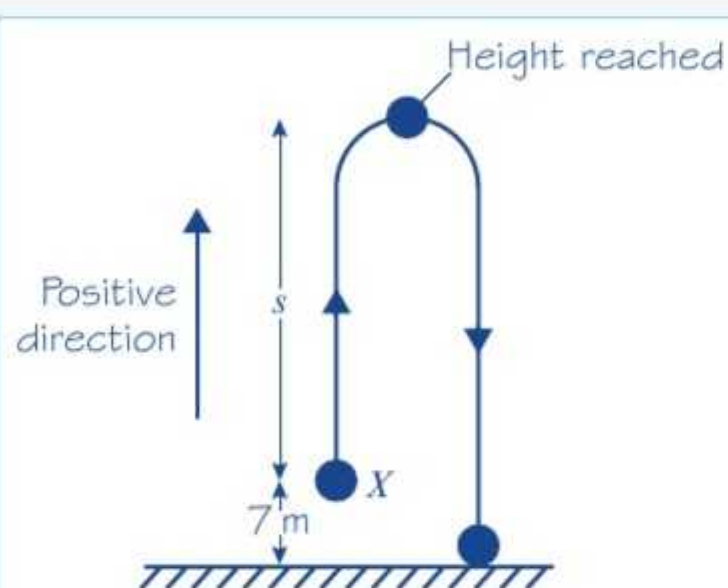
Use unrounded values in your calculations, but give your final answer correct to two significant figures.

Example 12

SKILLS PROBLEM-SOLVING

A ball is projected vertically upward, from a point X which is 7 m above the ground, with speed 21 m s^{-1} . Find:

- the greatest height above the ground reached by the ball
- the time of flight of the ball.



a

$$u = 21$$

$$v = 0$$

$$a = -9.8$$

$$s = ?$$

$$v^2 = u^2 + 2as$$

$$0^2 = 21^2 + 2 \times (-9.8) \times s = 441 - 19.6s$$

$$s = \frac{441}{19.6} = 22.5$$

$$(22.5 + 7) \text{ m} = 29.5 \text{ m}$$

Greatest height is 30 m (2 s.f.)

b

$$s = -7$$

$$u = 21$$

$$a = -9.8$$

$$t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$-7 = 21t - 4.9t^2$$

$$4.9t^2 - 21t - 7 = 0$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-21) \pm \sqrt{(-21)^2 - 4 \times 4.9 \times (-7)}}{2 \times 4.9}$$

$$= \frac{21 \pm \sqrt{578.2}}{9.8} \approx \frac{21 \pm 24.046}{9.8}$$

$$t \approx 4.5965,$$

$$\text{or } t \approx -0.3108$$

Time of flight is 4.6 s (2 s.f.)

Problem-solving

In this sketch the upward and downward motions have been sketched side by side. In reality they would be on top of one another, but drawing them separately makes it easier to see what is going on. Remember that X is 7 m above the ground, so mark this height on your sketch.

At its highest point, the ball is turning around. For an instant, it is neither going up nor down, so its velocity is zero.

22.5 m is the distance the ball has moved above X , but X is 7 m above the ground. You must add on another 7 m to get the greatest height above the ground reached by the ball.

The time of flight is the total time that the ball is in motion from the time that it is projected to the time that it stops moving. Here the ball will stop when it hits the ground. The point where the ball hits the ground is 7 m below the point from which it was projected so $s = -7$.

Online Use your calculator to check solutions to quadratic equations quickly.



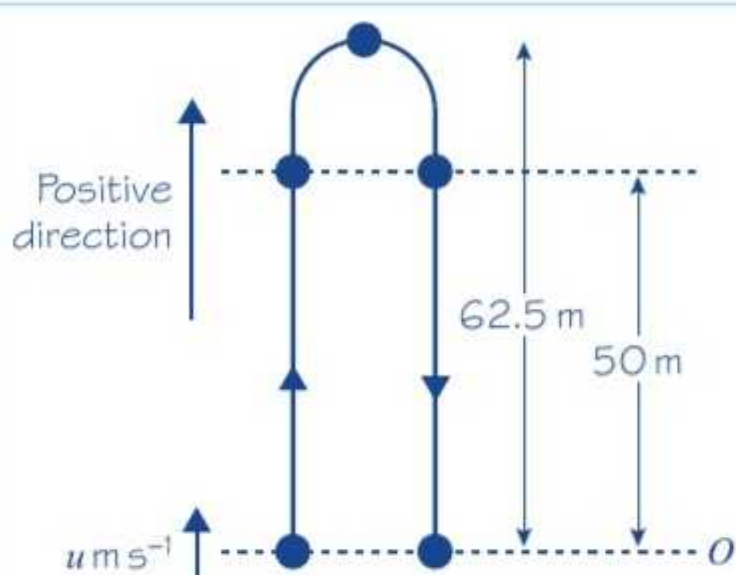
Rearrange the equation and use the quadratic formula.

Take the positive answer and round to two significant figures.

Example 13**SKILLS** PROBLEM-SOLVING

A particle is projected vertically upward from a point O with speed $u \text{ m s}^{-1}$. The greatest height reached by the particle is 62.5 m above O . Find:

- the value of u
- the total time for which the particle is 50 m or more above O .



a $v = 0$
 $s = 62.5$
 $a = -9.8$
 $u = ?$
 $v^2 = u^2 + 2as$
 $0^2 = u^2 + 2 \times (-9.8) \times 62.5$
 $u^2 = 1225$
 $u = \sqrt{1225} = 35 \text{ m s}^{-1}$

b $s = 50$
 $u = 35$
 $a = -9.8$
 $t = ?$
 $s = ut + \frac{1}{2}at^2$
 $50 = 35t - 4.9t^2$
 $4.9t^2 - 35t + 50 = 0$
 $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $= \frac{35 \pm \sqrt{(35)^2 - 4 \times 4.9 \times 50}}{9.8}$
 $= \frac{35 \pm \sqrt{245}}{9.8} \approx \frac{35 \pm 15.6525}{9.8}$

$t \approx 5.1686\dots$, or $t \approx 1.9742\dots$
 $(5.1686\dots) - (1.9742\dots) \approx 3.194$
 Particle is 50 m or more above O for 3.2 s
 (2 s.f.)

The particle will pass through the point 50 m above O twice: once on the way up and once on the way down.

There is no t , so you choose the formula without t .

In this part, you obtain an exact answer, so there is no need for approximation.

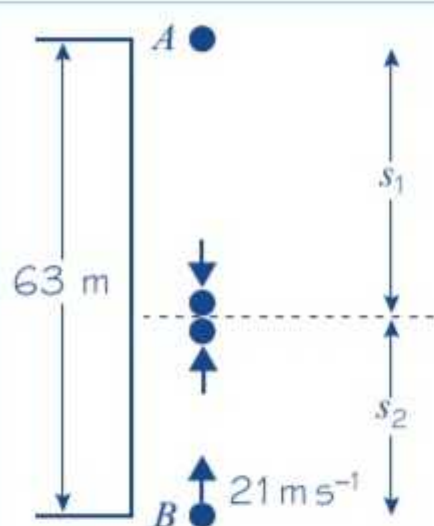
Two values of t need to be found: one on the way up and one on the way down.

Write the equation in the form $ax^2 + bx + c = 0$ and use the quadratic formula.

Between these two times the particle is always more than 50 m above O . You find the total time for which the particle is 50 m or more above O by finding the difference of these two values.

Example 14
SKILLS PROBLEM-SOLVING

A ball A falls vertically from rest at the top of a tower 63 m high. At the same time as A begins to fall, another ball B is projected vertically upward from the bottom of the tower with speed 21 m s^{-1} . The balls collide. Find the distance of the point where the balls collide from the bottom of the tower.



For A , the motion is downward

$$u = 0$$

$$a = 9.8$$

$$s = ut + \frac{1}{2}at^2$$

$$s_1 = 4.9t^2$$

For B , the motion is upward

$$u = 21$$

$$a = -9.8$$

$$s = ut + \frac{1}{2}at^2$$

$$s_2 = 21t - 4.9t^2$$

The height of the tower is 63 m.

$$s_1 + s_2 = 63$$

$$4.9t^2 + (21t - 4.9t^2) = 63$$

$$21t = 63$$

$$t = 3$$

$$s_2 = 21t - 4.9t^2$$

$$= 21 \times 3 - 4.9 \times 3^2 = 18.9$$

The balls collide 19 m from the bottom of the tower, to two significant figures.

Problem-solving

You must take special care with problems where objects are moving in different directions. Here A is moving downward and you will take the acceleration due to gravity as positive. However, B is moving upward so for B the acceleration due to gravity is negative.

You cannot find s_1 at this stage. You have to express it in terms of t .

As B is moving upward, the acceleration due to gravity is negative.

You now have expressions for s_1 and s_2 in terms of t .

Adding together the two distances gives the height of the tower. You can write this as an equation in t .

You have found t but you were asked for the distance from the bottom of the tower. Substitute your value for t into your equation for s_2 .

Exercise 2F
SKILLS PROBLEM-SOLVING

- A cliff diver jumps from a point 28 m above the surface of the water. Modelling the diver as a particle moving freely under gravity with initial velocity 0, find:
 - the time taken for the diver to hit the water
 - the speed of the diver when he hits the water.
- A particle is projected vertically upward with speed 20 m s^{-1} from a point on the ground. Find the time of flight of the particle.

3 VECTORS IN MECHANICS

2.1
2.2

Learning objectives

After completing this chapter you should be able to:

- Use vectors in two dimensions → pages 40–42
- Calculate the magnitude and direction of a vector → pages 43–44
- Understand and use position vectors → pages 46–49
- Understand vector magnitude and use vectors in speed and distance calculations → pages 46–49
- Use vectors to solve problems in context → pages 46–49

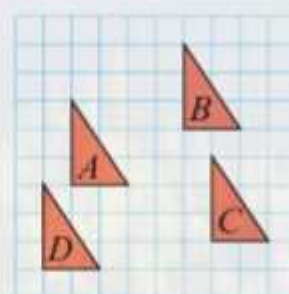
Prior knowledge check

1 Write the column vector for the translation of shape:

a A to B

b A to C

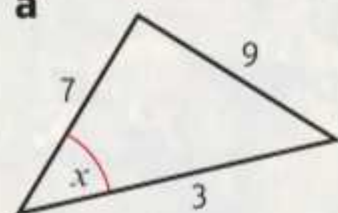
c A to D



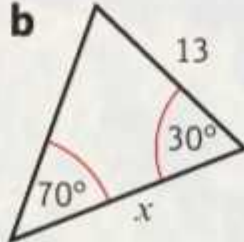
← International GCSE Mathematics

2 Find x to one decimal place:

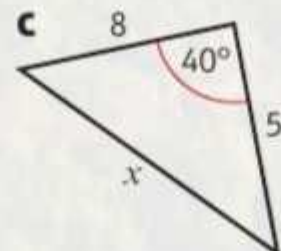
a



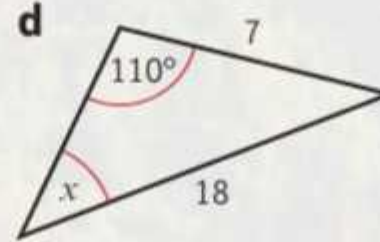
b



c

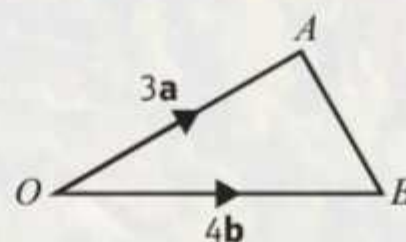


d



← International GCSE Mathematics

3 Given that $\vec{OA} = 3\mathbf{a}$ and $\vec{OB} = 4\mathbf{b}$, find vector $\vec{AB}$.
Hence find the magnitude of $\vec{AB}$.



← International GCSE Mathematics

Pilots use vector addition to work out the resultant vector for their speed and heading when an aeroplane encounters a strong cross-wind. Engineers also use vectors to work out the resultant forces acting on buildings in construction.



3.1 Working with vectors

- A vector is a quantity which has both magnitude and direction.

These are examples of **vector** quantities:

Quantity	Description	Unit
Displacement	distance in a particular direction	metre (m)
Velocity	rate of change of displacement	metres per second (m s^{-1})
Acceleration	rate of change of velocity	metres per second per second (m s^{-2})
Force/weight	described by magnitude, direction and point of application	newton (N)

- A scalar quantity has magnitude only.

These are examples of **scalar** quantities:

Quantity	Description	Unit
Distance	measure of length	metre (m)
Speed	measure of how quickly a body moves	metres per second (m s^{-1})
Time	measure of ongoing events taking place	second (s)
Mass	measure of the quantity of matter contained in an object	kilogram (kg)

Scalar quantities are always **positive**. When considering motion in a straight line (1-dimensional), **vector** quantities can be **positive** or **negative**.

Example 1

1

SKILLS

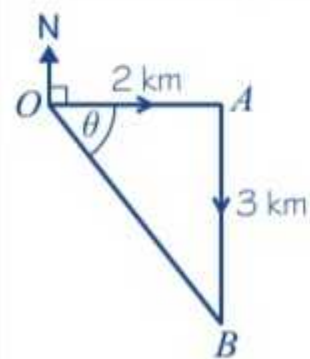
PROBLEM-SOLVING

A girl walks 2 km due east from a fixed point O to A , and then 3 km due south from A to B . Find the total distance walked and describe the displacement of B from O .

The distance the girl has walked is

$$2 \text{ km} + 3 \text{ km} = 5 \text{ km}$$

Representing the girl's journey on a diagram:



$$OB^2 = OA^2 + AB^2$$

$$OB^2 = 4 + 9 = 13$$

So the distance OB is 3.61 km (3 s.f.)

$$\tan \angle AOB = \frac{3}{2}$$

$$\angle AOB = 56.3^\circ$$

B is 3.61 km from O on a bearing of 146° .

Note that the distance of B from O is not the same distance as the girl has walked.

A diagram is essential and gives a clear representation of the displacement OB .

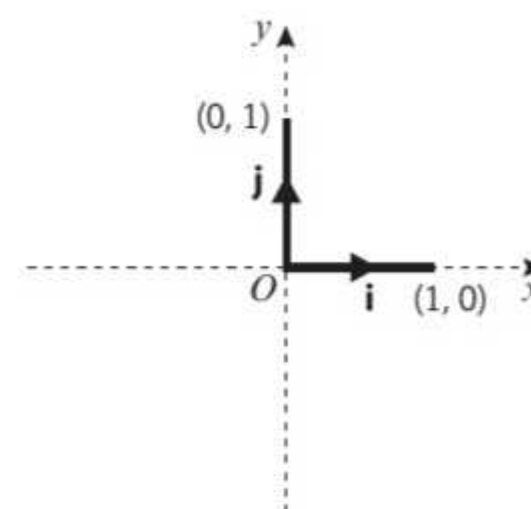
Using Pythagoras' theorem

If the question does not ask you to specify the direction in a particular form, then any method is acceptable.

A three figure bearing is always measured clockwise from north.

You can describe vectors using **i** and **j** notation.

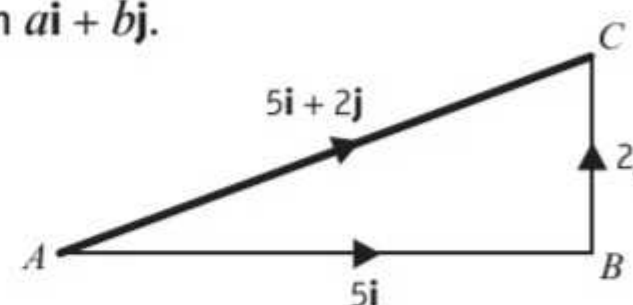
- A unit vector is a vector of length 1. The unit vectors along the Cartesian axes are usually denoted by **i** and **j** respectively.



- You can write any two-dimensional vector in the form $a\mathbf{i} + b\mathbf{j}$.

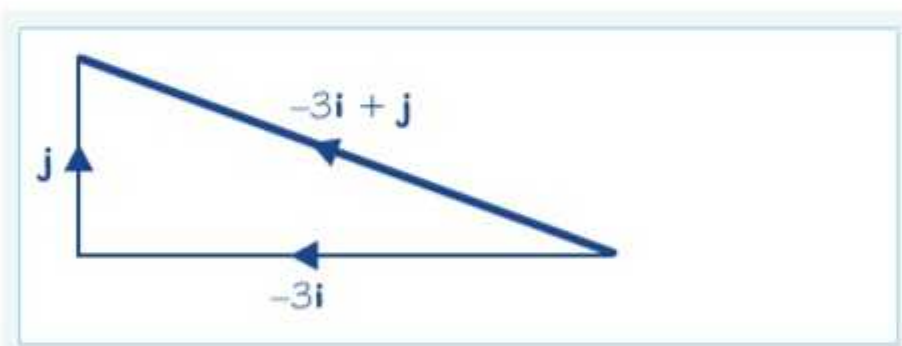
By the triangle law of addition:

$$\begin{aligned}\vec{AC} &= \vec{AB} + \vec{BC} \\ &= 5\mathbf{i} + 2\mathbf{j}\end{aligned}$$



Example 2

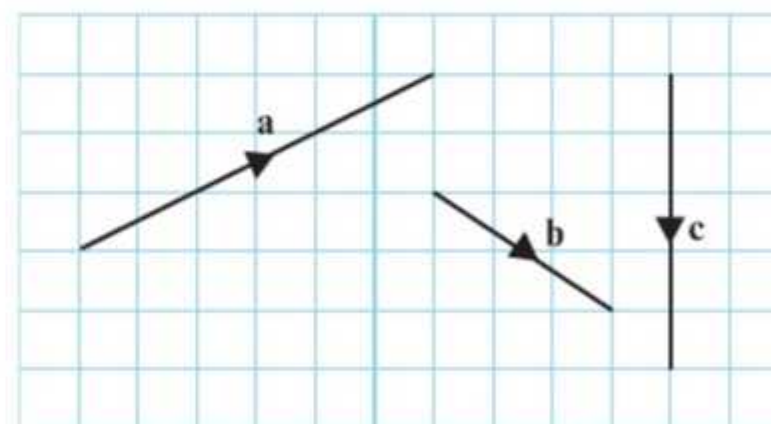
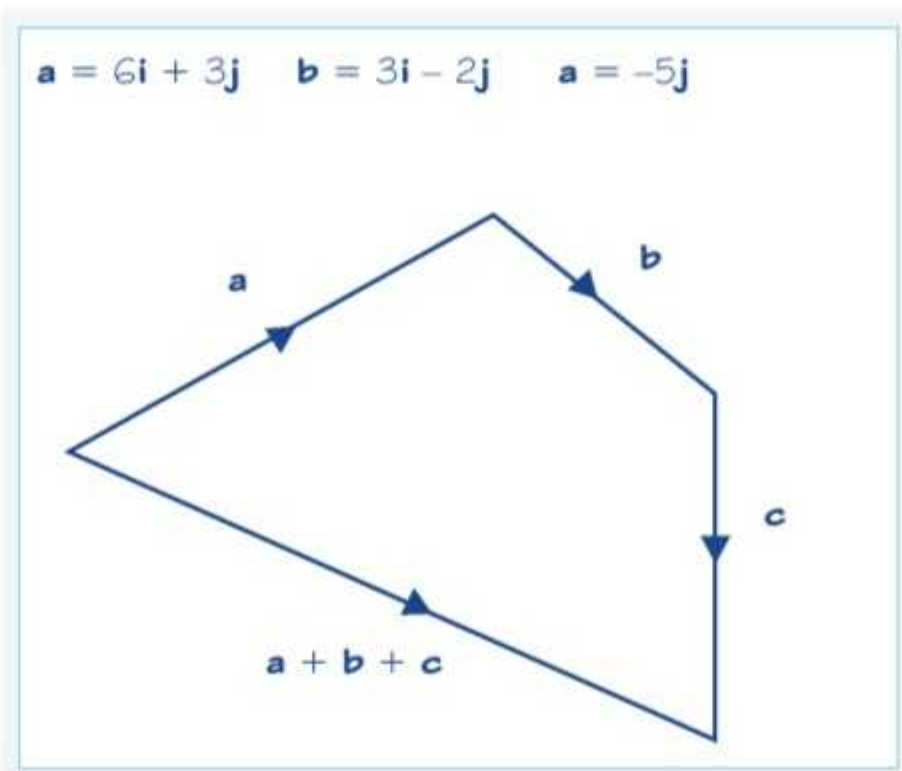
Draw a diagram to represent the vector $-3\mathbf{i} + \mathbf{j}$.



3 units in the direction of the unit vector $-\mathbf{i}$ and 1 unit in the direction of the unit vector $\mathbf{j}$.

Example 3

The diagram shows vectors **a**, **b** and **c**. Draw a diagram to illustrate the vector addition $\mathbf{a} + \mathbf{b} + \mathbf{c}$.



Add the vectors together and collect up **i** and **j** components.

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = 6\mathbf{i} + 3\mathbf{j} + 3\mathbf{i} - 2\mathbf{j} - 5\mathbf{j} = 9\mathbf{i} - 4\mathbf{j}$$

Exercise

3A

SKILLS

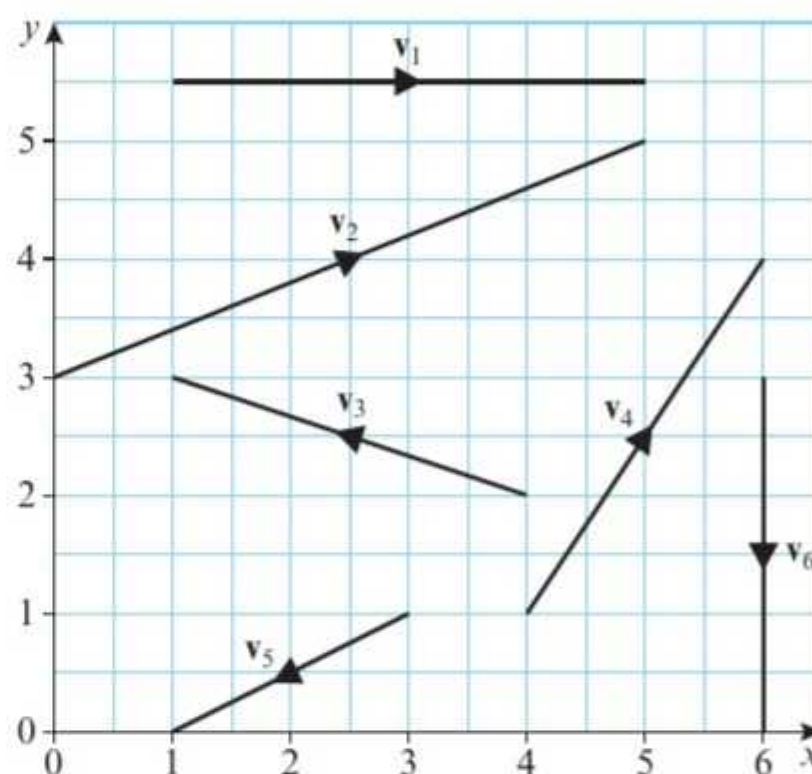
PROBLEM-SOLVING

- 1 A bird flies 5 km due north and then 7 km due east. How far is the bird from its original position, and in what direction?
- 2 A girl cycles 4 km due west and then 6 km due north. Calculate the total distance she has cycled, and her displacement from her starting point.

- 3 a Express the vectors $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_3$, $\mathbf{v}_4$, $\mathbf{v}_5$ and $\mathbf{v}_6$ using the $\mathbf{i}$ and $\mathbf{j}$ notation.

b Find:

- i $\mathbf{v}_1 + \mathbf{v}_2$
- ii $\mathbf{v}_4 + \mathbf{v}_5$
- iii $\mathbf{v}_6 + \mathbf{v}_1 + \mathbf{v}_5$



- 4 Find the magnitude of each of these vectors:

a $3\mathbf{i} + 4\mathbf{j}$	b $6\mathbf{i} - 8\mathbf{j}$	c $5\mathbf{i} + 12\mathbf{j}$	d $2\mathbf{i} + 4\mathbf{j}$
e $3\mathbf{i} - 5\mathbf{j}$	f $4\mathbf{i} + 7\mathbf{j}$	g $-3\mathbf{i} + 5\mathbf{j}$	h $-4\mathbf{i} - \mathbf{j}$
- 5 $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$, $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$ and $\mathbf{c} = 5\mathbf{i} - \mathbf{j}$. Find the exact value of the magnitude of:

a $\mathbf{a} + \mathbf{b}$	b $2\mathbf{a} - \mathbf{c}$	c $3\mathbf{b} - 2\mathbf{c}$
-----------------------------	------------------------------	-------------------------------
- 6 Find the angle that each of these vectors makes with the positive x -axis:

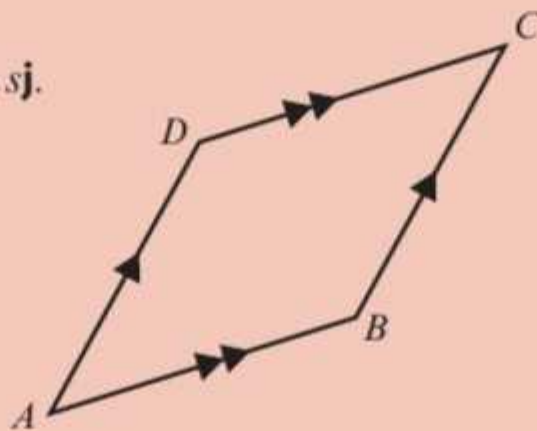
a $3\mathbf{i} + 4\mathbf{j}$	b $6\mathbf{i} - 8\mathbf{j}$	c $5\mathbf{i} + 12\mathbf{j}$	d $2\mathbf{i} + 4\mathbf{j}$
-------------------------------	-------------------------------	--------------------------------	-------------------------------
- 7 Find the angle that each of these vectors makes with the positive y -axis:

a $3\mathbf{i} - 5\mathbf{j}$	b $4\mathbf{i} + 7\mathbf{j}$	c $-3\mathbf{i} + 5\mathbf{j}$	d $-4\mathbf{i} - \mathbf{j}$
-------------------------------	-------------------------------	--------------------------------	-------------------------------

Challenge

In the diagram, $\overrightarrow{AB} = p\mathbf{i} + q\mathbf{j}$ and $\overrightarrow{AD} = r\mathbf{i} + s\mathbf{j}$. $ABCD$ is a parallelogram.

Prove that the area of $ABCD$ is $ps - qr$.



Problem-solving

Draw the parallelogram on a coordinate grid, and choose a position for the origin that will simplify your calculations.

3.2 Solving problems with vectors written using **i** and **j** notation

- When vectors are written in terms of the unit vectors **i** and **j** you can add them together by adding the terms in **i** and **j** separately. You can subtract vectors in a similar way.

Example 4

Given $\mathbf{p} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{q} = 5\mathbf{i} + \mathbf{j}$, find $\mathbf{p} + \mathbf{q}$ in terms of **i** and **j**.

$$\begin{aligned}\mathbf{p} + \mathbf{q} &= (2\mathbf{i} + 3\mathbf{j}) + (5\mathbf{i} + \mathbf{j}) \\ &= (2\mathbf{i} + 5\mathbf{i}) + (3\mathbf{j} + \mathbf{j}) \\ &= 7\mathbf{i} + 4\mathbf{j}\end{aligned}$$

Rearrange with the **i** terms together and the **j** terms together.

Simplify the answer.

Example 5

Given $\mathbf{a} = 5\mathbf{i} + 2\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j}$, find $2\mathbf{a} - \mathbf{b}$ in terms of **i** and **j**.

$$\begin{aligned}2\mathbf{a} &= 2(5\mathbf{i} + 2\mathbf{j}) \\ &= 10\mathbf{i} + 4\mathbf{j} \\ 2\mathbf{a} - \mathbf{b} &= (10\mathbf{i} + 4\mathbf{j}) - (3\mathbf{i} - 4\mathbf{j}) \\ &= (10\mathbf{i} - 3\mathbf{i}) + (4\mathbf{j} - (-4\mathbf{j})) \\ &= 7\mathbf{i} + 8\mathbf{j}\end{aligned}$$

When you double **a**, the **i** term and the **j** term are both doubled – just the same as when you multiply out a bracket in algebra.

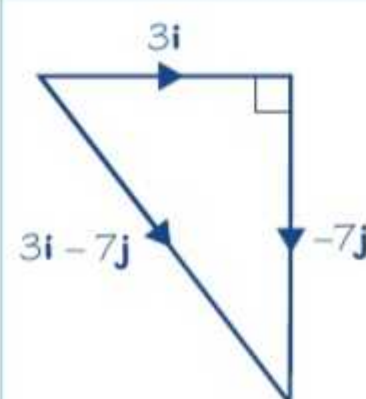
Rearrange with the **i** terms together and the **j** terms together.

Simplify the answer but take care when subtracting a negative term.

- When a vector is written in terms of the unit vectors **i** and **j** you can find the magnitude using Pythagoras' Theorem. The magnitude of vector **a** is written $|\mathbf{a}|$.

Example 6

Given $\mathbf{v} = 3\mathbf{i} - 7\mathbf{j}$, find the magnitude of **v**.



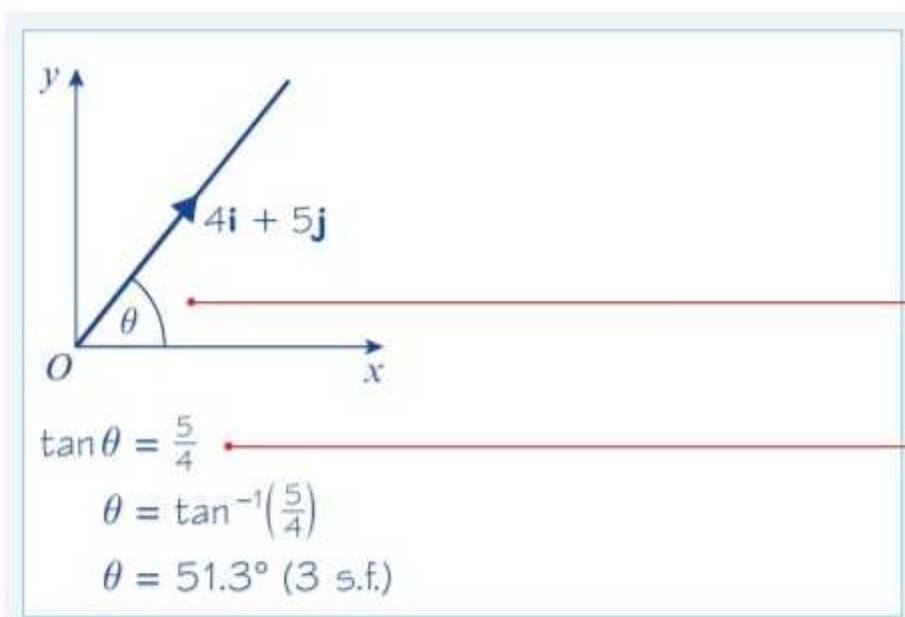
$$\begin{aligned}|\mathbf{v}| &= \sqrt{3^2 + (-7)^2} \\ &= \sqrt{58} \\ &= 7.62 \text{ (3 s.f.)}\end{aligned}$$

Because **i** and **j** are perpendicular vectors, you have a right-angled triangle.

$|\mathbf{v}|$ means the magnitude of **v**.

Example 7

Find the angle between the vector $4\mathbf{i} + 5\mathbf{j}$ and the positive x -axis.



Hint Always draw a diagram first.

Identify the angle you need to find.

We have a right-angled triangle, so finding the angle is straightforward.

Example 8

Given that $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}$ and $\mathbf{q} = 2\mathbf{i} - 6\mathbf{j}$, find λ if $\mathbf{p} + \lambda\mathbf{q}$ is parallel to the vector:

a $\mathbf{j}$ **b** $\mathbf{i} + \mathbf{j}$

a $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}, \mathbf{q} = 2\mathbf{i} - 6\mathbf{j}$
 $\mathbf{p} + \lambda\mathbf{q} = 4\mathbf{i} + 3\mathbf{j} + \lambda(2\mathbf{i} - 6\mathbf{j})$
 $\quad = 4\mathbf{i} + 2\lambda\mathbf{i} + 3\mathbf{j} - 6\lambda\mathbf{j}$
 $4\mathbf{i} + 2\lambda\mathbf{i} = \mathbf{0}$
 $2\lambda\mathbf{i} = -4\mathbf{i}$
 $\Rightarrow \lambda = -2$

b $\mathbf{p} = 4\mathbf{i} + 3\mathbf{j}$
 $\mathbf{q} = 2\mathbf{i} - 6\mathbf{j} \Rightarrow \lambda\mathbf{q} = 2\lambda\mathbf{i} - 6\lambda\mathbf{j}$
 $\mathbf{p} + \lambda\mathbf{q} = (4\mathbf{i} + 3\mathbf{j}) + (2\lambda\mathbf{i} - 6\lambda\mathbf{j})$
 $\quad = \mathbf{i}(4 + 2\lambda) + \mathbf{j}(3 - 6\lambda)$
 $4 + 2\lambda = 3 - 6\lambda$
 $\Rightarrow \lambda = -\frac{1}{8}$

Add together $\mathbf{p}$ and $\lambda\mathbf{q}$.

$\mathbf{p} + \lambda\mathbf{q}$ is parallel to $\mathbf{j}$, therefore the $\mathbf{i}$ -component of $\mathbf{p} + \lambda\mathbf{q}$ is 0.

We require the vector in the form $\mathbf{p} + \lambda\mathbf{q}$ so multiply $\mathbf{q}$ through by λ .

Rearrange with the $\mathbf{i}$ terms together and the $\mathbf{j}$ terms together.

The $\mathbf{i}$ and $\mathbf{j}$ components must be equal so equate them and solve for λ .

Exercise 3B

1 Given that $\mathbf{a} = 2\mathbf{i} + 3\mathbf{j}$ and $\mathbf{b} = 4\mathbf{i} - \mathbf{j}$, find these in terms of $\mathbf{i}$ and $\mathbf{j}$.

a $\mathbf{a} + \mathbf{b}$

b $3\mathbf{a} + \mathbf{b}$

c $2\mathbf{a} - \mathbf{b}$

d $2\mathbf{b} - \mathbf{a}$

e $3\mathbf{a} - 2\mathbf{b}$

f $\mathbf{b} - 3\mathbf{a}$

g $4\mathbf{b} - \mathbf{a}$

h $2\mathbf{a} - 3\mathbf{b}$

2 Given that $\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}$ and $\mathbf{b} = 3\mathbf{i} - \mathbf{j}$, find:

a λ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to the vector $\mathbf{i}$

b μ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to the vector $\mathbf{j}$.

- 3 Given that $\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{d} = \mathbf{i} - 2\mathbf{j}$, find:
- a λ if $\mathbf{c} + \lambda\mathbf{d}$ is parallel to $\mathbf{i} + \mathbf{j}$ b μ if $\mu\mathbf{c} + \mathbf{d}$ is parallel to $\mathbf{i} + 3\mathbf{j}$
 c s if $\mathbf{c} - s\mathbf{d}$ is parallel to $2\mathbf{i} + \mathbf{j}$ d t if $\mathbf{d} - t\mathbf{c}$ is parallel to $-2\mathbf{i} + 3\mathbf{j}$.
- 4 In this question, the horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed due east and due north respectively. Find the magnitude and bearing of these vectors:
- a $2\mathbf{i} + 3\mathbf{j}$ b $4\mathbf{i} - \mathbf{j}$ c $-3\mathbf{i} + 2\mathbf{j}$ d $-2\mathbf{i} - \mathbf{j}$

Challenge

Given that $\mathbf{a} = 3\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = \mathbf{i} + 4\mathbf{j}$, find the values of λ if $\mathbf{a} + \lambda\mathbf{b}$ is perpendicular to $3\mathbf{i} - 2\mathbf{j}$.

Comment on your answer.

SKILLS

ADAPTIVE LEARNING

SKILLS

EXECUTIVE FUNCTION

3.3 The velocity of a particle as a vector

- The velocity of a particle is a vector in the direction of motion. Its magnitude is the speed of the particle. The velocity is usually denoted by $\mathbf{v}$.

If a particle is moving with constant velocity $\mathbf{v} \text{ m s}^{-1}$, then after time t seconds it will have moved $\mathbf{vt} \text{ m}$. The displacement is parallel to the velocity. The magnitude of the displacement is the distance from the starting point.

Example 9

A particle is moving with constant velocity $\mathbf{v} = (3\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$. Find **a** the speed of the particle, **b** the distance moved every 4 seconds.

a $\mathbf{v} = (3\mathbf{i} + \mathbf{j})$,
 so speed $|\mathbf{v}| = \sqrt{3^2 + 1^2} = \sqrt{10}$
 $= 3.16 \text{ m s}^{-1}$ (3 s.f.)

The speed is the magnitude of the velocity.

b Method 1

Displacement $= \mathbf{vt} = (3\mathbf{i} + \mathbf{j}) \times 4$
 $= (12\mathbf{i} + 4\mathbf{j})$

Find the displacement of the particle in 4 seconds.

Therefore, the distance moved is given by

$|\mathbf{vt}| = \sqrt{12^2 + 4^2} = \sqrt{144 + 16} = \sqrt{160}$
 $= 12.6 \text{ m}$ (3 s.f.)

Displacement $= \mathbf{v} \times t$

Distance is magnitude of displacement.

Method 2

$\mathbf{v} = (3\mathbf{i} + \mathbf{j})$,
 so speed $= |\mathbf{v}| = \sqrt{3^2 + 1^2} = \sqrt{10}$
 $= 3.16 \text{ m s}^{-1}$

Find the speed of the particle.

Distance $= 3.16 \times 4 = 12.6 \text{ m}$ (3 s.f.)

Distance = speed $\times$ time

Use the unrounded value in your calculation.

Exercise

3C

SKILLS

EXECUTIVE FUNCTION

- Find the speed of a particle moving with these velocities:
a $(3\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ **b** $(24\mathbf{i} - 7\mathbf{j})\text{ km h}^{-1}$ **c** $(5\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ **d** $(-7\mathbf{i} + 4\mathbf{j})\text{ cm s}^{-1}$
- Find the distance moved by a particle which travels for:
a 5 hours at velocity $(8\mathbf{i} + 6\mathbf{j})\text{ km h}^{-1}$ **b** 10 seconds at velocity $(5\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$
c 45 minutes at velocity $(6\mathbf{i} + 2\mathbf{j})\text{ km h}^{-1}$ **d** 2 minutes at velocity $(-4\mathbf{i} - 7\mathbf{j})\text{ cm s}^{-1}$
- Find the speed and the distance travelled by a particle moving with:
a velocity $(-3\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ for 15 seconds **b** velocity $(2\mathbf{i} + 5\mathbf{j})\text{ m s}^{-1}$ for 3 seconds
c velocity $(5\mathbf{i} - 2\mathbf{j})\text{ km h}^{-1}$ for 3 hours **d** velocity $(12\mathbf{i} - 5\mathbf{j})\text{ km h}^{-1}$ for 30 minutes

3.4 Solving problems involving velocity and time using vectors

- If a particle starts from the point with position vector $\mathbf{r}_0$ and moves with constant velocity $\mathbf{v}$, then its displacement from its initial position at time t is $\mathbf{v}t$ and its position vector $\mathbf{r}$ is given by:

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$$

Example

10

SKILLS

PROBLEM-SOLVING

A particle starts from the point with position vector $(3\mathbf{i} + 7\mathbf{j})\text{ m}$ and moves with constant velocity $(2\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$. Find the position vector of the particle 4 seconds later.

$$\begin{aligned}\text{Displacement} &= \mathbf{v}t = 4(2\mathbf{i} - \mathbf{j}) = 8\mathbf{i} - 4\mathbf{j} \\ \text{Position vector } \mathbf{r} &= (3\mathbf{i} + 7\mathbf{j}) + (8\mathbf{i} - 4\mathbf{j}) \\ &= (3 + 8)\mathbf{i} + (7 - 4)\mathbf{j} \\ &= 11\mathbf{i} + 3\mathbf{j}\end{aligned}$$

$$\text{Distance} = \mathbf{v} \times t$$

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$$

Position vector after 4 seconds = position vector of starting point + displacement.

Example

11

A particle moving at a constant velocity is at the point with position vector $(2\mathbf{i} - 4\mathbf{j})\text{ m}$ at time $t = 0$. The particle is moving at a constant velocity. Five seconds later it is at the point with position vector $(12\mathbf{i} + 16\mathbf{j})\text{ m}$. Find the velocity of the particle.

$$\begin{aligned}\text{Displacement} &= (12\mathbf{i} + 16\mathbf{j}) - (2\mathbf{i} - 4\mathbf{j}) \\ &= (12 - 2)\mathbf{i} + (16 + 4)\mathbf{j} \\ &= 10\mathbf{i} + 20\mathbf{j}\end{aligned}$$

Travels $10\mathbf{i} + 20\mathbf{j}$ in 5 seconds, so

$$\mathbf{v} = \frac{1}{5}(10\mathbf{i} + 20\mathbf{j}) = (2\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$$

You need to use the same formula $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$, but this time you are using it to find $\mathbf{v}$.

Start by finding the displacement.

Travelling at constant velocity, so divide displacement by time taken to obtain $\mathbf{v}$.

Example 12

At time $t = 0$, a particle has position vector $4\mathbf{i} + 7\mathbf{j}$ and is moving with speed 15 m s^{-1} in the direction $3\mathbf{i} - 4\mathbf{j}$. Find its position vector after 2 seconds.

The magnitude of vector $3\mathbf{i} - 4\mathbf{j}$ is
 $\sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5$.
 So a unit vector in the direction of motion is
 $\frac{1}{5}(3\mathbf{i} - 4\mathbf{j})$, and the velocity is
 $15 \times \frac{1}{5}(3\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1} = (9\mathbf{i} - 12\mathbf{j}) \text{ m s}^{-1}$.
 New position $= (4\mathbf{i} + 7\mathbf{j}) + 2(9\mathbf{i} - 12\mathbf{j})$
 $= (4\mathbf{i} + 7\mathbf{j}) + (18\mathbf{i} - 24\mathbf{j})$
 $= (4 + 18)\mathbf{i} + (7 - 24)\mathbf{j}$
 $= 22\mathbf{i} - 17\mathbf{j}$

Start by using the speed and the direction of motion to find the velocity.

You need to start with a unit vector in the direction of motion, then multiply by the speed. This gives you a vector in the right direction with the required magnitude.

Now use $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t$

- The acceleration of a particle tells you how the velocity changes with time. Acceleration is a vector, usually denoted by $\mathbf{a}$. If a particle with initial velocity $\mathbf{u}$ moves with constant acceleration $\mathbf{a}$ then its velocity, $\mathbf{v}$ at time t is given by:

$$\mathbf{v} = \mathbf{u} + \mathbf{a}t$$

Example 13

A particle P has velocity $(-3\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ at time $t = 0$. The particle moves with constant acceleration $\mathbf{a} = (2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-2}$. Find the speed of the particle after 3 seconds.

After 3 seconds, the velocity of P is
 $\mathbf{v} = \mathbf{u} + \mathbf{a}t$
 $= (-3\mathbf{i} + \mathbf{j}) + 3 \times (2\mathbf{i} + 3\mathbf{j})$
 $= (-3\mathbf{i} + \mathbf{j}) + (6\mathbf{i} + 9\mathbf{j})$
 $= (3\mathbf{i} + 10\mathbf{j}) \text{ m s}^{-1}$
 So the speed of $P = \sqrt{3^2 + 10^2} = \sqrt{109}$
 $= 10.4 \text{ m s}^{-1}$ to 3 s.f.

Using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

The speed of P is the magnitude of its velocity.

- A force applied to a particle has both magnitude and direction, so force is also a vector. The force causes the particle to accelerate:

$\mathbf{F} = m\mathbf{a}$, where m is the mass of the particle.

Example 14

A constant force, $\mathbf{F} \text{ N}$, acts on a particle of mass 2 kg for 10 seconds. The particle is initially at rest, and 10 seconds later it has velocity $(10\mathbf{i} - 24\mathbf{j}) \text{ m s}^{-1}$. Find $\mathbf{F}$.

$(10\mathbf{i} - 24\mathbf{j}) = 10\mathbf{a}$
 So $\mathbf{a} = (\mathbf{i} - 2.4\mathbf{j}) \text{ m s}^{-2}$
 So $\mathbf{F} = 2 \times (\mathbf{i} - 2.4\mathbf{j}) = (2\mathbf{i} - 4.8\mathbf{j}) \text{ N}$

Use $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ to find the acceleration of the particle. $\mathbf{u} = 0$ as the particle is initially at rest.

Using $\mathbf{F} = m\mathbf{a}$

3 CENTRES OF MASS

2.1
2.2
2.3

Learning objectives

After completing this chapter you should be able to:

- Find the centre of mass of a set of particles arranged along a straight line → pages 51–52
- Find the centre of mass of a set of particles arranged in a plane → pages 53–57
- Find the positions of the centres of mass of standard uniform plane laminas → pages 57–61
- Find the positions of the centres of mass of composite laminas → pages 61–68
- Find the centre of mass of a framework → pages 68–72
- Solve problems involving a lamina in equilibrium → pages 72–79
- Solve problems involving a framework in equilibrium → pages 79–83
- Solve problems involving non-uniform laminas and frameworks → pages 83–87

Prior knowledge check

- 1 Work out the values of x and y :

$$8\begin{pmatrix} x \\ y \end{pmatrix} = 4\begin{pmatrix} 3 \\ 1 \end{pmatrix} + 6\begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad \leftarrow \text{International GCSE Mathematics}$$

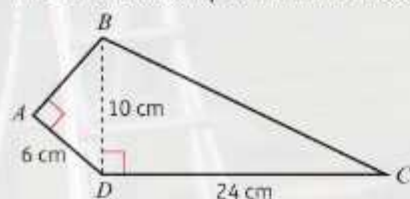
- 2 A uniform plank AB of length 6 m and mass 16 kg lies on the edge of a table. A mass of 4 kg is attached to one end of the plank at B , causing the plank to be on the point of tilting.



Find the distance AC .

← Mechanics 1 Section 8.5

- 3 Find the area of quadrilateral $ABCD$.



← International GCSE Mathematics

The centre of mass of large vehicles must be calculated, tested and sometimes adjusted, so that the vehicle does not topple over easily.

3.1 Centre of mass of a set of particles on a straight line

You can find the **centre of mass** of a set of particles arranged along a straight line by considering **moments**. You will use the fact that $\sum m_i x_i = \bar{x} \sum m_i$

Example 1

SKILLS

PROBLEM-SOLVING

A system of three particles with masses 2 kg, 5 kg and 3 kg are placed along the x -axis at the points (3, 0), (4, 0) and (6, 0) respectively. Find the centre of mass of the system.

$2g + 5g + 3g = Mg$
 $10 = M$

Taking moments about O :

$(2g \times 3) + (5g \times 4) + (3g \times 6) = Mg \times \bar{x}$
 $(2 \times 3) + (5 \times 4) + (3 \times 6) = 10\bar{x}$
 $6 + 20 + 18 = 10\bar{x}$
 $44 = 10\bar{x}$
 $4.4 = \bar{x}$

The centre of mass is at (4.4, 0).

Online Explore the centre of mass of systems of particles using GeoGebra.



Draw two diagrams, the first showing the **weights** of the three particles, the second showing the total weight Mg acting at the centre of mass G .

Compare the two systems vertically. Note that the g cancels.

Equate the **moments** of the two systems about O .

Substitute for M .

Example 2

A system of n particles with masses $m_1, m_2, \dots, m_n$ are placed along the x -axis at the points $(x_1, 0), (x_2, 0), \dots, (x_n, 0)$ respectively. Find the centre of mass of the system.

Then $M = m_1 + m_2 + \dots + m_n$
 and $m_1 x_1 + m_2 x_2 + \dots + m_n x_n = M \bar{x}$
 i.e. $m_1 x_1 + m_2 x_2 + \dots + m_n x_n = (m_1 + m_2 + \dots + m_n) \bar{x}$

or $\sum_{i=1}^n m_i x_i = \bar{x} \sum_{i=1}^n m_i$

So $\bar{x} = \frac{\sum_{i=1}^n m_i x_i}{\sum_{i=1}^n m_i}$

Draw two diagrams.

There is no need to include g as it cancels.

This is a key result and should be learned.

Note that this result holds for positive and negative coordinates.

- If a system of n particles with masses $m_1, m_2, \dots, m_n$ are placed along the x -axis at the points $(x_1, 0), (x_2, 0), \dots, (x_n, 0)$ respectively, then:

$$\sum_{i=1}^n m_i x_i = \bar{x} \sum_{i=1}^n m_i$$

where $(\bar{x}, 0)$ is the position of the centre of mass of the system.

Notation

This result could also be used for a system of particles placed along the y -axis:

$$\sum_{i=1}^n m_i y_i = \bar{y} \sum_{i=1}^n m_i$$

Exercise**3A****SKILLS****PROBLEM-SOLVING**

- Find the position of the centre of mass of four particles of masses 1 kg, 4 kg, 3 kg and 2 kg placed on the x -axis at the points (6, 0), (3, 0), (2, 0) and (4, 0) respectively.
- Three masses of 1 kg, 2 kg and 3 kg, are placed at the points with coordinates (0, 2), (0, 5) and (0, 1) respectively. Find the coordinates of G , the centre of mass of the three masses.
- Three particles of masses 2 kg, 3 kg and 5 kg, are placed at the points $(-1, 0)$, $(-4, 0)$ and $(5, 0)$ respectively. Find the coordinates of the centre of mass of the three particles.
- A light rod PQ of length 4 m has particles of masses 1 kg, 2 kg and 3 kg attached to it at the points P , Q and R respectively, where $PR = 2$ m. The centre of mass of the loaded rod is at the point G . Find the distance PG .
- Three particles of masses 5 kg, 3 kg and m kg lie on the y -axis at the points (0, 4), (0, 2) and (0, 5) respectively. The centre of mass of the system is at the point (0, 4). Find the value of m .
- P** A light rod PQ of length 2 m has particles of masses 0.4 kg and 0.6 kg fixed to it at the points P and R respectively, where $PR = 0.5$ m. Find the mass of the particle which must be fixed at Q so that the centre of mass of the loaded rod is at its **midpoint**.
- P** The centre of mass of four particles of masses $2m$, $3m$, $7m$ and $8m$, which are positioned at the points (0, a), (0, 2), (0, -1) and (0, 1) respectively, is the point G . Given that the coordinates of G are (0, 1), find the value of a .
- P** Particles of masses 3 kg, 2 kg and 1 kg lie on the y -axis at the points with coordinates (0, -2), (0, 7) and (0, 4) respectively. Another particle of mass 6 kg is added to the system so that the centre of mass of all four particles is at the origin. Find the position of this particle.
- E/P** Three particles A , B and C are placed along the x -axis. Particle A has mass 5 kg and is at the point (2, 0). Particle B has mass m_1 kg and is at the point (3, 0) and particle C has mass m_2 kg and is at the point $(-2, 0)$. The centre of mass of the three particles is at the point $G(1, 0)$. Given that the total mass of the three particles is 10 kg, find the values of m_1 and m_2 . **(3 marks)**
- E/P** Three particles of masses $(m - 1)$ kg, $(5 - m)$ kg and m kg lie on the y -axis at the points with coordinates (0, -1), (0, 1) and (0, 2) respectively. A fourth particle of mass $(m + 1)$ kg is added at the point (0, 0) so that the centre of mass of all four particles is at the point (0, 1). Show that $m = 0.5$ kg. **(3 marks)**

Challenge

Three particles, of masses 1 kg, 2 kg and 3 kg respectively, lie on the x -axis at points P , Q and R with $PQ:QR = 2:3$. The centre of mass of the particles is at G . Show that the ratio of the lengths $PQ:PG$ is 12:19.

3.2 Centre of mass of a set of particles arranged in a plane

You can use $\sum m_i x_i = \bar{x} \sum m_i$ and $\sum m_i y_i = \bar{y} \sum m_i$ to find the centre of mass of a set of point masses arranged in a plane by considering the x -coordinate and y -coordinate of the centre of mass separately.

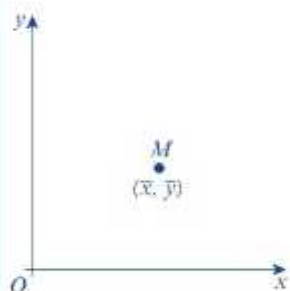
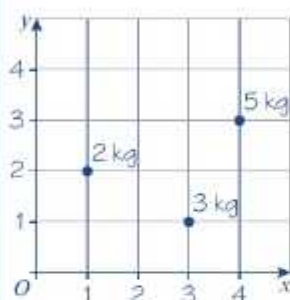
Example 3

SKILLS

PROBLEM-SOLVING

Find the coordinates of the centre of mass of the following system of particles:

2 kg at (1, 2); 3 kg at (3, 1); 5 kg at (4, 3)



$$\begin{aligned} 2g + 3g + 5g &= Mg \\ 10 &= M \end{aligned}$$

Method 1

Taking moments about the y -axis:

$$(2g \times 1) + (3g \times 3) + (5g \times 4) = Mg\bar{x}$$

$$\begin{aligned} (2 \times 1) + (3 \times 3) + (5 \times 4) &= (2 + 3 + 5)\bar{x} \\ 2 + 9 + 20 &= 10\bar{x} \\ 31 &= \bar{x} \end{aligned}$$

Taking moments about the x -axis:

$$\begin{aligned} (2g \times 2) + (3g \times 1) + (5g \times 3) &= (2 + 3 + 5)g\bar{y} \\ 2.2 &= \bar{y} \end{aligned}$$

The centre of mass is at (3.1, 2.2).

Online Explore the centre of mass of particles arranged in a plane using GeoGebra.



Draw two diagrams, the first showing the three particles, the second showing the total mass M placed at the centre of mass $(\bar{x}, \bar{y})$.

Equate the total weights.
Note that g cancels.

Equate the moments of the system about the y -axis.

Substitute for M .

Equate the moments of the system about the x -axis.

Method 2

$$2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3\begin{pmatrix} 3 \\ 1 \end{pmatrix} + 5\begin{pmatrix} 4 \\ 3 \end{pmatrix} = (2+3+5)\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 9 \\ 3 \end{pmatrix} + \begin{pmatrix} 20 \\ 15 \end{pmatrix} = \begin{pmatrix} 10\bar{x} \\ 10\bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 31 \\ 22 \end{pmatrix} = \begin{pmatrix} 10\bar{x} \\ 10\bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 3.1 \\ 2.2 \end{pmatrix} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

The centre of mass is at (3.1, 2.2).

Problem-solving

You can reduce the working by using position vectors.

The top line is

$$\sum m_i x_i = \bar{x} \sum m_i$$

and the bottom line is

$$\sum m_i y_i = \bar{y} \sum m_i$$

Divide both sides by 10.

- If a system consists of n particles: mass m_1 with position vector $\mathbf{r}_1$, mass m_2 with position vector $\mathbf{r}_2$, ..., mass m_n with position vector $\mathbf{r}_n$, then

$$\sum m_i \mathbf{r}_i = \bar{\mathbf{r}} \sum m_i$$

where $\bar{\mathbf{r}}$ is the position vector of the centre of mass of the system.

Notation

The position vector of a point can be written in terms of $\mathbf{i}$ and $\mathbf{j}$ or as a column vector. For example, the position vector of the point (3, 4) is $3\mathbf{i} + 4\mathbf{j}$ or $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$.

Example 4

Find the coordinates of the centre of mass of the following system of particles:

4 kg at (-1, 3); 2 kg at (-2, -4); 8 kg at (4, 0); 6 kg at (1, -3)

$$4\begin{pmatrix} -1 \\ 3 \end{pmatrix} + 2\begin{pmatrix} -2 \\ -4 \end{pmatrix} + 8\begin{pmatrix} 4 \\ 0 \end{pmatrix} + 6\begin{pmatrix} 1 \\ -3 \end{pmatrix} = (4+2+8+6)\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} -4 \\ 12 \end{pmatrix} + \begin{pmatrix} -4 \\ -8 \end{pmatrix} + \begin{pmatrix} 32 \\ 0 \end{pmatrix} + \begin{pmatrix} 6 \\ -18 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 30 \\ -14 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 1.5 \\ -0.7 \end{pmatrix} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

Centre of mass is at (1.5, -0.7).

The result applies with positive or negative coordinates.

Simplify the LHS.

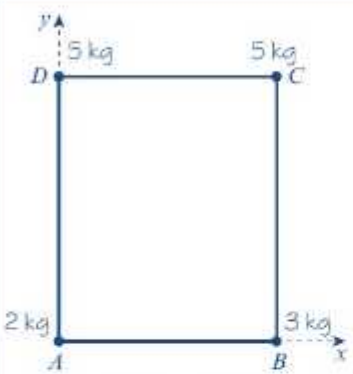
- If a question does not specify axes or coordinates you will need to choose your own axes and origin.

Example 5

A light rectangular plate $ABCD$ has $AB = 20$ cm and $AD = 50$ cm. Particles of masses 2 kg, 3 kg, 5 kg and 5 kg are attached to the plate at the points A , B , C and D respectively.

Find the distance of the centre of mass of the loaded plate from:

- a** AD **b** AB



$$2\begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 20 \\ 0 \end{pmatrix} + 5\begin{pmatrix} 20 \\ 50 \end{pmatrix} + 5\begin{pmatrix} 0 \\ 50 \end{pmatrix} = (2 + 3 + 5 + 5)\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 60 \\ 0 \end{pmatrix} + \begin{pmatrix} 100 \\ 250 \end{pmatrix} + \begin{pmatrix} 0 \\ 250 \end{pmatrix} = 15\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 160 \\ 500 \end{pmatrix} = 15\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} \frac{32}{3} \\ \frac{100}{3} \end{pmatrix} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

a $\frac{32}{3}$ cm
b $\frac{100}{3}$ cm

First draw a diagram.

Choose point A as the origin and put it in the bottom left-hand corner of your diagram.

Using AB and AD as 'axes', A is $(0, 0)$, B is $(20, 0)$, C is $(20, 50)$ and D is $(0, 50)$.

Using $\sum m_i \mathbf{r}_i = \mathbf{r} \sum m_i$

Simplify LHS.

Divide both sides by 15.

Watch out Check that your answers are sensible in the context of the question. The centre of mass should lie inside the rectangle.

Example 6

Particles of masses 4 kg, 3 kg, 2 kg and 1 kg are placed at the points (x, y) , $(3, 2)$, $(1, -5)$ and $(6, 0)$ respectively. Given that the centre of mass of the four particles is at the point $(2.5, -2)$, find the values of x and y .

$$4\begin{pmatrix} x \\ y \end{pmatrix} + 3\begin{pmatrix} 3 \\ 2 \end{pmatrix} + 2\begin{pmatrix} 1 \\ -5 \end{pmatrix} + 1\begin{pmatrix} 6 \\ 0 \end{pmatrix} = (4 + 3 + 2 + 1)\begin{pmatrix} 2.5 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 4x \\ 4y \end{pmatrix} + \begin{pmatrix} 9 \\ 6 \end{pmatrix} + \begin{pmatrix} 2 \\ -10 \end{pmatrix} + \begin{pmatrix} 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 25 \\ -20 \end{pmatrix}$$

$$\begin{pmatrix} 4x + 17 \\ 4y - 4 \end{pmatrix} = \begin{pmatrix} 25 \\ -20 \end{pmatrix}$$

$$\begin{aligned} 4x + 17 &= 25 \\ 4y - 4 &= -20 \end{aligned}$$

$x = 2, y = -4$

Using $\sum m_i \mathbf{r}_i = \mathbf{r} \sum m_i$

Equate the $\mathbf{i}$ and $\mathbf{j}$ components.

Solve the two equations for x and y .

Example 7

Three particles of masses 2 kg, 1 kg and m kg are situated at the points $(-1, 3)$, $(2, 9)$ and $(2, -1)$ respectively. Given that the centre of mass of the three particles is at the point $(1, \bar{y})$, find:

- a** the value of m
b the value of $\bar{y}$.

$$2\begin{pmatrix} -1 \\ 3 \end{pmatrix} + 1\begin{pmatrix} 2 \\ 9 \end{pmatrix} + m\begin{pmatrix} 2 \\ -1 \end{pmatrix} = (2 + 1 + m)\begin{pmatrix} 1 \\ \bar{y} \end{pmatrix}$$

Using $\sum m_i \mathbf{r}_i = \mathbf{r} \sum m_i$

$$\begin{pmatrix} -2 \\ 6 \end{pmatrix} + \begin{pmatrix} 2 \\ 9 \end{pmatrix} + \begin{pmatrix} 2m \\ -m \end{pmatrix} = (3 + m)\begin{pmatrix} 1 \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 2m \\ 15 - m \end{pmatrix} = \begin{pmatrix} 3 + m \\ (3 + m)\bar{y} \end{pmatrix}$$

$$2m = 3 + m$$

$$15 - m = (3 + m)\bar{y}$$

Equate the $\mathbf{i}$ and $\mathbf{j}$ components.

a $m = 3$

The first equation is easy to solve for m .

b $15 - 3 = (3 + 3)\bar{y}$

$$12 = 6\bar{y}$$

$$2 = \bar{y}$$

Substitute for m in the second equation and solve for $\bar{y}$.

Exercise 3B**SKILLS****PROBLEM-SOLVING**

- Two particles of equal mass are placed at the points $(1, -3)$ and $(5, 7)$ respectively. Find the centre of mass of the particles.
- Four particles of equal mass are situated at the points $(2, 0)$, $(-1, 3)$, $(2, -4)$ and $(-1, -2)$ respectively. Find the coordinates of the centre of mass of the particles.
- A system of three particles consists of 10 kg placed at $(2, 3)$, 15 kg placed at $(4, 2)$ and 25 kg placed at $(6, 6)$. Find the coordinates of the centre of mass of the system.
- Find the position vector of the centre of mass of three particles of masses 0.5 kg, 1.5 kg and 2 kg which are situated at the points with position vectors $(6\mathbf{i} - 3\mathbf{j})$, $(2\mathbf{i} + 5\mathbf{j})$ and $(3\mathbf{i} + 2\mathbf{j})$ respectively.
- Particles of masses m , $2m$, $5m$ and $2m$ are situated at $(-1, -1)$, $(3, 2)$, $(4, -2)$ and $(-2, 5)$ respectively. Find the coordinates of the centre of mass of the particles.
- A light rectangular metal plate $PQRS$ has $PQ = 4$ cm and $PS = 2$ cm. Particles of masses 3 kg, 5 kg, 1 kg and 7 kg are attached respectively to the corners P , Q , R and S of the plate. Find the distance of the centre of mass of the loaded plate from:
 - the side PQ
 - the side PS .

- (P) 7 Three particles of masses 1 kg, 2 kg and 3 kg are positioned at the points $(1, 0)$, $(4, 3)$ and (p, q) respectively. Given that the centre of mass of the particles is at the point $(2, 0)$, find the values of p and q .
- (P) 8 A system consists of three particles with masses $3m$, $4m$ and $5m$. The particles are situated at the points with coordinates $(-3, -4)$, $(0.5, 4)$ and $(0, -5)$ respectively. Find the coordinates of the position of a fourth particle of mass $7m$, given that the centre of mass of all four particles is at the origin.
- (E) 9 A light rectangular piece of card $ABCD$ has $AB = 8$ cm and $AD = 6$ cm. Four particles of masses 300 g, 200 g, 600 g and 100 g are fixed to the rectangle at the midpoints of the sides AB , BC , CD and DE respectively. Find the distance of the centre of mass of the loaded rectangle from the sides AB and AD . (4 marks)
- (E/P) 10 A light rectangular piece of card $ABCD$ has $AB = 8$ cm and $AD = 6$ cm. Three particles of masses 3 g, 2 g and 2 g are attached to the rectangle at the points A , B and C respectively.
- a Find the mass of a particle which must be placed at the point D for the centre of mass of the whole system of four particles to lie 3 cm from the line AB . (2 marks)
- b With this fourth particle in place, find the distance of the centre of mass of the system from the side AD . (4 marks)

Challenge

A light triangular piece of card ABC has sides $AB = 6$ cm, $AC = 5$ cm and $BC = 5$ cm. Three particles of masses m kg, 0.2 kg and 0.2 kg are fixed to the triangle at the midpoints of the sides AB , BC and AC respectively. The point P lies at the **intersection** of the lines joining each **vertex** of the triangle with the midpoint of the opposite side. Given that the centre of mass of the whole system lies at P , find the value of m .

3.3 Centres of mass of standard uniform plane laminas

You can find the positions of the centres of mass of standard **uniform** plane **laminas**, including a rectangle, a triangle and a semicircle.

- An object which has one dimension (thickness) very small compared with its other two (length and width) is modelled as a lamina. This means that it is regarded as being two-dimensional with area, but no volume.

Hint

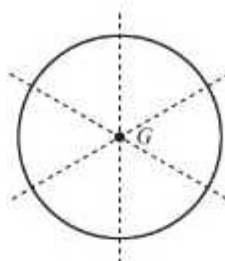
For example, a sheet of paper or a piece of card could be modelled as a lamina.

A lamina is **uniform** if its mass is evenly spread throughout its area.

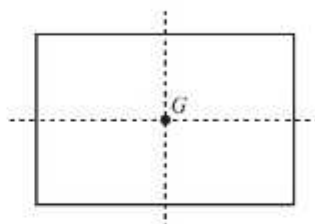
- If a uniform lamina has an axis of symmetry, then its centre of mass must lie on the axis of symmetry. If the lamina has more than one axis of symmetry, then it follows that the centre of mass must be at the point of intersection of the axes of symmetry.

Uniform circular disc

Since every diameter of the disc is a line of symmetry, the centre of mass of the disc is at their intersection. This is the centre of the disc.

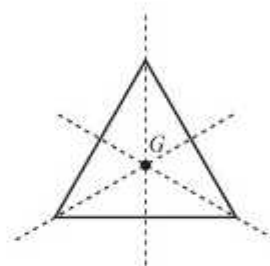
**Uniform rectangular lamina**

A uniform rectangular lamina has two lines of symmetry, each one joining the midpoints of a pair of opposite sides. The centre of mass is at the point where the two lines meet.

**Uniform triangular lamina**

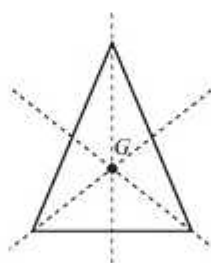
A uniform triangular lamina has axes of symmetry only if it is either equilateral or isosceles.

A uniform equilateral triangle has three axes of symmetry, each one joining a vertex to the midpoint of the opposite side. These three lines are called the **medians** of the triangle.

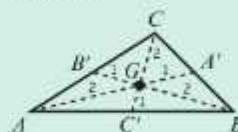


- The centre of mass of a uniform triangular lamina is at the intersection of the medians. This point is called the **centroid** of the triangle.

Note that the medians are not axes of symmetry of the triangle unless the triangle is equilateral (in which case all three medians are axes of symmetry) or isosceles (in which case one median is also an axis of symmetry).

**Hint**

It can be proved that the centroid G (and therefore the centre of mass) of any triangle is two-thirds of the way down each median from each vertex:



where A' is the midpoint of BC , B' is the midpoint of CA and C' is the midpoint of AB :

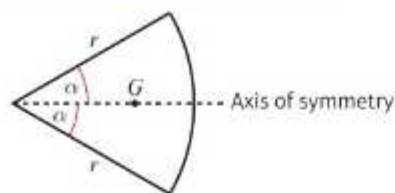
$$\text{i.e. } \frac{AG}{GA'} = \frac{BG}{GB'} = \frac{CG}{GC'} = \frac{2}{1}$$

- If the coordinates of the three vertices of a uniform triangular lamina are (x_1, y_1) , (x_2, y_2) and (x_3, y_3) then the coordinates of the centre of mass G are given by taking the average (mean) of the coordinates of the vertices:

$$G \text{ is the point } \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Hint

This is the two-dimensional version of a similar result for a uniform rod: if the ends of the rod are (x_1, y_1) and (x_2, y_2) then its centre of mass is its midpoint, $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$.

Uniform sector of a circle

A uniform sector of a circle of radius r and centre angle 2α , where α is measured in radians, has its centre of mass on the axis of symmetry at a distance $\frac{2r \sin \alpha}{3\alpha}$ from the centre.

Example 8

A uniform triangular lamina has vertices $A(1, 4)$, $B(3, 2)$ and $C(5, 3)$. Find the coordinates of its centre of mass.

$$G \text{ is the point } \left(\frac{1+3+5}{3}, \frac{4+2+3}{3} \right) = (3, 3)$$

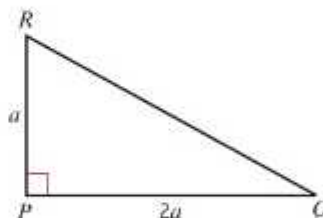
Online Explore centres of mass of standard uniform plane laminas using GeoGebra.



Find the mean of the vertices of the triangle. This is the **centroid** of the triangle.

Example 9

Find the centre of mass of the uniform triangular lamina shown.



Taking P as the origin and PQ and PR as axes:

P is $(0, 0)$; Q is $(2a, 0)$; R is $(0, a)$

G is the point

$$\left(\frac{0+2a+0}{3}, \frac{0+0+a}{3} \right) = \left(\frac{2a}{3}, \frac{a}{3} \right)$$

The centre of mass is $\frac{2a}{3}$ from PR and $\frac{a}{3}$ from PQ .

Here we need to choose our own axes and origin.

Write down the coordinates of each of the three vertices.

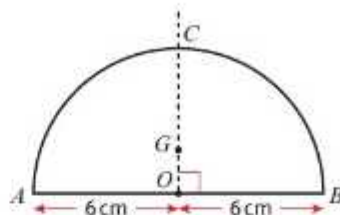
Find the mean of the three vertices.

Watch out When you choose your own axes you must not leave your answer in coordinate form.

Example 10 SKILLS PROBLEM-SOLVING

The diagram shows a uniform semicircular lamina of radius 6 cm with centre O .

Find the centre of mass of the lamina.



The centre of mass must lie on the line through O which is perpendicular to AB .

Let $OG = \bar{y}$. Then:

$$\bar{y} = \frac{2 \times 6 \times \sin \frac{\pi}{2}}{\frac{3\pi}{2}}$$

$$= \frac{12 \times 1}{\frac{3\pi}{2}}$$

$$= 12 \times \frac{2}{3\pi}$$

$$= \frac{8}{\pi}$$

The centre of mass of the lamina is on the line OC at a distance $\frac{8}{\pi}$ cm from O .

This is the axis of symmetry of the lamina.

Use the result for a sector which is in the formula booklet with $r = 6$ and $\alpha = \frac{\pi}{2}$.

$$\sin \frac{\pi}{2} = 1$$

You must give the angle in radians for this formula.

Simplify.

Exercise 3C SKILLS PROBLEM-SOLVING

1 Find the centre of mass of a uniform triangular lamina whose vertices are:

a (1, 2), (2, 6) and (3, 1)

b (-1, 4), (3, 5) and (7, 3)

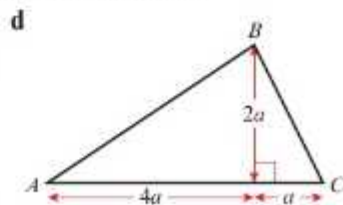
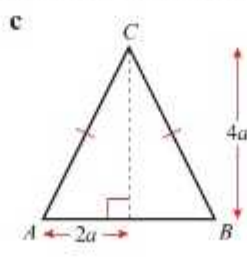
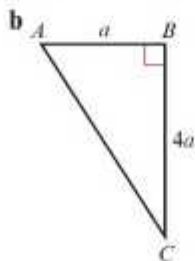
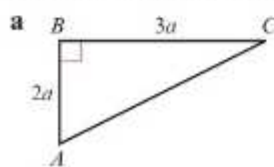
c (-3, 2), (4, 0) and (0, 1)

d (a , a), ($3a$, $2a$) and ($4a$, $6a$)

2 Find the position of the centre of mass of a uniform semicircular lamina of radius 4 cm and centre O .

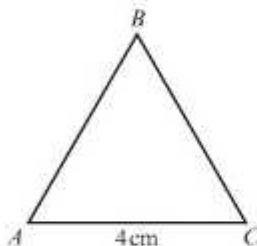
(P) 3 The centre of mass of a uniform triangular lamina ABC is at the point $(2, a)$. Given that A is the point $(4, 3)$, B is the point $(b, 1)$ and C is the point $(-1, 5)$, find the values of a and b .

4 Find the position of the centre of mass of the following uniform triangular laminas.



- (P) 5 A uniform triangular lamina is isosceles and has the line $y = 4$ as its axis of symmetry. One of the vertices of the triangle is the point $(2, 1)$. Given that the x -coordinate of the centre of mass of the lamina is -3 , find the coordinates of the other two vertices.
- (P) 6 A uniform rectangular lamina $ABCD$ is positioned such that AB lies on the line $y = 2x + 1$. Given that A is at the point $(0, 1)$ and C is at the point $(6, 7)$, find:
- the coordinates of the points B and D
 - the coordinates of the centre of mass of the lamina.
- (E/P) 7 A uniform triangular lamina ABC has coordinates $A(2, 1)$, $B(4, 1)$ and $C(x, y)$. The centre of mass of the lamina lies on the line $x = 3$. Given that the triangle ABC has an area of 4 cm^2 , work out:
- the possible values of x and y (3 marks)
 - the possible coordinates of the centre of mass. (4 marks)

- (E/P) 8 The diagram below shows an equilateral triangle ABC where AC is 4 cm .



Show that the centre of mass lies $\frac{4\sqrt{3}}{3} \text{ cm}$ from B .

(3 marks)

3.4 Centre of mass of a composite lamina

A **composite uniform lamina** consists of two or more standard uniform laminas joined together. You can find the centre of mass of a composite lamina by considering each part of the lamina as a particle positioned at its centre of mass. The masses of each part of the lamina will be proportional to their areas.

Example 11

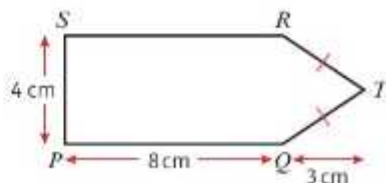
SKILLS

PROBLEM-SOLVING

A uniform lamina consists of a rectangle $PQRS$ joined to an isosceles triangle QRT , as shown in the diagram.

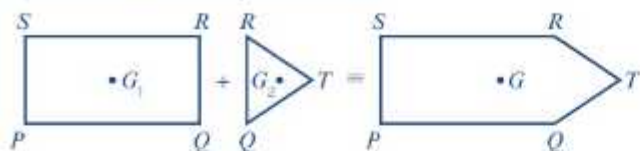
Find the distance of the centre of mass of the lamina from:

- PQ
- PS



Let the mass per unit area be m kg per cm^2 .

Split the lamina along the line QR :



Area of $PQRS = 8 \times 4 = 32 \text{ cm}^2$

So, mass of $PQRS = 32m$

Similarly, mass of $QRT = \frac{1}{2} \times 4 \times 3 \times m$
 $= 6m$

So, total mass of the lamina $= 32m + 6m = 38m$

Take P as the origin, and axes along PQ and PS .

The centre of mass of $PQRS$ is at the point $(4, 2)$.

The coordinates of Q are $(8, 0)$.

The coordinates of R are $(8, 4)$.

The coordinates of T are $(11, 2)$.

The centre of mass of $\triangle QRT$ will be

$$\left(\frac{8+8+11}{3}, \frac{0+4+2}{3} \right)$$

$$= (9, 2)$$

Replace the lamina by two particles:

$32m$ placed at $(4, 2)$

and $6m$ placed at $(9, 2)$

$$32m \begin{pmatrix} 4 \\ 2 \end{pmatrix} + 6m \begin{pmatrix} 9 \\ 2 \end{pmatrix} = 38m \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 128 \\ 64 \end{pmatrix} + \begin{pmatrix} 54 \\ 12 \end{pmatrix} = 38 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 182 \\ 76 \end{pmatrix} = 38 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\frac{91}{19} = \bar{x}$$

$$2 = \bar{y}$$

a Distance from PQ is 2 cm .

b Distance from PS is $\frac{91}{19} \text{ cm}$.

Since the lamina is uniform, the mass per unit area will be a constant.

You must always split up the lamina into standard shapes.

G_1 is the centre of mass of the rectangle.

G_2 is the centre of mass of the triangle.

Area of $\triangle$ is

$$\frac{1}{2} \times \text{base} \times \text{height}.$$

It's usually a good idea to take the origin at the bottom left-hand corner of your diagram.

This is the centre of the rectangle, G_1 .

Take the mean of the coordinates of the vertices of the triangle.

This is G_2 .

This is the key idea behind the method.

$\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$ is the position vector of the centre of mass.

Cancel the m s and simplify.

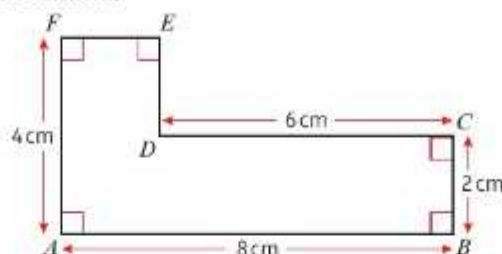
Problem-solving

Note that you could have got the answer to part **a** using the fact that the lamina has an axis of symmetry. You should always use this as it will considerably reduce the amount of working required.

- The centre of mass of a uniform plane lamina will always lie on an axis of symmetry.

Example 12

The diagram shows a uniform lamina.



Find the distance of the centre of mass of the lamina from:

- a** AF **b** AB

Hint

You can find the centre of mass in three different ways.

Method 1

Area	8	12	20
x	1	5	$\bar{x}$
y	2	1	$\bar{y}$

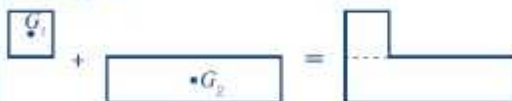
$$8\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 12\begin{pmatrix} 5 \\ 1 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 8 \\ 16 \end{pmatrix} + \begin{pmatrix} 60 \\ 12 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 68 \\ 28 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$3.4 = \bar{x}$$

$$1.4 = \bar{y}$$

Method 2

Area	4	16	20
x	1	4	$\bar{x}$
y	3	1	$\bar{y}$

$$4\begin{pmatrix} 1 \\ 3 \end{pmatrix} + 16\begin{pmatrix} 4 \\ 1 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 12 \end{pmatrix} + \begin{pmatrix} 64 \\ 16 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 68 \\ 28 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$3.4 = \bar{x}$$

$$1.4 = \bar{y}$$

You can summarise the area of each part of the shape and the positions of G_1 and G_2 in a table. Because the lamina is uniform you only need to know the area of each piece.

The centre of mass of the first rectangle is (1, 2).

The centre of mass of the second rectangle is at its centre (5, 1).

Using $\sum m_i r_i = r \sum m_i$

Simplify.

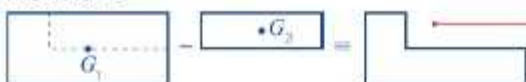
Solve for $\bar{x}$ and $\bar{y}$.

Split the shape using the dotted line shown. The centre of the square is (1, 3). The centre of the rectangle is (4, 1).

Using $\sum m_i r_i = r \sum m_i$

As before.

Method 3



Area	32	12	20
$\bar{x}$	4	5	$\bar{x}$
$\bar{y}$	2	3	$\bar{y}$

$$32 \begin{pmatrix} 4 \\ 2 \end{pmatrix} - 12 \begin{pmatrix} 5 \\ 3 \end{pmatrix} = 20 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 128 \\ 64 \end{pmatrix} - \begin{pmatrix} 60 \\ 36 \end{pmatrix} = 20 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 68 \\ 28 \end{pmatrix} = 20 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$3.4 = \bar{x}$$

$$1.4 = \bar{y}$$

a Distance from AF is 3.4 cm.

b Distance from AB is 1.4 cm.

You obtain the lamina by starting with a rectangle and removing another rectangle.

G_1 is at (4, 2).

G_2 is at (5, 3).

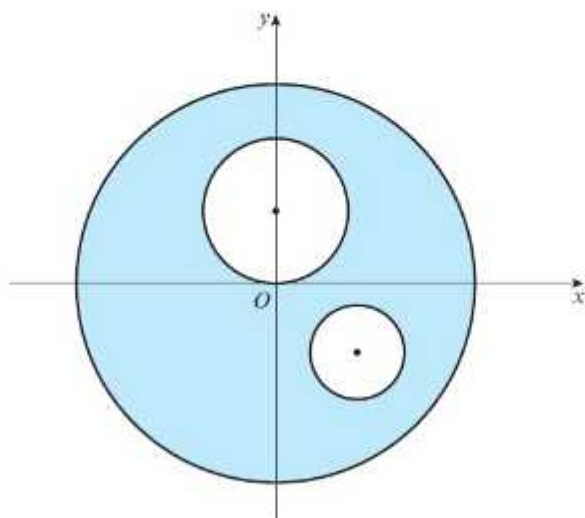
Note the subtraction, since you are removing, not adding, the second rectangle.

As before.

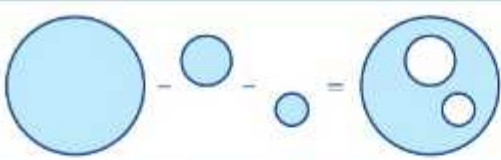
Remember to give your answers in the form asked for.

Example 13

A uniform circular disc, centre O , of radius 5 cm, has two circular holes cut in it, as shown below.



The larger hole has radius 2 cm and the smaller hole has radius 1 cm. The coordinates of the centres of the holes are (0, 2) and (2, -2) respectively. Find the coordinates of the centre of mass of the remaining lamina.



Area	$\pi \times 5^2$	$\pi \times 2^2$	$\pi \times 1^2$	$\pi(5^2 - 2^2 - 1^2)$
$\bar{x}$	0	0	2	$\bar{x}$
$\bar{y}$	0	2	-2	$\bar{y}$

$$\pi 5^2 \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \pi 2^2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} - \pi 1^2 \begin{pmatrix} 2 \\ -2 \end{pmatrix} = \pi(5^2 - 2^2 - 1^2) \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 8 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 20 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} -2 \\ -6 \end{pmatrix} = 20 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} -0.1 \\ -0.3 \end{pmatrix} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

The coordinates of the centre of mass of the lamina are $(-0.1, -0.3)$.

Problem-solving

When you remove part of a lamina, you can deal with the removed sections by adding sections with negative mass.

Setting out the key information in a table helps to clarify your working.

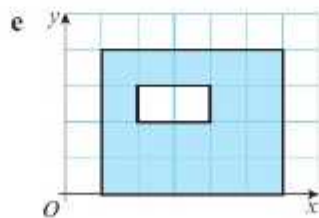
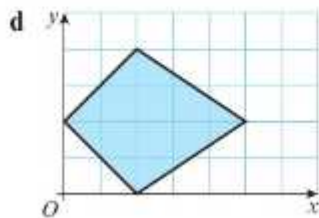
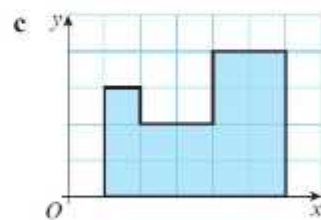
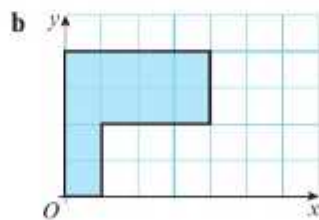
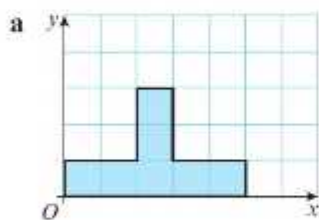
Note the subtraction signs for each removed section.

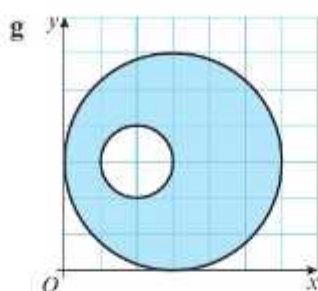
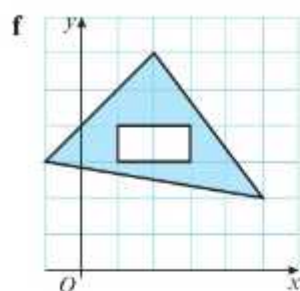
Cancel the π s and simplify.

Solve for $\bar{x}$ and $\bar{y}$.

Exercise**3D****SKILLS****PROBLEM-SOLVING**

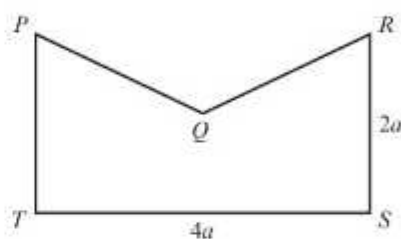
- 1 The following diagrams show uniform plane figures. Each one is drawn on a grid of unit squares. Find, in each case, the coordinates of the centre of mass.





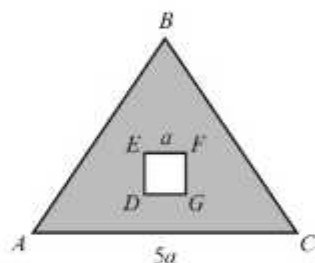
- E** 2 The uniform lamina $PQRST$ is formed by removing the triangle PQR from the rectangle $PRST$ with centre Q . The rectangle has sides of length $4a$ and $2a$. Find the distance of the centre of mass of $PQRST$ from Q .

(4 marks)



- E** 3 The uniform lamina shown in the diagram is formed by removing a square $DEFG$ of side length a from the equilateral triangle ABC of side length $5a$. The centre of mass of $DEFG$ lies on the perpendicular bisector (a line that divides something into two equal parts) of AC . Given that AC and DG are parallel and a distance a apart, work out the distance of the centre of mass of the whole lamina from B .

(6 marks)



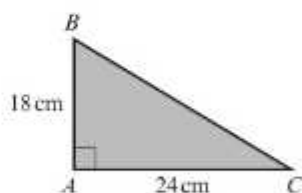
- E/P** 4 The diagram shows a metal template in the shape of a right-angled triangle ABC . The template is modelled as a uniform lamina.

- a** Find the distance of the centre of mass of the lamina from A .

(4 marks)

The mass of the template is 15 kg. A particle of mass 5 kg is attached to vertex C of the template.

- b** Find the position of the centre of mass of the template with the attached particle. (3 marks)



Problem-solving

When you attach a particle to a lamina, you can work out the new centre of mass by considering the lamina as a single particle whose weight acts at its centre of mass.

- E/P** 5 The diagram shows a uniform lamina formed from two rectangles. All the angles are right angles.

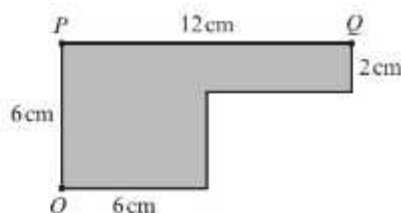
a Find the position of the centre of mass of the lamina. **(4 marks)**

The mass per unit area of the lamina is 30 g cm^{-2} .

Two particles of masses 200 g and 500 g are attached to points P and Q on the lamina respectively.

b Find the new centre of mass of the lamina, giving any lengths correct to 3 significant figures.

(3 marks)



- E/P** 6 The diagram shows a uniform piece of card $ABCDEFGH$ where $AB = 6 \text{ cm}$, $BC = 10 \text{ cm}$, $CD = 8 \text{ cm}$, $DE = 2 \text{ cm}$, $EF = 4 \text{ cm}$, $FG = 6 \text{ cm}$, $GH = 2 \text{ cm}$, $HA = 2 \text{ cm}$.

Assume that A is the origin and AH lies on the x -axis.

a Show that the centre of mass lies at the point $(\frac{69}{13}, \frac{41}{13})$.

(7 marks)

The template needs to be changed so that the centre of mass lies at the point $(\frac{69}{13}, \frac{41}{13})$. To achieve this, two squares of side length 3 cm are cut out of the card.

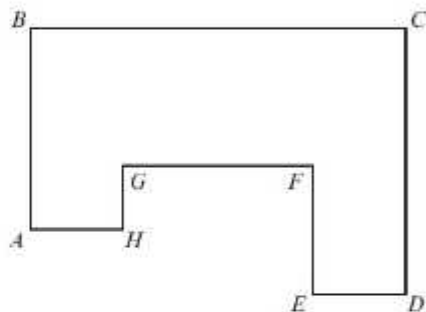
Their midpoints are $(a, \frac{41}{13})$ and $(5, \frac{41}{13})$.

b Explain how you can use symmetry to determine the value of a .

(2 marks)

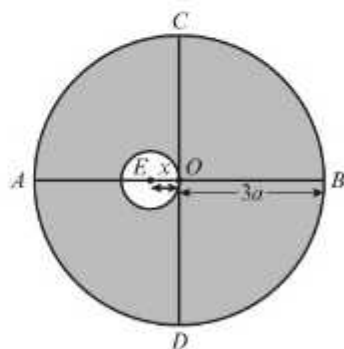
c Find the value of a .

(5 marks)



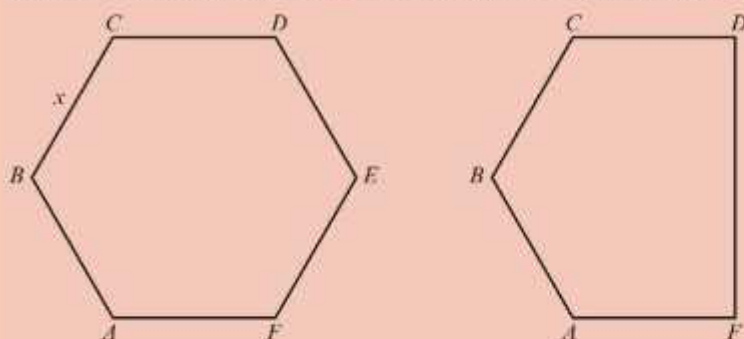
- E** 7 A uniform circular disc has centre O and radius $3a$. The lines AB and CD are perpendicular diameters of the disc. A circular hole of radius x is made in the disc, with the centre of the hole at the point E on AB and the edge of the hole touching O to form the lamina shown on the right. Given that the centre of mass of the lamina lies on the line AB a distance of $\frac{23}{8}a$ from the point B , find the value of x in terms of a .

(6 marks)



Challenge

A regular hexagon $ABCDEF$ of side length x has the triangle DEF removed to leave the irregular pentagon $ABCDF$ as shown in the diagram below.

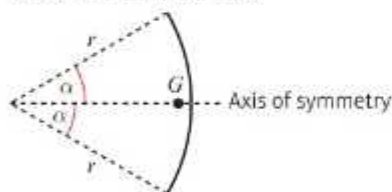


Given that the centre of mass of the hexagon is at the point M and that the centre of mass of the pentagon is at the point N , show that the length of MN is $\frac{2}{15}x$.

3.5 Centre of mass of a framework

You can find the centre of mass of a framework by using the centre of mass of each rod or wire which makes up the framework.

Links The centre of mass of a uniform straight rod is located at its midpoint.

Uniform circular arc

Hint This result can be found in the formula booklet.

A uniform circular arc of radius r and centre angle 2α , where α is measured in radians, has its centre of mass on the axis of symmetry at a distance $\frac{r \sin \alpha}{\alpha}$ from the centre.

- A **framework** consists of a number of rods joined together or a number of pieces of wire joined together.

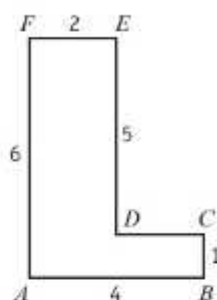
Provided that you can identify the position of the centre of mass of each of the rods or pieces of wire that make up a framework you can find the position of the centre of mass of the whole framework.

Example 14

A framework consists of a uniform length of wire which has been bent into the shape of a letter L, as shown.

Find the distance of the centre of mass of the framework from AB and AF .

Online Explore centres of mass of a framework using GeoGebra.



Since the wire is uniform, the mass of each edge will be proportional to its length. The centre of mass of each edge will be at its midpoint.

Taking A as the origin and axes along AB and AF :

$$4\begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1\begin{pmatrix} 4 \\ 0.5 \end{pmatrix} + 2\begin{pmatrix} 3 \\ 1 \end{pmatrix} + 5\begin{pmatrix} 2 \\ 3.5 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 6 \end{pmatrix} + 6\begin{pmatrix} 0 \\ 3 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 8 \\ 0 \end{pmatrix} + \begin{pmatrix} 4 \\ 0.5 \end{pmatrix} + \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \begin{pmatrix} 10 \\ 17.5 \end{pmatrix} + \begin{pmatrix} 2 \\ 12 \end{pmatrix} + \begin{pmatrix} 0 \\ 18 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 30 \\ 50 \end{pmatrix} = 20\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 1.5 \\ 2.5 \end{pmatrix} = \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

Distance from AF is 1.5 cm.

Distance from AB is 2.5 cm.

Each term on the LHS consists of the length of an edge multiplied by the position vector of the midpoint of the edge.

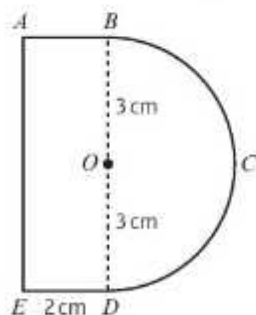
Simplify and collect terms.

This is $\bar{x}$.

This is $\bar{y}$.

Example 15

Find the position of the centre of mass of a framework constructed from a uniform piece of wire bent into the shape shown,



where the wire BCD is a semicircle, centre O , of radius 3 cm and wire $BAED$ forms three sides of a rectangle $ABDE$.

Take O as the origin and axes along OC and OB .

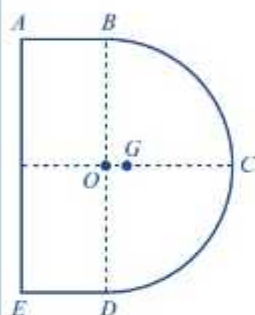
$$3\pi \begin{pmatrix} \frac{6}{\pi} \\ 0 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ -3 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 3 \end{pmatrix} + 6 \begin{pmatrix} -2 \\ 0 \end{pmatrix} = (10 + 3\pi) \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 18 \\ 0 \end{pmatrix} + \begin{pmatrix} -2 \\ -6 \end{pmatrix} + \begin{pmatrix} -2 \\ 6 \end{pmatrix} + \begin{pmatrix} -12 \\ 0 \end{pmatrix} = (10 + 3\pi) \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 0 \end{pmatrix} = (10 + 3\pi) \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$\bar{x} = \frac{2}{10 + 3\pi}$$

$$\bar{y} = 0$$



G is the centre of mass of the framework, on the axis of symmetry, a distance $\frac{2}{10 + 3\pi}$ cm to the right of O .

Use the result for the centre of mass of a uniform circular arc with $\alpha = \frac{\pi}{2}$ and $r = 3$ to find the centre of mass of the arc BCD .

3π is the length of the arc.

Simplify.

Collect terms.

You could have used the symmetry to deduce this without any working.

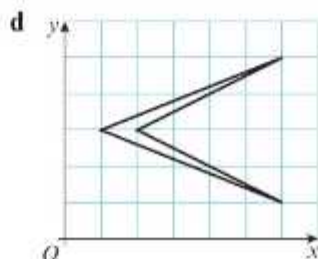
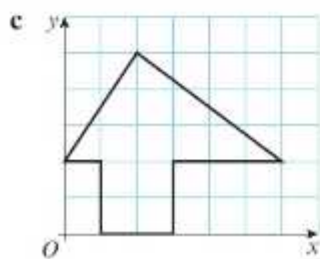
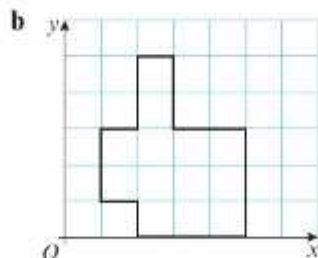
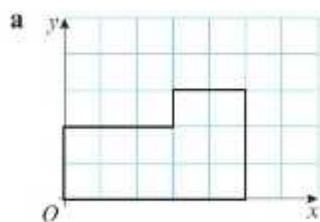
Exercise

3E

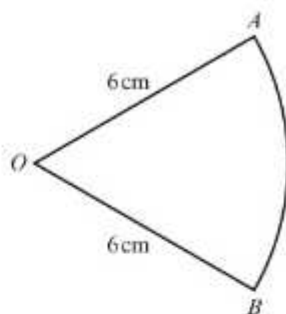
SKILLS

PROBLEM-SOLVING

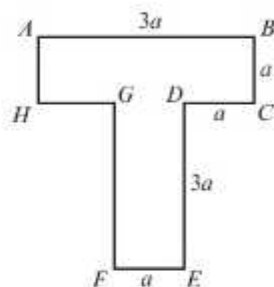
- 1 By regarding the shapes shown below as uniform plane wire frameworks, find the coordinates of the centre of mass of each shape.



- (E/P)** 2 Find the position of the centre of mass of the framework shown in the diagram, which is formed by bending a uniform piece of wire of total length $(12 + 2\pi)$ cm to form a sector of a circle, centre O , radius 6 cm. **(4 marks)**



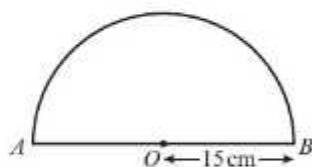
- (E)** 3 A framework consists of a uniform length of wire which has been bent into the shape of a letter T, as shown. Find the distance of the centre of mass of the framework from AB . **(6 marks)**



- (E/P)** 4 A uniform framework is constructed by bending wire into the shape of a semicircle and a diameter. The semicircle has radius 15 cm.

- a** Find the distance of the centre of mass of the framework from AB . **(4 marks)**

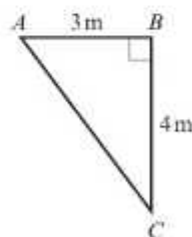
The metal used to form the framework has a mass of 8 grams per cm. Two identical particles of masses 100 g are attached to the framework, at A and B .



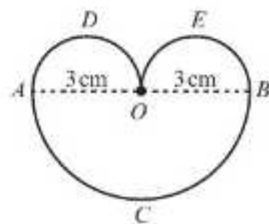
- b** Find:

- i** the total mass of the loaded framework
ii the distance of the centre of mass of the loaded framework from AB . **(4 marks)**

- (E/P)** 5 The diagram shows a triangular framework formed from uniform wire. Particles of masses 10 kg, 20 kg and 30 kg are attached to vertices A , B and C respectively. The mass of the unloaded framework is 15 kg. Find the position of the centre of mass of the loaded framework. **(8 marks)**



- (E/P)** 6 A uniform length of wire is bent to form the shape shown in the diagram. ACB is a semicircle of radius 3 cm, centre O . ADO and BEO are both semicircles of radius 1.5 cm. Find the position of the centre of mass of the framework. **(6 marks)**



- E/P** 7 A 3.5 m ladder is modelled as a framework made from uniform wire as shown in the diagram. The rungs are 50 cm wide and are 50 cm apart and the top and bottom rungs are 50 cm from the base and top of the ladder respectively. The base of the ladder rests on horizontal ground and the ladder stands vertically.

a Find the height of the centre of mass above the ground. (2 marks)

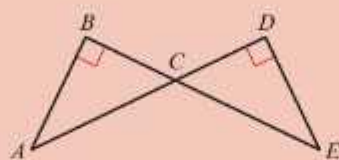
The bottom rung is removed from the ladder.

b Show that the height of the centre of mass of the ladder has increased by $\frac{5}{76}$ m. (4 marks)



Challenge

A metal framework $ABCDE$ is made from two congruent right-angled triangles such that ACD and BCE are straight lines, as shown in the diagram.

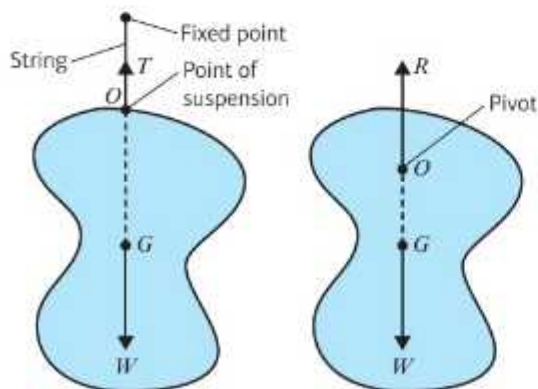


Given that $AB = 4$ cm and $CD = 3$ cm, work out the distance between C and the centre of mass of the framework.

3.6 Laminas in equilibrium

You can solve problems involving a lamina in **equilibrium**. A lamina can be **suspended** by means of a string attached to some point of the lamina, or can be allowed to **pivot** freely about a horizontal axis which passes through some point of the lamina.

- When a lamina is suspended freely from a fixed point, or pivots freely about a horizontal axis, it will rest in equilibrium in a vertical plane with its centre of mass vertically below the point of suspension or the pivot.



Hint

The lamina on the left is suspended from a fixed point. There are only two forces acting on it: the weight of the lamina and the **tension** in the string. Both forces pass through the point of suspension.

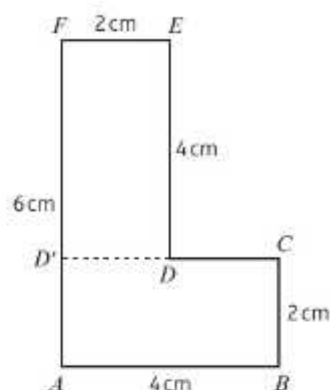
The lamina on the right is free to rotate about a fixed horizontal pivot. There are only two forces acting on it: the weight of the lamina and the reaction of the pivot on the lamina. Both pass through the pivot.

The **resultant** of the moments about O in both laminas is zero.

Example 16**SKILLS****PROBLEM-SOLVING**

Find the angle that the line AB makes with the vertical if this L-shaped uniform lamina is freely suspended from:

- a A
- b B
- c E



First find the centre of mass of the lamina.

Split the lamina along CD' .

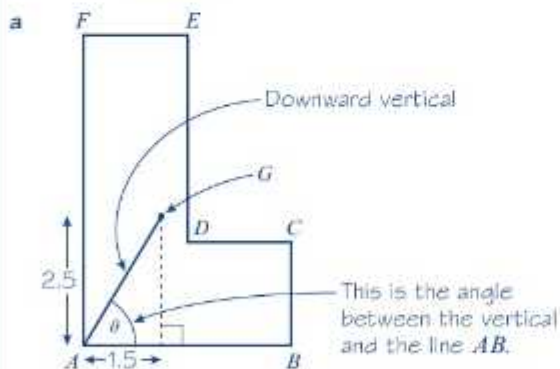
Take A as the origin and axes along AB and AF :

$$8\left(\frac{2}{1}\right) + 8\left(\frac{1}{4}\right) = 16\left(\frac{\bar{x}}{\bar{y}}\right)$$

$$\left(\frac{24}{40}\right) = 16\left(\frac{\bar{x}}{\bar{y}}\right)$$

$$\bar{x} = 1.5$$

$$\bar{y} = 2.5$$



$$\tan \theta = \frac{2.5}{1.5}$$

$$\Rightarrow \theta = 59.0^\circ \text{ (3 s.f.)}$$

Split the lamina into two rectangles.

Area $ABCD' = 8$

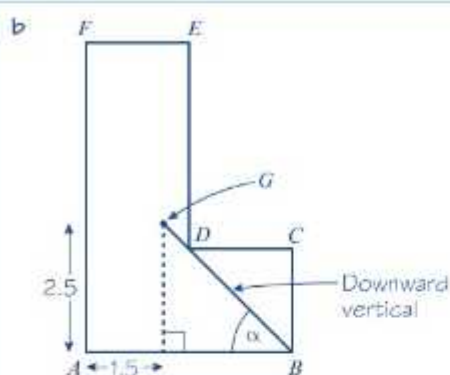
Area $DEFD' = 8$

Simplify.

Problem-solving

You do not need to draw the lamina hanging. Draw a line from the point of suspension to the centre of mass. Mark this in as the vertical.

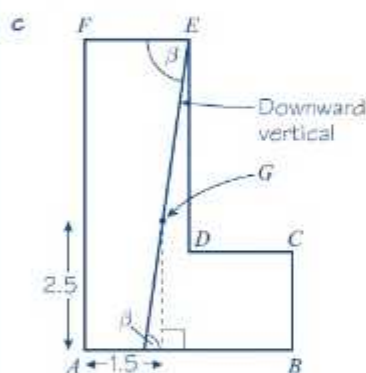
θ is the angle required.



$$\begin{aligned}\tan \alpha &= \frac{2.5}{4 - 1.5} \\ &= \frac{2.5}{2.5} \\ \Rightarrow \alpha &= 45^\circ\end{aligned}$$

This time draw a line from B to G and mark this as the vertical.

α is the angle required.



$$\begin{aligned}\tan \beta &= \frac{6 - 2.5}{2 - 1.5} \\ &= \frac{3.5}{0.5} \\ \Rightarrow \beta &= 81.9^\circ \text{ (3 s.f.)}\end{aligned}$$

This time draw a line from E to G and mark this as the vertical.

β is the angle required.

$\widehat{FEG} = \beta$ (alternate angles)

Using angle $\widehat{FEG}$:

$$\tan \beta = \frac{AF - \bar{y}}{EF - \bar{x}}$$

Example 17

The L-shaped lamina in Example 14 has mass M kg. Find the angle that FE makes with the vertical when a mass of $\frac{1}{10} M$ kg is attached to B and the lamina is freely suspended from F .

$$\begin{aligned}M \begin{pmatrix} 1.5 \\ 2.5 \end{pmatrix} + \frac{1}{10} M \begin{pmatrix} 4 \\ 0 \end{pmatrix} &= \frac{11}{10} M \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} \\ \begin{pmatrix} 1.9 \\ 2.5 \end{pmatrix} &= \frac{11}{10} \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} \\ \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} &= \begin{pmatrix} \frac{19}{11} \\ \frac{25}{11} \end{pmatrix}\end{aligned}$$

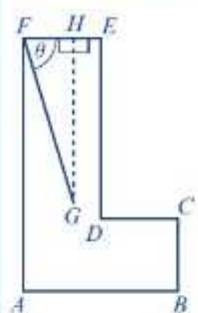
Problem-solving

Consider the lamina as a single particle whose weight acts at its centre of mass.

Recall from Example 14 that the centre of mass of the lamina is at $(1.5, 2.5)$.

The centre of mass of the added mass is at $(4, 0)$.

So the centre of mass of the loaded lamina is at the point $(\frac{19}{11}, \frac{25}{11})$.



$$\tan \theta = \frac{HG}{FH} = \frac{6 - \frac{25}{11}}{\frac{19}{11}} = \frac{41}{19}$$

$$\theta = 65.1^\circ \text{ (3 s.f.)}$$

Redraw the diagram showing the angle that you require.

G is the centre of mass and θ is the angle FE makes with the vertical when the lamina is suspended from F .

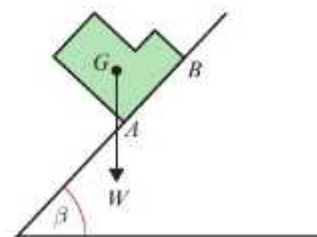
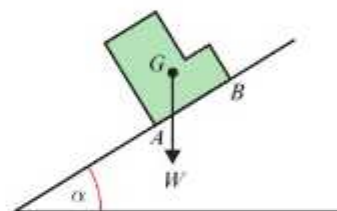
You can determine whether a body will remain in equilibrium or if equilibrium will be broken by sliding or toppling.

- If a lamina rests in equilibrium on a rough **inclined** plane then the line of action of the weight of the lamina must pass through the side of the lamina AB which is in contact with the plane.

The weight of the lamina produces a clockwise moment about A which keeps the lamina in contact with the plane.

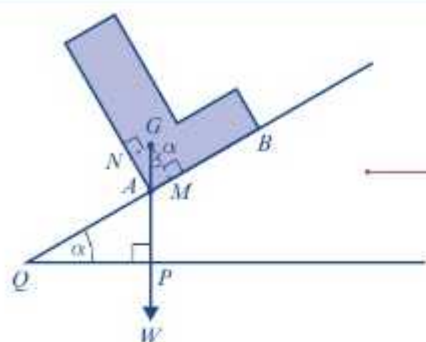
If the angle of the plane is increased so that the line of action of the weight passes outside the side AB then the weight produces an anticlockwise moment about A . The lamina will **topple** over.

You can usually assume that the coefficient of friction between the lamina and the plane is large enough to prevent the lamina from slipping down the plane.



Example 18

The L-shaped lamina from Example 14 is placed with AB in contact with a rough inclined plane. The angle of the plane is gradually increased. Assuming that the lamina does not slide down the plane, find the angle that the plane makes with the horizontal when the lamina is about to topple over.



When the lamina is about to topple its centre of mass G will be vertically above the point A .

$$AM = \bar{x} = 1.5; AN = \bar{y} = 2.5$$

$$\text{Angle } \hat{GAM} = 90^\circ - \alpha$$

$$\text{so, angle } \hat{AGM} = \alpha$$

$$\text{In } \triangle AMG, \tan \alpha = \frac{AM}{MG} = \frac{\bar{x}}{y}$$

$$= \frac{1.5}{2.5} = \frac{3}{5}$$

$$\alpha = 31^\circ \text{ (nearest degree)}$$

The plane must be rough enough to prevent the lamina from sliding down.

The moment of W about A is zero.

See Example 14.

$$\hat{GAM} = \hat{QAP}$$

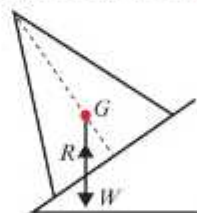
Since $\triangle AMG$ is right-angled.

Multiplying top and bottom by 2.

Hint

When a lamina on an inclined plane is about to topple, its centre of mass will be vertically above the lowest point of the lamina which is in contact with the plane.

A similar result applies for a **rigid body**.



The only forces acting on the body are its weight and the total reaction between the plane and the body. As the body is in equilibrium, these forces must be equal and opposite and act in the same vertical line.

- When a rigid body rests in equilibrium on a horizontal or rough inclined plane, then the line of action of the weight of the body must pass through the area of contact with the plane.

Exercise

3F

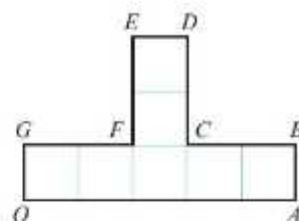
SKILLS

PROBLEM-SOLVING

- 1 a The diagram shows a uniform lamina.

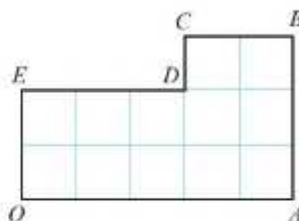
The lamina is freely suspended from the point O and hangs in equilibrium.

Find the angle between OA and the downward vertical.



- b The diagram shows a uniform lamina.

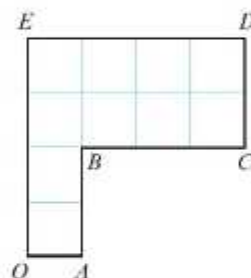
The lamina is freely suspended from the point O and hangs in equilibrium. Find the angle between OA and the downward vertical.



- c The diagram shows a uniform lamina.

The lamina is freely suspended from the point O and hangs in equilibrium.

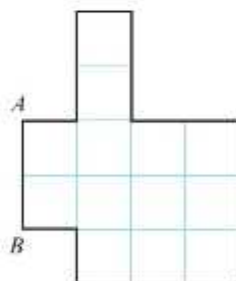
Find the angle between OA and the downward vertical.



- 2 The diagram shows a uniform lamina.

The lamina is freely suspended from the point A and hangs in equilibrium.

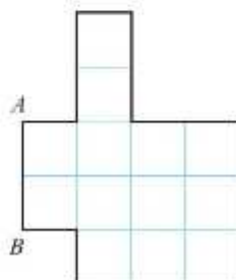
Find the angle between AB and the downward vertical.



- 3 The diagram shows a uniform lamina.

The lamina is free to rotate about a fixed smooth horizontal axis perpendicular to the plane of the lamina, passing through the point A , and hangs in equilibrium.

Find the angle between AB and the horizontal.



- E** 4 $PQRS$ is a uniform lamina.

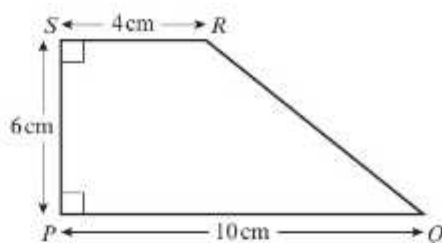
Find the distance of the centre of mass of the lamina from:

a PS (4 marks)

b PQ (4 marks)

The lamina is suspended from the point Q and allowed to hang in equilibrium.

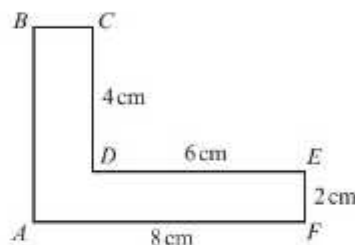
c Find the angle that PQ makes with the vertical.



(3 marks)

- E/P** 5 The L-shaped lamina shown has mass M kg.

Find the angle that BC makes with the vertical when a mass of $0.2M$ kg is attached to F and the lamina is freely suspended from C . (8 marks)



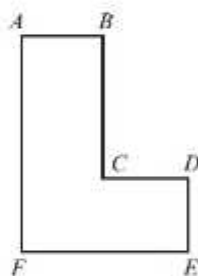
- 6 The L-shaped lamina $ABCDEF$ shown has sides of length $AB = 2$ cm, $BC = 4$ cm, $CD = 2$ cm, $DE = 2$ cm, $EF = 4$ cm and $FA = 6$ cm.

a Work out the horizontal distance of the centre of mass from FA .

b Work out the vertical distance of the centre of mass from EF .

The lamina is placed on a rough slope that is inclined at an angle of θ to the horizontal.

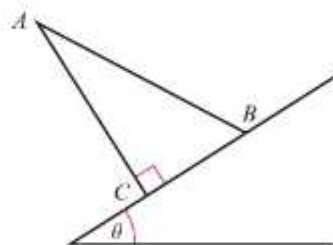
c Assuming that the lamina does not slide, work out the maximum value of θ that can be reached without the lamina toppling.



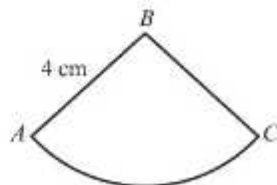
- 7 The right-angled triangular lamina ABC shown has mass M kg and sides of length $BC = 3$ cm and $AC = 6$ cm.

A mass of $0.2M$ kg is attached to the lamina at A .

Assuming that the lamina does not slide, work out the maximum value of θ that can be reached without the lamina toppling.



- E/P** 8 The lamina shown is a quarter circle with radius 4 cm and mass M kg. Find the angle that AB makes with the vertical when a mass of $0.5M$ kg is attached to C and the lamina is freely suspended from B . (6 marks)

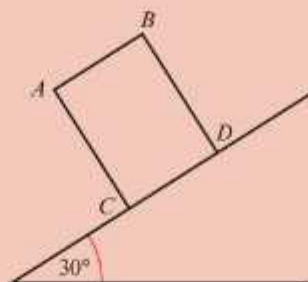


Challenge

The rectangular lamina $ABCD$ shown has mass M kg and sides of length $AB = CD = 4$ cm, $BC = DA = 6$ cm. A mass of kM kg is attached at A . The lamina is placed on a rough slope that is angled at 30° to the horizontal.

Assuming that the lamina does not slip, show that the maximum value of k that can be reached before the lamina topples

is $\frac{1}{\sqrt{3}} - \frac{1}{2}$.

**3.7 Frameworks in equilibrium**

Problems involving frameworks in equilibrium can be solved using the same methods that are used to solve problems involving laminas in equilibrium.

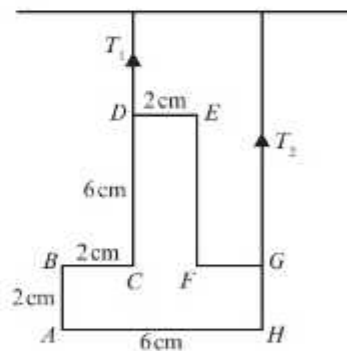
Example 19

The inverted T-shaped uniform framework of weight W shown in the diagram is freely suspended from D and G by two vertical strings, so that AH is horizontal. The strings and the framework lie in the same plane.

- a Find, in terms of W , the tensions T_1 and T_2 .

The string at D breaks and the framework hangs freely in equilibrium from G .

- b Find the angle that DE makes with the vertical when the framework comes to rest in equilibrium.



- a Let AH lie on the x -axis and AB lie on the y -axis with the point A at the origin.

$$2\begin{pmatrix} 0 \\ 1 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 6\begin{pmatrix} 2 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 3 \\ 8 \end{pmatrix} + 6\begin{pmatrix} 4 \\ 5 \end{pmatrix} + 2\begin{pmatrix} 5 \\ 2 \end{pmatrix} + 2\begin{pmatrix} 6 \\ 1 \end{pmatrix} + 6\begin{pmatrix} 3 \\ 0 \end{pmatrix} = 28\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

$$28\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = \begin{pmatrix} 84 \\ 88 \end{pmatrix}$$

$$\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = \begin{pmatrix} 3 \\ \frac{22}{7} \end{pmatrix}$$

Res (1) $T_1 + T_2 = W$

Taking moments about D gives

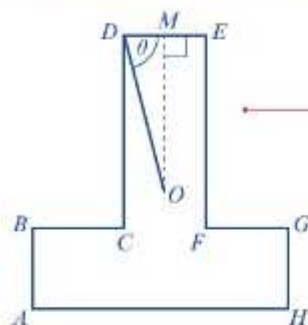
$$W = 4T_2 \Rightarrow T_2 = \frac{1}{4}W \text{ and } T_1 = \frac{3}{4}W$$

First find the centre of mass of the framework.

Problem-solving

The framework is in equilibrium, so the resultant moment must be 0. The centre of mass of the framework is a horizontal distance of 1 cm from D , so the clockwise moment is $1 \times W = W$. The anticlockwise moment is due to T_2 so is equal to $4T_2$.

- b The centre of mass lies $\frac{22}{7}$ cm from AH .



$$\tan \theta = \frac{OM}{DM} = \frac{8 - \frac{22}{7}}{1} = \frac{34}{7}$$

$$\theta = 78.4^\circ \text{ (3 s.f.)}$$

Redraw the diagram showing the angle that you require.

O is the centre of mass and θ is the angle DE makes with the vertical when suspended from D .

Example 20

The inverted T-shaped uniform framework used in Example 19 has mass W kg.

- a Find the location of the centre of mass when a mass of $\frac{1}{4}W$ kg is fixed at point H .
b Find the angle that DE makes with the vertical when the system is freely suspended from E .

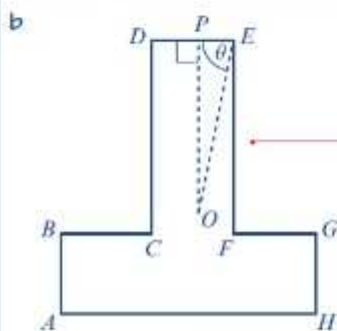
$$\begin{aligned} \text{a } W\left(\frac{3}{\frac{22}{7}}\right) + \frac{1}{4}W\left(\frac{6}{0}\right) &= \frac{5}{4}W\left(\frac{\bar{x}}{\bar{y}}\right) \\ \left(\frac{3}{\frac{22}{7}}\right) + \left(\frac{\frac{3}{2}}{0}\right) &= \frac{5}{4}\left(\frac{\bar{x}}{\bar{y}}\right) \\ \frac{5}{4}\left(\frac{\bar{x}}{\bar{y}}\right) &= \left(\frac{\frac{3}{2}}{\frac{22}{7}}\right) \\ \left(\frac{\bar{x}}{\bar{y}}\right) &= \left(\frac{\frac{18}{5}}{\frac{88}{35}}\right) \end{aligned}$$

So the centre of mass is at the point $\left(\frac{18}{5}, \frac{88}{35}\right)$.

Recall from Example 19 that the centre of mass of the lamina is at $\left(3, \frac{22}{7}\right)$.

The centre of mass of the added mass is at $(6, 0)$.

Find the centre of mass of the framework and the added mass.



$$\tan \theta = \frac{PO}{PE} = \frac{8 - \frac{88}{35}}{6 - \frac{18}{5}} = \frac{16}{7}$$

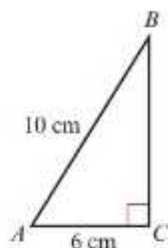
$$\theta = 66.4^\circ \text{ (3 s.f.)}$$

Redraw the diagram showing the angle that you require.

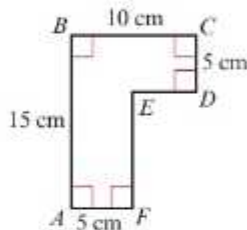
O is the centre of mass and θ is the angle DE makes with the vertical when the framework is suspended from E .

Exercise 3G

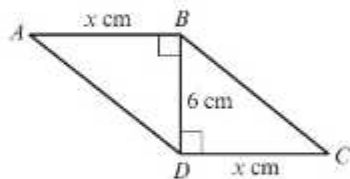
- 1 The uniform framework shown in the diagram is freely suspended from the point B and hangs in equilibrium. Find the angle between BC and the downward vertical.



- 2 The uniform framework shown in the diagram is freely suspended from the point D and hangs in equilibrium. Find the angle between CD and the downward vertical.

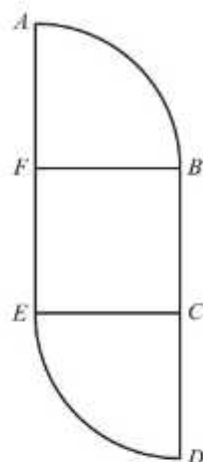


- E/P** 3 The uniform framework shown in the diagram is freely suspended from the point A and hangs in equilibrium.
- a Given that the angle between AB and the downward vertical is $\arctan \frac{3}{8}$, work out the value of x . **(4 marks)**
- The framework has mass M kg.

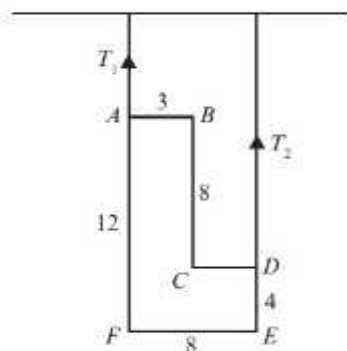


- b A particle of mass kM kg is attached to A and the framework is then freely suspended from the point B so that it hangs in equilibrium. Given that the downward vertical now makes an angle of $\arctan \frac{8}{15}$ with BD , work out the value of k . **(4 marks)**
- P** 4 The uniform framework shown in question 1 has mass M kg. A particle of mass $0.75M$ kg is attached to A and the framework is then freely suspended from the point B so that it hangs in equilibrium. Find the angle between BC and the downward vertical.
- P** 5 The uniform framework shown in question 2 has mass M kg. A particle of mass $0.15M$ kg is attached to F and the framework is then freely suspended from the point B so that it hangs in equilibrium. Find the angle between BC and the downward vertical.

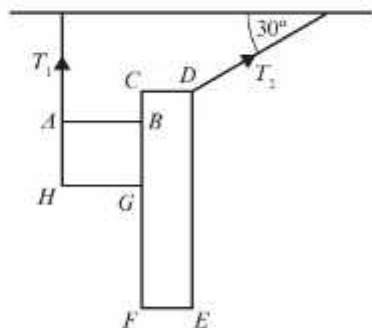
- E/P** 6 The uniform framework shown in the diagram is made of a square of side 4 cm and two quarter circles and has mass M kg. A particle of mass $0.1M$ kg is attached to C and the framework is then freely suspended from the point A so that it hangs in equilibrium. Find the angle between FE and the downward vertical. **(6 marks)**



- E/P** 7 The uniform framework of weight W shown in the diagram is suspended by two vertical strings from A and D such that FE is horizontal. The strings and the framework lie in the same plane. The lengths shown are in metres and all the angles are right angles.
- a** Work out, in terms of W , the tensions T_1 and T_2 . **(6 marks)**
- The string at A snaps and the framework hangs in equilibrium suspended from D .
- b** Find the angle that DE makes with the vertical. **(3 marks)**

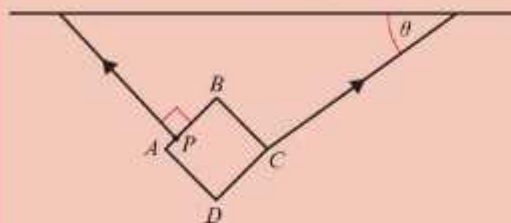


- E/P** 8 A uniform framework made from uniform rods is shown in the diagram. It has weight W and is suspended by one vertical string attached at A and one string angled at 60° to the vertical attached at D . The strings and the framework lie in the same plane. The framework hangs such that FE is horizontal. The lengths are $AB = 6$ cm, $AH = 4$ cm, $BC = 2$ cm, $CD = 3$ cm and $DE = 10$ cm. There is only one rod in the span BG .
- a** Work out, in terms of W , the tensions T_1 and T_2 . **(6 marks)**
- The string at D snaps.
- b** Find the angle that AB makes with the vertical when the framework comes to rest in equilibrium. **(3 marks)**



Challenge

A square uniform framework $ABCD$ is suspended by two light inextensible strings. The first string is perpendicular to AB and is attached at the point P , which lies on AB such that $AP:PB = 1:3$. The second string is attached at C and makes an angle of θ with the horizontal, as shown in the diagram.



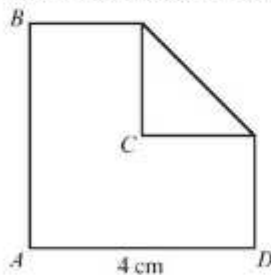
Given that the framework rests in equilibrium with AC horizontal, find θ , giving your answer in degrees, correct to one decimal place.

3.8 Non-uniform composite laminas and frameworks

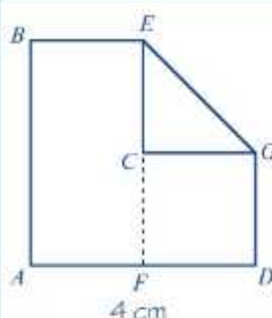
You can use the methods developed in this chapter to solve problems involving non-uniform laminas and frameworks.

Example 21

The lamina shown is made from a square piece of cardboard $ABCD$ that has had the corner C folded to the centre of the square. Find the position of the centre of mass of the lamina.

**Problem-solving**

You can model this situation as a composite lamina made up of three different uniform laminas. The folded section of card is modelled as a lamina with **twice** the density of the other sections.



Redraw the diagram, splitting the lamina into its composite parts.

Let A be the origin so that AD lies on the x -axis and AB lies on the y -axis.

The coordinates of the centre of mass of the rectangle $ABEF$ are $(1, 2)$.

The coordinates of the centre of mass of the square $CGDF$ are $(3, 1)$.

The coordinates of the centre of mass of the triangle CEG are

$$\left(\frac{2+2+4}{3}, \frac{2+4+2}{3} \right) = \left(\frac{8}{3}, \frac{8}{3} \right)$$

Centre of mass of a triangle with vertices at (x_1, y_1) , (x_2, y_2) , (x_3, y_3) is found at:

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Rectangle $ABEF$ has area 8 cm^2 , square $CGDF$ has area 4 cm^2 . Triangle CEG has area 2 cm^2 , but this material is twice the density of the other material.

Remember that the card is folded over so the triangle CEG is made of two layers.

$$8 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 4 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} \frac{8}{3} \\ \frac{8}{3} \end{pmatrix} = 16 \begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

Use 4 for the mass of the triangular section.

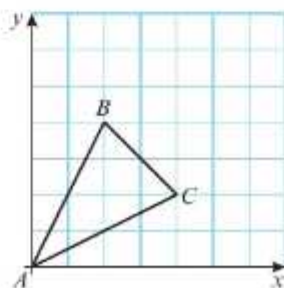
$$\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = \begin{pmatrix} \frac{23}{12} \\ \frac{23}{12} \end{pmatrix}$$

So the centre of mass of the lamina is at

$$\left(\frac{23}{12}, \frac{23}{12} \right).$$

Example 22

The triangular framework shown is made from three pieces of uniform wire, AB , AC and BC of mass $2M$, $3M$ and $5M$ respectively. Find the coordinates of the centre of mass of the framework.



$$2M\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3M\begin{pmatrix} 2 \\ 1 \end{pmatrix} + 5M\begin{pmatrix} 3 \\ 3 \end{pmatrix} = 10M\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix}$$

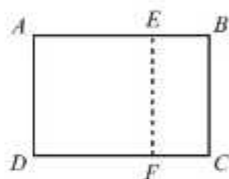
$$\begin{pmatrix} \bar{x} \\ \bar{y} \end{pmatrix} = \begin{pmatrix} 2.3 \\ 2.2 \end{pmatrix}$$

So the coordinates of the centre of mass are at $(2.3, 2.2)$.

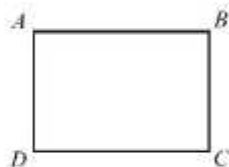
Multiply the mass of each piece of wire by the position of its centre of mass.

Exercise 3H

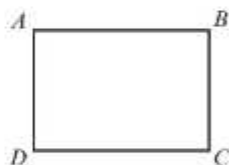
- (P) 1 The rectangular lamina shown is made from a rectangular piece of cardboard $ABCD$ where AB is 6 cm and BC is 4 cm. The rectangular lamina is folded along the line EF to make the square $AEFD$. Find the position of the centre of mass of the square lamina.



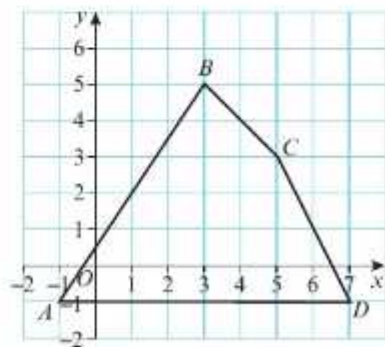
- (P) 2 The rectangular lamina shown is made from a rectangular piece of cardboard $ABCD$ where AB is 10 cm and BC is 6 cm. The lamina is folded so that the side AD lies along the side AB . Find the position of the centre of mass of the resulting lamina.



- (E/P) 3 The diagram shows a rectangular lamina $ABCD$, where $AB = 10$ cm and $BC = 6$ cm. The lamina is folded so that D lies exactly on top of B . It is then suspended freely from A . Find the angle that AD makes with the vertical. (8 marks)



- (P) 4 The framework shown is made from four pieces of uniform wire, AB , BC , CD and DA , of masses $2M$, $3M$, M and $5M$ respectively. Find the coordinates of the centre of mass of the framework.



4 DYNAMICS OF A PARTICLE MOVING IN A STRAIGHT LINE

Learning objectives

After completing this chapter you should be able to:

- Draw force diagrams and calculate resultant forces → pages 55–57
- Understand and use Newton's first law of motion → pages 55–57
- Calculate resultant forces by adding vectors → pages 58–60
- Understand and use Newton's second law of motion, $F = ma$ → pages 60–64
- Apply Newton's second law to vector forces and acceleration → pages 64–67
- Solve problems involving connected particles → pages 67–70
- Understand and use Newton's third law of motion → pages 68–70
- Solve problems involving pulleys → pages 71–75

4.1
4.2

Prior knowledge check

1 Calculate:

a $(2\mathbf{i} + \mathbf{j}) + (3\mathbf{i} - 4\mathbf{j})$

b $(-\mathbf{i} + 3\mathbf{j}) - (3\mathbf{i} - \mathbf{j})$

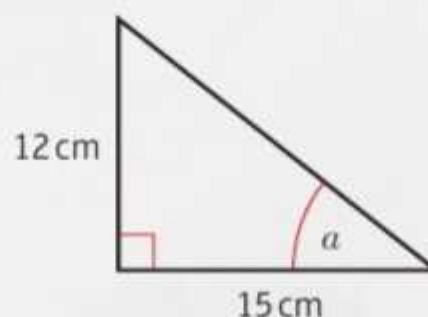
← International GCSE Mathematics

2 The diagram shows a right-angled triangle.

Work out:

a the length of the hypotenuse

b the size of the angle a .



Give your answers correct to 1 d.p. ← International GCSE Mathematics

3 A car starts from rest and accelerates at a constant rate of 1.5 m s^{-2} .

a Work out the velocity of the car after 12 seconds.

After 12 seconds, the driver brakes, causing the car to decelerate at a constant rate of 1 m s^{-2} .

b Calculate the distance the car travels from the instant the driver brakes until the car comes to rest.

← Mechanics 1, Chapter 2

The weight of an air-sea rescue crew man is balanced by the tension in the cable. By modelling the forces in this situation, you can calculate how strong the cable needs to be.

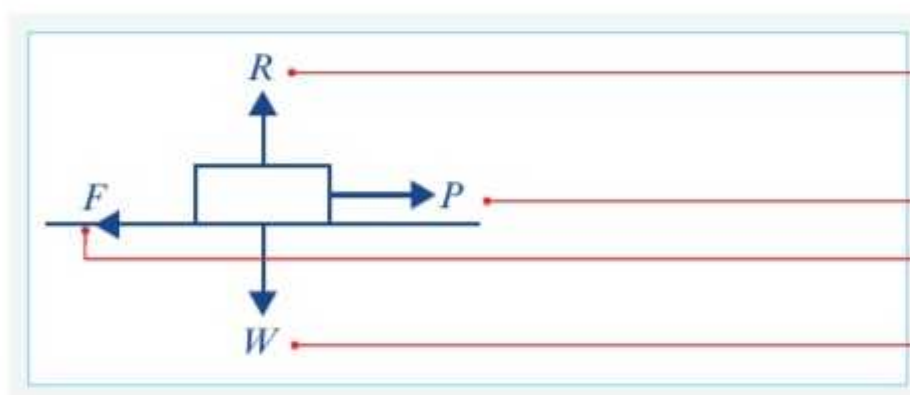
4.1 Force diagrams

A force diagram is a diagram showing all the forces acting on an object. Each force is shown as an arrow pointing in the direction in which the force acts. Force diagrams are used to model problems involving forces.

Example 1

SKILLS INTERPRETATION

A block of weight W is being pulled to the right by a force, P , across a rough horizontal plane. Draw a force diagram to show all the forces acting on the block.



R is the normal reaction of the rough horizontal plane on the block. ← Mechanics 1 Section 1.3

P is the force pulling the block.

F is the resistance due to friction between the block and the plane.

W is the weight of the block.

When the forces acting upon an object are balanced, the object is said to be in **equilibrium**.

- **Newton's first law of motion** states that an object at rest will stay at rest, and that an object moving with constant velocity will continue to move with constant velocity unless an unbalanced force acts on the object.

Watch out Constant velocity means that neither the speed nor the direction is changing.

When there is more than one force acting on an object you can resolve the forces in a certain direction to find the resultant force in that direction. The direction you are resolving in becomes the positive direction. You add forces acting in this direction and subtract forces acting in the opposite direction.

In your answers, you can use the letter R , together with an arrow, $R(\uparrow)$, to indicate the direction in which you are resolving the forces.

In this section you will only resolve forces that are horizontal or vertical.

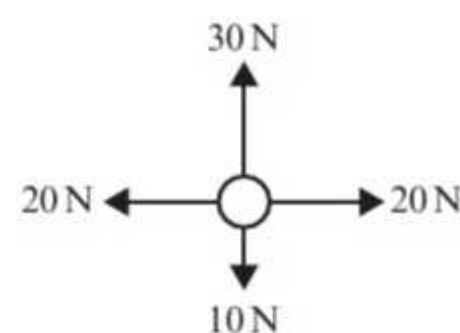
- A resultant force acting on an object will cause the object to accelerate in the same direction as the resultant force.

Example 2

SKILLS INTERPRETATION

The diagram shows the forces acting on a particle.

- Draw a force diagram to represent the resultant force.
- Describe the motion of the particle.



a 20 N



b The particle is accelerating upward.

$R(\rightarrow): 20 - 20 = 0$, so the horizontal forces are balanced.

$R(\uparrow): 30 - 10 = 20$, so the resultant force is 20 N upward.

Exercise

4A

SKILLS

INTERPRETATION

- 1 A box is at rest on a horizontal table. Draw a force diagram to show all the forces acting on the box.
- 2 A trapeze bar is suspended motionless from the ceiling by two ropes. Draw a force diagram to show the forces acting on the ropes and the trapeze bar.
- 3 Ignoring air resistance, draw a diagram to show the forces acting on an apple as it falls from a tree.
- 4 A car's engine applies a force parallel to the surface of a horizontal road that causes the car to move with constant velocity. Considering the resistance to motion, draw a diagram to show the forces acting on the car.
- 5 An air-sea rescue crew member is suspended motionless from a helicopter. Ignoring air resistance, show all the forces acting on him.

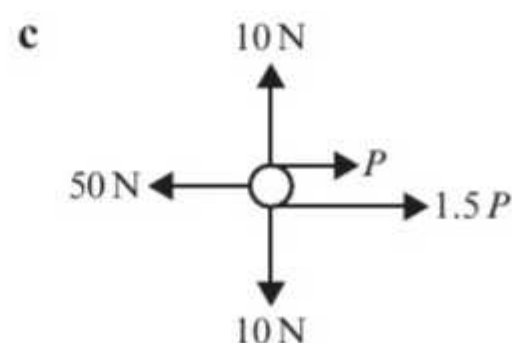
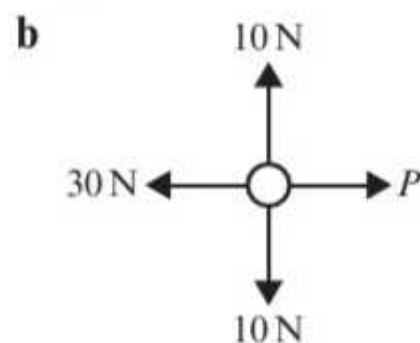
- (P) 6 A satellite orbits the Earth at constant speed. State, with a reason, whether any resultant force is acting on the satellite.

Problem-solving

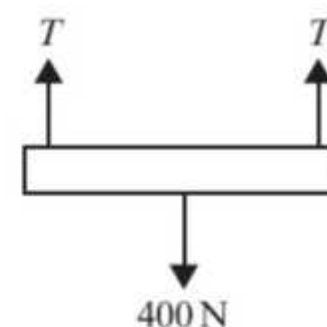
Consider the velocity of the satellite as it orbits the Earth.

- 7 A particle of weight 5 N sits at rest on a horizontal plane. State the value of the normal reaction acting on the particle.

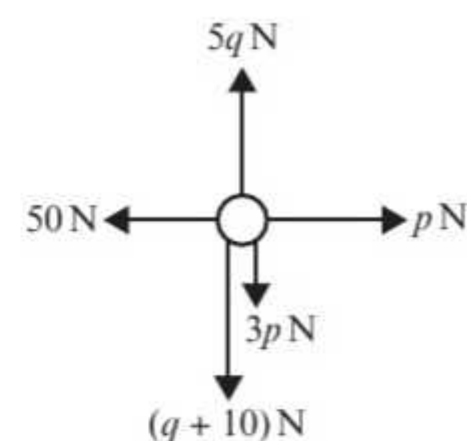
- 8 Given that each of the particles is stationary, work out the value of P :



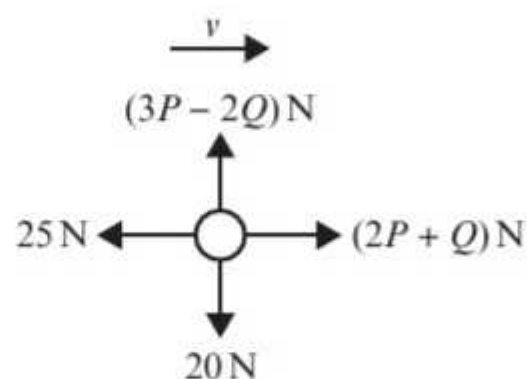
- 9 A platform is lifted vertically at constant velocity as shown in the diagram.
 - a Ignoring air resistance, work out the tension, T in each rope. The tension in each rope is reduced by 50 N.
 - b Describe the resulting motion of the platform.



- 10 The diagram shows a particle acted on by a set of forces. Given that the particle is at rest, find the value of p and the value of q .



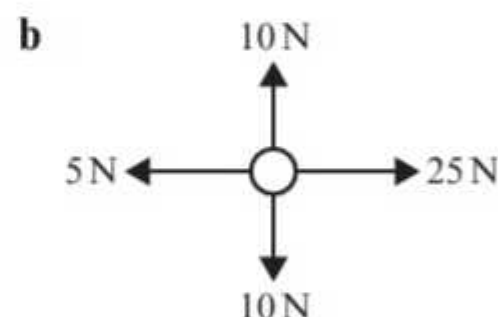
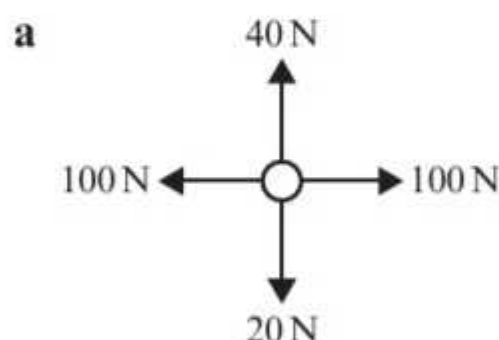
- 11 Given that the particle in this diagram is moving with constant velocity, v , find the values of P and Q .

**Problem-solving**

Set up two simultaneous equations.

- 12 Each diagram below shows the forces acting on a particle.

- Work out the size and direction of the resultant force.
- Describe the motion of the particle.



- 13 A truck is moving along a horizontal level road. The truck's engine provides a forward thrust of 10 000 N. The total resistance is modelled as a constant force of magnitude 1600 N.

- Modelling the truck as a particle, draw a force diagram to show the forces acting on the truck.
- Calculate the resultant force acting on the truck.

- 14 A car is moving along a horizontal level road. The car's engine provides a constant driving force. The motion of the car is opposed by a constant resistance.
- Modelling the car as a particle, draw a force diagram to show the forces acting on the car.
 - Given that the resultant force acting on the car is 4200 N in the direction of motion, and that the magnitude of the driving force is eight times the magnitude of the resistance force, calculate the magnitude of the resistance.

Problem-solving

Use algebra to describe the relationship between the driving force and the resistance.

4.2 Forces as vectors

You can write forces as vectors using $\mathbf{i}$ – $\mathbf{j}$ notation or as column vectors.

- You can find the resultant of two or more forces given as vectors by adding the vectors.

Links If two forces $(p\mathbf{i} + q\mathbf{j})$ N and $(r\mathbf{i} + s\mathbf{j})$ N are acting on a particle, the resultant force will be $((p + r)\mathbf{i} + (q + s)\mathbf{j})$ N. ← Mechanics 1 Section 3.2

When a particle is in equilibrium the resultant vector force will be $0\mathbf{i} + 0\mathbf{j}$.

Example 3 SKILLS PROBLEM-SOLVING

The forces $2\mathbf{i} + 3\mathbf{j}$, $4\mathbf{i} - \mathbf{j}$, $-3\mathbf{i} + 2\mathbf{j}$ and $a\mathbf{i} + b\mathbf{j}$ act on an object which is in equilibrium. Find the values of a and b .

$$\begin{aligned}(2\mathbf{i} + 3\mathbf{j}) + (4\mathbf{i} - \mathbf{j}) + (-3\mathbf{i} + 2\mathbf{j}) + (a\mathbf{i} + b\mathbf{j}) &= \mathbf{0} \\ (2 + 4 - 3 + a)\mathbf{i} + (3 - 1 + 2 + b)\mathbf{j} &= \mathbf{0} \\ \Rightarrow 3 + a = 0 \quad \text{and} \quad 4 + b = 0 \\ \Rightarrow a = -3 \quad \text{and} \quad b = -4\end{aligned}$$

If an object is in equilibrium then the resultant force will be zero.

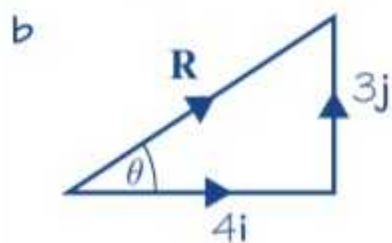
You can consider the $\mathbf{i}$ and $\mathbf{j}$ components separately because they are perpendicular.

Example 4 SKILLS PROBLEM-SOLVING

In this question, $\mathbf{i}$ represents the unit vector due east, and $\mathbf{j}$ represents the unit vector due north. A particle begins at rest at the origin. It is acted on by three forces $(2\mathbf{i} + \mathbf{j})$ N, $(3\mathbf{i} - 2\mathbf{j})$ N and $(-\mathbf{i} + 4\mathbf{j})$ N.

- Find the resultant force in the form $p\mathbf{i} + q\mathbf{j}$.
- Work out the magnitude and bearing of the resultant force.
- Describe the motion of the particle.

a $(2\mathbf{i} + \mathbf{j}) + (3\mathbf{i} - 2\mathbf{j}) + (-\mathbf{i} + 4\mathbf{j}) = 4\mathbf{i} + 3\mathbf{j}$



$$\mathbf{R} = (4\mathbf{i} + 3\mathbf{j}) \text{ N}$$

Therefore the magnitude of $\mathbf{R}$ is given by

$$|\mathbf{R}| = \sqrt{4^2 + 3^2} = \sqrt{25} = 5 \text{ N}$$

$$\tan \theta = \frac{3}{4}$$

$$\theta = 36.9^\circ \text{ (1 d.p.)}$$

$$\text{Bearing} = 90^\circ - 36.9^\circ = 053.1^\circ$$

- c The particle accelerates in the direction of the resultant force.

Add together the $\mathbf{i}$ -components and the $\mathbf{j}$ -components.

Notation The unit vector $\mathbf{i}$ is usually taken to be due east or the positive x -direction. The unit vector $\mathbf{j}$ is usually taken to be due north or the positive y -direction. Questions involving finding bearings will often specify this.

Use Pythagoras' theorem to find the magnitude of the resultant.

$$\text{Use } \tan \theta = \frac{\text{opp}}{\text{adj}}$$

Bearings are measured clockwise from north so subtract θ from 90° .

Exercise 4B

SKILLS

PROBLEM-SOLVING

- 1 In each part of the question a particle is acted upon by the forces given. Work out the resultant force acting on the particle.

a $(-\mathbf{i} + 3\mathbf{j})$ N and $(4\mathbf{i} - \mathbf{j})$ N

b $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$ N and $\begin{pmatrix} -3 \\ -6 \end{pmatrix}$ N

Notation $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$ N is the same as $(5\mathbf{i} + 3\mathbf{j})$ N.

c $(\mathbf{i} + \mathbf{j})$ N, $(5\mathbf{i} - 3\mathbf{j})$ N and $(-2\mathbf{i} - \mathbf{j})$ N d $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ N, $\begin{pmatrix} 6 \\ 0 \end{pmatrix}$ N and $\begin{pmatrix} -2 \\ -7 \end{pmatrix}$ N

- 2 An object is in equilibrium at O under the action of three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$. Find $\mathbf{F}_3$ in these cases:

a $\mathbf{F}_1 = (2\mathbf{i} + 7\mathbf{j})$ and $\mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j})$

b $\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j})$ and $\mathbf{F}_2 = (2\mathbf{i} + 3\mathbf{j})$

- (P) 3 The forces $\begin{pmatrix} a \\ 2b \end{pmatrix}$ N, $\begin{pmatrix} -2a \\ -b \end{pmatrix}$ N and $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ N act on an object which is in equilibrium. Find the values of a and b .

- 4 For each force find:

i the magnitude of the force

ii the angle the force makes with $\mathbf{i}$.

a $(3\mathbf{i} + 4\mathbf{j})$ N

b $(5\mathbf{i} - \mathbf{j})$ N

c $(-2\mathbf{i} + 3\mathbf{j})$ N

d $\begin{pmatrix} -1 \\ -1 \end{pmatrix}$ N

- 5 In this question, $\mathbf{i}$ represents the unit vector due east, and $\mathbf{j}$ represents the unit vector due north. A particle is acted upon by forces of:

a $(-2\mathbf{i} + \mathbf{j})$ N, $(5\mathbf{i} + 2\mathbf{j})$ N and $(-\mathbf{i} - 4\mathbf{j})$ N

b $(-2\mathbf{i} + \mathbf{j})$ N, $(2\mathbf{i} - 3\mathbf{j})$ N and $(3\mathbf{i} + 6\mathbf{j})$ N

Work out:

i the resultant vector

ii the magnitude of the resultant vector

iii the bearing of the resultant vector.

- (P) 6 The forces $(a\mathbf{i} - b\mathbf{j})$ N, $(b\mathbf{i} + a\mathbf{j})$ N and $(-4\mathbf{i} - 2\mathbf{j})$ N act on an object which is in equilibrium. Find the values of a and b .

Problem-solving

Use the $\mathbf{i}$ components and the $\mathbf{j}$ components to set up and solve two simultaneous equations.

- (P) 7 The forces $(2a\mathbf{i} + 2b\mathbf{j})$ N, $(-5b\mathbf{i} + 3a\mathbf{j})$ N and $(-11\mathbf{i} - 7\mathbf{j})$ N act on an object which is in equilibrium. Find the values of a and b .

- 8 Three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle. $\mathbf{F}_1 = (-3\mathbf{i} + 7\mathbf{j})$ N, $\mathbf{F}_2 = (\mathbf{i} - \mathbf{j})$ N and $\mathbf{F}_3 = (p\mathbf{i} + q\mathbf{j})$ N.

a Given that this particle is in equilibrium, determine the value of p and the value of q .

The resultant of the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ is $\mathbf{R}$.

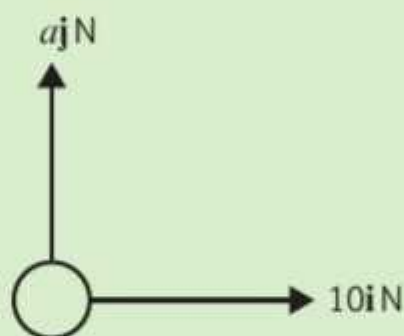
b Calculate, in N, the magnitude of $\mathbf{R}$.

c Calculate, to the nearest degree, the angle between the line of action of $\mathbf{R}$ and the vector $\mathbf{j}$.

- E/P** 9 A particle is acted upon by two forces $\mathbf{F}_1$ and $\mathbf{F}_2$, given by $\mathbf{F}_1 = (3\mathbf{i} - 2\mathbf{j})$ N and $\mathbf{F}_2 = (a\mathbf{i} + 2a\mathbf{j})$ N, where a is a positive constant.
- a Find the angle between $\mathbf{F}_2$ and $\mathbf{i}$. (2 marks)
- The resultant of $\mathbf{F}_1$ and $\mathbf{F}_2$ is $\mathbf{R}$.
- b Given that $\mathbf{R}$ is parallel to $13\mathbf{i} + 10\mathbf{j}$, find the value of a . (4 marks)
- E** 10 Three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ acting on a particle P are given by the vectors $\mathbf{F}_1 = \begin{pmatrix} -7 \\ -4 \end{pmatrix}$ N, $\mathbf{F}_2 = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ N and $\mathbf{F}_3 = \begin{pmatrix} a \\ b \end{pmatrix}$ N, where a and b are constants.
- a Given that P is in equilibrium, find the value of a and the value of b . (3 marks)
- b The force $\mathbf{F}_1$ is now removed. The resultant of $\mathbf{F}_2$ and $\mathbf{F}_3$ is $\mathbf{R}$. Find:
- i the magnitude of $\mathbf{R}$ (2 marks)
- ii the angle, to the nearest degree, that the direction of $\mathbf{R}$ makes with the horizontal. (3 marks)

Challenge

An object is acted upon by a horizontal force of $10\mathbf{i}$ N and a vertical force $a\mathbf{j}$ N as shown in the diagram. The resultant of the two forces acts in the direction 60° to the horizontal. Work out the value of a and the magnitude of the resultant force.



4.3 Forces and acceleration

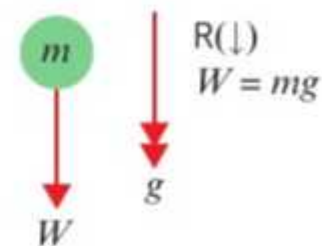
A non-zero resultant force that acts on a particle will cause the particle to accelerate in the direction of the resultant force.

- **Newton's second law of motion** states that the force needed to accelerate a particle is equal to the product of the mass of the particle and the acceleration produced: $F = ma$.

A force of 1 N will accelerate a mass of 1 kg at a rate of 1 m s^{-2} . If a force F N acts on a particle of mass m kg causing it to accelerate at $a \text{ m s}^{-2}$, the **equation of motion** for the particle is $F = ma$.

Gravity is the force between any object and the Earth. The force due to gravity acting on an object is called the **weight** of the object, and it acts vertically downward. A body falling freely experiences an acceleration of $g = 9.8 \text{ m s}^{-2}$. Using the relationship $F = ma$ you can write the equation of motion for a body of mass m kg with weight W N.

- $W = mg$



Example 5**SKILLS** PROBLEM-SOLVING

Find the acceleration when a particle of mass 1.5 kg is acted on by a resultant force of 6 N.

$$F = ma$$

$$6 = 1.5a$$

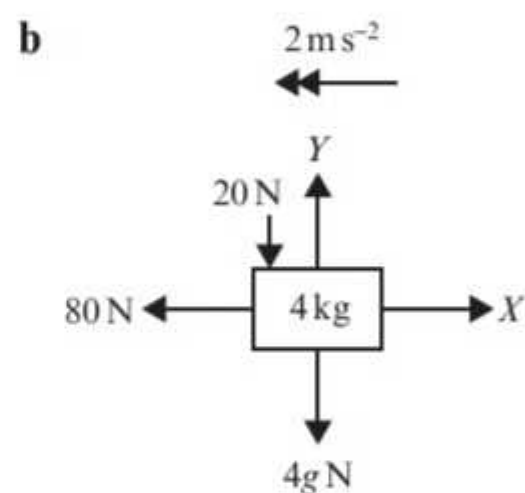
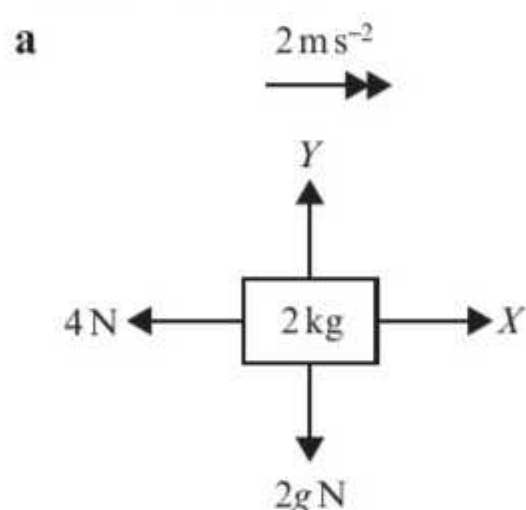
$$a = 4$$

The acceleration is 4 m s^{-2} .

Substitute the values you know and solve the equation to find a .

Example 6**SKILLS** PROBLEM-SOLVING

In each of these diagrams the body is accelerating as shown. Find the magnitudes of the unknown forces X and Y .



a $R(\rightarrow), X - 4 = 2 \times 2$
 $X = 8 \text{ N}$

$R(\uparrow), Y - 2g = 2 \times 0$
 $Y = 2g$
 $= 19.6 \text{ N}$

b $R(\leftarrow), 80 - X = 4 \times 2$
 $X = 72 \text{ N}$

$R(\uparrow), Y - 20 - 4g = 4 \times 0$
 $Y = 20 + (4 \times 9.8)$
 $= 59.2 \text{ N (3 s.f.)}$

$R(\rightarrow)$ means that you are finding the resultant force in the horizontal direction, in the direction of the arrow. The arrow is in the positive direction.

This resultant force causes an acceleration of 2 m s^{-2} . Use $F = ma$.

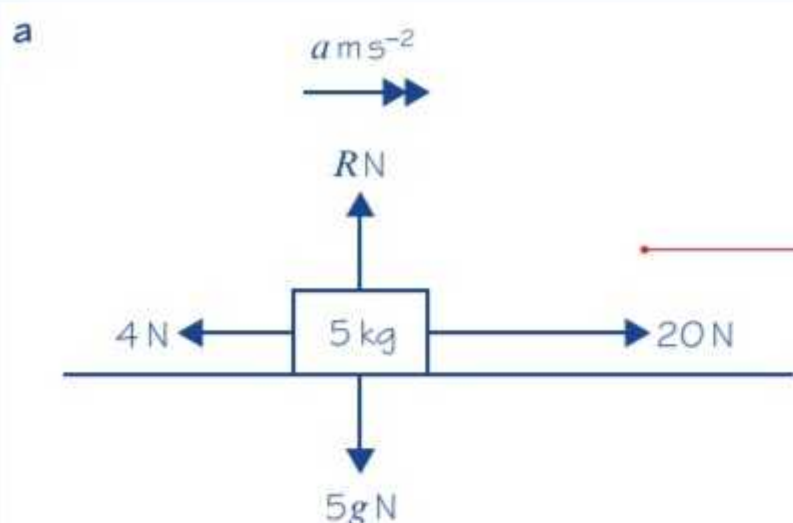
It is usually easier to take the positive direction as the direction of the acceleration.

There is no vertical acceleration, so $a = 0$.

Example 7**SKILLS** PROBLEM-SOLVING

A body of mass 5 kg is pulled along a rough horizontal table by a horizontal force of magnitude 20 N against a constant friction force of magnitude 4 N. Given that the body is initially at rest, find:

- the acceleration of the body
- the distance travelled by the body in the first 4 seconds
- the magnitude of the normal reaction between the body and the table.



$$R(\rightarrow), 20 - 4 = 5a$$

$$a = \frac{16}{5} = 3.2$$

The body accelerates at 3.2 m s^{-2} .

b $s = ut + \frac{1}{2}at^2$

$$s = (0 \times 4) + \frac{1}{2} \times 3.2 \times 4^2$$

$$= 25.6$$

The body moves a distance of 25.6 m.

c $R(\uparrow), R - 5g = 5 \times 0 = 0$

$$R = 5g = 5 \times 9.8 = 49 \text{ N}$$

The normal reaction has magnitude 49 N.

Draw a diagram showing all the forces and the acceleration.

Resolve horizontally, taking the positive direction as the direction of the acceleration, and write down an equation of motion for the body.

Since the acceleration is constant.

Substitute in the values.

Resolve vertically. Since the body is moving horizontally $a = 0$, so the right-hand side of the equation of motion is 0.

Exercise

4C

SKILLS

PROBLEM-SOLVING

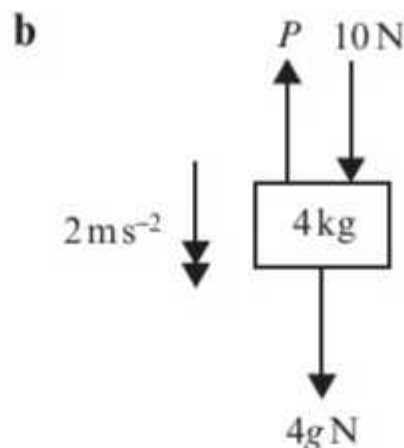
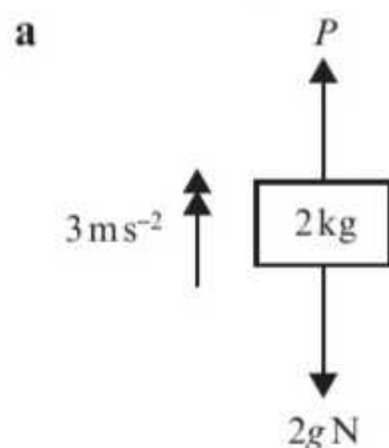
- Find the acceleration when a particle of mass 400 kg is acted on by a resultant force of 120 N.
- Find the weight in newtons of a particle of mass 4 kg.
- An object moving on a rough surface experiences a constant frictional force of 30 N which decelerates it at a rate of 1.2 m s^{-2} . Find the mass of the object.

- (P) 4 An astronaut weighs 735 N on the Earth and 120 N on the moon. Work out the value of acceleration due to gravity on the moon.

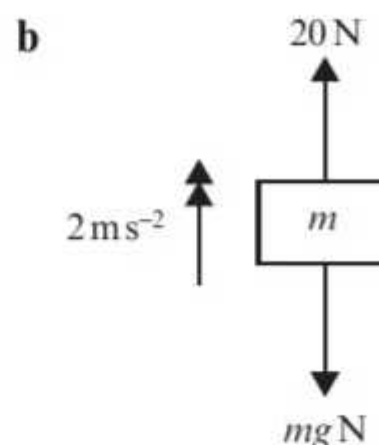
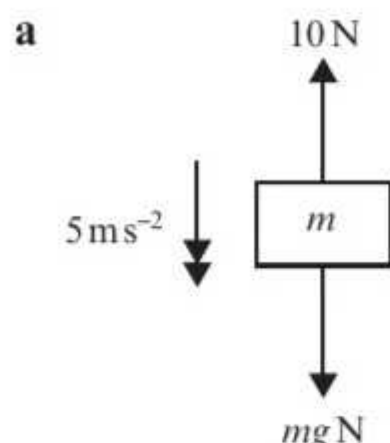
Problem-solving

Start by finding the mass of the astronaut.

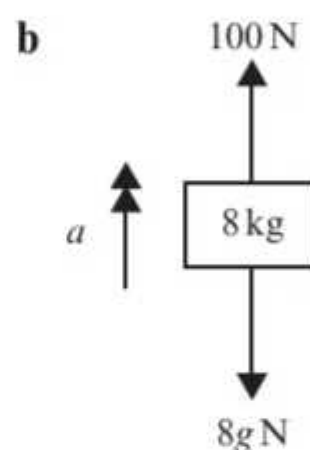
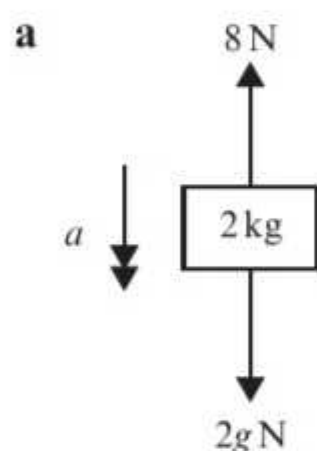
- In each scenario, the forces acting on the body cause it to accelerate as shown. Find the magnitude of the unknown force.



- 6 In each scenario, the forces acting on the body cause it to accelerate as shown. In each case find the mass of the body, m .



- 7 In each scenario, the forces acting on the body cause it to accelerate with magnitude $a\text{ m s}^{-2}$. In each case find the value of a .



- (P) 8 A horizontal force of 10 N acts upon a particle of mass 3 kg causing it to accelerate at 2 m s^{-2} along a rough horizontal plane. Calculate the value of the force due to friction.

Problem-solving

Draw a force diagram showing all the forces acting on the particle.

- (E/P) 9 A lift of mass 500 kg is lowered or raised by means of a metal cable attached to its top. The lift contains passengers whose total mass is 300 kg . The lift starts from rest and accelerates at a constant rate, reaching a speed of 3 m s^{-1} after moving a distance of 5 m . Find:
- a the acceleration of the lift (3 marks)
 - b the tension in the cable if the lift is moving vertically downward (2 marks)
 - c the tension in the cable if the lift is moving vertically upward. (2 marks)
- (E) 10 A trolley of mass 50 kg is pulled from rest in a straight line along a horizontal path by means of a horizontal rope attached to its front end. The trolley accelerates at a constant rate and after 2 s its speed is 1 m s^{-1} . As it moves, the trolley experiences a resistance to motion of magnitude 20 N . Find:
- a the acceleration of the trolley (3 marks)
 - b the tension in the rope. (2 marks)
- (E/P) 11 The engine of a van of mass 400 kg cuts out when it is moving along a straight horizontal road with speed 16 m s^{-1} . The van comes to rest without the brakes being applied. In a model of the scenario it is assumed that the van is subject to a resistive force which has constant magnitude of 200 N .

Hint

Use $v^2 = u^2 + 2as$.

- a Find how long it takes the van to stop. (3 marks)
- b Find how far the van travels before it stops. (2 marks)
- c Comment on the suitability of the modelling assumption. (1 mark)

Challenge

A small stone of mass 400 g is projected vertically upward from the bottom of a pond full of water with speed 10 m s^{-1} . As the stone moves through the water it experiences a constant resistance of magnitude 3 N. Assuming that the stone does not reach the surface of the pond, find:

- a the greatest height above the bottom of the pond that the stone reaches
- b the speed of the stone as it hits the bottom of the pond on its return
- c the total time taken for the stone to return to its initial position on the bottom of the pond.

4.4 Motion in two dimensions

- You can use $F = ma$ to solve problems involving vector forces acting on particles.

Notation

In this version of the equation of motion, F and a are vectors.

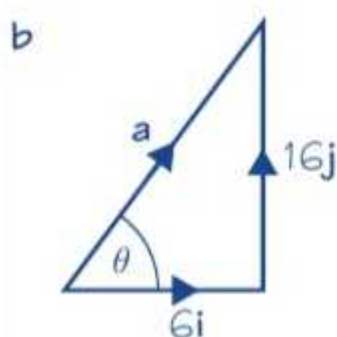
You can write acceleration as a 2D vector in the form $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-2}$ or $\begin{pmatrix} p \\ q \end{pmatrix} \text{ m s}^{-2}$.

Example**8****SKILLS****PROBLEM-SOLVING**

In this question, $\mathbf{i}$ represents the unit vector due east, and $\mathbf{j}$ represents the unit vector due north. A resultant force of $(3\mathbf{i} + 8\mathbf{j}) \text{ N}$ acts upon a particle of mass 0.5 kg.

- a Find the acceleration of the particle in the form $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-2}$.
- b Find the magnitude and bearing of the acceleration of the particle.

a $F = ma$
 $(3\mathbf{i} + 8\mathbf{j}) = 0.5 \times a$
 $a = (6\mathbf{i} + 16\mathbf{j}) \text{ m s}^{-2}$



$$|a| = \sqrt{6^2 + 16^2} = 2\sqrt{73} \text{ N}$$

$$= 17.1 \text{ m s}^{-2} \text{ (1 d.p.)}$$

$$\tan \theta = \frac{16}{6} \text{ so } \theta = 69.4^\circ \text{ (1 d.p.)}$$

So the bearing of the acceleration is
 $90^\circ - 69.4^\circ = 020.6^\circ$

Write the vector equation of motion.

To divide $(3\mathbf{i} + 8\mathbf{j})$ by 0.5, you divide each component by 0.5.

Draw a diagram to represent the acceleration vector.

Use Pythagoras' Theorem to work out the magnitude of the acceleration vector.

Remember bearings are always measured clockwise from north.

Example 9**SKILLS****PROBLEM-SOLVING**

Forces $\mathbf{F}_1 = (2\mathbf{i} + 4\mathbf{j})$ N, $\mathbf{F}_2 = (-5\mathbf{i} + 4\mathbf{j})$ N, and $\mathbf{F}_3 = (6\mathbf{i} - 5\mathbf{j})$ N act on a particle of mass 3 kg. Find the acceleration of the particle.

Resultant force

$$\begin{aligned} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \\ &= (2\mathbf{i} + 4\mathbf{j}) + (-5\mathbf{i} + 4\mathbf{j}) + (6\mathbf{i} - 5\mathbf{j}) \\ &= 3\mathbf{i} + 3\mathbf{j} \end{aligned}$$

$$3\mathbf{i} + 3\mathbf{j} = 3\mathbf{a} \Rightarrow \mathbf{a} = (\mathbf{i} + \mathbf{j}) \text{ m s}^{-2}$$

Add the vectors to find the resultant force.

Use $\mathbf{F} = m\mathbf{a}$.**Example 10****SKILLS****PROBLEM-SOLVING**

A boat is modelled as a particle of mass 60 kg being acted on by three forces:

$$\mathbf{F}_1 = \begin{pmatrix} 80 \\ 50 \end{pmatrix} \text{ N} \quad \mathbf{F}_2 = \begin{pmatrix} 10p \\ 20q \end{pmatrix} \text{ N} \quad \mathbf{F}_3 = \begin{pmatrix} -75 \\ 100 \end{pmatrix} \text{ N}$$

Given that the boat is accelerating at a rate of $\begin{pmatrix} 0.8 \\ -1.5 \end{pmatrix} \text{ m s}^{-2}$, find the values of p and q .

Resultant force

$$\begin{aligned} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \\ &= \begin{pmatrix} 80 \\ 50 \end{pmatrix} + \begin{pmatrix} 10p \\ 20q \end{pmatrix} + \begin{pmatrix} -75 \\ 100 \end{pmatrix} \\ &= \begin{pmatrix} 5 + 10p \\ 150 + 20q \end{pmatrix} \text{ N} \end{aligned}$$

 $\mathbf{F} = m\mathbf{a}$

$$\begin{pmatrix} 5 + 10p \\ 150 + 20q \end{pmatrix} = 60 \times \begin{pmatrix} 0.8 \\ -1.5 \end{pmatrix} = \begin{pmatrix} 48 \\ -90 \end{pmatrix}$$

$$\begin{aligned} \text{So } 5 + 10p &= 48 \Rightarrow p = 4.3 \\ \text{and } 150 + 20q &= -90 \Rightarrow q = -12 \end{aligned}$$

Find the resultant force acting on the boat in terms of p and q .Use $\mathbf{F} = m\mathbf{a}$. Remember that you need to multiply each component in the acceleration by 60.Solve separate equations for each component to find the values of p and q .**Exercise 4D****SKILLS****PROBLEM-SOLVING**

In all the questions in this exercise, $\mathbf{i}$ represents the unit vector due east, and $\mathbf{j}$ represents the unit vector due north.

- A resultant force of $(\mathbf{i} + 4\mathbf{j})$ N acts upon a particle of mass 2 kg.
 - Find the acceleration of the particle in the form $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-2}$.
 - Find the magnitude and bearing of the acceleration of the particle.
- A resultant force of $(4\mathbf{i} + 3\mathbf{j})$ N acts on a particle of mass m kg causing it to accelerate at $(20\mathbf{i} + 15\mathbf{j}) \text{ m s}^{-2}$. Work out the mass of the particle.

- 3 A particle of mass 3 kg is acted on by a force $\mathbf{F}$.
Given that the particle accelerates at $(7\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$:
- find an expression for $\mathbf{F}$ in the form $(p\mathbf{i} + q\mathbf{j}) \text{ N}$
 - find the magnitude and bearing of $\mathbf{F}$.
- 4 Two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$, act on a particle of mass m . Find the acceleration of the particle, $\mathbf{a} \text{ m s}^{-2}$, given that:
- $\mathbf{F}_1 = (2\mathbf{i} + 7\mathbf{j}) \text{ N}$, $\mathbf{F}_2 = (-3\mathbf{i} + \mathbf{j}) \text{ N}$, $m = 0.25 \text{ kg}$
 - $\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j}) \text{ N}$, $\mathbf{F}_2 = (2\mathbf{i} + 3\mathbf{j}) \text{ N}$, $m = 6 \text{ kg}$
 - $\mathbf{F}_1 = (-40\mathbf{i} - 20\mathbf{j}) \text{ N}$, $\mathbf{F}_2 = (25\mathbf{i} + 10\mathbf{j}) \text{ N}$, $m = 15 \text{ kg}$
 - $\mathbf{F}_1 = 4\mathbf{j} \text{ N}$, $\mathbf{F}_2 = (-2\mathbf{i} + 5\mathbf{j}) \text{ N}$, $m = 1.5 \text{ kg}$

Notation

You are asked to find the acceleration as a vector, $\mathbf{a}$. You can give your answer as a column vector or using $\mathbf{i}$ - $\mathbf{j}$ notation.

- 5 A particle of mass 8 kg is at rest. It is acted on by three forces, $\mathbf{F}_1 = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \text{ N}$, $\mathbf{F}_2 = \begin{pmatrix} 2 \\ -5 \end{pmatrix} \text{ N}$ and $\mathbf{F}_3 = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \text{ N}$.
- Find the magnitude and direction of the acceleration of the particle, $\mathbf{a} \text{ m s}^{-2}$.
 - Find the time taken for the particle to travel a distance of 20 m.

Hint

Use $s = ut + \frac{1}{2}at^2$ with $s = 20$ and $u = 0$.

- E/P** 6 Two forces, $(2\mathbf{i} + 3\mathbf{j}) \text{ N}$ and $(p\mathbf{i} + q\mathbf{j}) \text{ N}$, act on a particle P . The resultant of the two forces is $\mathbf{R}$. Given that $\mathbf{R}$ acts in a direction which is parallel to the vector $(-\mathbf{i} + 4\mathbf{j})$, show that $4p + q + 11 = 0$. **(4 marks)**

Problem-solving

You can write $\mathbf{R}$ in the form $(-k\mathbf{i} + 4k\mathbf{j}) \text{ N}$ for some constant k .

- E** 7 A particle of mass 4 kg starts from rest and is acted upon by a force $\mathbf{R}$ of $(6\mathbf{i} + b\mathbf{j}) \text{ N}$. $\mathbf{R}$ acts on a bearing of 045° .
- Find the value of b . **(1 mark)**
 - Calculate the magnitude of $\mathbf{R}$. **(2 marks)**
 - Work out the magnitude of the acceleration of the particle. **(2 marks)**
 - Find the total distance travelled by the particle during the first 5 seconds of its motion. **(3 marks)**
- P** 8 Three forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ act on a particle. $\mathbf{F}_1 = (-3\mathbf{i} + 7\mathbf{j}) \text{ N}$, $\mathbf{F}_2 = (\mathbf{i} - \mathbf{j}) \text{ N}$ and $\mathbf{F}_3 = (p\mathbf{i} + q\mathbf{j}) \text{ N}$.
- Given that this particle is in equilibrium, determine the value of p and the value of q .
Force $\mathbf{F}_2$ is removed.
 - Given that in the first 10 seconds of its motion the particle travels a distance of 12 m, find the exact mass of the particle in kg.
- P** 9 A particle of mass $m \text{ kg}$ is acted upon by forces of $(5\mathbf{i} + 6\mathbf{j}) \text{ N}$, $(2\mathbf{i} - 2\mathbf{j}) \text{ N}$ and $(-\mathbf{i} - 4\mathbf{j}) \text{ N}$, causing it to accelerate at 7 m s^{-2} . Work out the mass of the particle. Give your answer correct to 2 d.p.

- E/P** 10 Two forces, $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$ N and $\begin{pmatrix} p \\ q \end{pmatrix}$ N, act on a particle P of mass m kg. The resultant of the two forces is $\mathbf{R}$.
- a Given that $\mathbf{R}$ acts in a direction which is parallel to the vector $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, show that $2p + q + 9 = 0$. (4 marks)
- b Given also that $p = 1$ and that P moves with an acceleration of magnitude $15\sqrt{5} \text{ m s}^{-2}$, find the value of m . (7 marks)

Challenge

A particle of mass 0.5 kg is acted on by two forces:

$$\mathbf{F}_1 = -4\mathbf{i} \text{ N} \quad \mathbf{F}_2 = (k\mathbf{i} + 2k\mathbf{j}) \text{ N}$$

where k is a positive constant.

Given that the particle is accelerating at a rate of $8\sqrt{17} \text{ m s}^{-2}$, find the value of k .

4.5 Connected particles

If a system involves the motion of more than one particle, the particles may be considered separately. However, if all parts of the system are moving in the **same straight line**, then you can also treat the whole system as a single particle.

- You can solve problems involving connected particles by considering the particles separately or, if they are moving in the same straight line, as a single particle.

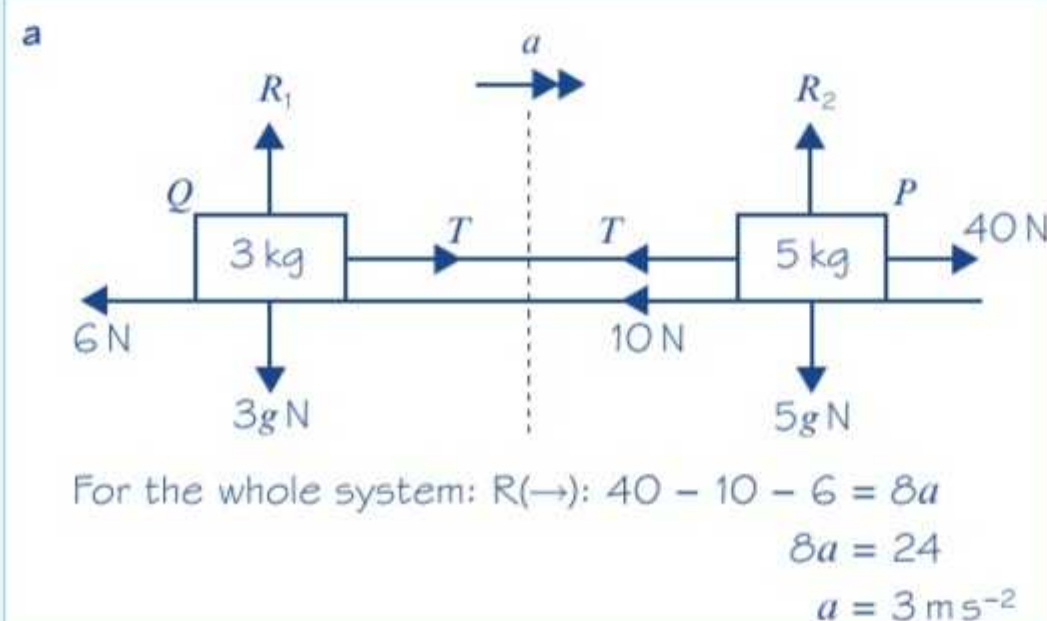
Watch out

Particles need to remain in contact, or be connected by an inextensible rod or taut string to be considered as a single particle.

Example 11**SKILLS****PROBLEM-SOLVING**

Two particles, P and Q , of masses 5 kg and 3 kg respectively, are connected by a light inextensible string. Particle P is pulled by a horizontal force of magnitude 40 N along a rough horizontal plane. Particle P experiences a frictional force of 10 N and particle Q experiences a frictional force of 6 N.

- a Find the acceleration of the particles.
b Find the tension in the string.
c Explain how the modelling assumptions that the string is light and inextensible have been used.

**Problem-solving**

In part **a**, by considering the system as a single particle you eliminate the need to find the tension in the string. Otherwise you would need to set up two simultaneous equations involving a and T .

In part **b**, the particles need to be considered separately to find the tension in the string.

b For P : $R(\rightarrow): 40 - T - 10 = 5 \times 3$
 $T = 15 \text{ N}$

c Inextensible $\Rightarrow$ acceleration of masses is the same.
 light $\Rightarrow$ tension is the same throughout the length of the string and the mass of the string is negligible.

You could also have chosen particle Q to find the tension. Check to see that it gives the same answer.

■ **Newton's third law of motion** states that for every action there is an equal and opposite reaction.

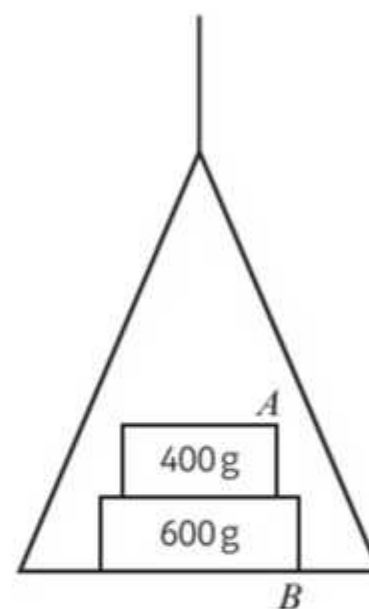
Newton's third law means that when two bodies A and B are in contact, if body A exerts a force on body B , then body B exerts a force on body A that is equal in magnitude and acts in the opposite direction.

Example 12

SKILLS PROBLEM-SOLVING

A light scale-pan is attached to a vertical light inextensible string. The scale-pan carries two masses A and B . The mass of A is 400 g and the mass of B is 600 g. A rests on top of B , as shown in the diagram. The scale-pan is raised vertically, using the string, with acceleration 0.5 m s^{-2} .

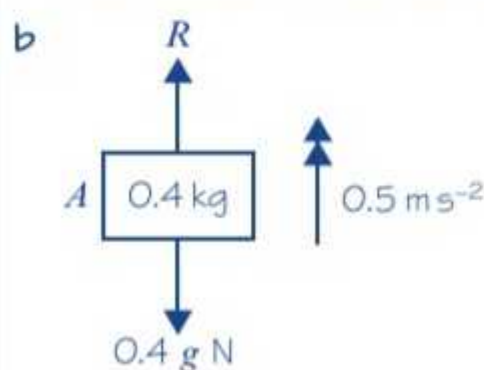
- Find the tension in the string.
- Find the force exerted on mass B by mass A .
- Find the force exerted on mass B by the scale-pan.



a For the whole system:
 $R(\uparrow): T - 0.4g - 0.6g = (0.4 + 0.6)a$

So,
 $T - g = 1 \times 0.5$
 $T = 1.20.3 \text{ N}$

The tension in the string is 10 N (2 s.f.)



For A only:
 $R(\uparrow): R - 0.4g = 0.4 \times 0.5$
 $R = 4.12 \text{ N}$

So the force exerted on B by A is
 4.1 N (2 s.f.) downward.

You can use this since all parts of the system are moving in the same straight line.

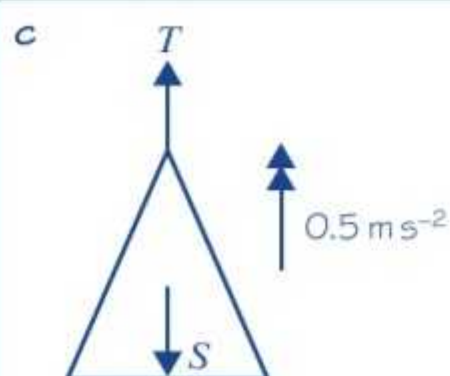
Note that you must convert 400 g to 0.4 kg and 600 g to 0.6 kg.

$a = 0.5$

Simplify.

Find the force exerted on A by B and then use Newton's 3rd law to say that the force exerted on B by A will have the same magnitude but is in the opposite direction.

You have used $g = 9.8 \text{ m s}^{-2}$ so give your final answer correct to two significant figures.



For scale-pan only:

$$R(\uparrow) \quad T - S = 0 \times 0.5 \\ = 0$$

$$\text{So,} \quad T = S = 10.3 \text{ N}$$

So, the force exerted on *B* by the scale-pan is 10 N (2 s.f.) upward.

Problem-solving

It's easier to find the force exerted on the scale-pan by *B* and then use Newton's 3rd law to say that the force exerted on *B* by the scale-pan has the same magnitude but is in the opposite direction.

The scale-pan is light, i.e. its mass is 0.

From part a.

Use Newton's 3rd law.

Exercise

4E

SKILLS

PROBLEM-SOLVING

- Two particles *P* and *Q*, of masses 8 kg and 2 kg respectively, are connected by a light inextensible string. The particles are on a smooth horizontal plane. A horizontal force of magnitude *F* is applied to *P* in a direction away from *Q* and when the string is taut the particles move with acceleration 0.4 m s^{-2} .
 - Find the value of *F*.
 - Find the tension in the string.
 - Explain how the modelling assumptions that the string is light and inextensible are used.
- Two particles *P* and *Q* of masses 20 kg and *m* kg respectively, are connected by a light inextensible rod. The particles lie on a smooth horizontal plane. A horizontal force of 60 N is applied to *Q* in a direction toward *P*, causing the particles to move with acceleration 2 m s^{-2} .
 - Find the mass, *m*, of *Q*.
 - Find the thrust in the rod.
- Two particles *P* and *Q*, of masses 7 kg and 8 kg respectively, are connected by a light inextensible string. The particles are on a smooth horizontal plane. A horizontal force of 30 N is applied to *Q* in a direction away from *P*. When the string is taut the particles move with acceleration *a* m s^{-2} .
 - Find the acceleration, *a*, of the system.
 - Find the tension in the string.
- Two boxes *A* and *B*, of masses 110 kg and 190 kg respectively, sit on the floor of a lift of mass 1700 kg. Box *A* rests on top of box *B*. The lift is supported by a light inextensible cable and is descending with constant acceleration 1.8 m s^{-2} :
 - Find the tension in the cable.
 - Find the force exerted by box *B*
 - on box *A*
 - on the floor of the lift.

Hint

For part b, consider *P* on its own.

- P** 5 A lorry of mass m kg is towing a trailer of mass $3m$ kg along a straight horizontal road. The lorry and trailer are connected by a light inextensible tow-bar. The lorry exerts a driving force of 50 000 N causing the lorry and trailer to accelerate at 5 m s^{-2} . The lorry and trailer experience resistances of 4000 N and 10 000 N respectively.
- Find the mass of the lorry and hence the mass of the trailer.
 - Find the tension in the tow-bar.
 - Explain how the modelling assumptions that the tow-bar is light and inextensible affect your calculations.
- E** 6 Two particles A and B , of masses 10 kg and 5 kg respectively, are connected by a light inextensible string. Particle B hangs directly below particle A . A force of 180 N is applied to A vertically upward causing the particles to accelerate.
- Find the magnitude of the acceleration. (3 marks)
 - Find the tension in the string. (2 marks)
- E/P** 7 Two particles A and B , of masses 6 kg and m kg respectively, are connected by a light inextensible string. Particle B hangs directly below particle A . A force of 118 N is applied to A vertically upward causing the particles to accelerate at 2 m s^{-2} .
- Find the mass, m , of particle B . (3 marks)
 - Find the tension in the string. (2 marks)
- E/P** 8 A train engine of mass 6400 kg is pulling a carriage of mass 1600 kg along a straight horizontal railway track. The engine is connected to the carriage by a shunt which is parallel to the direction of motion of the coupling. The shunt is modelled as a light rod. The engine provides a constant driving force of 12 000 N. The resistances to the motion of the engine and the carriage are modelled as constant forces of magnitude R N and 2000 N respectively.
- Given that the acceleration of the engine and the carriage is 0.5 m s^{-2} :
- find the value of R (3 marks)
 - show that the tension in the shunt is 2800 N. (2 marks)
- E** 9 A car of mass 900 kg pulls a trailer of mass 300 kg along a straight horizontal road using a light tow-bar which is parallel to the road. The horizontal resistances to motion of the car and the trailer have magnitudes 200 N and 100 N respectively. The engine of the car produces a constant horizontal driving force on the car of magnitude 1200 N.
- Show that the acceleration of the car and trailer is 0.75 m s^{-2} . (2 marks)
 - Find the magnitude of the tension in the tow-bar. (3 marks)
- The car is moving along the road when the driver sees a set of traffic lights have turned red. He reduces the force produced by the engine to zero and applies the brakes. The brakes produce a force on the car of magnitude F newtons and the car and trailer decelerate.
- Given that the resistances to motion are unchanged and the magnitude of the thrust in the tow-bar is 100 N, find the value of F . (7 marks)

4.6 Pulleys

In this section you will see how to model systems of connected particles involving pulleys. The problems you answer will assume that particles are connected by a light, inextensible string, which passes over a **smooth pulley**. This means that the tension in the string will be the same **on both sides** of the pulley. The parts of the string on each side of the pulley will be either horizontal or vertical. More complicated systems will be considered in Chapter 5.

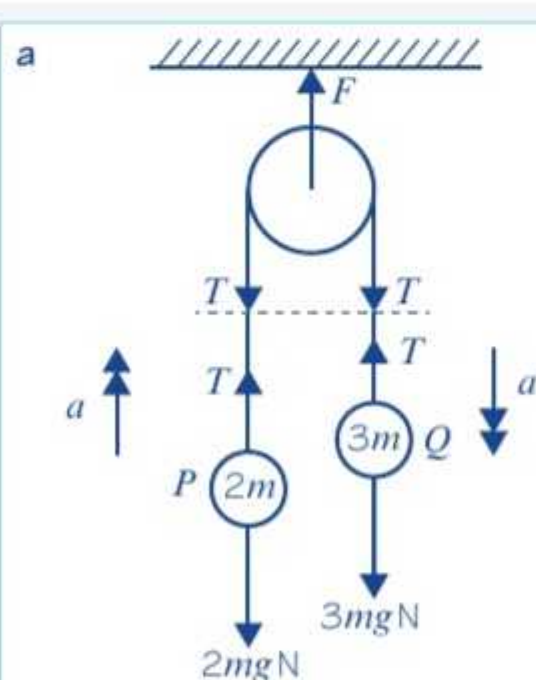
Watch out You cannot treat a system involving a pulley as a single particle. This is because the particles are moving in different directions.

Example 13

SKILLS PROBLEM-SOLVING

Particles P and Q , of masses $2m$ and $3m$, are attached to the ends of a light inextensible string. The string passes over a small smooth fixed pulley and the masses hang with the string taut. The system is released from rest.

- Write down an equation of motion for P .
 - Write down an equation of motion for Q .
- Find the acceleration of each mass.
- Find the tension in the string.
- Find the force exerted on the pulley by the string.
- Find the distance moved by Q in the first 4 s, assuming that P does not reach the pulley.
- State how you have used the fact that the pulley is smooth in your calculations.



i For P , $R(\uparrow)$: $T - 2mg = 2ma$ (1)

ii For Q , $R(\downarrow)$: $3mg - T = 3ma$ (2)

b Adding equations (1) and (2):

$$3mg - \cancel{T} + \cancel{T} - 2mg = 3ma + 2ma$$

$$mg = 5ma$$

$$\frac{1}{5}g = a$$

The acceleration of each mass is $\frac{1}{5}g$.

Problem-solving

Resolve vertically for both P and Q . This will give you simultaneous equations involving the tension T and the acceleration a which can then be solved.

Draw a diagram showing all the forces acting on each mass and the pulley, and the acceleration.

Now resolve for each mass separately, in the direction of its acceleration.

Add the equations to eliminate T .

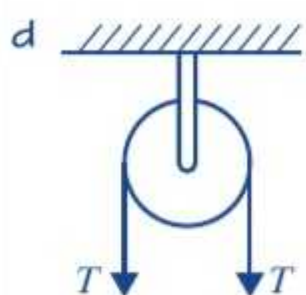
Simplifying.

You could also give your final answer as 1.96 m s^{-2} (3 s.f.).

c From (1): $T - 2mg = 2m \times \frac{1}{5}g$

$$T = \frac{12mg}{5} \text{ N}$$

The tension in the string is $\frac{12mg}{5}$ N.



The force exerted on the pulley by the string is $2TN$ downward or $\frac{24mg}{5}$ N.

e $u = 0$, $a = \frac{1}{5}g$, $t = 4$, $s = ?$

$$s = ut + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2} \times 1.96 \times 4^2$$

$$= 15.68 \text{ m}$$

$$= 15.7 \text{ m (3 s.f.)}$$

Q moves through a distance of 15.7 m (3 s.f.)

f The tension in the string is the same at P as at Q .

Substitute for a .

Collect terms.

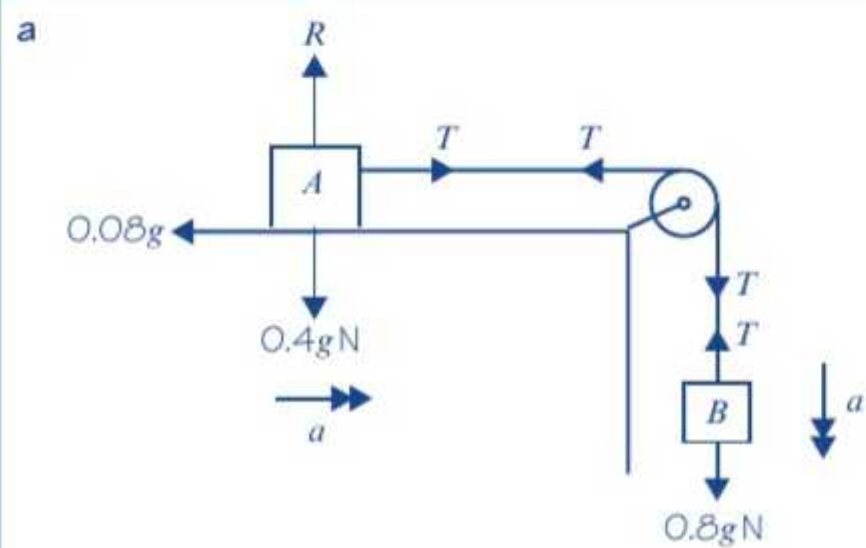
Since a is a constant we can use any of the formulae for constant acceleration.

Example 14

SKILLS PROBLEM-SOLVING

Two particles A and B , of masses 0.4 kg and 0.8 kg respectively, are connected by a light inextensible string. Particle A lies on a rough horizontal table 4.5 m from a small smooth pulley which is fixed at the edge of the table. The string passes over the pulley and B hangs freely, with the string taut, 0.5 m above horizontal ground. A frictional force of magnitude $0.08g$ opposes the motion of particle A . The system is released from rest. Find:

- the acceleration of the system
- the time taken for B to reach the ground
- the total distance travelled by A before it first comes to rest.



Problem-solving

Draw a diagram showing all the forces and the accelerations. The pulley is smooth so the tension in the string is the same on each side of the pulley.

For A only: $R(\rightarrow), T - 0.08g = 0.4a$ (1)

For B only: $R(\downarrow), 0.8g - T = 0.8a$ (2)

Add (1) and (2):

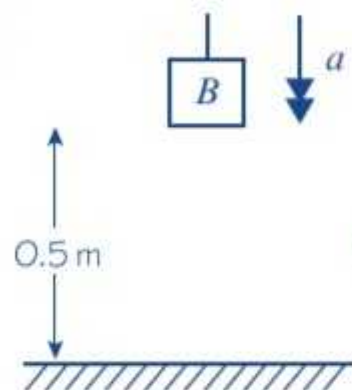
$$0.8g - \cancel{T} + \cancel{T} - 0.08g = 0.8a + 0.4a$$

$$0.72g = 1.2a$$

$$0.6g = a$$

The acceleration of the system is $0.6g$.

b



$$u = 0, s = 0.5,$$

$$a = 5.88, t = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$0.5 = 0 + \frac{1}{2} \times 5.88 \times t^2$$

$$t = 0.412 \text{ (3 s.f.)}$$

The time taken for B to hit the ground is 0.412 s (3 s.f.)

c Find the speed of B when it hits the ground.

$$u = 0, a = 5.88, t = 0.41239, v = ?$$

$$v = u + at$$

$$v_B = 0 + 5.88 \times 0.41239 = 2.42487 \text{ m s}^{-1}.$$

Speed of A on the table is 2.42487 m s^{-1} .

Once B hits the ground the string will go slack and A will begin to decelerate as it slides against the friction on the table.

$$\text{From (1): } -0.08g = 0.4a'$$

$$a' = -0.2g$$

$$u_A = 2.42487, v = 0, a' = -0.2g, s = ?$$

$$v^2 = u^2 + 2as$$

$$0^2 = 2.42487^2 - 0.4gs$$

$$s = 1.50 \text{ m (3 s.f.)}$$

A slides a further 1.50 m along the table before it comes to rest.

$\therefore$ Total distance moved by A is

$$0.5 + 1.50 = 2.00 \text{ m (3 s.f.)}$$

Write equations of motion for each of A and B separately.

Eliminate the T terms.

You could also give your answer as 5.88 m s^{-2} (3 s.f.).

Use an unrounded value of the acceleration.

The acceleration is constant.

Use an unrounded value for t .

Using surds, $v_B = \sqrt{\frac{3g}{5}}$

Since the string is inextensible.

Put $T = 0$ in equation (1) as the string is now slack.

This is the new acceleration of A along the table.

Exercise

4F

SKILLS

PROBLEM-SOLVING

- 1 Two particles A and B , of masses 4 kg and 3 kg respectively, are connected by a light inextensible string which passes over a small smooth fixed pulley. The particles are released from rest with the string taut.

a Find the tension in the string.

5 FORCES AND FRICTION

4.4

Learning objectives

After completing this chapter you should be able to:

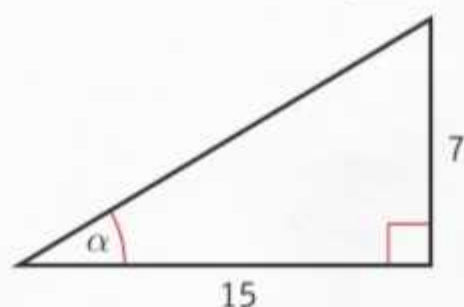
- Resolve forces into components → pages 85–90
- Use the triangle law to find a resultant force → pages 87–90
- Solve problems involving smooth or rough inclined planes → pages 90–93
- Understand friction and the coefficient of friction → pages 94–99
- Use $F \leq \mu R$ → pages 94–99

Prior knowledge check

- 1** A particle of mass 5 kg is acted on by two forces:
 $\mathbf{F}_1 = (8\mathbf{i} + 2\mathbf{j})$ N and $\mathbf{F}_2 = (-3\mathbf{i} + 8\mathbf{j})$ N.
Find the acceleration of the particle in the form
 $(p\mathbf{i} + q\mathbf{j}) \text{ m s}^{-2}$.

← Mechanics 1 Section 3.6

- 2** In the diagram below, calculate:
a the length of the hypotenuse
b the size of α .
Give your answers correct to 2 d.p.



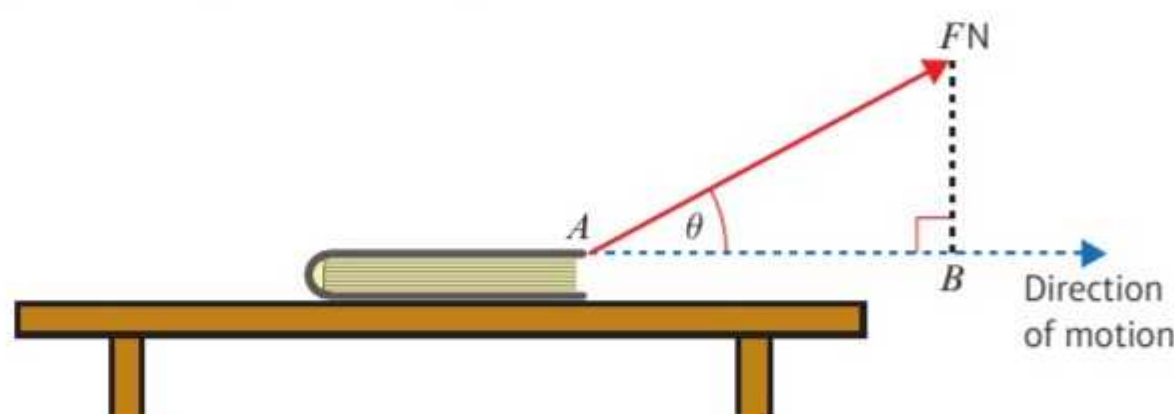
← International GCSE Mathematics

A car's braking distance is determined by its speed, the effectiveness of the brakes and the frictional force between the car's wheels and the road. In wet or icy conditions, friction is reduced so the braking distance is increased.

5.1 Resolving forces

- If a force is applied at an angle to the direction of motion you can resolve it to find the component of the force that acts in the direction of motion.

The book shown below is being dragged along the table by means of a force of magnitude F . The book is moving horizontally, and the angle between the force and the direction of motion is θ .



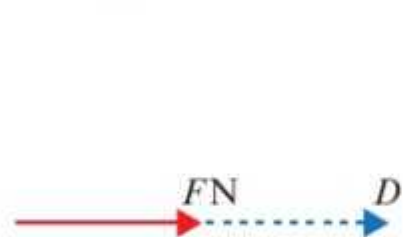
The effect of the force in the direction of motion is the length of AB . This is called the **component of the force in the direction of motion**. Using the rule for a right-angled triangle

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}},$$

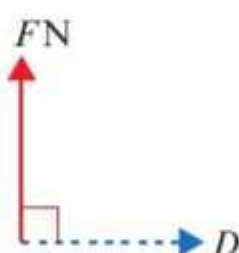
you can see that the magnitude of the force in the direction

of AB is $F \cos \theta$. Finding this value is called **resolving** the force in the direction of motion.

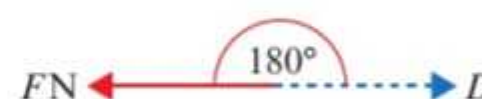
- The component of a force of magnitude F in a certain direction is $F \cos \theta$, where θ is the size of the angle between the force and the direction.



If F acts in the direction D , then the component of F in that direction is:
 $F \cos 0^\circ = F \times 1 = F$



If F acts at a right angle to D , then the component of F in that direction is:
 $F \cos 90^\circ = F \times 0 = 0$

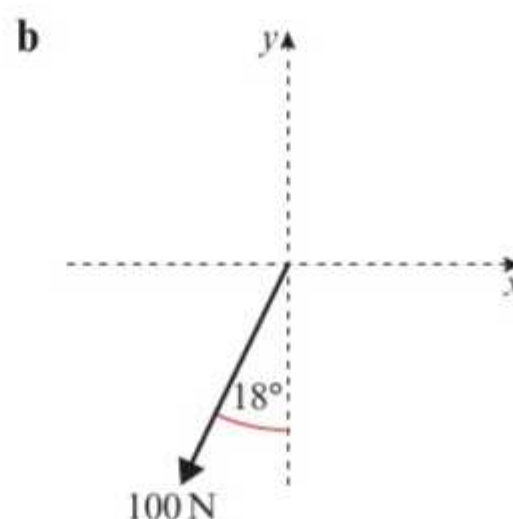
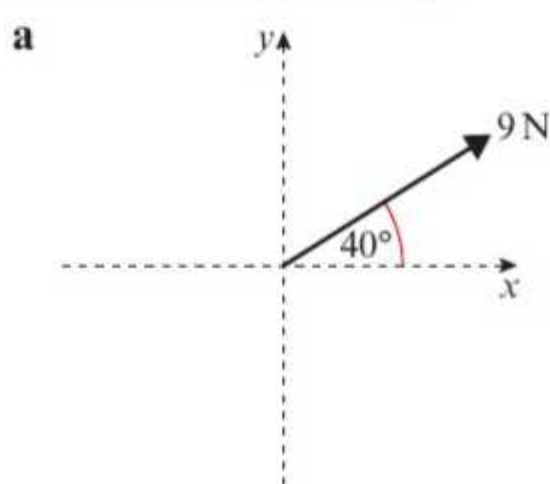


If F acts in the opposite direction to D , then the component of F in that direction is:
 $F \cos 180^\circ = F \times -1 = -F$

Example 1

SKILLS PROBLEM-SOLVING

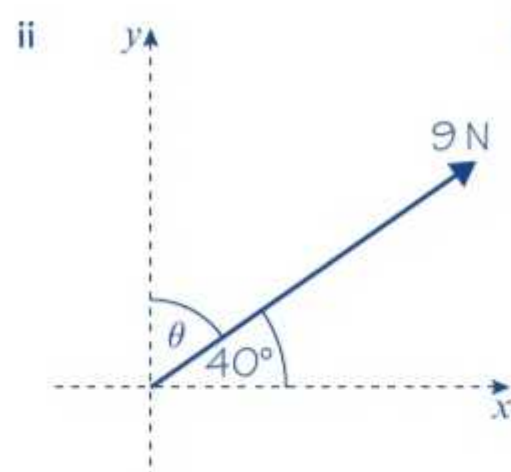
- Find the component of each force in: **i** the x -direction **ii** the y -direction.
iii Hence, write each force in the form $p\mathbf{i} + q\mathbf{j}$ where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the x and y directions respectively.



a i $\theta = 40^\circ$

$$\begin{aligned}\text{Component in } x\text{-direction} &= F \cos \theta \\ &= 9 \times \cos 40^\circ \\ &= 6.89 \text{ N (3 s.f.)}\end{aligned}$$

Give your answers correct to three significant figures.



$$\begin{aligned}\theta &= 90^\circ - 40^\circ \\ &= 50^\circ\end{aligned}$$

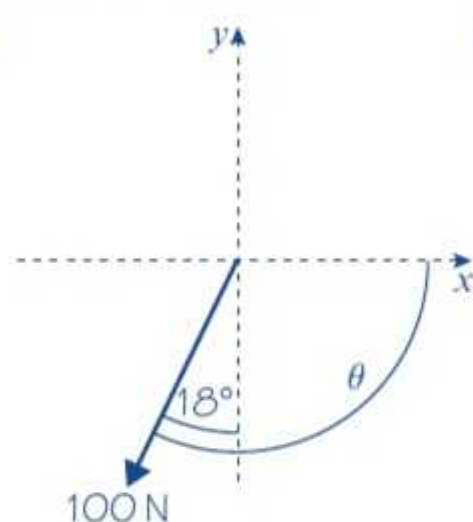
Make sure you find the angle between the force and the direction you are resolving in.

$$\begin{aligned}\text{Component in } y\text{-direction} &= F \cos \theta \\ &= 9 \times \cos 50^\circ \\ &= 5.79 \text{ N (3 s.f.)}\end{aligned}$$

You could also use $F \sin 40^\circ$ as $\sin 40^\circ = \cos (90^\circ - 40^\circ) = \cos 50^\circ$

iii $(6.89\mathbf{i} + 5.79\mathbf{j}) \text{ N}$

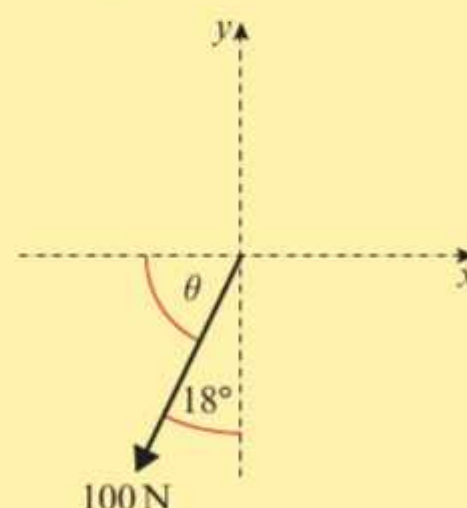
b i



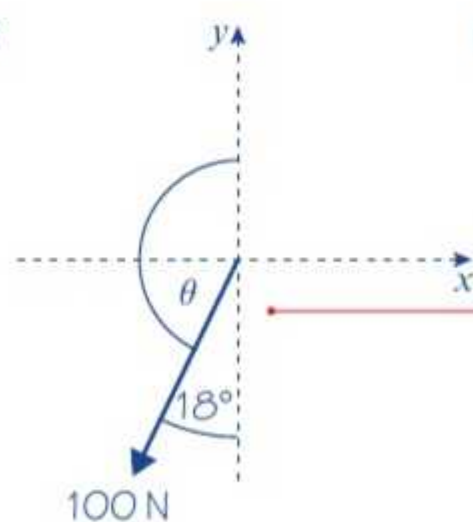
$$\begin{aligned}\theta &= 90^\circ + 18^\circ \\ &= 108^\circ\end{aligned}$$

$$\begin{aligned}\text{Component in } x\text{-direction} &= F \cos \theta \\ &= 100 \times \cos 108^\circ \\ &= -30.9 \text{ N (3 s.f.)}\end{aligned}$$

You get a negative answer because you are resolving in the positive x -direction. You could also resolve in the negative x -direction using $\theta = 90^\circ - 18^\circ = 72^\circ$, then change the sign of your answer from positive to negative:



ii



$$\begin{aligned}\theta &= 180^\circ - 18^\circ \\ &= 162^\circ\end{aligned}$$

$$\begin{aligned}\text{Component in } y\text{-direction} &= F \cos \theta \\ &= 100 \times \cos 162^\circ \\ &= -95.1 \text{ N (3 s.f.)}\end{aligned}$$

You could use $\theta = 18^\circ$ then change the sign of your answer from positive to negative: $-100 \cos 18^\circ = -95.1 \text{ N (3 s.f.)}$.

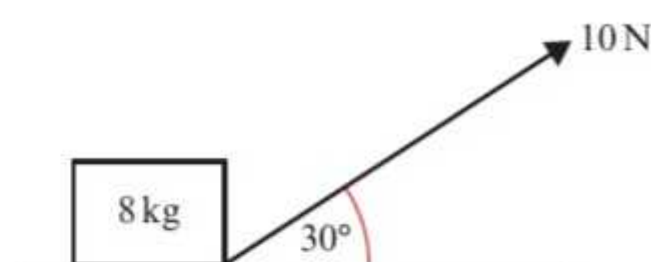
You can measure θ in either the clockwise or the anticlockwise direction since $\cos \theta = \cos (360^\circ - \theta)$.

iii $(-30.9\mathbf{i} - 95.1\mathbf{j}) \text{ N}$

Example 2**SKILLS** PROBLEM-SOLVING

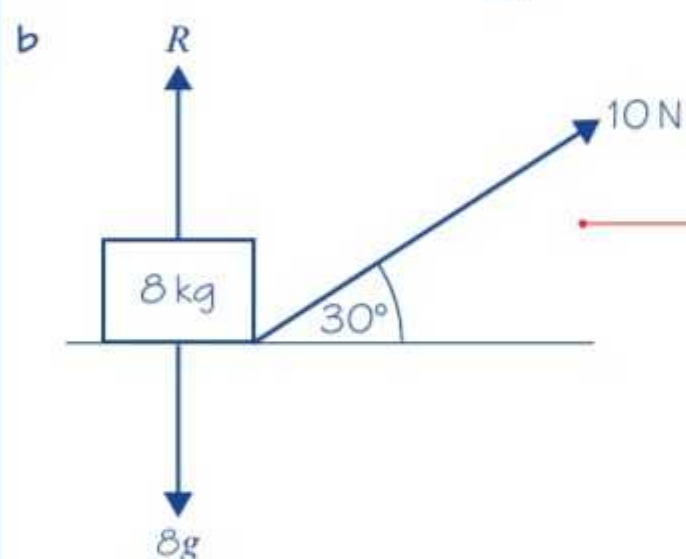
A box of mass 8 kg lies on a smooth horizontal floor. A force of 10 N is applied at an angle of 30° causing the box to accelerate horizontally along the floor.

- Work out the acceleration of the box.
- Calculate the normal reaction between the box and the floor.



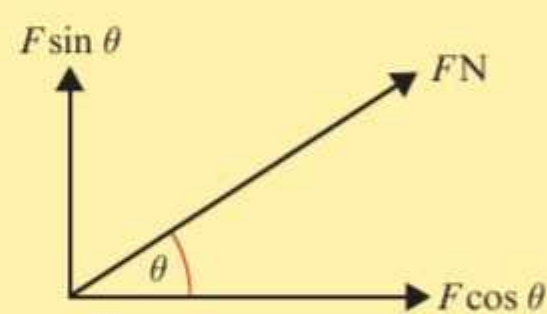
a $R(\rightarrow), 10 \cos 30^\circ = 8a$
 $8a = 5\sqrt{3}$
 $a = \frac{5\sqrt{3}}{8} \text{ m s}^{-2}$

Resolve the force horizontally and write an equation of motion for the box.



Add the weight of the box and the normal reaction to the force diagram.

$R(\uparrow), R + 10 \sin 30^\circ = 8g$
 $R = 78.4 - 5$
 $= 73.4 \text{ N (3 s.f.)}$

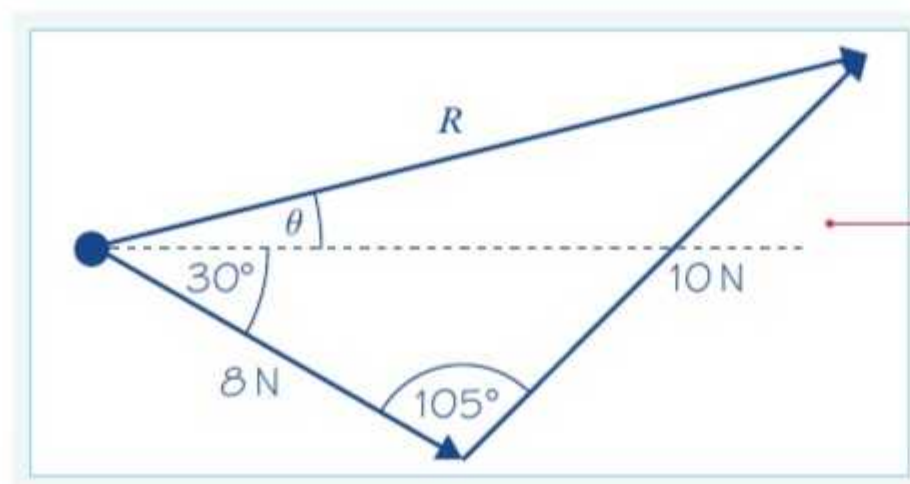
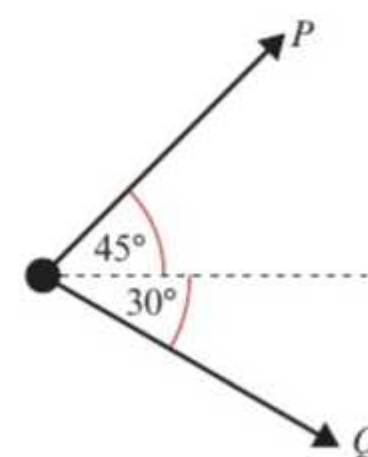


The component in the y -direction is $F \cos(90^\circ - \theta) = F \sin \theta$

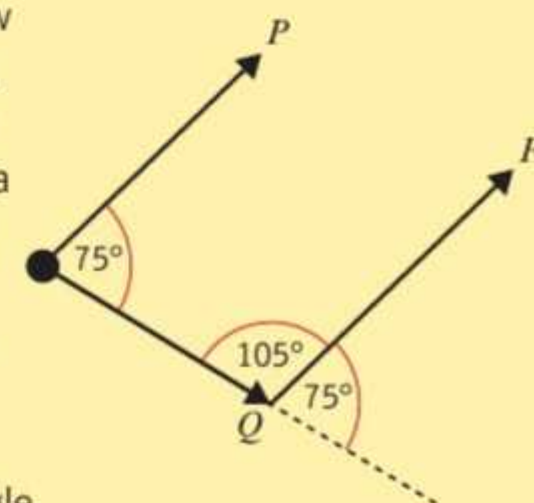
You can use the **triangle law** of vector addition to find the resultant of two forces acting at an angle without resolving them into components.

Example 3**SKILLS** CRITICAL THINKING

Two forces P and Q act on a particle as shown. P has a magnitude of 10 N and Q has a magnitude of 8 N. Work out the magnitude and direction of the resultant force.



Use the triangle law for vector addition. The resultant force is the third side of a triangle formed by forces P and Q . You might need to use geometry to work out missing angles in the triangle.



$$R^2 = 8^2 + 10^2 - 2 \times 8 \times 10 \cos 105^\circ$$

$$= 164 - 160 \cos 105^\circ = 205.411\dots$$

$$R = 14.332\dots = 14.3 \text{ N (3 s.f.)}$$

$$\frac{\sin(\theta + 30^\circ)}{10} = \frac{\sin 105^\circ}{14.332\dots}$$

$$\sin(\theta + 30^\circ) = \frac{10 \sin 105^\circ}{14.332\dots} = 0.673\dots$$

$$\theta + 30^\circ = 42.373\dots^\circ$$

$$\theta = 12.4^\circ \text{ (3 s.f.)}$$

The resultant force R has a magnitude of 14.3 N and acts at an angle of 12.4° above the horizontal.

Use the cosine rule to calculate the magnitude of R .

Use the sine rule to work out θ .

Remember to use unrounded values in your calculations, then round your final answer.

Use your diagram to check that your answer makes sense.

Online

Explore the resultant of two forces using GeoGebra.



Example 4

4

SKILLS

PROBLEM-SOLVING

Three forces act upon a particle as shown. Given that the particle is in equilibrium, calculate the magnitude of P .

$$R(\rightarrow), \quad 100 \cos 30^\circ + P \cos \theta = 140 \cos 45^\circ$$

$$P \cos \theta = 12.392\dots \quad (1)$$

$$R(\uparrow), \quad 100 \sin 30^\circ + 140 \sin 45^\circ = P \sin \theta$$

$$P \sin \theta = 148.994\dots \quad (2)$$

$$\frac{P \sin \theta}{P \cos \theta} = \frac{148.994\dots}{12.392\dots}$$

$$\tan \theta = 12.023\dots$$

$$\theta = 85.245\dots^\circ$$

$$P \cos 85.2454\dots^\circ = 12.392\dots$$

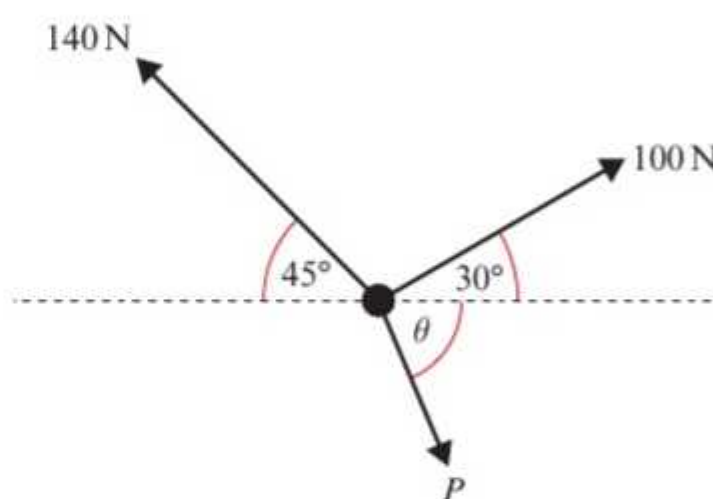
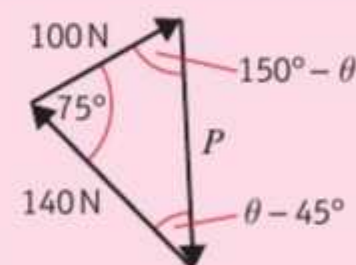
$$P = 150 \text{ N (3 s.f.)}$$

Resolve horizontally and vertically. You can solve these two equations simultaneously by dividing to eliminate P .

Problem-solving

You could also solve this problem by drawing a **triangle of forces**.

The particle is in equilibrium, so the three forces will form a closed triangle.



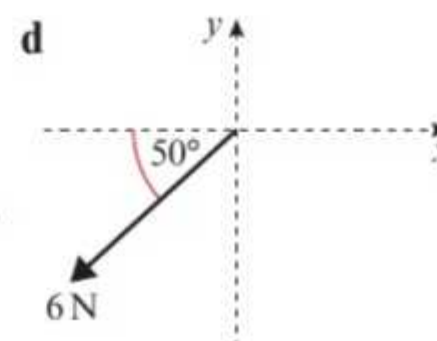
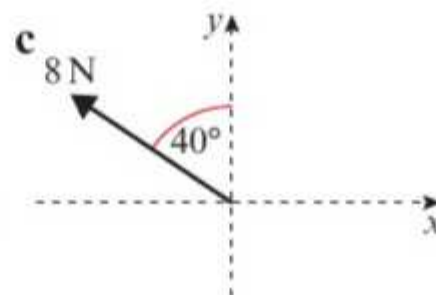
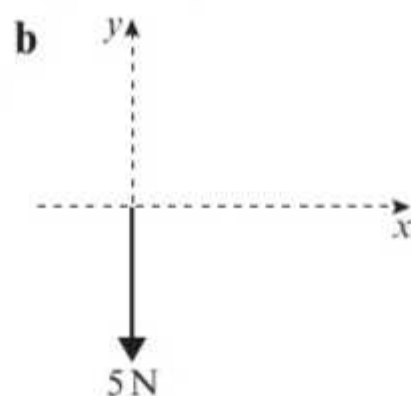
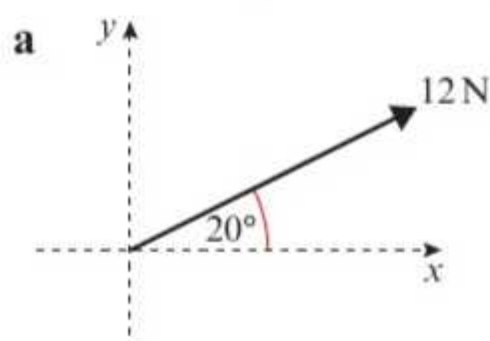
Exercise 5A

5A

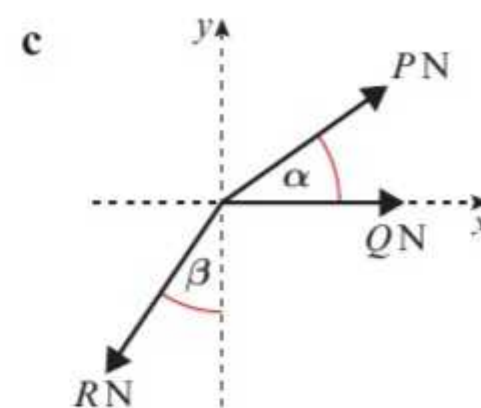
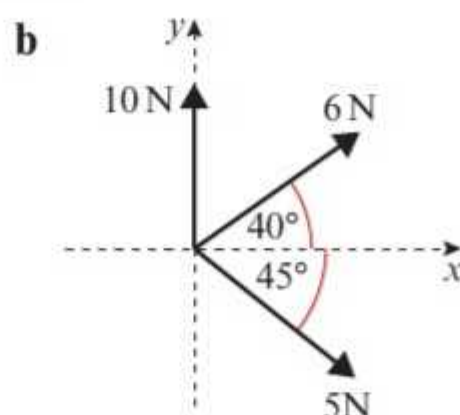
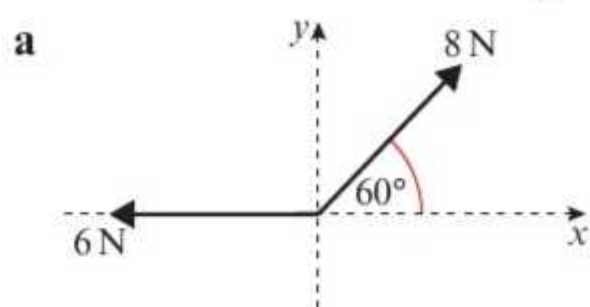
SKILLS

PROBLEM-SOLVING

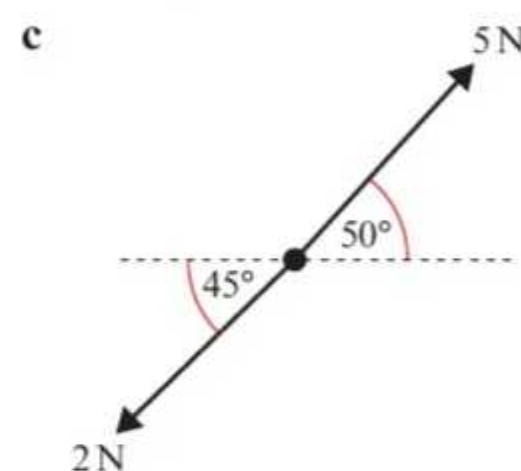
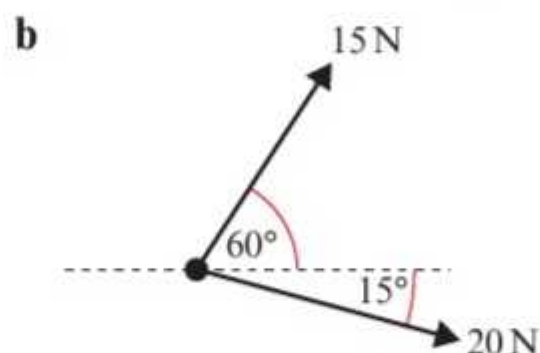
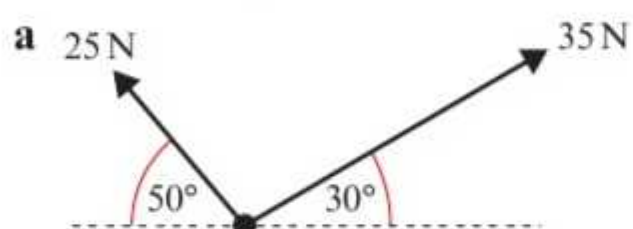
- 1 Find the component of each force in:
 - i the x -direction
 - ii the y -direction.
 - iii Hence write each force in the form $p\mathbf{i} + q\mathbf{j}$ where $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors in the x and y directions respectively.



- 2 For each of the following systems of forces, find the sum of the components in:
 i the x -direction ii the y -direction.

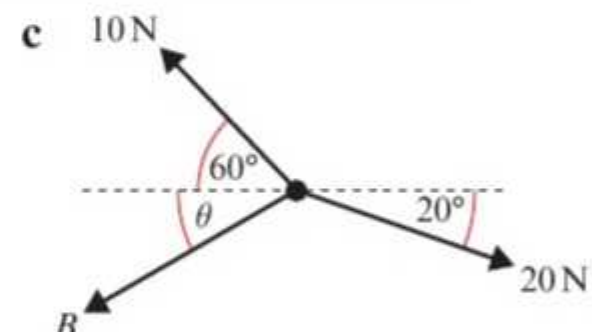
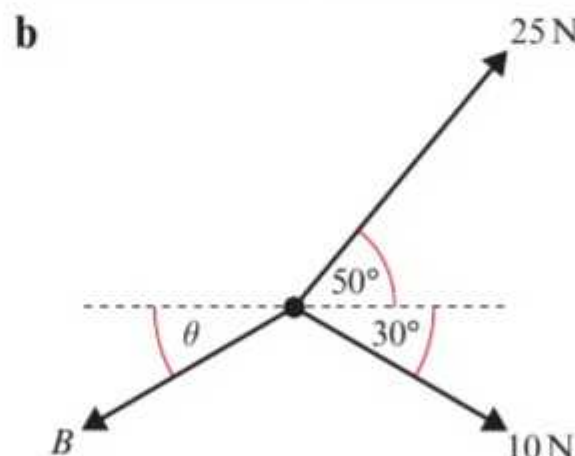
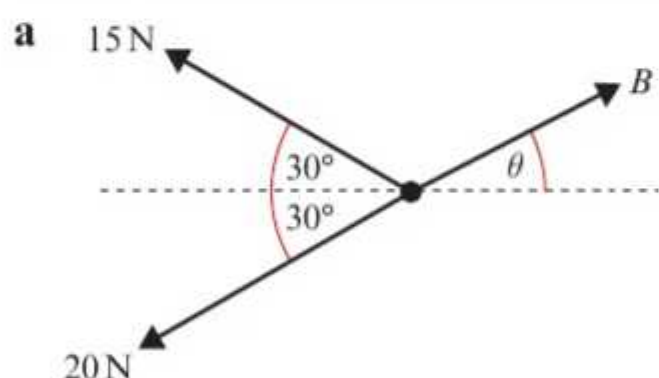


- 3 Find the magnitude and direction of the resultant force acting on each of the particles shown below.



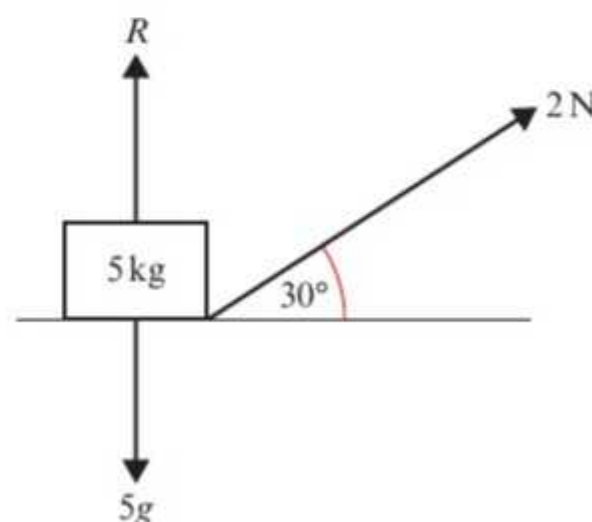
- (P) 4 Three forces act upon a particle as shown in the diagrams below.

Given that each particle is in equilibrium, calculate the magnitude of B and the value of θ .



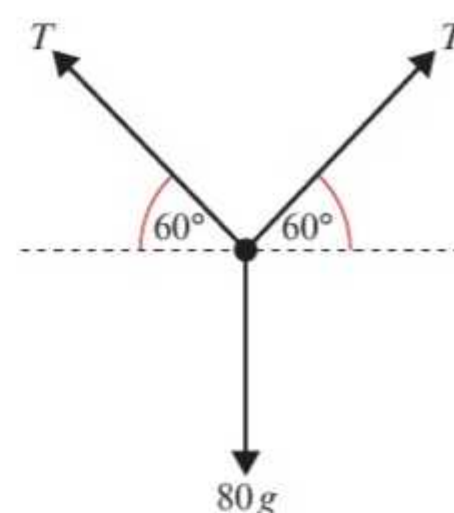
- 5 A box of mass 5 kg lies on a smooth horizontal floor. The box is pulled by a force of 2 N applied at an angle of 30° to the horizontal, causing the box to accelerate horizontally along the floor.

- a Work out the acceleration of the box.
 b Work out the normal reaction of the box with the floor.



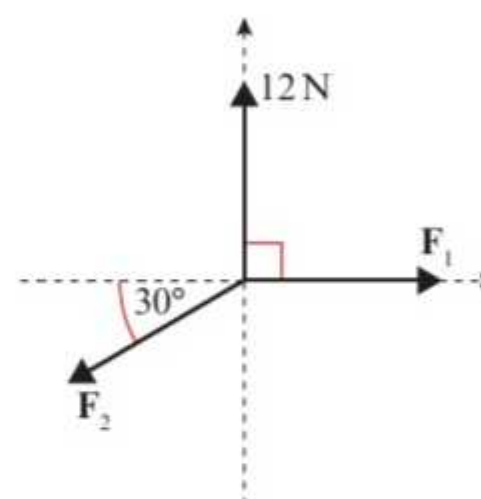
- (E) 6 A force P is applied to a box of mass 10 kg, causing the box to accelerate at 2 m s^{-2} along a smooth, horizontal plane. Given that the force causing the acceleration is applied at 45° to the plane, work out the value of P . (3 marks)
- (E) 7 A force of 20 N is applied to a box of mass m kg, causing the box to accelerate at 0.5 m s^{-2} along a smooth, horizontal plane. Given that the force causing the acceleration is applied at 25° to the plane, work out the value of m . (3 marks)

- E/P** 8 A parachutist of mass 80 kg is attached to a parachute by two lines, each with tension T . The parachutist is falling with constant velocity, and experiences a resistance to motion due to air resistance equal to one quarter of her weight. Show that the tension in each line, T , is $20\sqrt{3}\text{ g N}$.



(3 marks)

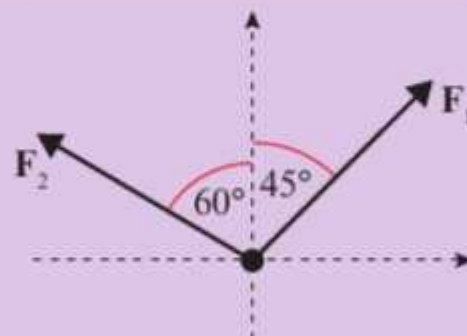
- E/P** 9 A system of forces act upon a particle as shown in the diagram. The resultant force on the particle is $(2\sqrt{3}\mathbf{i} + 2\mathbf{j})\text{ N}$. Calculate the magnitudes of $\mathbf{F}_1$ and $\mathbf{F}_2$.



(3 marks)

Challenge

Two forces act upon a particle as shown in the diagram. The resultant force on the particle is $(3\mathbf{i} + 5\mathbf{j})\text{ N}$. Calculate the magnitudes of $\mathbf{F}_1$ and $\mathbf{F}_2$.



5.2 Inclined planes

Force diagrams may be used to model situations involving objects on **inclined** planes.

- To solve problems involving inclined planes, it is usually easier to resolve forces parallel to and at right angles to the plane.

Example 5

SKILLS PROBLEM-SOLVING

A block of mass 10 kg slides down a smooth slope angled at 15° to the horizontal.

- Draw a force diagram to show all the forces acting on the block.
- Calculate the magnitude of the normal reaction of the slope on the block.
- Find the acceleration of the block.

a

b $R(\perp), \quad R = 10g \cos 15^\circ$
 $= 94.7 \text{ N (3 s.f.)}$

c $R(\parallel), \quad 10g \cos 75^\circ = 10a$
 $a = 2.54 \text{ m s}^{-2} \text{ (3 s.f.)}$

Watch out The normal reaction force acts at right angles to the plane, not vertically.

Your working will be easier if you resolve at right angles to the plane. The weight of the block acts at an angle of 15° to this direction.

Notation The diagonal arrows, $R(\perp)$ and $R(\parallel)$, show that you are resolving down the slope and perpendicular to the slope. You can also use $R(\parallel)$ to show resolution parallel to the slope and $R(\perp)$ to show resolution perpendicular to the slope.

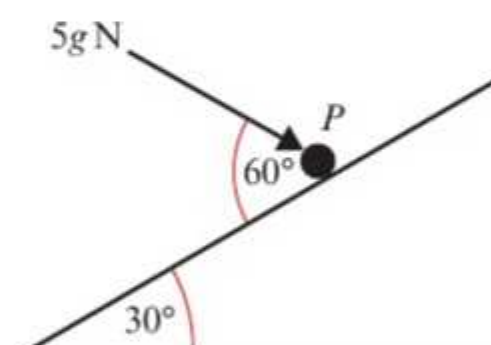
Resolve down the slope and use $F = ma$.

Example 6

6

SKILLS PROBLEM-SOLVING

A particle of mass m is pushed up a smooth slope by a force of magnitude $5g \text{ N}$ acting at an angle of 60° to the slope, causing the particle to accelerate up the slope at 0.5 m s^{-2} . Find the mass of the particle.



$R(\perp), \quad 5g \cos 60^\circ - mg \sin 30^\circ = 0.5m$
 $2.5g - 0.5mg = 0.5m$
 $2.5g = 0.5m + 0.5mg$
 $5g = m + mg$
 $5g = m(1 + g)$

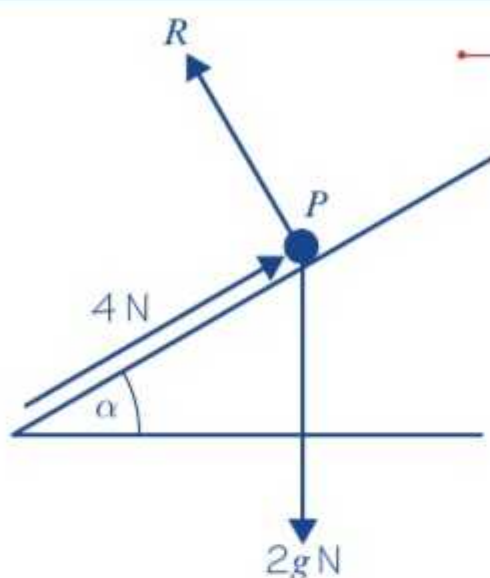
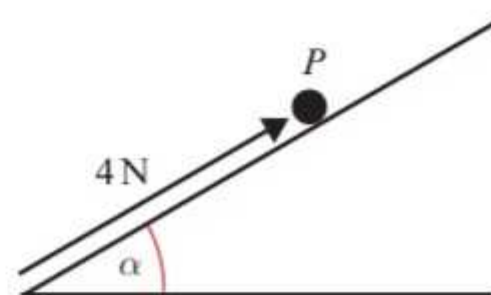
$m = \left(\frac{5g}{1 + g} \right) = 4.54 \text{ kg (3 s.f.)}$

Draw a diagram to show all the forces acting on the particle.

Resolve up the slope, in the direction of the acceleration, and write an equation of motion for the particle.

Example 7**SKILLS** PROBLEM-SOLVING

A particle P of mass 2 kg is moving on a smooth slope and is being acted on by a force of 4 N that acts parallel to the slope, as shown. The slope is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$. Work out the acceleration of the particle.



$$\begin{aligned}
 R(\perp), \quad 4 - 2g \sin \alpha &= 2a \\
 4 - 2 \times 9.8 \times \frac{3}{5} &= 2a \\
 2a &= -7.76 \\
 a &= -3.88 \text{ m s}^{-2} \text{ (3 s.f.)}
 \end{aligned}$$

The particle accelerates down the slope at 3.88 m s^{-2} .

Draw a diagram to show all the forces acting on the particle.

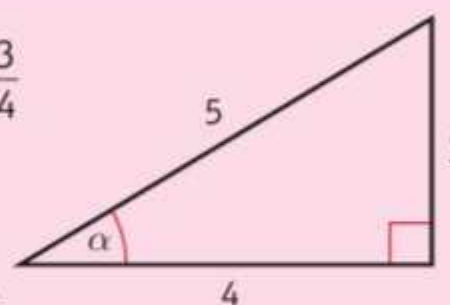
Resolve up the slope, in the direction of the acceleration, and use Newton's second law.

Problem-solving

You know that $\tan \alpha = \frac{3}{4}$ so you can draw a triangle to work out $\sin \alpha$ and $\cos \alpha$.

$$\sin \alpha = \frac{3}{5} \text{ and } \cos \alpha = \frac{4}{5}$$

You can use these exact values in your calculations.



You resolved up the slope and the acceleration is negative, so the particle is accelerating down the slope.

Exercise 5B**SKILLS** PROBLEM-SOLVING

- A particle of mass 3 kg slides down a smooth plane that is inclined at 20° to the horizontal.
 - Draw a force diagram to represent all the forces acting on the particle.
 - Work out the normal reaction between the particle and the plane.
 - Find the acceleration of the particle.
- A force of 50 N is pulling a particle of mass 5 kg up a smooth plane that is inclined at 30° to the horizontal. Given that the force acts parallel to the plane,
 - draw a force diagram to represent all the forces acting on the particle
 - work out the normal reaction between the particle and the plane
 - find the acceleration of the particle.
- A particle of mass 0.5 kg is held at rest on a smooth slope that is inclined at an angle α to the horizontal. The particle is released. Given that $\tan \alpha = \frac{3}{4}$, calculate:
 - the normal reaction between the particle and the plane
 - the acceleration of the particle.

- (E)** 4 A force of 30 N is pulling a particle of mass 6 kg up a rough slope that is inclined at 15° to the horizontal. The force acts in the direction of motion of the particle and the particle experiences a constant resistance due to friction.

a Draw a force diagram to represent all the forces acting on the particle. (4 marks)

Given that the particle is moving with constant speed,

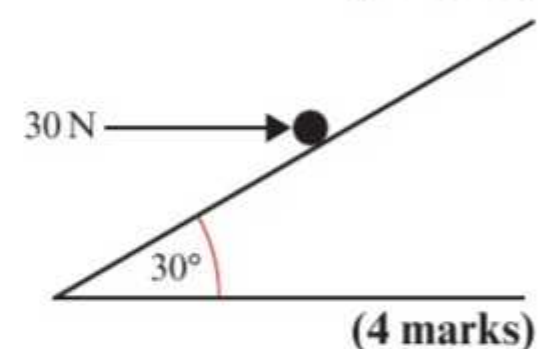
b calculate the magnitude of the resistance due to friction. (5 marks)

- (E)** 5 A particle of mass m kg is sliding down a smooth slope that is angled at 30° to the horizontal. The normal reaction between the plane and the particle is 5 N.

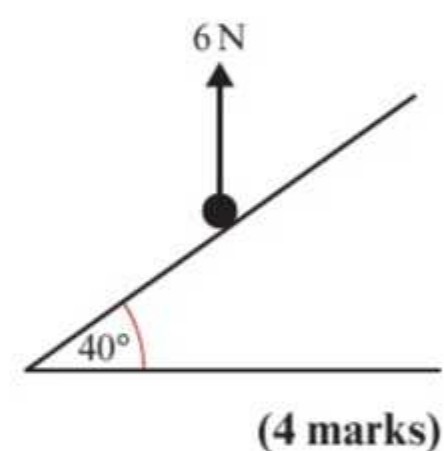
a Calculate the mass m of the particle. (3 marks)

b Calculate the acceleration of the particle. (3 marks)

- (E/P)** 6 A force of 30 N acts horizontally on a particle of mass 5 kg that rests on a smooth slope that is inclined at 30° to the horizontal as shown in the diagram. Find the acceleration of the particle.

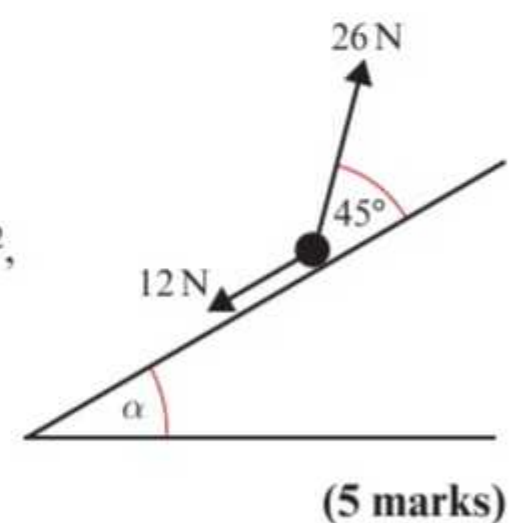


- (E/P)** 7 A particle of mass 3 kg is moving on a rough slope that is inclined at 40° to the horizontal. A force of 6 N acts vertically upon the particle. Given that the particle is moving at a constant velocity, calculate the value of F , the constant resistance due to friction.



- (E/P)** 8 A particle of mass m kg is pulled up a rough slope by a force of 26 N that acts at an angle of 45° to the slope. The particle experiences a constant frictional force of magnitude 12 N.

Given that $\tan \alpha = \frac{1}{\sqrt{3}}$ and that the acceleration of the particle is 1 m s^{-2} , show that $m = 1.08 \text{ kg}$ (3 s.f.).



Challenge

A particle is sliding down a smooth slope inclined at an angle θ to the horizontal, where $0 < \theta < 30^\circ$. The angle of inclination of the slope is increased by 60° , and the magnitude of the acceleration of the particle increases from a to $4a$.

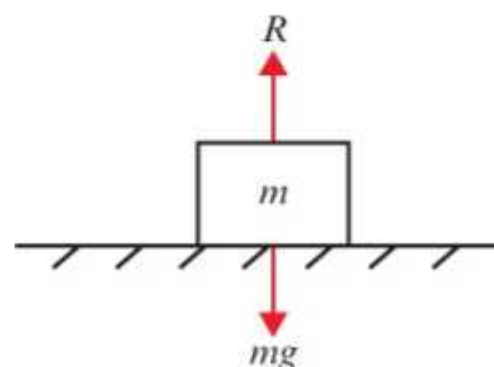
a Show that $\tan \theta = \frac{\sqrt{3}}{7}$.

b Hence find θ , giving your answer to 3 significant figures.

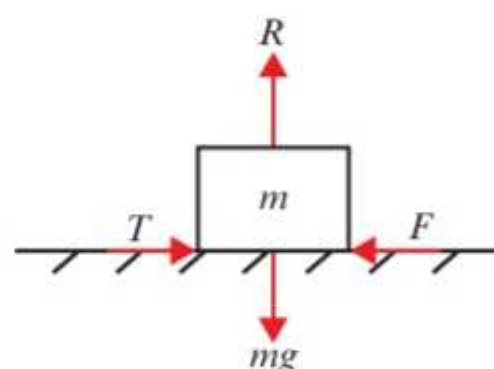
5.3 Friction

Friction is a force which opposes motion between two rough surfaces. It occurs when the two surfaces are moving relative to one another, or when there is a **tendency** (i.e. where something is likely to behave in a certain way) for them to move relative to one another.

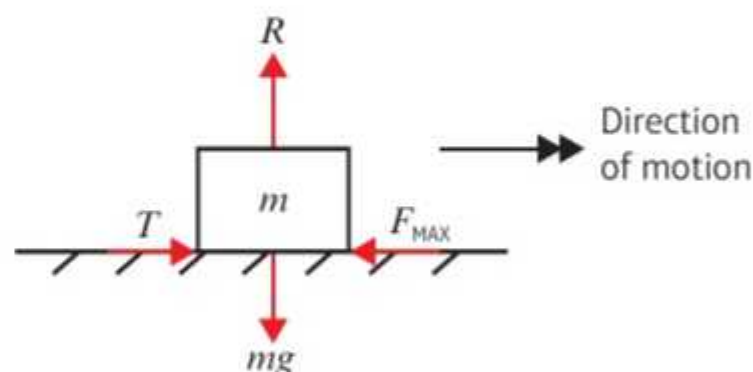
This block is stationary. There is no horizontal force being applied, so there is no tendency for the block to move. There is no frictional force acting on the block.



This block is also stationary. There is a horizontal force being applied which is not sufficient to move the block. There is a tendency for the block to move, but it doesn't because the force of friction is equal and opposite to the force being applied.



As the applied force increases, the force of friction increases to prevent the block from moving. If the magnitude of the applied force exceeds a certain **maximum** or **limiting value**, the block will move. While the block moves, the force of friction will remain constant at its maximum value.



- When a particle is on the point of moving it is said to be in limiting equilibrium.

The limiting value of the friction depends on two things:

- the normal reaction R between the two surfaces in contact
- the roughness of the two surfaces in contact.

You can measure roughness using the **coefficient of friction**, which is represented by the letter μ (pronounced *myoo*). The rougher the two surfaces, the larger the value of μ . For smooth surfaces there is no friction and $\mu = 0$.

- The maximum or limiting value of the friction between two surfaces, F_{MAX} , is given by

$$F_{\text{MAX}} = \mu R$$

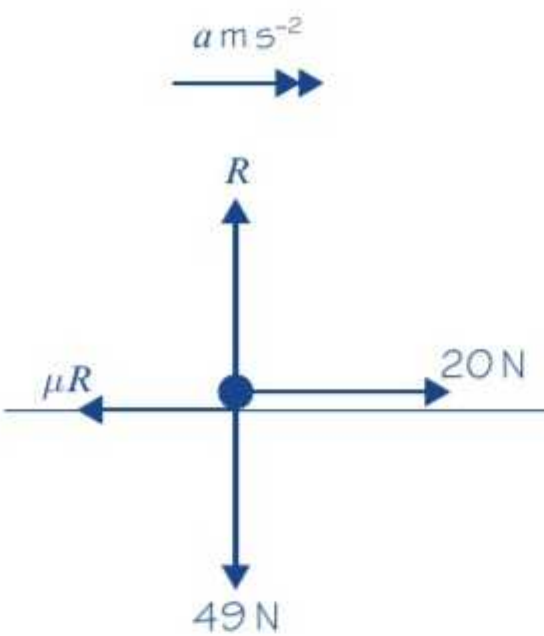
where μ is the coefficient of friction and R is the normal reaction between the two surfaces.

Example 8

SKILLS PROBLEM-SOLVING

A particle of mass 5 kg is pulled along a rough horizontal surface by a horizontal force of magnitude 20 N. The coefficient of friction between the particle and the floor is 0.2. Calculate:

- the magnitude of frictional force
- the acceleration of the particle.



$a \text{ m s}^{-2}$
 R
 μR 20 N
 49 N

a $R(\uparrow),$ $R = 49$
 $F = \mu R = 0.2 \times 49 = 9.8 \text{ N}$

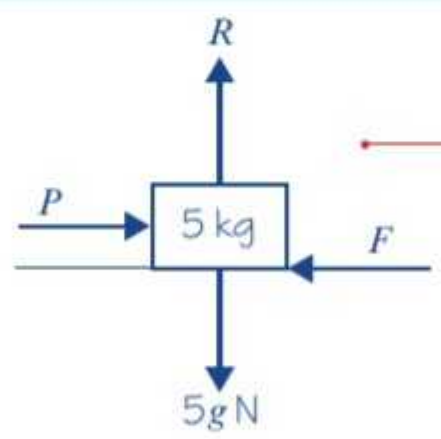
b $R(\rightarrow),$ $20 - \mu R = 5a$
 $5a = 20 - 9.8$
 $= 10.2$
 $a = 2.04 \text{ m s}^{-2} \text{ (3 s.f.)}$

For a moving particle, $F = F_{\text{MAX}}$ so you can use $F = \mu R$ to find the magnitude of the frictional force.

Write an equation of motion for the particle, resolving horizontally. Note that the frictional force always acts in a direction so as to **oppose** the motion of the particle.

Example**9****SKILLS****PROBLEM-SOLVING**

A block of mass 5 kg lies on rough horizontal ground. The coefficient of friction between the block and the ground is 0.4. A horizontal force P is applied to the block. Find the magnitude of the frictional force acting on the block and the acceleration of the block when the magnitude of P is:

a 10 N**b** 19.6 N**c** 30 N.


R
 P 5 kg F
 $5g \text{ N}$

$R(\uparrow),$ $R = 5g = 49 \text{ N}$
 So $F_{\text{MAX}} = \mu R = 0.4 \times 49$
 $= 19.6 \text{ N}$

The maximum available frictional force is 19.6 N.

a When $P = 10 \text{ N}$, the friction will only need to be 10 N to prevent the block from sliding and the block will remain at rest in equilibrium.

First draw a diagram showing all the forces acting on the block.

The normal reaction will equal the weight as the force P has no vertical component and there is no vertical acceleration.

You then need to calculate the maximum possible frictional force for this situation.

Do not round this value as you will need to use it in your calculations.

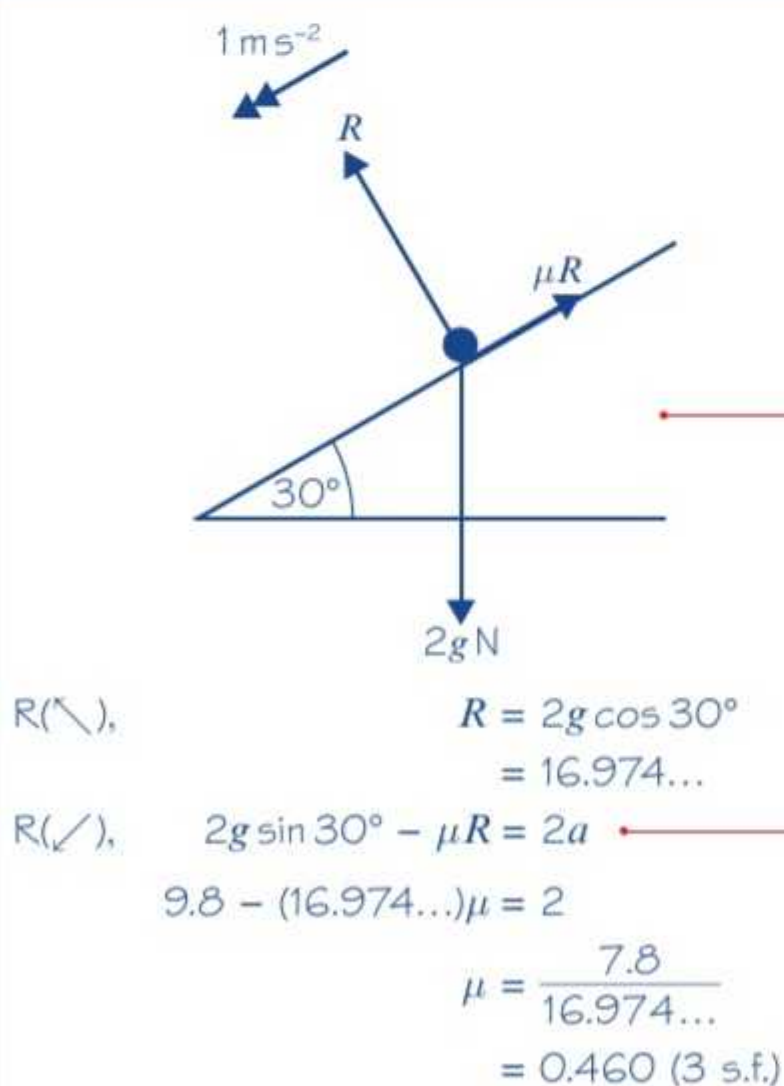
- b** When $P = 19.6 \text{ N}$, the friction will need to be at its maximum value of 19.6 N to prevent the block from sliding, and the block will remain at rest in **limiting equilibrium**.
- c** When $P = 30 \text{ N}$, the friction will be unable to prevent the block from sliding, and it will remain at its maximum value of 19.6 N . The block will accelerate from rest along the plane in the direction of P with acceleration a , where
- $$30 - 19.6 = 5a$$
- $$a = 2.08 \text{ m s}^{-2} \text{ (3 s.f.)}$$

Watch out An object in limiting equilibrium can either be at rest or moving with constant velocity.

Example 10

SKILLS PROBLEM-SOLVING

A particle of mass 2 kg is sliding down a rough slope that is inclined at 30° to the horizontal. Given that the acceleration of the particle is 1 m s^{-2} , find the coefficient of friction, μ , between the particle and the slope.



Online Explore this problem with different masses, slopes and frictional coefficients using GeoGebra.



Draw a diagram showing the weight, the frictional force and the normal reaction.

Use $F = ma$ to write an equation of motion for the particle.

Watch out Make sure you use the normal reaction, not the weight, when substituting into $F_{\text{MAX}} = \mu R$.

6 MOMENTUM AND IMPULSE

4.3

Learning objectives

After completing this chapter you should be able to:

- Calculate the momentum of a particle and the impulse of a force
- Solve problems involving collisions using the principle of conservation of momentum

→ pages 102–104

→ pages 104–109

Prior knowledge check

- 1** The forces $\mathbf{F}_1 = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{F}_2 = 5\mathbf{i} + 4\mathbf{j}$ act on a particle. Find the magnitude and direction of the resultant force.

← Mechanics 1 Section 3.4

- 2** A particle moves in a straight line with constant acceleration.
Given s = displacement in m, u = initial velocity in m s^{-1} , v = final velocity in m s^{-1} , a = acceleration in m s^{-2} and t = time in seconds, find:

a v when $u = 3$, $a = 0.5$, $t = 5$

b s when $u = 4.5$, $a = -1.5$, $t = 2$

← Mechanics 1 Sections 2.4, 2.5

- 3** A body of mass 2 kg is acted on by a force $\mathbf{F}$ N. The body starts from rest and moves in a straight line. After 5 seconds, the displacement of the body is 20 m. Find the magnitude of $\mathbf{F}$.

← Mechanics 1 Section 4.3

Newton's cradle demonstrates the **principle of conservation of momentum**. When the first ball collides with the second, the first ball stops, but its momentum is transferred to the second ball, then the third, then the fourth, until it reaches the very last ball.

6.1 Momentum in one dimension

You can calculate the momentum of a particle and the impulse of a force.

- The momentum of a body of mass m which is moving with velocity v is mv .

If m is in kg and v is in m s^{-1} then the units of momentum will be kg m s^{-1} .

However, since $\text{kg m s}^{-1} = (\text{kg m s}^{-2}) \text{s}$ and kg m s^{-2} are the units of force ($F = ma$) you can also measure momentum in newton seconds (Ns).

Notation Velocity is a vector quantity and mass is a scalar, so momentum is a vector quantity.

Example 1

SKILLS PROBLEM-SOLVING

Find the magnitude of the momentum of:

- a a cricket ball of mass 400 g moving at 18 m s^{-1}
- b a lorry of mass 5 tonnes moving at 0.3 m s^{-1} .

a Momentum = mass $\times$ velocity

$$\begin{aligned}\text{Magnitude of momentum} &= \frac{400}{1000} \times 18 \\ &= 7.2 \text{ Ns}\end{aligned}$$

The mass must be in kg, so divide by 1000.

The units can be Ns or kg m s^{-1} .

b Momentum = mass $\times$ velocity

$$\begin{aligned}\text{Magnitude of momentum} &= (5 \times 1000) \times 0.3 \\ &= 1500 \text{ kg m s}^{-1}\end{aligned}$$

1 tonne = 1000 kg.

- If a constant force F acts for time t then we define the impulse of the force to be Ft .

If F is in N and t is in s then the units of impulse will be Ns.

Notation Force is a vector quantity and time is a scalar, so impulse is a vector quantity.

Examples of an impulse include a bat hitting a ball, a snooker ball hitting another ball or a jerk in a string when it suddenly goes tight. In all these cases the time for which the force acts is very small but the force is quite large. This means that the product of the two, which gives the impulse, is of reasonable size. However, there is no theoretical limit on the size of t .

Suppose a body of mass m is moving with an initial speed u and is then acted upon by a force F for time t . This results in its final speed being v .

Its acceleration is given by $a = \frac{v - u}{t}$

Substituting into $F = ma$: $F = m\left(\frac{v - u}{t}\right)$

$$\begin{aligned}Ft &= m(v - u) \\ &= mv - mu\end{aligned}$$

The impulse I of the force is given by $I = Ft$.

- $I = mv - mu$

Impulse = final momentum – initial momentum

Impulse = change in momentum

This is called the **impulse-momentum principle**.

Watch out This is a vector equation, so for motion in a straight line a positive direction must be chosen and each value must be given the correct sign.

Example 2**SKILLS** PROBLEM-SOLVING

A body of mass 2 kg is initially at rest on a smooth horizontal plane. A horizontal force of magnitude 4.5 N acts on the body for 6 s. Find:

- the magnitude of the impulse given to the body by the force
- the final speed of the body.

a Magnitude of the impulse = force $\times$ time
 $= 4.5 \times 6$
 $= 27 \text{ N s}$

b Impulse = Final momentum – Initial momentum
 $27 = 2v - 0$
 $v = 13.5 \text{ m s}^{-1}$

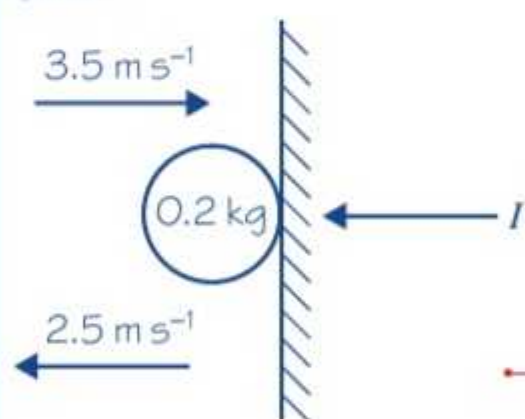
The units can be N s or kg m s^{-1} .

The body is at rest initially.

Example 3**SKILLS** PROBLEM-SOLVING

A ball of mass 0.2 kg hits a fixed vertical wall at right angles with speed 3.5 m s^{-1} . The ball rebounds with speed 2.5 m s^{-1} . Find the magnitude of the impulse exerted on the wall by the ball.

The diagram shows the initial and final direction and speed of the ball and the impulse acting on it.



($\leftarrow$): $I = (0.2 \times 2.5) - (0.2 \times (-3.5))$
 $= 0.5 + 0.7$
 $= 1.2 \text{ N s}$

Therefore, by Newton's 3rd law, the magnitude of the impulse exerted on the wall by the ball is 1.2 N s.

Problem-solving

Because the wall is fixed you cannot apply the impulse–momentum principle to it. Find the magnitude of the impulse exerted on the ball by the wall and then use Newton's 3rd law to deduce that the magnitude of the impulse exerted on the wall by the ball will be the same.

Note that this is a plan view of the situation.

Choose a positive direction ($\leftarrow$) and apply the impulse–momentum principle to the ball.

The initial speed is in the negative direction.

Exercise 6A**SKILLS** PROBLEM-SOLVING

- A ball of mass 0.5 kg is at rest when it is struck by a bat and receives an impulse of 15 N s. Find its speed immediately after it is struck.
- A ball of mass 0.3 kg moving along a horizontal surface hits a fixed vertical wall at right angles with speed 3.5 m s^{-1} . The ball rebounds at right angles to the wall. Given that the magnitude of the impulse exerted on the ball by the wall is 1.8 N s, find the speed of the ball just after it rebounds.

- 3 A toy car of mass 0.2 kg is pushed from rest along a smooth horizontal floor by a horizontal force of magnitude 0.4 N for 1.5 s . Find its speed at the end of the 1.5 s .
- E** 4 A ball of mass 0.2 kg , moving along a horizontal surface, hits a fixed vertical wall at right angles. The ball rebounds at right angles to the wall with speed 3.5 m s^{-1} . Given that the magnitude of the impulse exerted on the ball by the wall is 2 N s , find the speed of the ball just before it hits the wall. **(3 marks)**
- E/P** 5 A ball of mass 0.2 kg is dropped from a height of 2.5 m above horizontal ground. After hitting the ground it bounces to a height of 1.8 m above the ground. Find the magnitude of the impulse received by the ball from the ground. **(4 marks)**

6.2 Conservation of momentum

You can solve problems involving collisions using the **principle of conservation of momentum**.

By Newton's 3rd law, when two bodies collide, each one exerts an equal and opposite force on the other. They are in contact for the same time, so they each exert an impulse on the other of equal magnitude but in opposite directions.

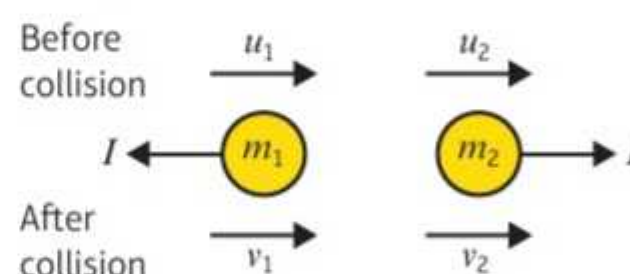


By the impulse–momentum principle, the changes in momentum of each body are equal but opposite in direction. Thus, these changes in momentum cancel each other out, and the momentum of the whole system is unchanged. This is called the **principle of conservation of momentum**.

■ Total momentum before impact = total momentum after impact

You can write this in symbols for two masses m_1 and m_2 with speeds u_1 and u_2 respectively before the collision, and speeds v_1 and v_2 respectively after the collision:

$$\mathbf{m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2}$$



When solving problems involving collisions, always:

- draw a diagram showing the speeds before and after the collision with arrows
- if appropriate, include the impulses on your diagram with arrows
- choose a positive direction and apply the impulse–momentum principle and/or the principle of conservation of momentum.

Example

4

SKILLS

PROBLEM-SOLVING

A particle P of mass 2 kg is moving with speed 3 m s^{-1} on a smooth horizontal plane. Particle Q of mass 3 kg is at rest on the plane. Particle P collides with particle Q and after the collision Q moves off with speed $\frac{7}{3}\text{ m s}^{-1}$. Find:

- the speed and direction of motion of P after the collision
- the magnitude of the impulse received by P in the collision.

Online

Explore particle collisions using GeoGebra.



a

Conservation of momentum: ($\rightarrow$)

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$(2 \times 3) + (3 \times 0) = 2v + \left(3 \times \frac{7}{3}\right)$$

$$6 = 2v + 7$$

$$v = -\frac{1}{2}$$

The direction of motion of P is reversed by the collision and its speed is $\frac{1}{2} \text{ m s}^{-1}$.

b For Q : ($\rightarrow$) $I = 3\left(\frac{7}{3} - 0\right)$

$$= 7 \text{ N s}$$

Alternatively, for P : ($\leftarrow$)

$$I = 2((-v) - (-3))$$

$$= 2\left(\frac{1}{2} + 3\right)$$

$$= 7 \text{ N s}$$

So the impulse received by P has magnitude 7 N s .

Draw a diagram showing the speeds before and after the collision (with arrows) and the impulses (with arrows).

Choose a positive direction and apply the principle of conservation of momentum.

Since v is negative, P moves in the opposite direction after the collision.

Watch out The direction of motion of P in your answer must be with reference to the original direction of motion of P . Do not use the words left or right.

To find the impulse, consider one particle and apply the impulse-momentum principle. Here it is easier to consider Q .

Since each particle receives an impulse of equal magnitude, the magnitude of the impulse received by P is also 7 N s .

Example 5

SKILLS PROBLEM-SOLVING

Two particles A and B , of masses 2 kg and 4 kg respectively, are moving toward each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide. Before the collision, the speeds of A and B are 3 m s^{-1} and 2 m s^{-1} respectively. After the collision, the direction of motion of A is reversed and its speed is 2 m s^{-1} . Find:

- the speed and direction of B after the collision
- the magnitude of the impulse given by A to B in the collision.

a

Conservation of momentum: ($\rightarrow$)

$$(2 \times 3) + (4 \times (-2)) = (2 \times (-2)) + 4v$$

$$6 - 8 = -4 + 4v$$

$$2 = 4v$$

$$0.5 = v$$

B has speed 0.5 m s^{-1} and its direction of motion is reversed by the collision.

Online Explore collisions with two moving particles using GeoGebra.



You need to 'guess' the direction of B after the collision. If it is moving in the other direction the answer will be negative.

This defines the positive direction.

Each speed must be given the correct sign.

As the value of v is positive the 'guess' for the direction of B after the collision was correct.

b For A : ($\leftarrow$)

impulse–momentum principle

$$I = 2(2 - (-3))$$

$$= 10 \text{ N s}$$

The magnitude of the impulse given by A to B is 10 N s .

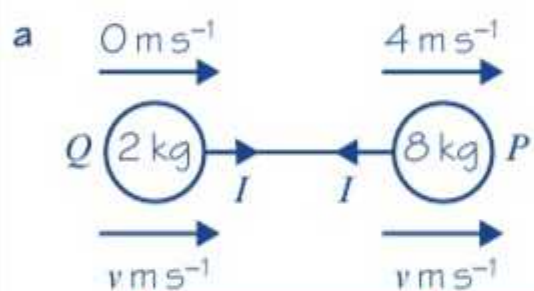
Although we could consider either particle, it is safer to consider A since its initial and final speed were given in the question.

The magnitude of the impulse given by A to B is the same as the magnitude of the impulse given by B to A .

Example**6****SKILLS****PROBLEM-SOLVING**

Two particles P and Q , of masses 8 kg and 2 kg respectively, are connected by a light inextensible string. The particles are at rest on a smooth horizontal plane with the string slack. Particle P is projected directly away from Q with speed 4 m s^{-1} .

- Find the common speed of the particles after the string goes taut.
- Find the magnitude of the impulse transmitted through the string when it goes taut.



Using conservation of momentum ($\rightarrow$):

$$(2 \times 0) + (8 \times 4) = 2v + 8v$$

$$32 = 10v$$

$$3.2 = v$$

The common speed of the particles is 3.2 m s^{-1} .

- b For Q ($\rightarrow$):

$$I = 2(v - 0)$$

$$= 2 \times 3.2$$

$$= 6.4 \text{ N s}$$

The magnitude of the impulse transmitted through the string (the 'jerk') is 6.4 N s .

The string is inextensible so these are the same.

This must be applied to the whole system.

To find the impulse we must consider one of the particles and apply the impulse–momentum principle. It is easier to consider Q .

Example**7****SKILLS****PROBLEM-SOLVING**

Two particles A and B , of masses 2 kg and 4 kg respectively, are moving toward each other in opposite directions along the same straight line on a smooth horizontal surface. The particles collide. Before the collision, the speeds of A and B are 3 m s^{-1} and 2 m s^{-1} respectively. Given that the magnitude of the impulse due to the collision is 7 N s , find:

- the velocity of A after the collision
- the velocity of B after the collision.

For A : ($\leftarrow$)

$$7 = 2(v_1 - (-3))$$

$$7 = 2(v_1 + 3)$$

$$3.5 = v_1 + 3$$

$$0.5 = v_1$$

For B : ($\rightarrow$)

$$7 = 4(v_2 - (-2))$$

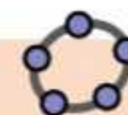
$$1.75 = v_2 + 2$$

$$-0.25 = v_2$$

a The direction of motion of A is reversed and its speed is 0.5 m s^{-1} .

b The direction of motion of B is unchanged and its speed is 0.25 m s^{-1} .

Online Explore particle collisions with known impulse using GeoGebra.



The diagram should show the speeds and impulses with arrows.

Again you need to guess which way the particles go after the collision. There are three sensible possibilities: the one shown, $\overleftarrow{v_1} \overrightarrow{v_2}$ or $\overleftarrow{v_1} \overleftarrow{v_2}$.

To find either v_1 or v_2 , consider one particle only, choose a positive direction and apply the impulse–momentum principle.

As v_1 is positive, the guess for the direction of A was correct.

As v_2 is negative, the guess for the direction of B was incorrect and it travels in the opposite direction to that shown on the diagram.

Watch out The question asks for the **velocities** so you need to state the speed and the direction of motion.

Exercise

6B

SKILLS

PROBLEM-SOLVING

- A particle P of mass 2 kg is moving on a smooth horizontal plane with speed 4 m s^{-1} . It collides with a second particle Q of mass 1 kg which is at rest. After the collision P has speed 2 m s^{-1} and it continues to move in the same direction. Find the speed of Q after the collision.
- A railway truck of mass 25 tonnes moving at 4 m s^{-1} collides with a stationary truck of mass 20 tonnes . As a result of the collision the trucks couple together. Find the common speed of the trucks after the collision.
- Particles A and B have masses 0.5 kg and 0.2 kg respectively. They are moving with speeds 5 m s^{-1} and 2 m s^{-1} respectively in the same direction along the same straight line on a smooth horizontal surface when they collide. After the collision, A continues to move in the same direction with speed 4 m s^{-1} . Find the speed of B after the collision.
- A particle of mass 2 kg is moving on a smooth horizontal plane with speed 4 m s^{-1} . It collides with a second particle of mass 1 kg which is at rest. After the collision the particles join together.
 - Find the common speed of the particles after the collision.
 - Find the magnitude of the impulse in the collision.
- Two particles A and B , of masses 2 kg and 5 kg respectively, are moving toward each other along the same straight line on a smooth horizontal surface. The particles collide. Before the collision the speeds of A and B are 6 m s^{-1} and 4 m s^{-1} respectively. After the collision the direction of motion of A is reversed and its speed is 1.5 m s^{-1} . Find:
 - the speed and direction of B after the collision (3 marks)
 - the magnitude of the impulse given by A to B in the collision. (3 marks)

6 STATICS OF RIGID BODIES

6.1

Learning objectives

- Understand the moment of a force
- Solve problems involving ladders resting against smooth or rough vertical walls and on smooth or rough ground

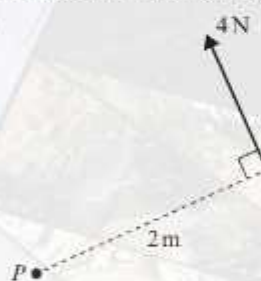
→ pages 153–157

→ pages 153–157

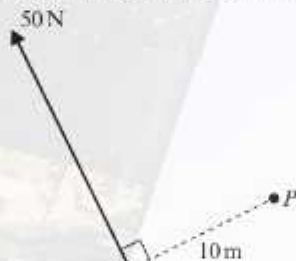
Prior knowledge check

- 1 Calculate the moment about P of each force acting on a lamina.

a



b



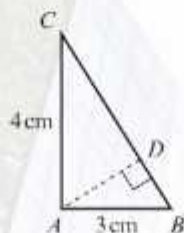
← Mechanics 1 Section 8.1

- 2 A uniform rod PQ of mass $m\text{ kg}$ and length 6 m rests on two pivots A and B as shown below, with $PA = 1\text{ m}$ and $BQ = 2\text{ m}$. Work out the normal reaction acting on each pivot.



← Mechanics 1 Section 8.2

- 3 In the triangle ABC , work out the length AD .



← International GCSE Mathematics

When designing and building the Torre Mare Nostrum building in Barcelona, the engineers would have needed to use moments to calculate the forces acting on the building.

6.1 Static rigid bodies

If you need to consider the rotational forces acting on an object you can model it as a **rigid body**.

■ For a rigid body in **static equilibrium**:

- the body is stationary
- the resultant force in any direction is zero
- the resultant moment about any given point is zero.

You will sometimes need to consider the moments acting on a body and the resultant force acting on the body separately.

Links The moment of a force of magnitude F N about a point P is Fd , where d is the **perpendicular distance** from the line of action of the force to P .

← Mechanics 1 Section 8.1

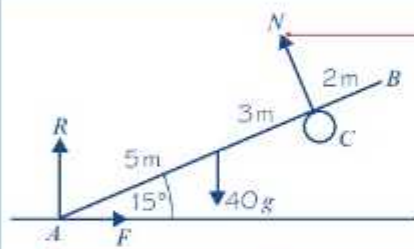
Example 1

SKILLS

PROBLEM-SOLVING

A uniform rod AB of mass 40 kg and length 10 m rests with the end A on rough horizontal ground. The rod rests against a smooth peg C where $AC = 8$ m. The rod is in limiting equilibrium at an angle of 15° to the horizontal. Find:

- the magnitude of the reaction at C
- the coefficient of friction between the rod and the ground.



Start with a diagram showing all the forces.

N , the reaction at C , is perpendicular to the rod. The peg is smooth, so there is no friction here.

At A there is a normal reaction and a frictional force. The peg is smooth so the only force stopping the rod from sliding is the frictional force at A .

a Taking moments about A :

$$40g \times 5 \cos 15^\circ = N \times 8$$

$$N = \frac{200g \cos 15^\circ}{8}$$

$$= 25g \cos 15^\circ$$

$$= 236.65 \dots \text{ N}$$

The reaction at C has magnitude 240 N (2 s.f.).

b $R(\rightarrow)$, $F = N \cos 75^\circ = 61.25 \text{ N}$

$$R(\uparrow), \quad R + N \cos 15^\circ = 40g$$

$$R = 40g - N \cos 15^\circ = 163.41 \dots \text{ N}$$

The rod is in limiting equilibrium, so

$$F = \mu R, \quad \mu = \frac{F}{R} = 0.37 \text{ (2 s.f.)}$$

Problem-solving

Solve part **a** by taking moments. If you take moments about A you can ignore the frictional force. For part **b** you can resolve forces horizontally and vertically for the whole body.

Resolve horizontally and vertically.

It is important to consider whether the rigid body is in limiting equilibrium. Only if this is the case does the frictional force take its maximum value μR where μ is the coefficient of friction. Otherwise you can only assume $F \leq \mu R$.

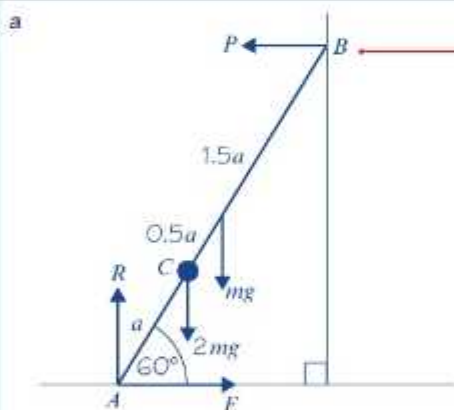
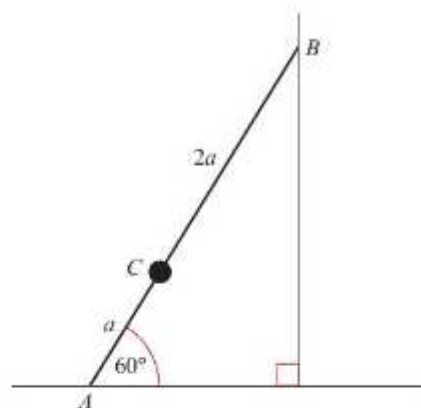
Example 2

SKILLS

PROBLEM-SOLVING

A ladder AB , of mass m and length $3a$, has one end A resting on rough horizontal ground. The other end B rests against a smooth vertical wall. A load of mass $2m$ is fixed on the ladder at the point C , where $AC = a$. The ladder is modelled as a uniform rod in a vertical plane perpendicular to the wall and the load is modelled as a particle. The ladder rests in limiting equilibrium at an angle of 60° with the ground.

- Find the coefficient of friction between the ladder and the ground.
- State how you have used the assumption that the ladder is uniform in your calculations.



$$R(\rightarrow), \quad F = P$$

$$R(\uparrow), \quad R = 2mg + mg = 3mg$$

Taking moments about B :

$$2mg \times 2a \cos 60^\circ + mg \times 1.5a \cos 60^\circ$$

$$+ F \times 3a \sin 60^\circ = R \times 3a \cos 60^\circ$$

$$5.5mg \cos 60^\circ + 3F \sin 60^\circ$$

$$= 3R \cos 60^\circ$$

$$2.75mg + \frac{3\sqrt{3}}{2} F = 1.5R$$

Since $R = 3mg$, and $F = \mu R$ (as the ladder is in limiting equilibrium)

$$2.75mg + \frac{3\sqrt{3}}{2} \mu \times 3mg = 1.5 \times 3mg$$

$$\mu = \frac{4.5 - 2.75}{\left(\frac{9\sqrt{3}}{2}\right)} = \frac{7}{18\sqrt{3}} = 0.225 \text{ (3 s.f.)}$$

The reaction at B is perpendicular to the wall. The wall is smooth, so there is no friction at B .

Watch out The reactions at the wall and the floor are different and so must be labelled with different letters.

Online Explore the forces in this question in a more detailed diagram using GeoGebra.



Resolve vertically and horizontally for the whole system.

You want to find $\mu = \frac{F}{R}$ so take moments at B . Remember to use the **perpendicular distance** from each force to B .

You can divide both sides by a .

Divide through by mg and solve to find μ . The answer is independent of g so round to 3 s.f.

- b The assumption that the ladder is uniform allows you to assume that its weight acts at its midpoint.

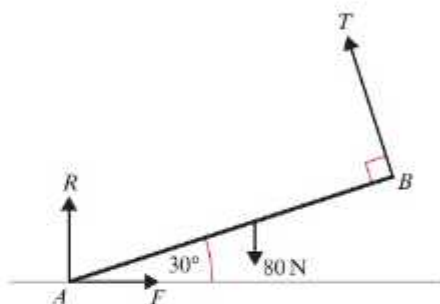
Problem-solving

There are other options for points to take moments about. For example, if you were to take moments about the point where the lines of action of R and P meet you would eliminate R and P from your working and simplify your calculation.

Exercise**6A****SKILLS****PROBLEM-SOLVING**

Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$.

- 1 A uniform rod AB of weight 80 N rests with its lower end A on a rough horizontal floor. A string attached to end B keeps the rod in equilibrium. The string is held at 90° to the rod. The tension in the string is T . The coefficient of friction between the rod and the ground is μ . R is the normal reaction at A and F is the frictional force at A . Find the magnitudes of T , R and F , and the least possible value of μ .



- 2 A uniform ladder of mass 10 kg and length 5 m rests against a smooth vertical wall with its lower end on rough horizontal ground. The ladder rests in equilibrium at an angle of 65° to the horizontal. Find:
 - a the magnitude of the normal reaction S at the wall
 - b the magnitude of the normal reaction R at the ground and the frictional force at the ground
 - c the least possible value of the coefficient of friction between the ladder and the ground.
 - d State how you have used the assumption that the ladder is uniform in your calculations.

- P** 3 A uniform ladder AB of mass 20 kg rests with its top A against a smooth vertical wall and its base B on rough horizontal ground. The coefficient of friction between the ladder and the ground is $\frac{3}{4}$. A mass of 10 kg is attached to the ladder. Given that the ladder is about to slip, find the inclination of the ladder to the horizontal,

- a if the 10 kg mass is attached at A
- b if the 10 kg mass is attached at B .
- c State how you have used the assumption that the wall is smooth in your calculations.

- E** 4 A uniform ladder of mass 20 kg and length 8 m rests against a smooth vertical wall with its lower end on rough horizontal ground. The coefficient of friction between the ground and the ladder is 0.3 . The ladder is inclined at an angle θ to the horizontal, where $\tan \theta = 2$. A boy of mass 30 kg climbs up the ladder. By modelling the ladder as a uniform rod, the boy as a particle and the wall as smooth and vertical,

- a find how far up the ladder the boy can climb before the ladder slips. (8 marks)
- b Criticise this model with respect to:
 - i the ladder
 - ii the wall. (2 marks)

7 STATICS OF A PARTICLE

5.1
5.2
5.3

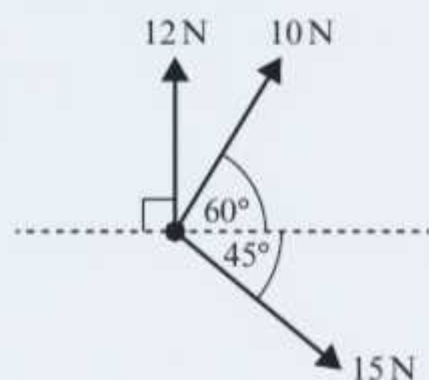
Learning objectives

After completing this chapter you should be able to:

- Find an unknown force when a system is in equilibrium → pages 113–116
- Solve statics problems involving weight, tension and pulleys → pages 117–121
- Understand and solve problems involving limiting equilibrium → pages 121–126

Prior knowledge check

- 1 A particle of mass 2 kg sits on a rough plane that is inclined at 45° to the horizontal. A force of 10 N acts parallel to and up the plane. Given that the particle is on the point of moving, work out the coefficient of friction μ . ← Mechanics 1 Section 5.2
- 2 A particle of negligible mass has three forces acting upon it as shown in the diagram below. Work out the magnitude and direction of the resultant force relative to the horizontal dashed line.



← Mechanics 1 Section 5.1

A tightrope walker uses a mathematical model to calculate the tension in his wire. This allows him to make sure that the wire is strong enough to hold his weight safely.

7.1 Static particles

- A particle or rigid body is in **static** equilibrium if it is at rest and the resultant force acting on the particle or rigid body is zero.

To solve problems in statics you should:

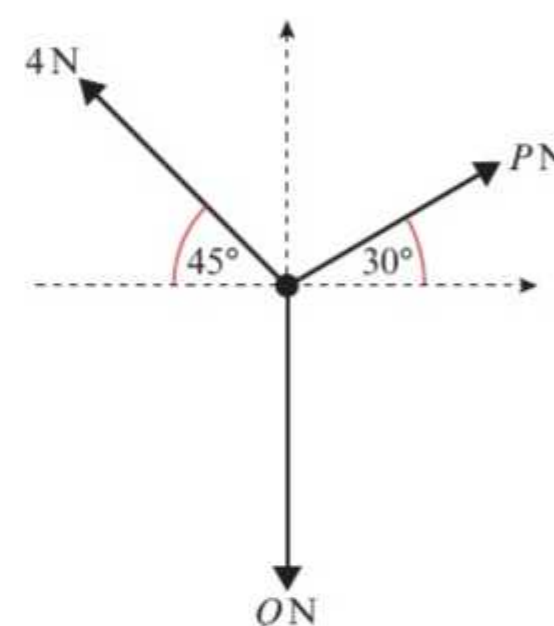
- Draw a diagram showing clearly the forces acting on the particle.
- Resolve the forces into horizontal and vertical components or, if the particle is on an inclined plane, into components parallel and perpendicular to the plane.
- Set the sum of the components in each direction equal to zero.
- Solve the resulting equations to find the unknown force(s).

Example 1

SKILLS PROBLEM-SOLVING

The diagram shows a particle in equilibrium under the forces shown. By resolving horizontally and vertically, find the magnitudes of the forces P and Q .

Hint The particle is not accelerating, so $a = 0$.
 $F = ma = 0$



Method 1:

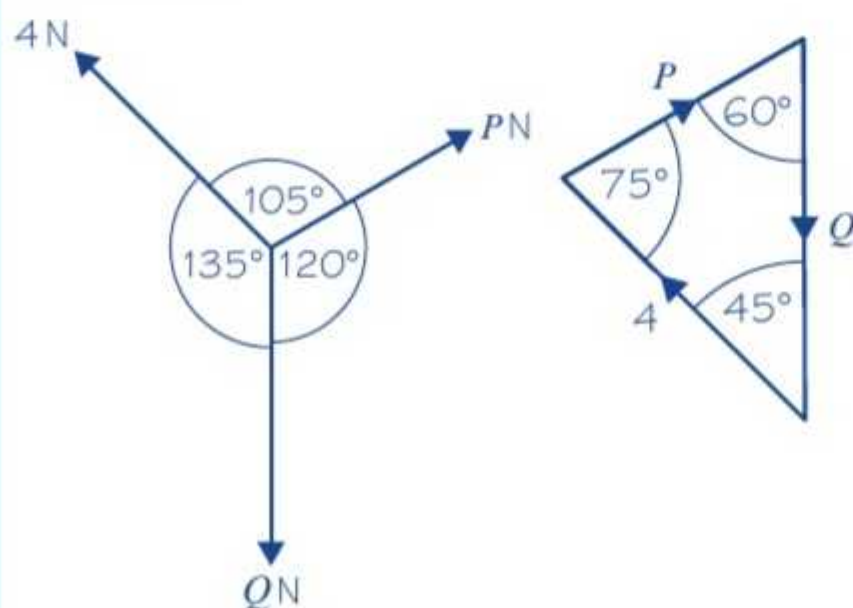
$$R(\rightarrow), \quad P \cos 30^\circ - 4 \cos 45^\circ = 0$$

$$R(\uparrow), \quad P \sin 30^\circ + 4 \sin 45^\circ - Q = 0$$

$$P = \frac{4 \cos 45^\circ}{\cos 30^\circ} = 3.27 \text{ (3 s.f.)}$$

$$Q = P \sin 30^\circ + 4 \sin 45^\circ = 4.46 \text{ (3 s.f.)}$$

Method 2:



Resolve horizontally and vertically. Equate the sum of the forces to zero as there is no acceleration (the particle is in equilibrium).

Solve the first equation to find P (as there is only one unknown quantity), and then use your value for P in the second equation to find Q .

If exact answers are required these would be

$$P = \frac{4\sqrt{6}}{3} \text{ and } Q = \frac{2(\sqrt{6} + 3\sqrt{2})}{3}$$

Problem-solving

You can use a vector diagram to solve equilibrium problems involving three forces. Because the particle is in equilibrium, the three forces will form a **closed triangle**.

If the angle between forces on the force diagram is θ , the angle between those forces on the **triangle of forces** is $180^\circ - \theta$.

The length of each side of the triangle is the magnitude of the force.

$$\text{So } \frac{4}{\sin 60^\circ} = \frac{P}{\sin 45^\circ} = \frac{Q}{\sin 75^\circ}$$

$$P = \frac{4 \sin 45^\circ}{\sin 60^\circ} = \frac{4\sqrt{6}}{3} \text{ N}$$

$$Q = \frac{4 \sin 75^\circ}{\sin 60^\circ} = \frac{2(\sqrt{6} + 3\sqrt{2})}{3} \text{ N}$$

Use the sine rule.

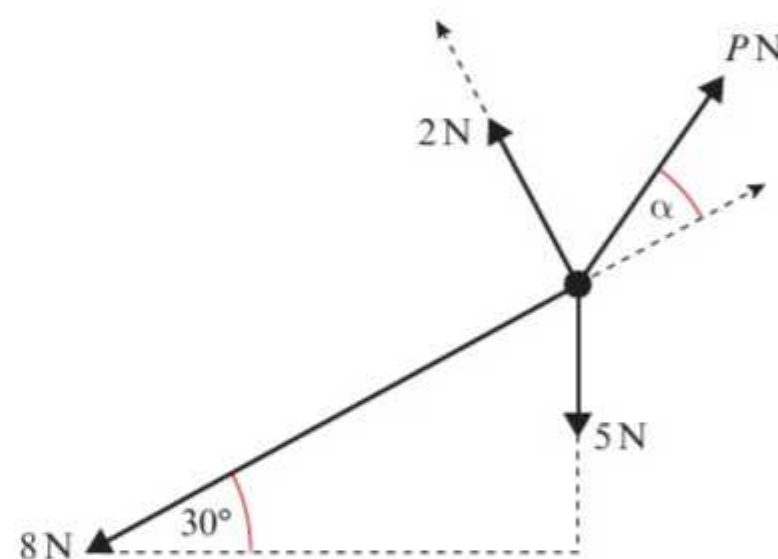
← Pure Maths 1 Section 6.2

Watch out This method works only for a particle in equilibrium. If the resultant force is not zero, the vector diagram will not be a closed triangle.

Example 2**SKILLS** PROBLEM-SOLVING

The diagram shows a particle in equilibrium on an inclined plane under the forces shown.

Find the magnitude of the force P and the size of the angle α .



$$\begin{aligned} R(\nearrow), \quad P \cos \alpha - 8 - 5 \sin 30^\circ &= 0 \\ \therefore P \cos \alpha &= 8 + 5 \sin 30^\circ \quad (1) \end{aligned}$$

$$\begin{aligned} R(\nwarrow), \quad P \sin \alpha + 2 - 5 \cos 30^\circ &= 0 \\ \therefore P \sin \alpha &= 5 \cos 30^\circ - 2 \quad (2) \end{aligned}$$

Divide equation (2) by equation (1) to give:

$$\tan \alpha = \frac{5 \cos 30^\circ - 2}{8 + 5 \sin 30^\circ} = \frac{2.330}{10.5} = 0.222$$

$$\therefore \alpha = 12.5^\circ \text{ (3 s.f.)}$$

Substitute into equation (1):

$$P \sin 12.5^\circ = 5 \cos 30^\circ - 2$$

$$\therefore P = 10.8 \text{ (3 s.f.)}$$

Resolve parallel to the plane. Take the direction up the plane as positive.

Rearrange the equation to make $P \cos \alpha$ the subject.

Resolve perpendicular to the plane. Rearrange the second equation to make $P \sin \alpha$ the subject.

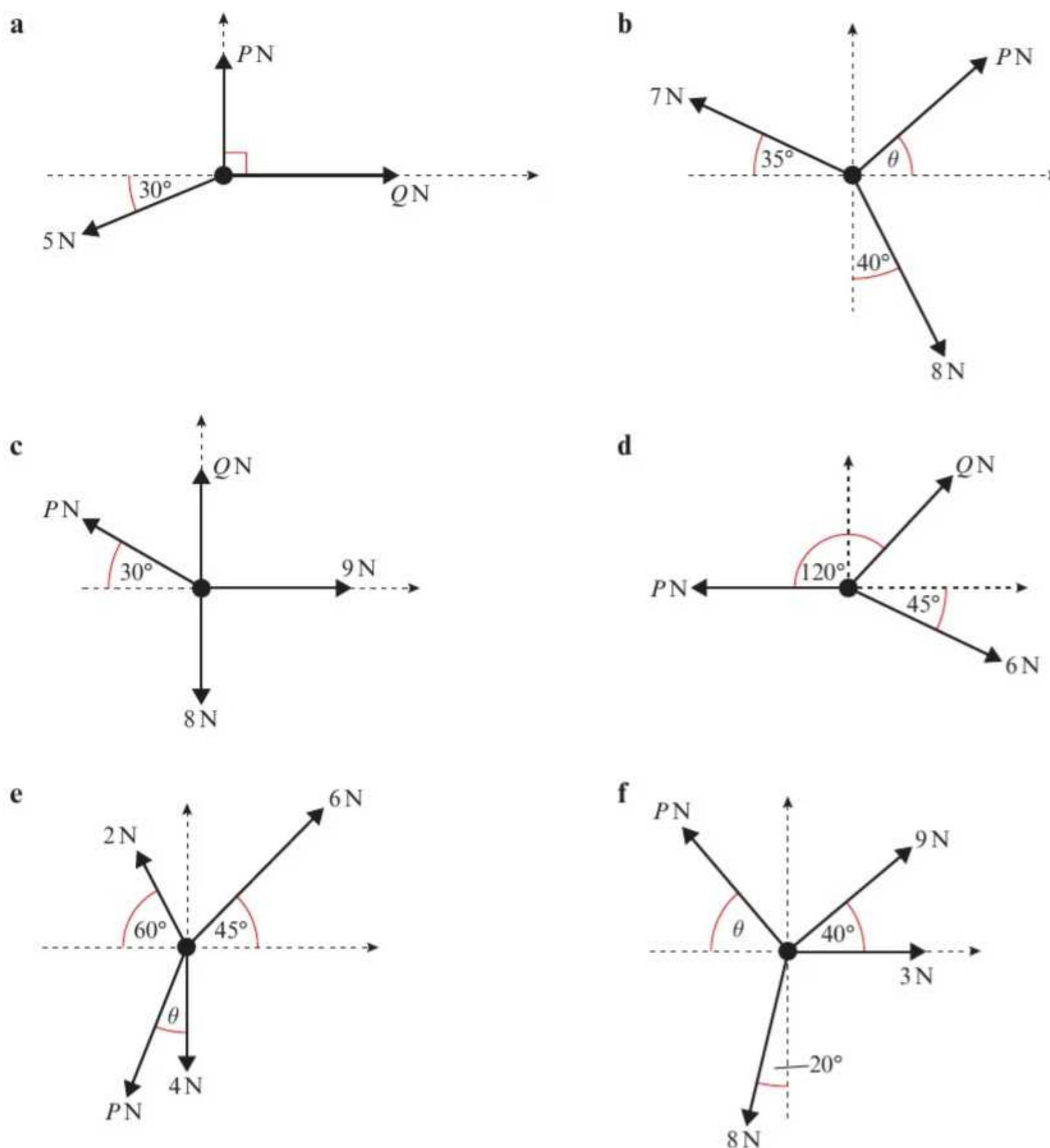
After division use $\frac{\sin \alpha}{\cos \alpha} \equiv \tan \alpha$.

Use $\tan^{-1}$ and give your answer to three significant figures.

You could check your answers by substituting into equation (2).

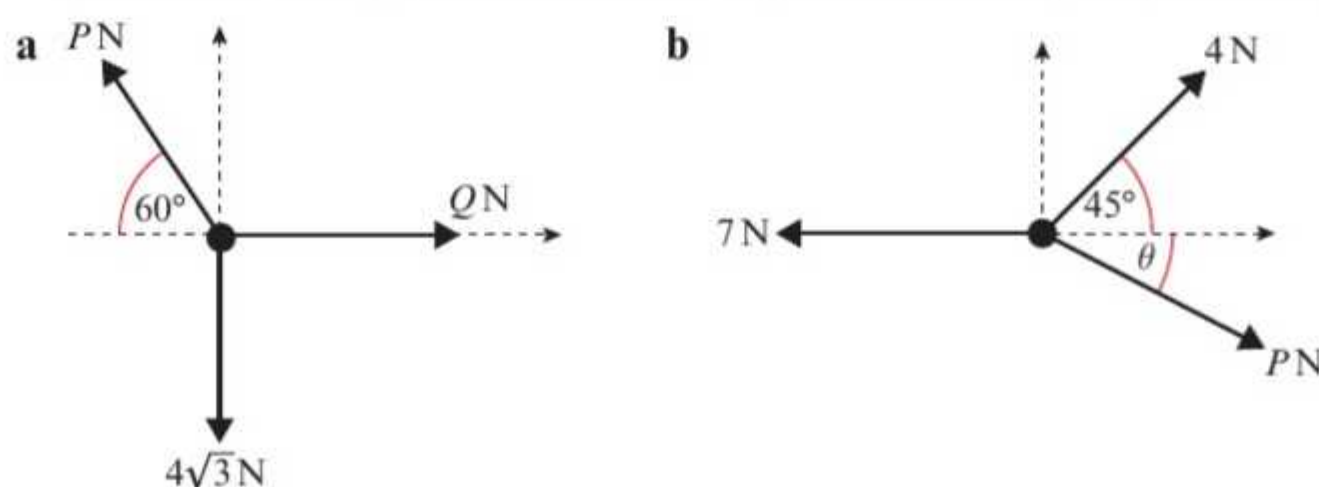
Exercise 7A**SKILLS** PROBLEM-SOLVING

- 1 Each of the following diagrams shows a particle in static equilibrium. For each particle:
 - i resolve the components in the x -direction
 - ii resolve the components in the y -direction
 - iii find the magnitude of any unknown forces (marked P and Q) and the size of any unknown angles (marked θ).

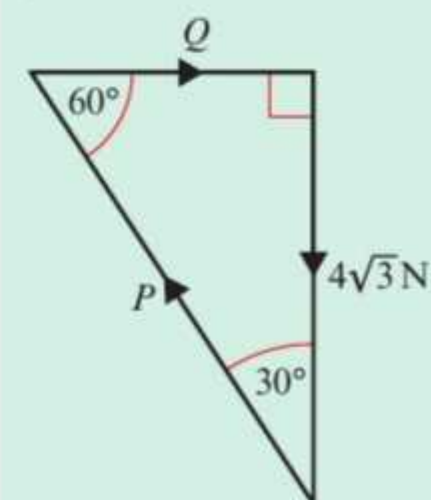


2 For each of the following particles in static equilibrium:

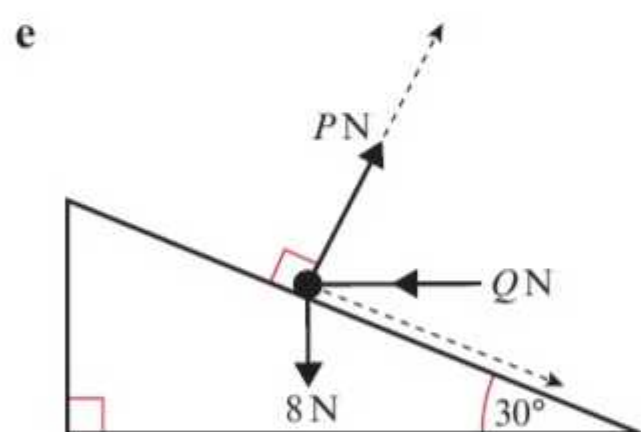
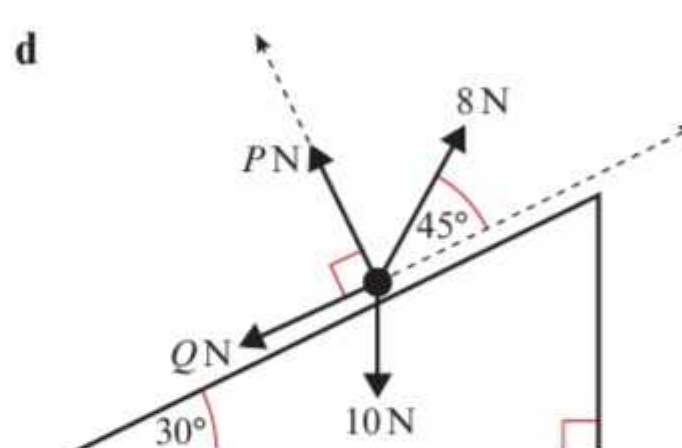
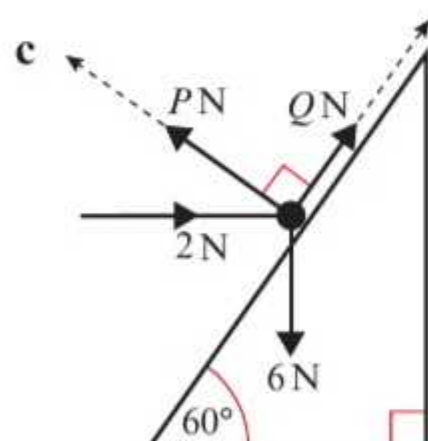
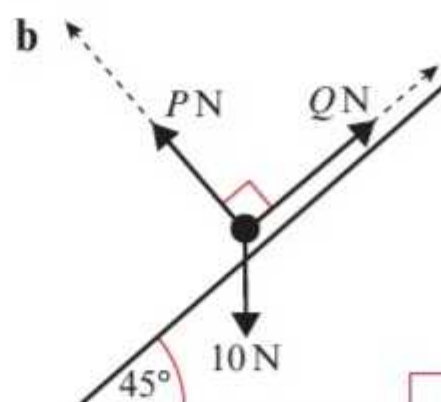
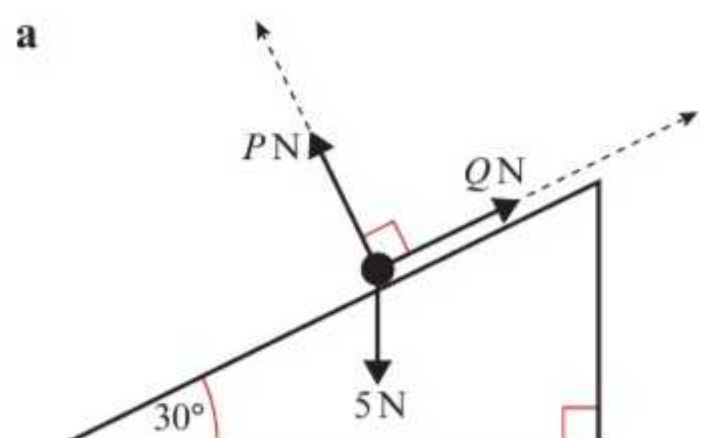
- draw a triangle of forces diagram
- use trigonometry to find the magnitude of any unknown forces (marked P and Q) and the size of any unknown angles (marked θ).



Hint The triangle of forces diagram for part **a** is:

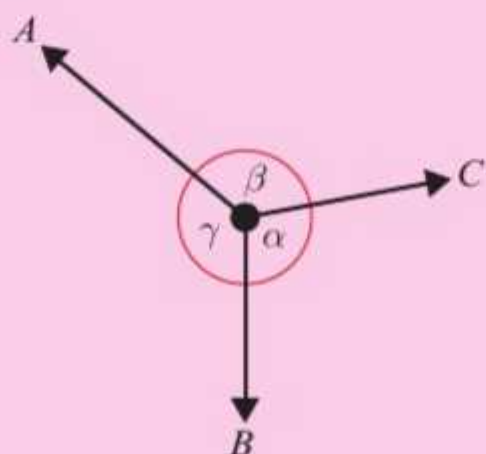


- 3 Each of these particles rests in equilibrium on an inclined plane under the forces shown. In each case, find the magnitude of forces P and Q .



Challenge

The diagram shows three **coplanar** forces of A , B and C acting on a particle in equilibrium.



Show that $\frac{A}{\sin \alpha} = \frac{B}{\sin \beta} = \frac{C}{\sin \gamma}$.

Notation This result is known as **Lami's Theorem**.

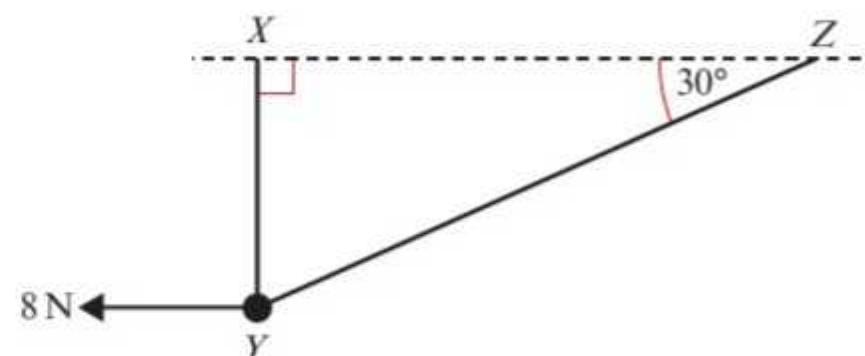
7.2 Modelling with statics

You can use force diagrams to model objects in static equilibrium, and to solve problems involving weight, tension and pulleys.

Example 3

SKILLS PROBLEM-SOLVING

A smooth bead Y is threaded on a light inextensible string. The ends of the string are attached to two fixed points, X and Z , on the same horizontal level. The bead is held in equilibrium by a horizontal force of magnitude 8 N acting parallel to ZX . The bead Y is vertically below X and $\angle XZY = 30^\circ$ as shown in the diagram.



Find the tension in the string and the weight of the bead.

$R(\rightarrow), \quad T \cos 30^\circ - 8 = 0$
 $\therefore T = \frac{8}{\cos 30^\circ}$
 $= \frac{16}{3}\sqrt{3} = 9.24 \text{ (3 s.f.)}$
 $R(\uparrow), \quad T + T \sin 30^\circ - W = 0$
 $\therefore W = T(1 + \sin 30^\circ)$
 $= \frac{16}{3}\sqrt{3} \left(1 + \frac{1}{2}\right)$
 $= 8\sqrt{3} = 13.9 \text{ (3 s.f.)}$

Draw a forces diagram.

Notation The bead is **smooth** so the tension in the string will be the same on either side of the bead.

This angle is 30° (alternate angles).

Let the weight be $W\text{ N}$.

Resolve horizontally and make T the subject of the formula.

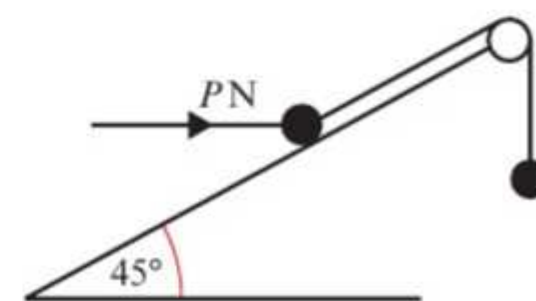
Give your answer to three significant figures, as an approximation for g has not been used.

Resolve vertically. Make W the subject of the formula and substitute for T .

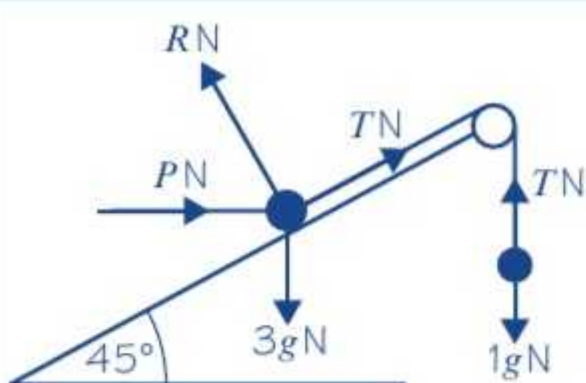
Example 4

SKILLS PROBLEM-SOLVING

A mass of 3 kg rests on the surface of a smooth plane which is inclined at an angle of 45° to the horizontal. The mass is attached to a cable which passes up the plane along the line of greatest slope and then passes over a smooth pulley at the top of the plane. The cable carries a mass of 1 kg freely suspended at the other end. The masses are modelled as particles, and the cable as a light inextensible string. There is a force of $P\text{ N}$ acting horizontally on the 3 kg mass and the system is in equilibrium.



- Calculate: **a** the magnitude of P **b** the normal reaction between the mass and the plane.
c State how you have used the assumption that the pulley is smooth in your calculations.



- a** Consider the 1 kg mass:

$$R(\uparrow), \quad T - 1g = 0$$

$$\therefore T = g = 9.8$$

Consider the 3 kg mass:

$$R(\nearrow), \quad T + P \cos 45^\circ - 3g \sin 45^\circ = 0$$

$$\therefore P \cos 45^\circ = 3g \sin 45^\circ - T$$

$$\text{But } T = g$$

$$\therefore P \cos 45^\circ = 3g \sin 45^\circ - g$$

$$P = 3g - \frac{g}{\cos 45^\circ}$$

$$= 3g - g\sqrt{2} = 16 \text{ (2 s.f.)}$$

$$\text{b } R(\searrow), \quad R - P \sin 45^\circ - 3g \cos 45^\circ = 0$$

$$\therefore R = P \sin 45^\circ + 3g \cos 45^\circ$$

$$= 6g \frac{\sqrt{2}}{2} - g = 32 \text{ (2 s.f.)}$$

- c** The pulley is smooth so the tension in the string will be the same on both sides of the pulley.

Draw a diagram showing the forces acting on each particle. The tension, T , will be the same throughout the string. The normal reaction, R , acts perpendicular to the plane. Show the weights $3g$ N and $1g$ N.

Resolve vertically to obtain T .

Resolve up the plane.

R has no component in this direction as R is perpendicular to the plane.

Substitute the value for T you found earlier.

Divide this equation by $\cos 45^\circ$ and use the fact that $\frac{\sin 45^\circ}{\cos 45^\circ} = \tan 45^\circ = 1$.

Use the result that $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}}$.

Resolve perpendicular to the plane.

Substitute the value of P which you have found to evaluate R .

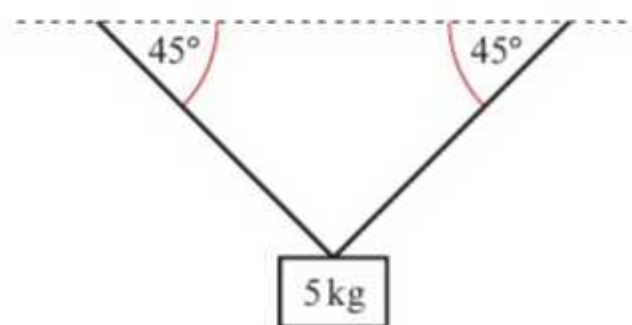
Exercise

7B

SKILLS

PROBLEM-SOLVING

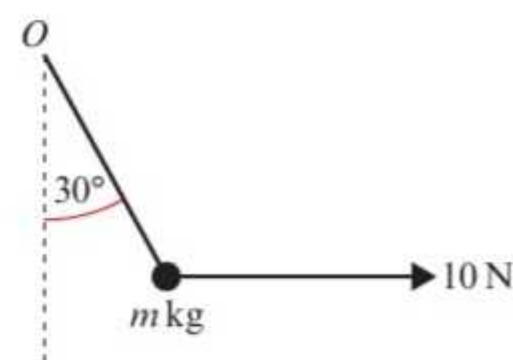
- P** 1 A picture of mass 5 kg is suspended by two light inextensible strings, each inclined at 45° to the horizontal as shown. By modelling the picture as a particle, find the tension in the strings when the system is in equilibrium.



Problem-solving

This is a three-force problem involving an object in static equilibrium, so you could use a triangle of forces.

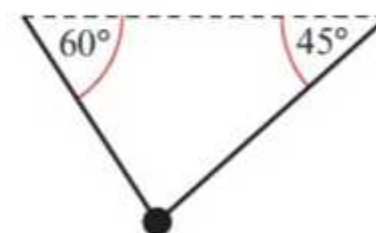
- 2 A particle of mass m kg is suspended by a single light inextensible string. The string is inclined at an angle of 30° to the vertical and the other end of the string is attached to a fixed point O . Equilibrium is maintained by a horizontal force of magnitude 10 N which acts on the particle, as shown in the diagram. Find:
- a the tension in the string b the value of m .



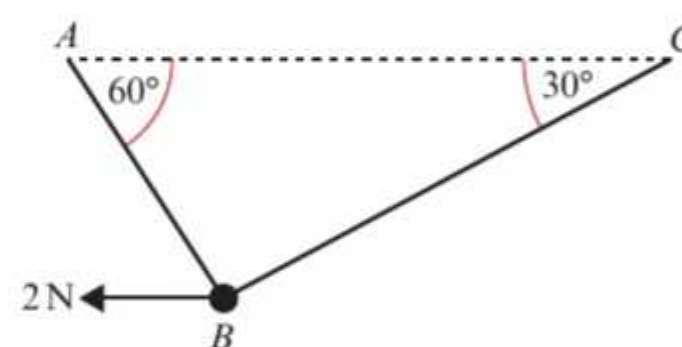
- 3 A particle of weight 12 N is suspended by a light inextensible string from a fixed point O . A horizontal force of 8 N is applied to the particle, and the particle remains in equilibrium with the string at an angle θ to the vertical. Find:
- a the angle θ b the tension in the string.

- 4 A particle of mass 6 kg hangs in equilibrium, suspended by two light inextensible strings, inclined at 60° and 45° to the horizontal, as shown. Find the tension in each of the strings.

Hint The particle is attached **separately** to each string, so the tension in the two strings can be different.



- (E)** 5 A smooth bead B is threaded on a light inextensible string. The ends of the string are attached to two fixed points, A and C , on the same horizontal level. The bead is held in equilibrium by a horizontal force of magnitude 2 N acting parallel to CA . The sections of string make angles of 60° and 30° with the horizontal. Find:
- a the tension in the string
b the mass of the bead.
c State how you have used the modelling assumption that the bead is smooth in your calculations.



(3 marks)

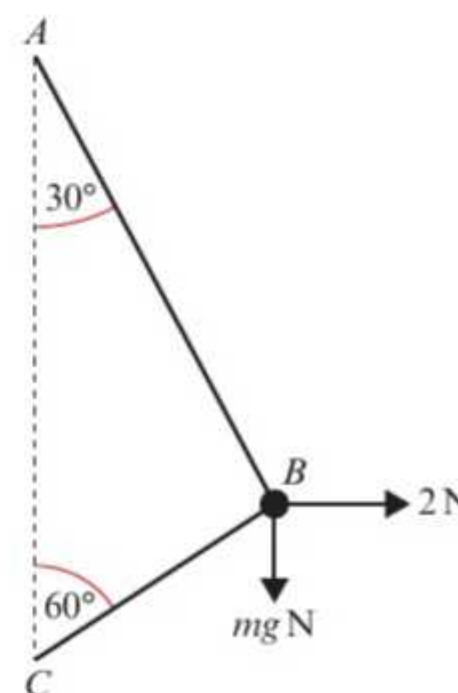
(4 marks)

(1 mark)

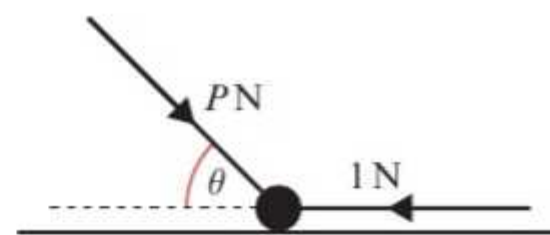
- (E)** 6 A smooth bead B is threaded on a light inextensible string. The ends of the string are attached to two fixed points A and C where A is vertically above C . The bead is held in equilibrium by a horizontal force of magnitude 2 N. The sections AB and BC of the string make angles of 30° and 60° with the vertical respectively. Find:
- a the tension in the string
b the mass of the bead, giving your answer to the nearest gram.

(3 marks)

(4 marks)



- (E)** 7 A particle of weight 2 N rests on a smooth horizontal surface and remains in equilibrium under the action of the two external forces shown in the diagram. One is a horizontal force of magnitude 1 N and the other is a force of magnitude P N which acts at an angle θ to the horizontal, where $\tan \theta = \frac{12}{5}$. Find:



a the value of P

(3 marks)

b the normal reaction between the particle and the surface.

(2 marks)

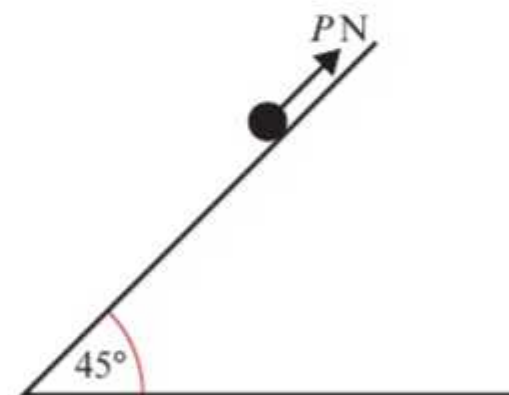
- (P)** 8 A particle A of mass m kg rests on a smooth horizontal table. The particle is attached by a light inextensible string to another particle B of mass $2m$ kg, which hangs over the edge of the table. The string passes over a smooth pulley, which is fixed at the edge of the table so that the string is horizontal between A and the pulley and then is vertical between the pulley and B . A horizontal force F N applied to A maintains equilibrium. The normal reaction between A and the table is R N.

a Find the values of F and R in terms of m .

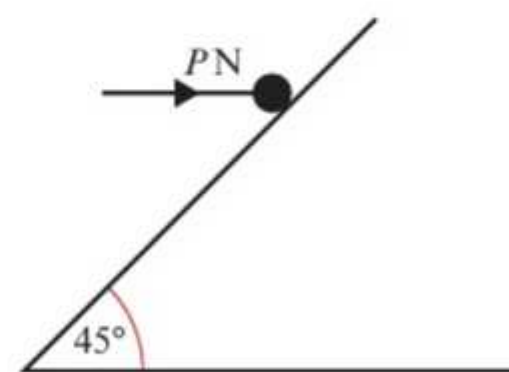
The pulley is now raised to a position above the edge of the table so that the string is inclined at 30° to the horizontal between A and the pulley. The string still hangs vertically between the pulley and B . A horizontal force F' N applied to A maintains equilibrium in this new situation. The normal reaction between A and the table is now R' N.

b Find, in terms of m , the values of F' and R' .

- 9 A particle of mass 2 kg rests on a smooth inclined plane, which makes an angle of 45° with the horizontal. The particle is maintained in equilibrium by a force P N acting up the line of greatest slope of the inclined plane, as shown in the diagram. Find the value of P .

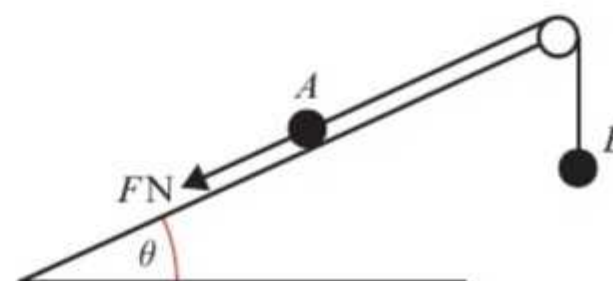


- 10 A particle of mass 4 kg is held in equilibrium on a smooth plane which is inclined at 45° to the horizontal by a horizontal force of magnitude P N, as shown in the diagram. Find the value of P .



- (E/P)** 11 A particle A of mass 2 kg rests in equilibrium on a smooth inclined plane. The plane makes an angle θ with the horizontal, where $\tan \theta = \frac{3}{4}$.

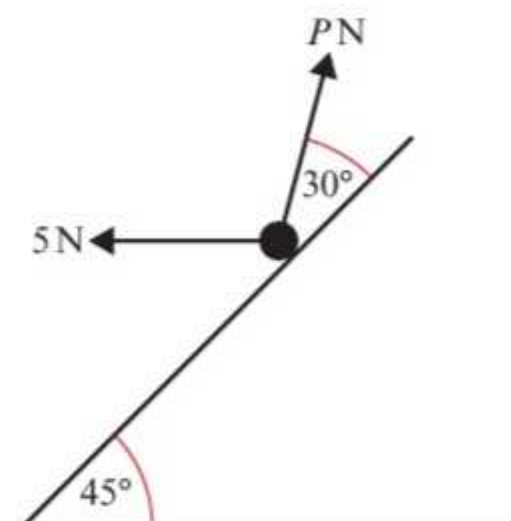
The particle is attached to one end of a light inextensible string which passes over a smooth pulley, as shown in the diagram. The other end of the string is attached to a particle B of mass 5 kg. Particle A is also acted upon by a force of magnitude F N down the plane, along the line of greatest slope.



Find:

- the magnitude of the normal reaction between A and the plane (5 marks)
- the value of F . (3 marks)
- State how you have used the fact that the pulley is smooth in your calculations. (1 mark)

- E/P** 12 A particle of weight 20 N rests in equilibrium on a smooth inclined plane. It is maintained in equilibrium by the application of two external forces as shown in the diagram. One of the forces is a horizontal force of 5 N, the other is a force P N acting at an angle of 30° to the plane, as shown in the diagram. Find the magnitude of the normal reaction between the particle and the plane. (8 marks)



7.3 Friction and static particles

When a body is in static equilibrium under the action of a number of forces, including friction, you need to consider whether or not the body is on the point of moving.

In many cases the force of friction will be less than μR , as a smaller force is sufficient to prevent motion and to maintain static equilibrium. In these situations the equilibrium is not limiting.

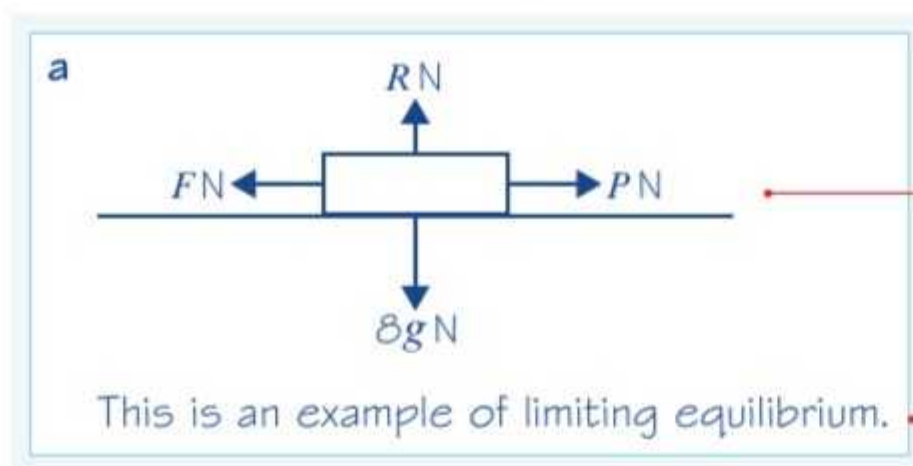
- The maximum value of the frictional force $F_{\text{MAX}} = \mu R$ is reached when the body you are considering is on the point of moving. The body is then said to be in limiting equilibrium.
- In general, the force of friction F is such that $F \leq \mu R$, and the direction of the frictional force is opposite to the direction in which the body would move if the frictional force were absent.

Example 5

SKILLS PROBLEM-SOLVING

A mass of 8 kg rests on a rough horizontal plane. The mass may be modelled as a particle, and the coefficient of friction between the mass and the plane is 0.5. Find the magnitude of the maximum force P N which acts on this mass without causing it to move if:

- the force P is horizontal
- the force P acts at an angle 60° above the horizontal.



Draw a diagram showing the weight $8g$ N, the normal reaction R N, the force P N and the friction F N. The friction is in the opposite direction to the force P N.

The question asks you for the maximum force before movement takes place.

$$R(\uparrow), \quad R - 8g = 0$$

$$\therefore R = 8g$$

As friction is limiting, $F = \mu R$

$$\therefore F = 0.5 \times 8g \\ = 39.2 \text{ N}$$

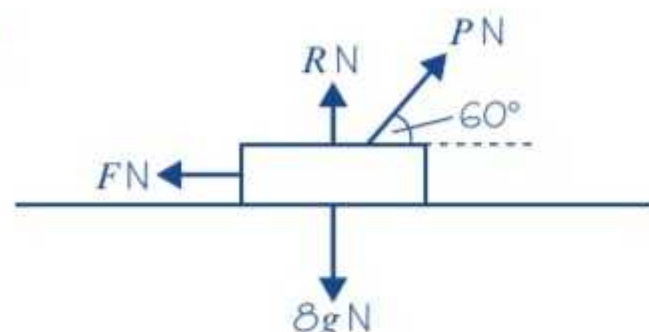
$$R(\rightarrow), \quad P - F = 0$$

$$\therefore P = F = 39 \text{ N (2 s.f.)}$$

For an object in limiting equilibrium, $F = F_{\text{MAX}}$.

Give your answer to two significant figures.

b



Draw another diagram showing P at 60° above the horizontal.

Again this is limiting equilibrium.

$$R(\uparrow), \quad R + P \sin 60^\circ - 8g = 0$$

$$\therefore R = 8g - P \sin 60^\circ$$

As friction is limiting, $F = \mu R$

$$\therefore F = 0.5 (8g - P \sin 60^\circ)$$

$$R(\rightarrow), \quad P \cos 60^\circ - F = 0$$

$$\therefore P \cos 60^\circ = 0.5 (8g - P \sin 60^\circ)$$

$$\therefore P \cos 60^\circ + 0.5 P \sin 60^\circ = 0.5 \times 8g$$

$$\therefore P (\cos 60^\circ + 0.5 \sin 60^\circ) = 4g$$

$$\therefore P = \frac{4g}{\cos 60^\circ + 0.5 \sin 60^\circ}$$

$$P = 42.0 \text{ N (3 s.f.)}$$

Express R in terms of P .

Use $F = \mu R$ with $\mu = 0.5$.

As $F = P \cos 60^\circ$ eliminate F from the previous equation.

Collect the terms in P and factorise to make P the subject.

Example**6****SKILLS****PROBLEM-SOLVING**

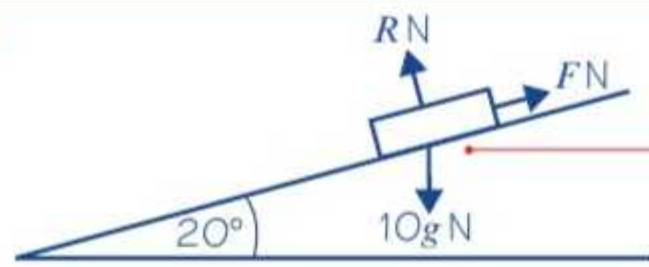
A box of mass 10 kg rests in limiting equilibrium on a rough plane inclined at 20° above the horizontal.

a Find the coefficient of friction between the box and the plane.

A horizontal force of magnitude P N is applied to the box.

b Given that the box remains in equilibrium, find the maximum possible value of P .

a



Model the box as a particle and draw a diagram showing the weight, the normal reaction and the force of friction.

The friction acts up the plane, as it acts in an opposite direction to the motion that would take place if there was no friction.

$$R(\uparrow), \quad R - 10g \cos 20^\circ = 0$$

$$\therefore R = 92.089 \dots \text{N}$$

$$R(\rightarrow), \quad F - 10g \sin 20^\circ = 0$$

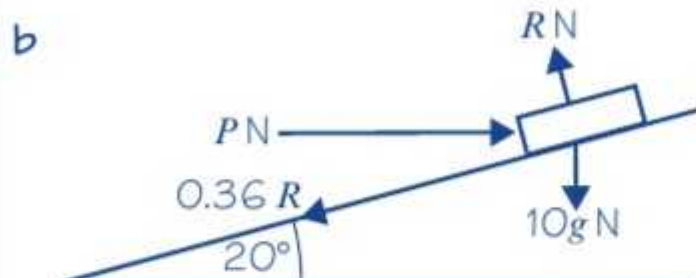
$$\therefore F = 33.517 \dots \text{N}$$

As the friction is limiting,

$$F = \mu R$$

$$\therefore 33.517 \dots = \mu \times 92.089 \dots$$

$$\therefore \mu = \frac{33.517 \dots}{92.089 \dots} = 0.36 \text{ (to 2 s.f.)}$$



When P is at its maximum value,

$$F = \mu R = 0.36R$$

$$R(\rightarrow), \quad P \cos 20^\circ - 0.36R - 10g \sin 20^\circ = 0$$

(1)

$$R(\uparrow), \quad R - 10g \cos 20^\circ - P \sin 20^\circ = 0 \quad (2)$$

$$\text{From (1): } R = \frac{P \cos 20^\circ - 10g \sin 20^\circ}{0.36}$$

$$\text{From (2) } R = P \sin 20^\circ + 10g \cos 20^\circ$$

$$\text{So } \frac{P \cos 20^\circ - 10g \sin 20^\circ}{0.36}$$

$$= P \sin 20^\circ + 10g \cos 20^\circ$$

$$P = \frac{3.6g \cos 20^\circ + 10g \sin 20^\circ}{\cos 20^\circ - 0.36 \sin 20^\circ}$$

$$= 82 \text{ N (2 s.f.)}$$

Resolve perpendicular and parallel to the plane.

Online Use the **STO** function to store exact values on your calculator.



Find R and F , then use $F = \mu R$ to find μ .

Give your answer to two significant figures and note that $\mu = \tan 20^\circ$.

Watch out For the maximum possible value of P , the box will be on the point of moving up the slope, so the friction will act down the slope.

Eliminate R to find P .

You have used $g = 9.8 \text{ m s}^{-2}$ in your calculations, so round your final answer to 2 significant figures.

Exercise

7C

SKILLS

PROBLEM-SOLVING

- 1 A book of mass 2 kg rests on a rough horizontal table. When a force of magnitude 8 N acts on the book, at an angle of 20° to the horizontal in an upward direction, the book is on the point of slipping.

Calculate, to three significant figures, the value of the coefficient of friction between the book and the table.

- 2 A block of mass 4 kg rests on a rough horizontal table. When a force of 6 N acts on the block, at an angle of 30° to the horizontal in a downward direction, the block is on the point of slipping. Find the value of the coefficient of friction between the block and the table.

Hint

'On the point of slipping' means that the book is in limiting equilibrium.

8 MOMENTS

6.1

Learning objectives

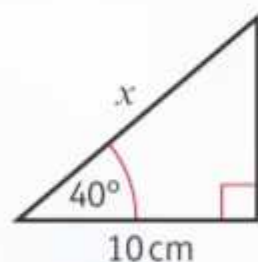
After completing this chapter you should be able to:

- Calculate the turning effect of a force applied to a rigid body → pages 130–131
- Calculate the resultant moment of a set of forces acting on a rigid body → pages 132–133
- Solve problems involving uniform rods in equilibrium → pages 133–136
- Solve problems involving non-uniform rods → pages 136–139
- Solve problems involving rods on the point of tilting → pages 139–141

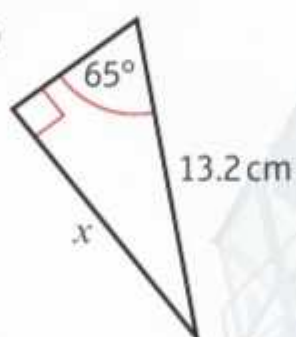
Prior knowledge check

1 Find the value of x in each of the following:

a

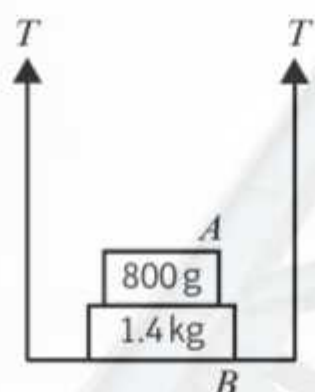


b



← International GCSE Mathematics

2 Masses A and B rest on a light scale-pan supported by two strings, each with tension T .



Find:

- a the value of T
- b the normal reaction of the scale-pan on mass B
- c the normal reaction of mass B on mass A .

← Mechanics 1 Section 4.5

Moments measure the turning effect of a force. Engineers use moments to work out how much load can be applied safely to a crane.

8.1 Moments

So far you have looked mostly at situations involving particles. This means you can ignore rotational effects. In this chapter, you begin to model objects as **rigid bodies**. This allows you to consider the size of the object as well as where forces are applied.

The **moment** of a force measures the turning effect of the force on a rigid body.

It is the product of the magnitude of the force and the perpendicular distance from the axis of rotation.

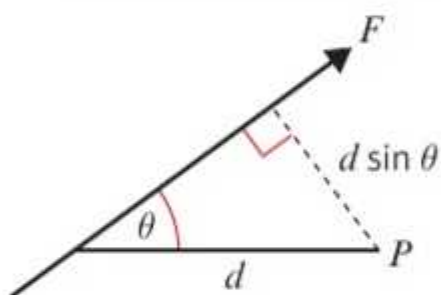
- Clockwise moment of F about $P = |F| \times d$



Watch out When you describe a moment, you need to give the direction of rotation.

In the diagram above, the distance given is perpendicular to the line of action of the force. When this is not the case, you need to use trigonometry to find the perpendicular distance.

- Clockwise moment of F about $P = |F| \times d \sin \theta$



Notation A moment is a force multiplied by a distance, so its units are **newton metres (Nm)** or **newton centimetres (Ncm)**.

Online Explore the moment of a force acting about a point using GeoGebra.



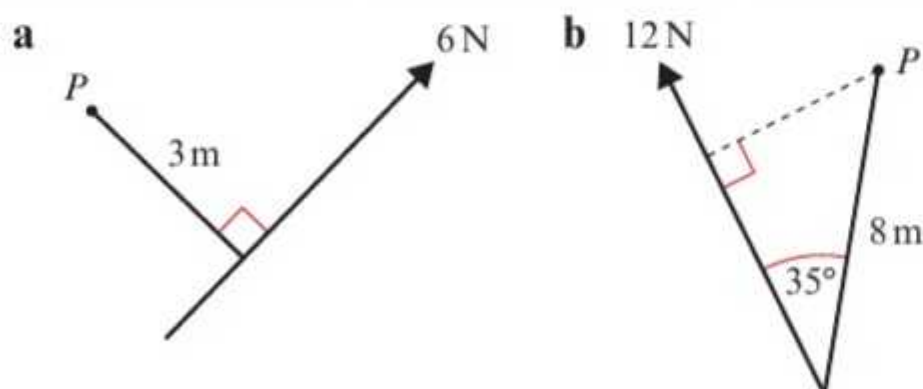
Example 1

1

SKILLS

PROBLEM-SOLVING

Find the moment of each force about the point P .



- a** Moment of the 6N force about P
 = magnitude of force $\times$ perpendicular distance
 = $6 \times 3 = 18 \text{ Nm}$ anticlockwise
- b** Moment of the 12N force about P
 = magnitude of force $\times$ perpendicular distance
 = $12 \times 8 \sin 35^\circ = 55.1 \text{ Nm}$ clockwise (3 s.f.)

The distance given on the diagram is the perpendicular distance, so you can substitute the given values directly into the formula.

Don't forget to include the direction of the rotation when you describe the moment of the force.

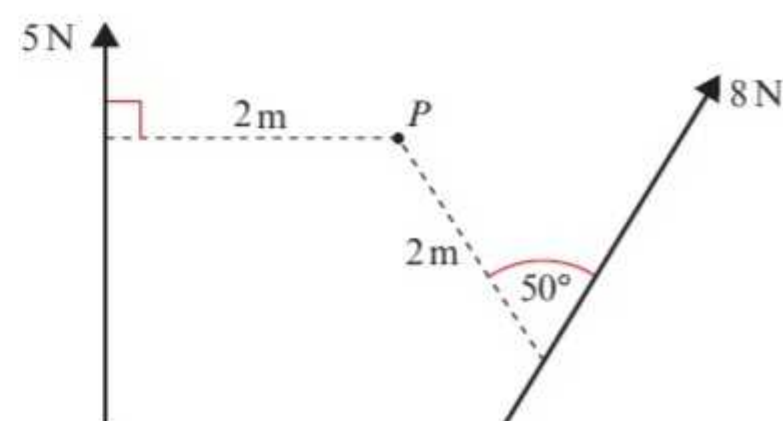
This time you need to use the perpendicular distance $8 \sin 35^\circ$.

Example 2**SKILLS** PROBLEM-SOLVING

The diagram shows two forces acting on a lamina. Find the moment of each of the forces about the point P .

Moment of the 5 N force
 = magnitude of force $\times$ perpendicular distance
 = $5 \times 2 = 10 \text{ Nm}$ clockwise

Moment of the 8 N force
 = magnitude of force $\times$ perpendicular distance
 = $8 \times 2 \sin 50^\circ = 12.3 \text{ Nm}$ anticlockwise (3 s.f.)

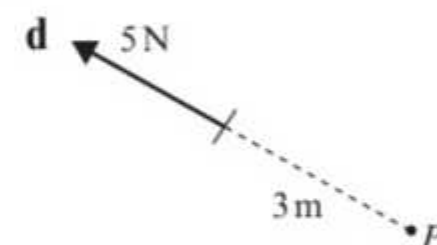
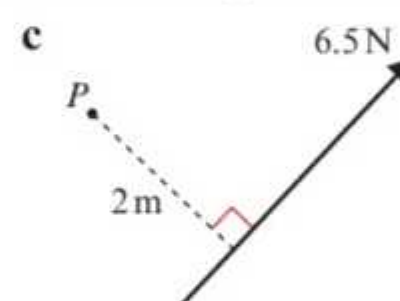
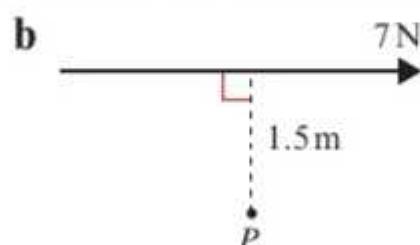
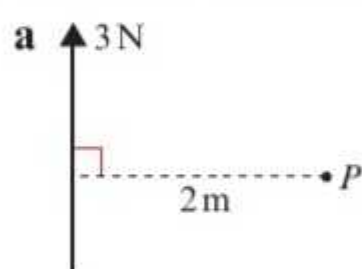


Notation A lamina is a 2D object whose thickness can be ignored.

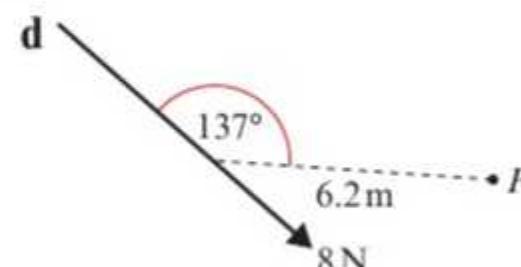
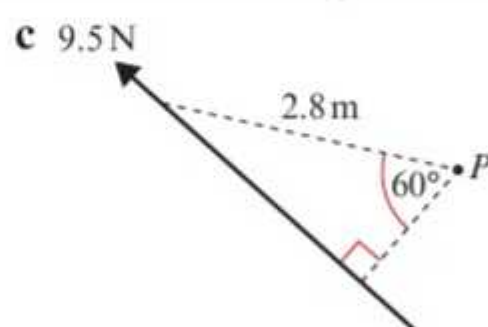
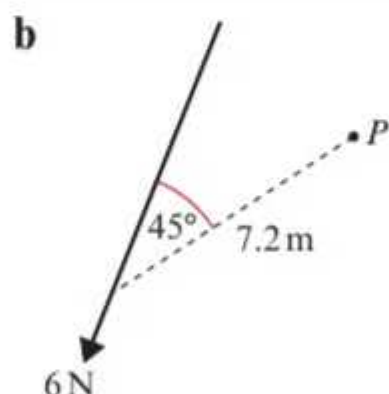
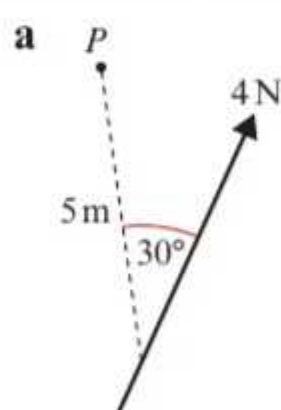
The moments act in opposite directions.

Exercise 8A**SKILLS** PROBLEM-SOLVING

1 Calculate the moment about P of each of these forces acting on a lamina.



2 Calculate the moment about P of each of these forces acting on a lamina.

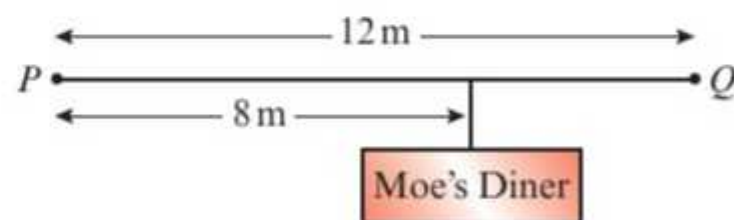


3 The diagram shows a sign hanging from a wooden beam. The sign has a mass of 4 kg.

a Calculate the moment of the weight of the mass:

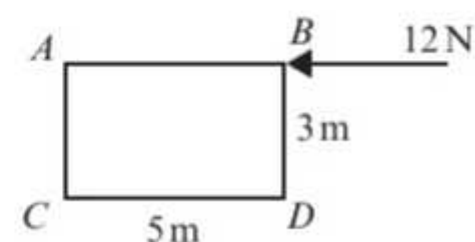
i about P **ii** about Q .

b Comment on any modelling assumptions you have made.

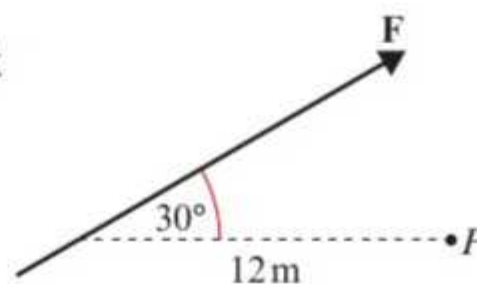


(P) 4 $ABCD$ is a rectangular lamina. A force of 12 N acts horizontally at B , as shown in the diagram. Find the moment of this force about:

a A **b** B **c** C **d** D



(P) 5 In the diagram, the force F produces a moment of 15 Nm clockwise about the pivot P . Calculate the magnitude of F .



8.2 Resultant moments

When you have several **coplanar** forces acting on a body, you can determine the turning effect around a given point by choosing a positive direction (clockwise or anticlockwise) and then finding the sum of the moments produced by each force.

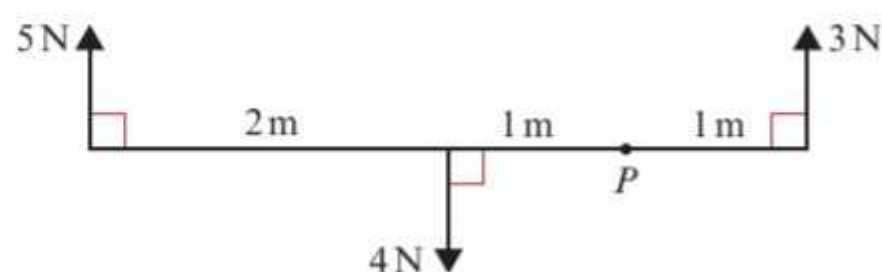
Notation **Coplanar forces**
are forces that act in the same plane.

- The sum of the moments acting on a body is called the resultant moment.

Example 3

SKILLS PROBLEM-SOLVING

The diagram shows a set of forces acting on a light rod. Calculate the resultant moment about the point P .



The moment of the 5 N force is $5 \times (2 + 1) = 15 \text{ N m}$ clockwise.

The moment of the 4 N force is $4 \times 1 = 4 \text{ N m}$ anticlockwise.

The moment of the 3 N force is $3 \times 1 = 3 \text{ N m}$ anticlockwise.

Choosing clockwise as positive:

Resultant moment $= 15 + (-4) + (-3) = 8 \text{ N m}$
 $\therefore$ resultant moment is 8 N m clockwise.

Your positive direction is clockwise, so the anticlockwise moments are negative.

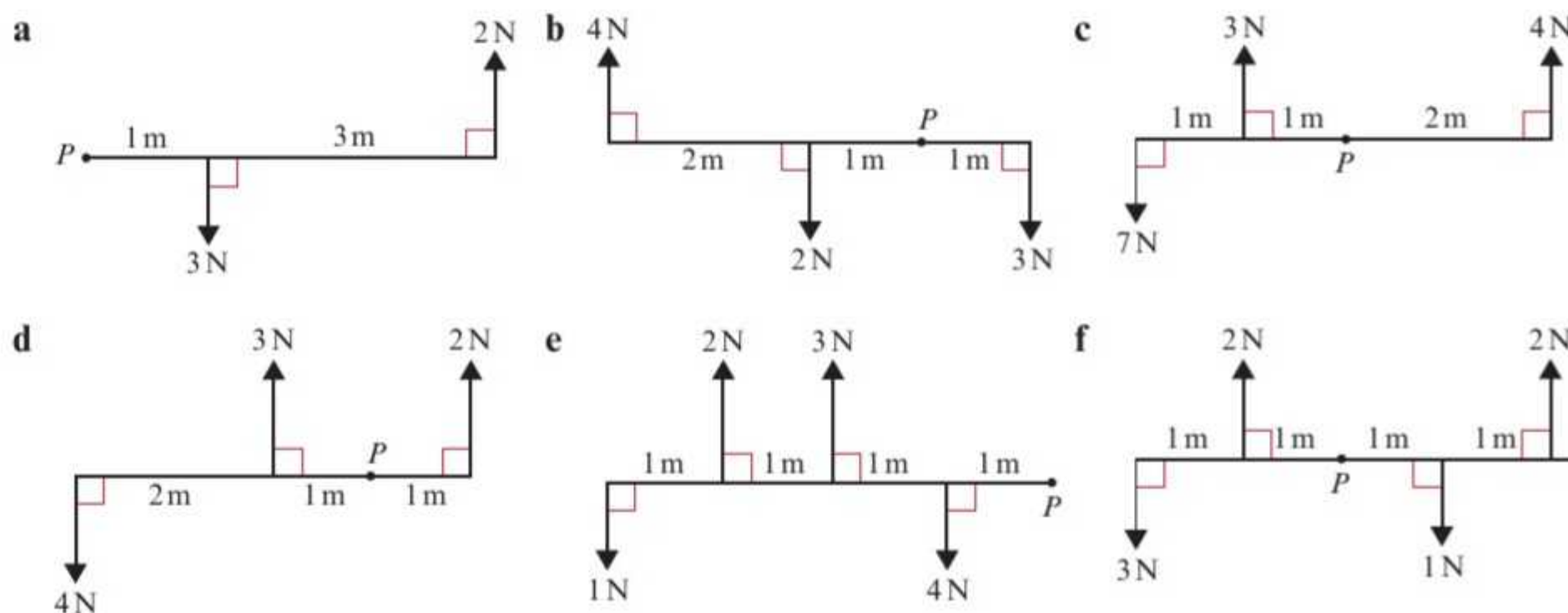
Problem-solving

You could also solve this problem by considering the clockwise and anticlockwise moments separately.
 Sum of clockwise moments $= 15 \text{ N m}$
 Sum of anticlockwise moments $= 3 + 4 = 7 \text{ N m}$
 Resultant moment $= 15 - 7 = 8 \text{ N m}$ clockwise

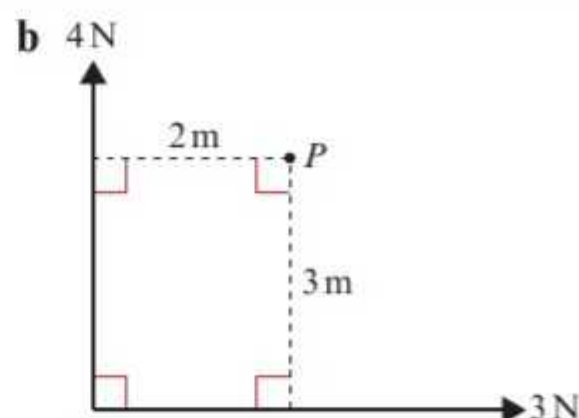
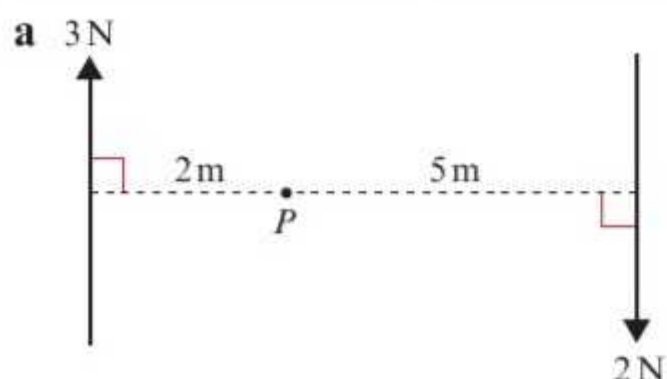
Exercise 8B

SKILLS PROBLEM-SOLVING

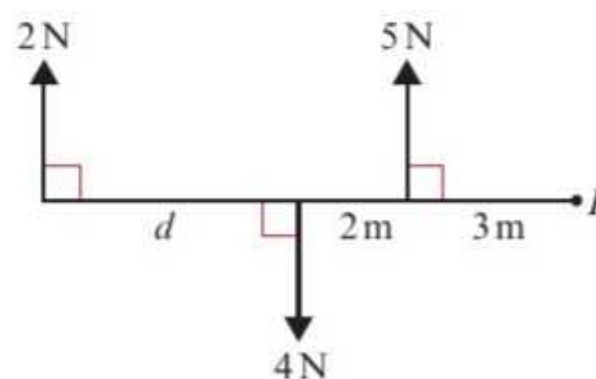
- 1 These diagrams show sets of forces acting on a light rod. In each case, calculate the resultant moment about P .



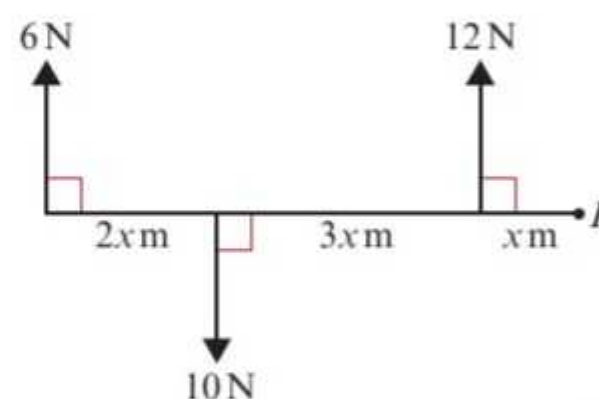
- 2 These diagrams show forces acting on a lamina. In each case, find the resultant moment about P .



- (P)** 3 The diagram shows a set of forces acting on a light rod. The resultant moment about P is 17 N m clockwise. Find the length d .



- (E/P)** 4 The diagram shows a set of forces acting on a light rod. The resultant moment about P is 12.8 N m clockwise. Find the value of x .



(3 marks)

8.3 Equilibrium

- When a rigid body is in equilibrium, the resultant force in any direction is 0 N and the resultant moment about any point is 0 N m .

You can simplify many problems involving rigid bodies by choosing which point(s) to take moments about. When you take moments at a given point, you can ignore the rotational effect of any forces acting at that point.

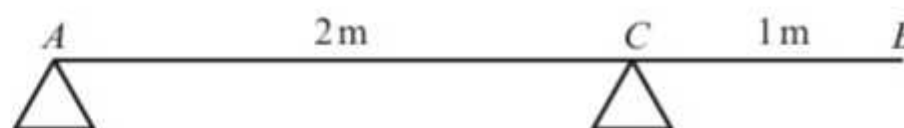
Hint If the resultant moment is zero then the sum of the clockwise moments equals the sum of the anticlockwise moments.

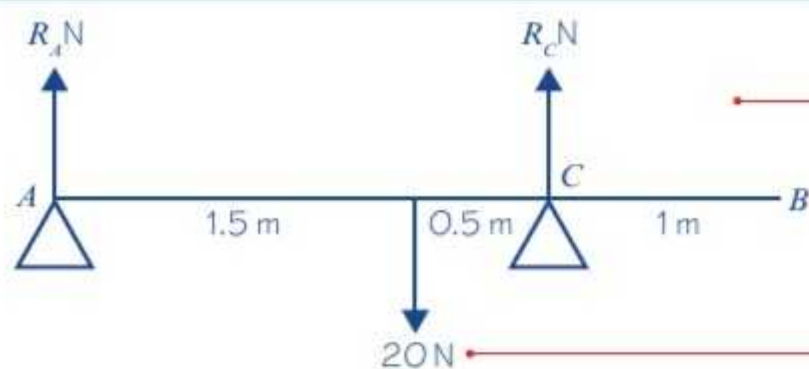
Example 4

SKILLS PROBLEM-SOLVING

The diagram shows a uniform rod AB , of length 3 m and weight 20 N , resting horizontally on supports at A and C , where $AC = 2 \text{ m}$.

Calculate the magnitude of the reaction at each of the supports.





The rod is in equilibrium.

Resolving vertically: $R_A + R_C = 20$

Considering the moments about point A:

$$20 \times 1.5 = R_C \times (1.5 + 0.5)$$

$$30 = 2R_C$$

$$15 = R_C$$

$$R_A + 15 = 20$$

$$R_A = 5$$

Therefore the reaction at A is 5 N and the reaction at C is 15 N.

Draw a diagram showing all the forces acting.

The weight of the rod acts at its centre of mass. You are told that this is a uniform rod, so the weight acts at the midpoint of the rod.

Total of forces acting upward = total of forces acting downward.

Clockwise moment = anticlockwise moment

Problem-solving

Take moments about the point that makes the algebra as simple as possible. Taking moments about A results in an equation with just one unknown.

Substituting the value of R_C into the first equation.

Example 5

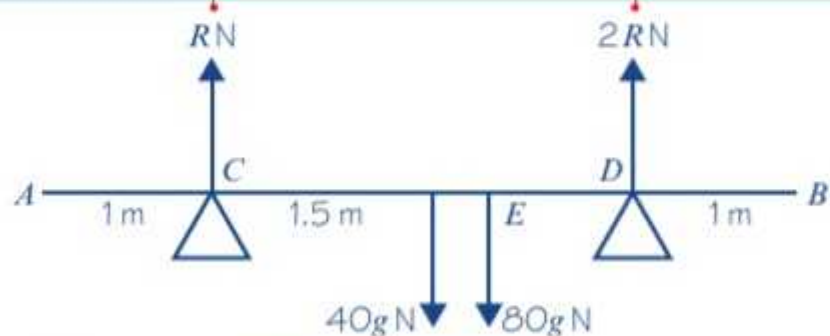
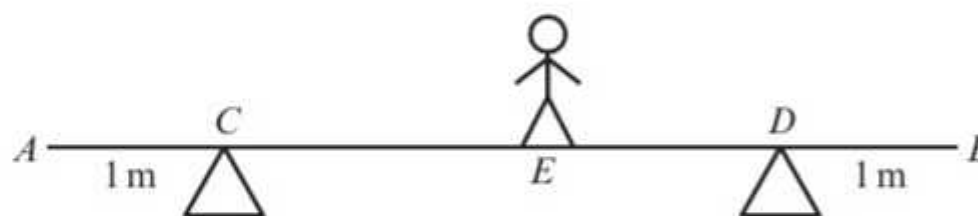
SKILLS PROBLEM-SOLVING

A uniform beam AB, of mass 40 kg and length 5 m, rests horizontally on supports at C and D, where $AC = DB = 1$ m.

When a man of mass 80 kg stands

on the beam at E the magnitude of the reaction at D is twice the magnitude of the reaction at C.

By modelling the beam as a rod and the man as a particle, find the distance AE.



Resolving vertically:

$$R + 2R = 40g + 80g$$

$$3R = 120g$$

$$R = 40g$$

Let the distance AE be x m.

Taking moments about A:

$$40g \times 2.5 + 80g \times x = 40g \times 1 + 80g \times 4$$

$$100g + 80g \times x = 360g$$

$$80g \times x = 260g$$

$$\Rightarrow x = \frac{260g}{80g} = \frac{26}{8} = 3.25$$

Distance AE = 3.25 m

Draw a diagram showing the forces.

Because you are told a relationship between the reaction at C and the reaction at D you can use this on your diagram.

The rod is in equilibrium so there is no resultant force.

Clockwise moment = anticlockwise moment

Problem-solving

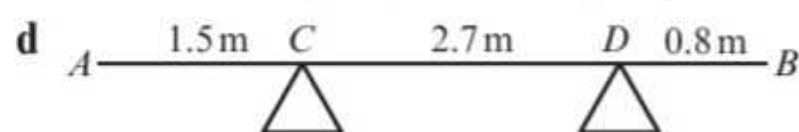
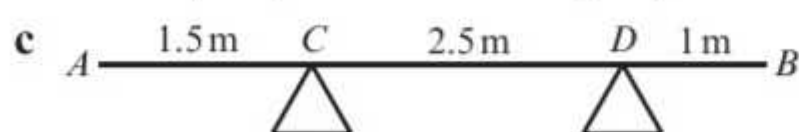
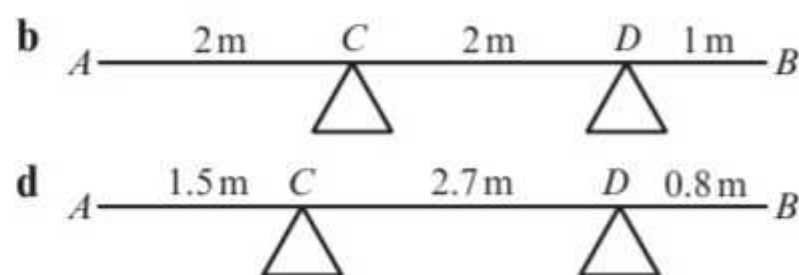
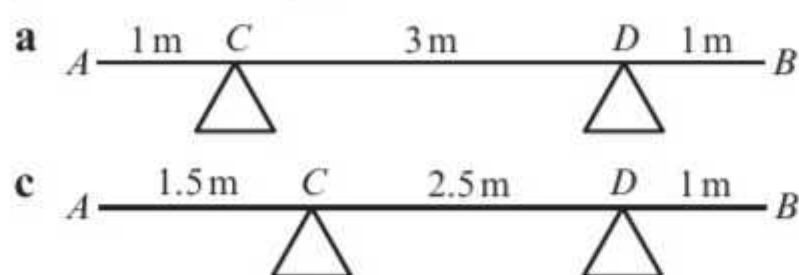
How have you used the modelling assumptions in the question?

- Since the beam is a rod, it does not bend.
- Since the man is a particle, his weight acts at the point E.

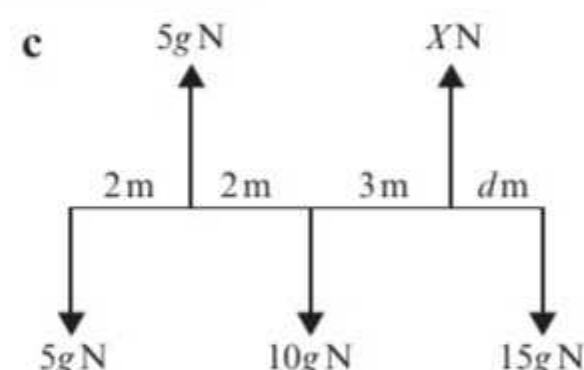
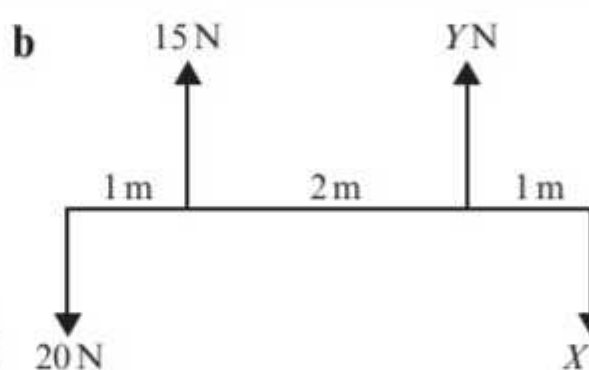
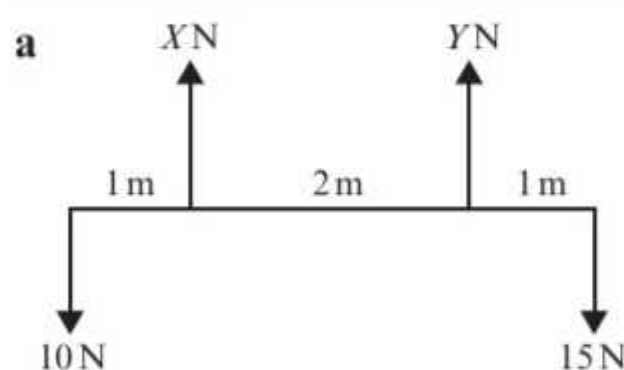
Exercise 8C

SKILLS PROBLEM-SOLVING

- 1 AB is a uniform rod of length 5 m and weight 20 N. In these diagrams AB is resting in a horizontal position on supports at C and D . In each case, find the magnitudes of the reactions at C and D .



- 2 Each of these diagrams shows a light rod in equilibrium in a horizontal position under the action of a set of forces. Find the values of the unknown forces and distances.

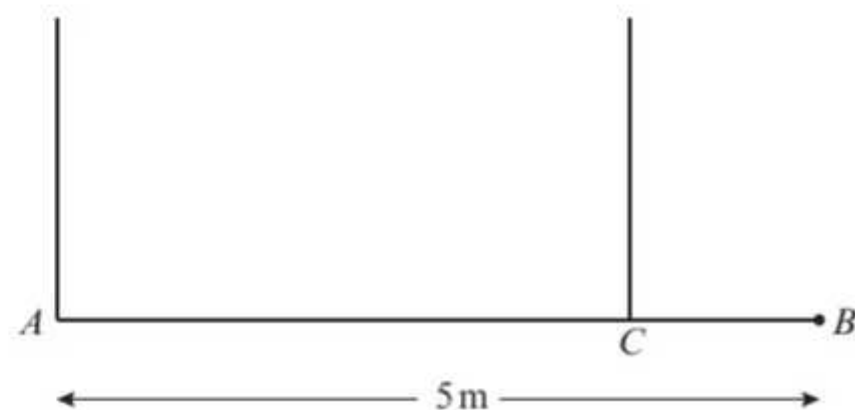


- 3 Jack and Jill are playing on a seesaw made from a uniform plank AB , of length 5 m pivoted at M , the midpoint of AB . Jack has mass 35 kg and Jill has mass 28 kg. Jill sits at A and Jack sits at a distance x m from B . The plank is in equilibrium. Find the value of x .
- 4 A uniform rod AB , of length 3 m and mass 12 kg, is pivoted at C , where $AC = 1$ m. A vertical force F applied at A maintains the rod in horizontal equilibrium. Calculate the magnitude of F .
- 5 A broom consists of a broomstick of length 130 cm and mass 5 kg, and a broomhead of mass 5.5 kg attached at one end. By modelling the broomstick as a uniform rod and the broomhead as a particle, find where a support should be placed so that the broom will balance horizontally.

- (P)** 6 A uniform rod AB , of length 4 m and weight 20 N, is suspended horizontally by two vertical strings attached at A and at B . A particle of weight 10 N is attached to the rod at point C , where $AC = 1.5$ m.
- a** Find the magnitudes of the tensions in the two strings.
- The particle is moved so that the rod remains in horizontal equilibrium with the tension in the string at B 1.5 times the tension in the string at A .
- b** Find the new distance of the particle from A .

- (E/P)** 7 A uniform beam AB , of length 5 m and mass 60 kg, has a load of 40 kg attached at B . It is then held horizontally in equilibrium by two vertical wires attached at A and C . The tension in the wire at C is four times the tension in the wire at A . By modelling the beam as a uniform rod and the load as a particle, calculate:

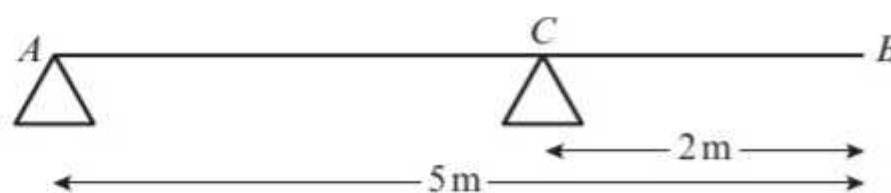
- a** the tension in the wire at C
- b** the distance CB .



(2 marks)

(5 marks)

- E** 8 A uniform plank AB has length 5 m and mass 15 kg. The plank is held in equilibrium horizontally by two smooth supports A and C as shown in the diagram, where $BC = 2$ m.



a Find the reaction on the plank at C .

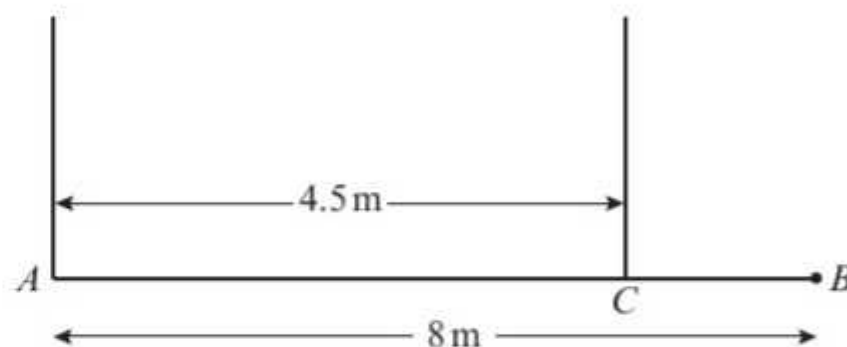
(3 marks)

A person of mass 45 kg stands on the plank at the point D and it remains in equilibrium. The reactions on the plank at A and C are now equal.

b Find the distance AD .

(7 marks)

- E/P** 9 A uniform beam AB has weight W N and length 8 m. The beam is held in a horizontal position in equilibrium by two vertical light inextensible wires attached to the beam at the points A and C where $AC = 4.5$ m, as shown in the diagram. A particle of weight 30 N is attached to the beam at B .



a Show that the tension in the wire attached to the beam at C is $(\frac{8}{9}W + \frac{160}{3})$ N.

(4 marks)

b Find, in terms of W , the tension in the wire attached to the beam at A .

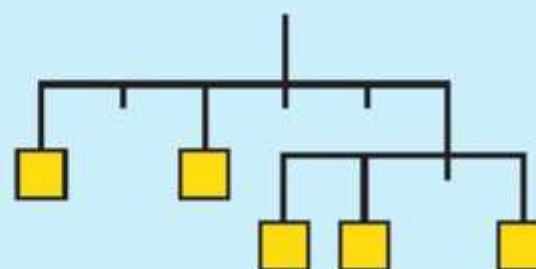
(3 marks)

c Given that the tension in the wire attached to the beam at C is twelve times the tension in the wire attached to the beam at A , find the value of W .

(3 marks)

Challenge

The diagram shows a kinetic sculpture made from hanging rods. The distances between the points marked on each rod are equal. Arrange 1 kg, 2 kg, 3 kg, 4 kg and 5 kg weights onto the marked squares, using each weight once, so that the sculpture hangs in equilibrium with the rods horizontal.



8.4 Centres of mass

So far you have considered only **uniform** rods, where the centre of mass is always at the midpoint. If a rod is **non-uniform** the centre of mass is not necessarily at the midpoint of the rod.

You might need to consider the moment due to the weight of a non-uniform rod, or find the position of its centre of mass.

Example

6

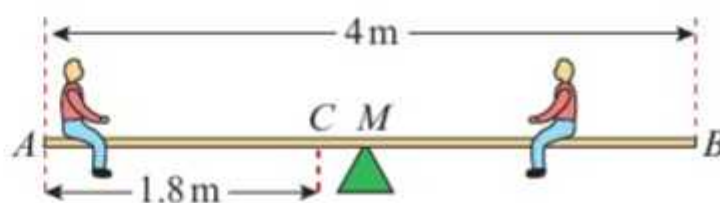
SKILLS

PROBLEM-SOLVING

Sam and Tamsin are sitting on a non-uniform plank AB of mass 25 kg and length 4 m. The plank is pivoted at M , the midpoint of AB . The centre of mass of AB is at C where AC is 1.8 m. Sam has mass 35 kg.

Tamsin has mass 25 kg and sits at A .

Where must Sam sit for the plank to be horizontal?



Taking moments about M :

$$25g \times 2 + 25g \times 0.2 = 35g \times (x - 2)$$

$$50 + 5 = 35x - 70$$

$$35x = 125$$

$$x = 3.57$$

Sam should sit 3.57 m from end A (or 0.43 m from end B).

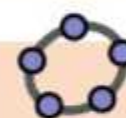
Model the plank as a rod and the children as particles. Then draw a diagram to represent the situation.

Suppose that Sam sits at a point x m from A .

Take moments about M to eliminate the reaction at M from your calculations.

Divide both sides of the equation by g .

Online Explore the moment acting about pivot M using GeoGebra.



Example 7

SKILLS PROBLEM-SOLVING

A non-uniform rod AB is 3 m long and has weight 20 N. It is in a horizontal position resting on supports at points C and D , where $AC = 1$ m and $AD = 2.5$ m. The magnitude of the reaction at C is three times the magnitude of the reaction at D . Find the distance of the centre of mass of the rod from A .

Suppose that the centre of mass acts at a point x m from A .

Resolving vertically, $3R + R = 20$

$$R = 5$$

Taking moments about A :

$$20 \times x = 15 \times 1 + 5 \times 2.5$$

$$20x = 27.5$$

$$x = 1.38 \text{ (3 s.f.)}$$

The centre of mass is 1.38 m from A , to 3 s.f.

Draw a diagram. Make sure that you have used all the information given in the question.

Whichever point you choose to take moments about, you are going to need to know the magnitude of R .

Use your value of R .

Exercise 8D

SKILLS PROBLEM-SOLVING

- 1 A non-uniform rod AB , of length 4 m and weight 6 N, rests horizontally on two supports, A and B . Given that the centre of mass of the rod is 2.4 m from A , find the reactions at the two supports.

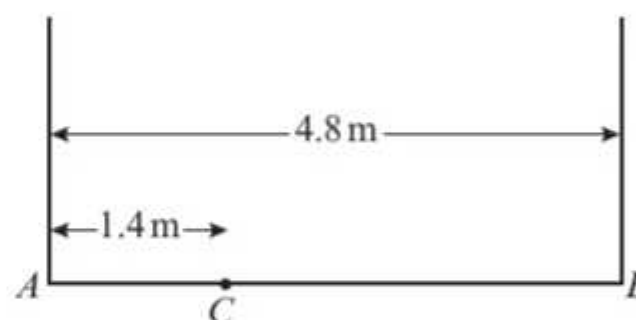
- 2 A non-uniform bar AB , of length 5 m, is supported horizontally on supports, A and B . The reactions at these supports are $3g$ N and $7g$ N respectively.
- State the weight of the bar.
 - Find the distance of the centre of mass of the bar from A .
- 3 A non-uniform plank AB , of length 4 m and weight 120 N, is pivoted at its midpoint. The plank is in equilibrium in a horizontal position with a child of weight 200 N sitting at A and a child of weight 300 N sitting at B . By modelling the plank as a rod and the two children as particles, find the distance of the centre of mass of the plank from A .

- (P)** 4 A non-uniform rod AB , of length 5 m and mass 15 kg, rests horizontally suspended from the ceiling by two vertical strings attached to C and D , where $AC = 1$ m and $AD = 3.5$ m.
- Given that the centre of mass is at E , where $AE = 3$ m, find the magnitudes of the tensions in the strings.

When a particle of mass 9 kg is attached to the rod at F , the tension in the string at D is twice the tension in the string at C .

- Find the distance AF .

- (E/P)** 5 A plank AB has mass 24 kg and length 4.8 m. A load of mass 15 kg is attached to the plank at the point C , where $AC = 1.4$ m. The loaded plank is held in equilibrium, with AB horizontal, by two vertical ropes, one attached at A and the other attached at B , as shown in the diagram. The plank is modelled as a uniform rod, the load as a particle, and the ropes as light inextensible strings.



- Find the tension in the rope attached at B .

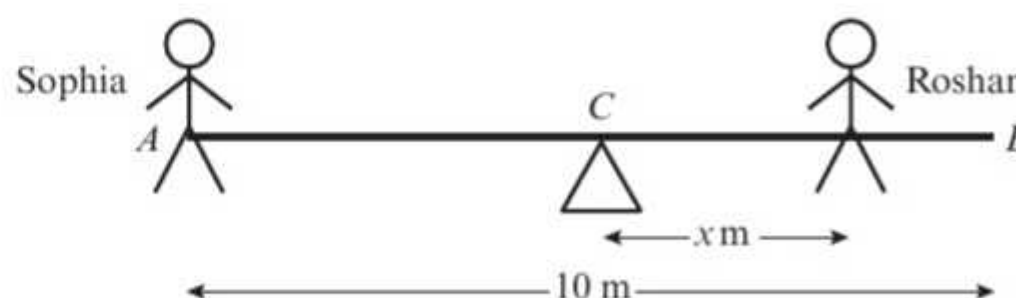
(4 marks)

The plank is now modelled as a non-uniform rod. With the new model, the tension in the rope attached at A is 25 N greater than the tension in the rope attached at B .

- Find the distance of the centre of mass of the plank from A .

(6 marks)

- (E)** 6 A seesaw in a playground consists of a beam AB of length 10 m which is supported by a smooth pivot at its centre C . Sophia has mass 30 kg and sits on the end A . Her brother Roshan has mass 50 kg and sits at a distance x metres from C , as shown in the diagram. The beam is initially modelled as a uniform rod.



- Using this model, find the value of x for which the seesaw can rest in equilibrium in a horizontal position.

(3 marks)

- State what is implied by the modelling assumption that the beam is uniform.

(1 mark)

Roshan finds he must sit at a distance 4 m from C for the seesaw to rest horizontally in equilibrium. The beam is now modelled as a non-uniform rod of mass 25 kg.

- Using this model, find the distance of the centre of mass of the beam from C .

(4 marks)

- E/P** 7 A non-uniform rod AB , of length 25 m and weight 80 N, rests horizontally in equilibrium on supports C and D as shown in the diagram. The centre of mass of the rod is 10 m from A .



A particle of weight W newtons is attached to the rod at a point E , where E is x metres from A . The rod remains in equilibrium and the magnitude of the reaction at C is five times the magnitude of the reaction at D .

Show that $W = \frac{400}{25 - 3x}$.

(5 marks)

8.5 Tilting

You need to be able to answer questions involving rods that are on the point of **tilting**.

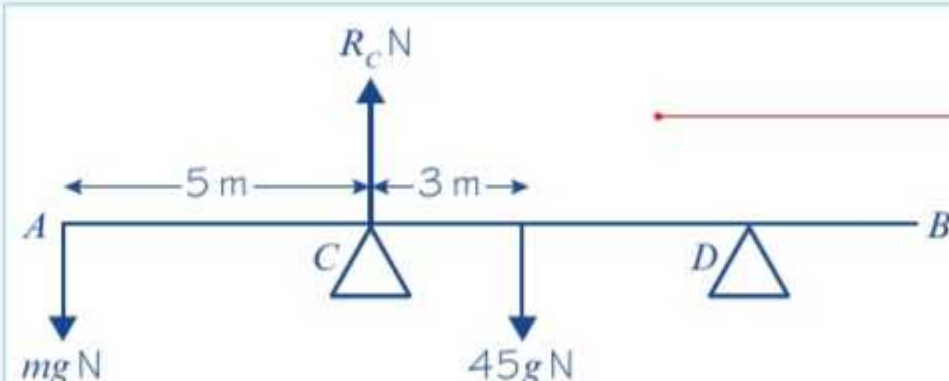
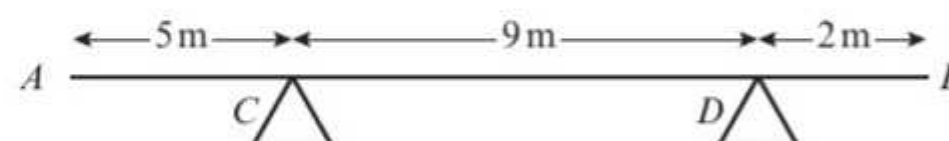
- When a rigid body is on the point of tilting about a pivot, the reaction at any other support (or the tension in any other wire or string) is zero.

Example 8

SKILLS

PROBLEM-SOLVING

A uniform beam AB , of mass 45 kg and length 16 m, rests horizontally on supports C and D where $AC = 5$ m and $CD = 9$ m. When a child stands at A , the beam is on the point of tilting about C . Find the mass of the child.



Taking moments about C :

$$mg \times 5 = 3 \times 45g$$

$$5mg = 135g$$

$$m = \frac{135g}{5g} = 27$$

The mass of the child is 27 kg.

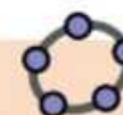
Draw a diagram showing the forces. Remember, as the beam is 'on the point of tilting', C is a pivot and the reaction force at D is zero.

Take moments about C . This means you can ignore the reaction R_C .

Anticlockwise moment = Clockwise moment

Online

See the point at which the beam starts to tilt due to the weight of the child and explore the problem with different forces and distances using GeoGebra.



Example

9

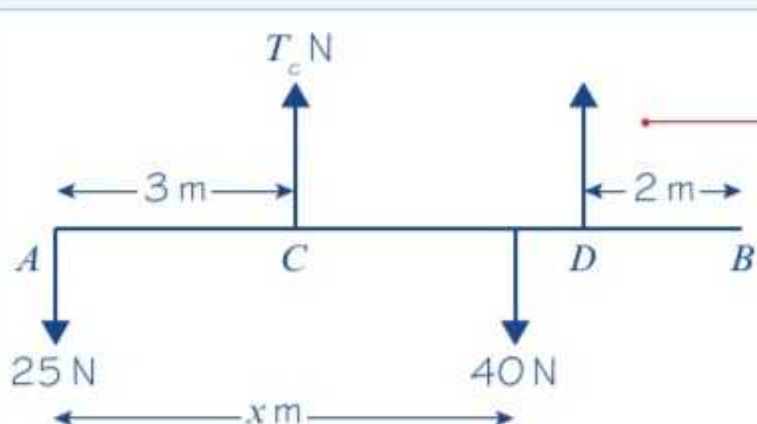
SKILLS

PROBLEM-SOLVING

A non-uniform rod AB , of length 10 m and weight 40 N, is suspended from a pair of light cables attached to C and D where $AC = 3$ m and $BD = 2$ m.

When a weight of 25 N is hung from A the rod is on the point of rotating.

Find the distance of the centre of mass of the rod from A .



Taking moments about C :

$$25 \times 3 = 40 \times (x - 3)$$

$$75 = 40x - 120$$

$$40x = 195$$

$$x = 4.875$$

Distance of the centre of mass from A is 4.875 m.

Draw a diagram showing the forces about C . As the rod is 'on the point of rotating', C is a pivot and the tension force at D is zero. As the rod is non-uniform you don't know the distance from A to the centre of mass, so you can label it as x .

You don't require the tension force, T_C , so take moments about C .

Problem-solving

The distance from C to the centre of mass is $(x - 3)$ m. Equate the clockwise and anticlockwise moments then solve the equation to find the value of x .

Exercise

8E

SKILLS

PROBLEM-SOLVING

- 1 A uniform rod AB has length 4 m and mass 8 kg. It is resting in a horizontal position on supports at points C and D where $AC = 1$ m and $AD = 2.5$ m. A particle of mass m kg is placed at point E where $AE = 3.3$ m. Given that the rod is about to tilt about D , calculate the value of m .
- 2 A uniform bar AB , of length 6 m and weight 40 N, is resting in a horizontal position on supports at points C and D where $AC = 2$ m and $AD = 5$ m. When a particle of weight 30 N is attached to the bar at point E , the bar is on the point of tilting about C . Calculate the distance AE .
- 3 A plank AB , of mass 12 kg and length 3 m, is in equilibrium in a horizontal position resting on supports at C and D where $AC = 0.7$ m and $DB = 1.1$ m. A boy of mass 32 kg stands on the plank at point E . The plank is about to tilt about D . By modelling the plank as a uniform rod and the boy as a particle, calculate the distance AE .
- 4 A uniform rod AB has length 5 m and weight 20 N. The rod is resting on supports at points C and D where $AC = 2$ m and $BD = 1$ m.
 - a Find the magnitudes of the reactions at C and D .

A particle of weight 12 N is placed on the rod at point A .

 - b Show that this causes the rod to tilt about C .

A second particle of weight 100 N is placed on the rod at E to hold it in equilibrium.

 - c Find the minimum and maximum possible distances of E from A .